

Project Data Page:





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DESIGN CALCULATION

In Accordance with ASME Section VIII Division 1

ASME Code Version : 2025

Analysis Performed by : SL User

Job File : ...\\170-C-AK-Final.pvdb

Date of Analysis : Sep 9, 2026 9:41am

PV Elite 28, January 2026



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FileName : 170-C-AK-Final

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Vessel Design Summary:

Step: 34

9:41am

Sep 9, 2026

Vessel Design Summary:

ASME Code, Section VIII Division 1, 2025

Diameter Spec : 770.000 mm ID

Vessel Design Length, Tangent to Tangent 12504.35 mm

Specified Datum Line Distance 60.00 mm

Shell Side Design Temperature 65 °C

Channel Side Design Temperature 65 °C

Shell Side Design Pressure 2250.000 KPa.

Channel Side Design Pressure 5850.000 KPa.

Shop Shell Side Test Pressure 6376.309 KPa.

Field Shell Side Test Pressure 2925.000 KPa.

Shop Channel Side Test Pressure 9032.466 KPa.

Field Channel Side Test Pressure 7605.000 KPa.

Wind Design Code ASCE/SEI 7-16

Earthquake Design Code ASCE/SEI 7-16

Materials of Construction:

Component Type	Material	Class	Thickness	UNS #	Material Treatment Options
Shell	SA-516 70	...	...	K02700	
Head	SA-516 70	...	...	K02700	
Flange	SA-350 LF2	2	...	K03011	+IT
Nozzle	SA-350 LF2	2	...	K03011	+IT
Nozzle Flg	SA-105	...	...	K03504	
Tubes	SA-334 6	...	...	K03006	+IT
Tubesheet	SA-350 LF2	2	...	K03011	+IT
Flg Bolting	SA-193 B7	...	<= 64	G41400	
Hrz Bolting	SA-193 B7	...	<= 64	G41400	
Saddles	SA-516 70	...	...	K02700	

+FG - Fine Grained, +R - rolled, +T - Tempered, +N - Normalized

+Q - quenched, +CBV - Cb & V added, +IT - Impact tested

Element Pressures and MAWP (KPa. & mm):

Element Description or Type	Design Pressure + Stat. head	Ext. Press.	Element M.A.W.P	Total Corrosion Allowance	Str. Flg. Gov.	In Creep Range
LHS Bonnet	5857.577	103.42	5895.600	3.0000	No	No
LHS ch shell	5857.577	103.42	6556.800	3.0000	N/A	No
LHS ch.flg	5857.547	103.42	6556.800	3.0000	N/A	No
main shell-1	2257.577	103.42	3839.100	3.0000	N/A	No
main shell-2	2257.577	103.42	3839.100	3.0000	N/A	No
main shell-3	2257.577	103.42	3839.100	3.0000	N/A	No
main shell-4	2257.577	103.42	3839.100	3.0000	N/A	No
RHS ch Flg	5857.646	103.42	6556.700	3.0000	N/A	No
RHS Ch.Shell	5857.577	103.42	6556.800	3.0000	N/A	No
RHS Bonnet	5857.577	103.42	5895.600	3.0000	No	No

Liquid Level: 780.00 mm Dens.: 0.001 kg./cm³ Sp. Gr.: 1.000

Element Types and Properties:



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Vessel Design Summary:

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Element Type	"To" Elev mm	Element Length mm	Nominal Thickness mm	Finished Thickness mm	Reqd Thk Internal mm	Reqd Thk External mm	Long Eff	Circ Eff
Ellipse	0.0	60.0	24.0	19.5	19.4	5.1	1.00	1.00
Cylinder	1000.0	1000.0	22.0	22.0	19.9	6.2	1.00	1.00
Body Flg	1192.2	189.0	153.0	145.0	140.9	140.9	1.00	1.00
Cylinder	3772.2	2420.0	14.0	14.0	9.4	9.9	1.00	1.00
Cylinder	6192.2	2420.0	14.0	14.0	9.4	9.9	1.00	1.00
Cylinder	8612.2	2420.0	14.0	14.0	9.4	9.9	1.00	1.00
Cylinder	11032.2	2420.0	14.0	14.0	9.4	9.9	1.00	1.00
Body Flg	11384.3	189.0	153.0	145.0	140.9	140.9	1.00	1.00
Cylinder	12384.3	1000.0	22.0	22.0	19.9	6.2	1.00	1.00
Ellipse	12444.3	60.0	24.0	19.5	19.4	5.1	1.00	1.00

Local Stress Analysis Results:

Description	Analysis Type	Max Stress Ratio	High Stress Location	Pass Fail
N1-Tube Inlet	WRC-297	0.884	Shell	Passed
N2-Tube outlet	WRC-297	0.884	Shell	Passed
N3-Shell inlet	WRC-297	0.976	Shell	Passed
N4-Shell outle	WRC-297	0.976	Shell	Passed

Foundation/Support Summary

Saddle Parameters:

Saddle Width	281.000 mm
Saddle Bearing Angle	120.000 deg.
Centerline Dimension	915.000 mm
Wear Pad Width	425.000 mm
Wear Pad Thickness	10.000 mm
Wear Pad Bearing Angle	132.000 deg.
Distance from Saddle to Tangent	1940.000 mm
Baseplate Length	780.000 mm
Baseplate Thickness	16.000 mm
Baseplate Width	310.000 mm
Number of Ribs (including outside ribs)	3
Rib Thickness	10.000 mm
Web Thickness	10.000 mm
Height of Center Web	525.000 mm
Number of Bolts in Baseplate	4

Baseplate Sketch



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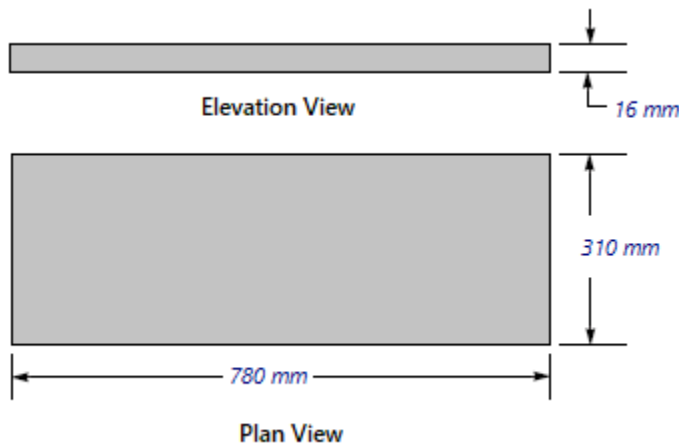
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Vessel Design Summary:

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Sep 9, 2026

**Summary of Maximum Saddle Loads, Operating Case, Un-Factored:**

Maximum Vertical Saddle Load	12976.22	Kgf
Maximum Transverse Saddle Shear Load	951.59	Kgf
Maximum Longitudinal Saddle Shear Load	1173.26	Kgf

Summary of Maximum Saddle Loads, Pressure Test Case:

Maximum Vertical Saddle Load	10473.20	Kgf
Maximum Transverse Saddle Shear Load	172.19	Kgf
Maximum Longitudinal Saddle Shear Load	42.75	Kgf

Weights:

Fabricated - Bare w/o removable internals	13179.8	kg.
Shop Test - Fabricated + water (full)	19087.9	kg.
Shipping - Fab. + rem. intls.+ shipping app.	13179.8	kg.
Erected - Fab. + rem. intls.+ insul. (etc)	13679.8	kg.
Empty - Fab. + intls. + details + wghts.	14848.3	kg.
Operating - Empty + operating liquid (no ca)	20256.4	kg.
Field Test - Empty weight + test medium (full)	19549.6	kg.

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Nozzle Summary:

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Nozzle Calculation Summary (KPa. & mm):

Description	MAWP	Ext	MAPNC	UG-45	[tr]	Weld Path	Areas or Stresses
N1-Tube Inlet	5951	OK	...	OK	11.10	OK	Passed
N2-Tube outlet	5943.4	OK	...	OK	11.10	OK	Passed
N3-Shell inlet	3839.1	OK	...	OK	9.39	OK	Passed
N4-Shell outlet	3839.1	OK	...	OK	9.41	OK	Passed

Nozzle MAWP Summary:

Minimum MAWP Nozzles : 3839.1 Nozzle : N3-Shell inlet [Shellside]
 Minimum MAWP Nozzles : 5943.4 Nozzle : N2-Tube outlet [Tubeside]

Note: MAWPs (Internal Case) shown above are at the High Point.

Multiple output lines for the same nozzle indicates required Code calculations in both the longitudinal and circumferential planes of reinforcement where applicable.

Check the spatial relationship between the nozzles:

From Node	Nozzle Description	X Coordinate mm	Layout Angle deg	Dia. Limit mm
20	N1-Tube Inlet	560.000	0.000	498.000
20	N2-Tube outlet	560.000	180.000	498.000
40	N3-Shell inlet	2612.175	0.000	498.000
70	N4-Shell outlet	9872.176	180.000	498.000

The nozzle spacing is computed by the following:

$$= \sqrt{ll^2 + lc^2} \text{ where}$$

ll - Arc length along the inside vessel surface in the long. direction.

lc - Arc length along the inside vessel surface in the circ. direction

If any interferences/violations are found, they will be noted below.

No interference violations have been detected!

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Nozzle Schedule:

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Nozzle Schedule:

Description	Nominal or Actual Size	Schd or FVC Type	Flg Type	Nozzle O/Dia mm	Wall Thk mm	Reinforcing Diameter	Pad Thk mm	Cut Length mm	Flg Class
N1-Tube Inlet	243 mm	Actual	WNF	273.0	15.000	...	...	312	600
N2-Tube outlet	243 mm	Actual	WNF	273.0	15.000	...	...	312	600
N3-Shell inlet	243 mm	Actual	WNF	273.0	15.000	...	...	384	300
N4-Shell outlet	243 mm	Actual	WNF	273.0	15.000	...	...	384	300

General Notes for the above table:

The cut length is the outside projection + inside projection + drop + in-plane shell thickness. This value does not include weld gaps, nor does it account for shrinkage.

In the case of oblique nozzles, the outside diameter must be increased. The re-pad width around the nozzle is calculated as follows:

Width of Pad = (Pad Outside Dia. (per above) - Nozzle Outside Dia.)/2

For hub nozzles, the thickness and diameter shown are those of the smaller and thinner section.

Nozzle Material and Weld Fillet Leg Size Details (mm):

Description	Material	Shl Grve Weld	Noz Shl/Pad Weld	Pad OD Weld	Pad Grve Weld	Inside Weld
N1-Tube Inl	SA-350 LF2	20.000	20.000	...	...	...
N2-Tube out	SA-350 LF2	20.000	20.000	...	...	...
N3-Shell in	SA-350 LF2	14.000	20.000	...	...	...
N4-Shell ou	SA-350 LF2	14.000	20.000	...	...	...

Note: The Outside projections below do not include the flange thickness.

Nozzle Miscellaneous Data:

Description	Elev/Distance From Datum mm	Layout Angle deg	Proj Outside mm	Proj Inside mm	Installed in Component
N1-Tube Inlet	500.000	0.0	250.00	0.00	LHS ch shell
N2-Tube outlet	500.000	180.0	250.00	0.00	LHS ch shell
N3-Shell inlet	2552.175	0.0	300.00	0.00	main shell-1
N4-Shell outlet	9812.175	180.0	300.00	0.00	main shell-4

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Bill of Material:

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Bill of Materials:

QTY	DESCRIPTION	MATERIAL
2	ELLIPTICAL HEAD: 2.0 X 1, 24.0mm THK X 770.0mm ID X 60.0mm	SA-516 70
2	CYLINDER: 22.0mm THK X 770.0mm ID X 1000.0mm	SA-516 70
2	WELD NECK FLANGE: 153.0mm THK X 1040.0mm OD	SA-350 LF2
4	CYLINDER: 14.0mm THK X 770.0mm ID X 2420.0mm	SA-516 70
2	INSULATION: 59mm X 70mm THK	Cold Insulation
2	INSULATION: 1000mm X 70mm THK	Cold Insulation
2	INSULATION: 155mm X 70mm THK	Cold Insulation
45	BAFFLES: 775.2mm DIA X 8.0mm THK	SA-516 70
6	TIE RODS: 12mm DIA X 9560mm	SA-193 B7M
2	TUBESHEET: 1170.0mm X 160.0mm THK	SA-350 LF2
282	TUBES: 10000.0mm X 25.4mm DIA X 2.1mm THK	SA-334 6
2	SADDLE: 281.0mm X 120 DEG	SA-516 70
3	INSULATION: 2420mm X 100mm THK	Insulation
1	INSULATION: 2420mm X 100mm THK	Hot Insulation
2	GASKET: 828.0mm OD X 802.0mm ID	...
72	BODY FLANGE BOLTS: 38.1mm DIA	SA-193 B7
72	NUTS FOR BODY FLANGE BOLTS: 38.1mm DIA	...
1	NAMEPLATE	...

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MDMT Summary:

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Minimum Design Metal Temperature Results Summary (°C):

Description	Curve	Basic MDMT	Reduced MDMT	UG-20(f) MDMT	Thickness ratio	Gov Thk mm	E*	PWHT reqd
Notes								
main shell-1 [8]	B	-18	-18	-29	0.998	14.000	1.00	Yes
main shell-2 [8]	B	-18	-18	-29	0.998	14.000	1.00	Yes
main shell-3 [8]	B	-18	-18	-29	0.998	14.000	1.00	Yes
main shell-4 [8]	B	-18	-18	-29	0.998	14.000	1.00	Yes
Left Saddle f[24]	B	-46	-46		0.998	10.000	1.00	Yes
Right Saddle [24]	B	-46	-46		0.998	10.000	1.00	Yes
N3-Shell inlet[1]	B	-18	-42	-29	0.581	14.000	1.00	Yes
Nozzle Flg [5]	A	-18	-48					
N4-Shell outlet[1]	B	-18	-42	-29	0.583	14.000	1.00	Yes
Nozzle Flg [5]	A	-18	-48					
Tubesheet: SS[13]	B	-5	-19		0.754	40.000	1.00	Yes
Warmest MDMT:		-5	-18					
LHS ch.flg [11]	!	-18	-18	-29	0.773	22.000	1.00	Yes
RHS ch Flg [11]	!	-18	-18	-29	0.773	22.000	1.00	Yes
LHS Bonnet [10]	B	-8	-8	-29	0.999	19.500	1.00	Yes
LHS Bonnet [7]	B	-2	-13	-29	0.810	24.000	1.00	Yes
LHS ch shell [8]	B	-5	-10	-29	0.895	22.000	1.00	Yes
RHS Ch.Shell [8]	B	-5	-10	-29	0.895	22.000	1.00	Yes
RHS Bonnet [10]	B	-8	-8	-29	0.999	19.500	1.00	Yes
RHS Bonnet [7]	B	-2	-13	-29	0.810	24.000	1.00	Yes
N1-Tube Inlet [1]	B	-5	-29	-29	0.888	22.000	1.00	Yes
Nozzle Flg [5]	A	-18	-42					
N2-Tube outlet[1]	B	-5	-29	-29	0.890	22.000	1.00	Yes
Nozzle Flg [5]	A	-18	-42					
Tubesheet: CS[14]	!	-18	-18	-29	0.754	40.000	1.00	Yes
Bolting [21]		-48						
Bolting [21]		-48						
Warmest MDMT:		-2	-8					

Exchanger Side	Computed MDMT	Required MDMT	Pass/Fail
Shell	-19.0	-33.0	Fail
Channel/Tube	-29.0	-33.0	Fail

Notes:

- [!] - This was an impact tested material.
- [1] - Governing Nozzle Weld.
- [4] - ASME Flange MDMT Calcs; Thickness ratio per UCS-66(b)(1)(-c).
- [5] - ASME Flange MDMT Calcs; Thickness ratio per UCS-66(b)(1)(-b).
- [6] - MDMT Calculations at the Shell/Head Joint.
- [7] - MDMT Calculations for the Straight Flange.
- [8] - Cylinder/Cone/Flange Junction MDMT.
- [9] - Calculations in the Spherical Portion of the Head.
- [10] - Calculations in the Knuckle Portion of the Head.
- [11] - Calculated (Body Flange) Flange MDMT.
- [12] - Calculated Flat Head MDMT per UCS-66.3
- [13] - Tubesheet MDMT, shell side, if applicable
- [14] - Tubesheet MDMT, tube side, if applicable
- [15] - Nozzle Material
- [16] - Shell or Head Material
- [17] - Impact Testing required
- [18] - Impact Testing not required, see UCS-66(b)(3)
- [20] - Cylinder/Cone Junction MDMT based on Longitudinal Stress considerations
- [21] - Body Flange Bolting Material



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MDMT Summary:

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[22] - Nozzle Flange Bolting Material
[23] - Stiffening Ring to Shell Weld
[24] - Saddle to Shell Weld

UG-84(b)(2) was not considered.

UCS-66(g) was not considered.

UCS-66(i) was not considered.

Notes:

Impact test temps were not entered in and not considered in the analysis.

UCS-66(i) applies to impact tested materials not by specification and

UCS-66(g) applies to materials impact tested per UG-84.1 General Note (c).

The Basic MDMT includes the (30F) PWHT credit if applicable.

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Warnings and Errors:

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Class From To : Basic Element Checks.

Class From To: Check of Additional Element Data

There were no geometry errors or warnings.

PV Elite performs all calculations internally in Imperial Units to remain compliant with the ASME Code and any built in assumptions in the ASME Code formulas. The finalized results are reflected to show the set of selected units for this analysis.

Information Regarding Nozzle Loads

Please note that nozzle loadings, if included, are assumed to be local in nature and will not contribute to or create a net section bending moment. Therefore, the addition of nozzle loads will not affect the support load calculation. If you wish to create a load on the support(s) from nozzle loads, you can enable that feature in tools->configuration->Job Specific Settings.



One or more nozzles in this model was specified using the actual thickness basis and a standard flange is also specified. In many cases a matching nominal diameter cannot be determined and the weight of the flange and optional blind flange cannot be determined. When actual thickness basis is specified, you must calculate the weight of the assembly (by hand) and enter it into the 'Overriding Weight' field in the nozzle dialog when there is a standard flange specified. However, for FVC nozzles that have their nominal diameter specified in the FVC dialog, entering the overriding weight for that nozzle is not required.



The ASME Code Section VIII Division 1 does not explicitly require stresses on elements to be computed under test conditions. Please see interpretation VIII-1-10-12 for more information. PV Elite also computes weight due to test fluid and uses the weight in various calculations such as saddle and leg loads. PV Elite also computes static pressure due to operating and test liquids. The static pressure is used in conjunction with the design and test pressures to determine a total pressure in the determination of required thickness, MAWP etc. This ensures compliance with UG-22 requirements.

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Input Echo:

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Units used in this Analysis (MM):

Name	System Unit	Constant	User Unit
Length	Feet	304.8000	mm
Force	Pounds	0.4536	Kgf
Mass	Pounds	0.4536	kg.
Area	sq. inches	6.4516	cm ²
Moment	ft. lbs.	0.1383	Kg-m.
Stress	lbs./sq.in.	0.0069	N./mm ²
Temperature	Degrees F	0.5556	°C
Pressure	psig	6.8946	KPa.
Elast. Modulus	lbs./sq.in.	6.8946	KPa.
Pipe Density	lbs./cu.in.	0.0277	kg./cm ³
Ins. Density	lbs./cu.ft.	0.1600E-04	kg./cm ³
Fluid Density	lbs./cu.ft.	0.1600E-04	kg./cm ³
Wind Speed	miles/hr	1.6093	Km/hr
Tray Weight	lbs./sq.ft.	0.0005	Kgs/cm ²
Inertia	in.4	41.6231	cm4
G Load	G's	1.0000	g's
Wind Load	lbs./sq.ft.	4.8825	Kgs/m ²
Elevation	Feet	304.8000	mm
Volume	in.3	16.3900	cm ³
Diameter	inches	25.4000	mm
Thickness	inches	25.4000	mm

PV Elite Vessel Analysis Program: Input Data

Exchanger Design Pressures and Temperatures

Shell Side Design Pressure	2250	KPa.
Channel Side Design Pressure	5850	KPa.
Shell Side Design Temperature	65.0	°C
Channel Side Design Temperature	65.0	°C
Radiography, Shell Side	RT-1	
Radiography, Channel Side	RT-1	
Service Type, Shell Side	Lethal	
Service Type, Channel Side	Lethal	
MDMT (CET), Shell Side	-33.0	°C
MDMT (CET), Tube Side	-33.0	°C
User defined MAWP, Shell Side	0	KPa.
User defined MAWP, Channel Side	0	KPa.
User defined MAPnc, Shell Side	0	KPa.
User defined MAPnc, Channel Side	0	KPa.
User defined Test Pres., Shell Side	0	KPa.
User defined Test Pres., Channel Side	0	KPa.

Projection of Nozzle from Vessel Top	0	mm
Projection of Nozzle from Vessel Bottom	0	mm
Type of Construction	Welded	
Use Higher Longitudinal Stresses (Flag)	Y	
Select t for Internal Pressure (Flag)	N	
Select t for External Pressure (Flag)	N	
Select t for Axial Stress (Flag)	N	
Select Location for Stiff. Rings (Flag)	N	
Consider Vortex Shedding	N	

Shop Pressure Test:

Type of Pressure Test	UG-99(c)
Pressure Test Position	Horizontal
Test Performed in Corroded Condition	Yes



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Input Echo:

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Field Pressure Test:

Type of Pressure Test UG-99(b) Note [35]
 Pressure Test Position Horizontal
 Test Performed in Corroded Condition Yes

Load Case 1	NP+EW+WI+FW+BW
Load Case 2	NP+EW+EE+FS+BS
Load Case 3	NP+OW+WI+FW+BW
Load Case 4	NP+OW+EQ+FS+BS
Load Case 5	NP+HW+HI
Load Case 6	NP+HW+HE
Load Case 7	IP+OW+WI+FW+BW
Load Case 8	IP+OW+EQ+FS+BS
Load Case 9	EP+OW+WI+FW+BW
Load Case 10	EP+OW+EQ+FS+BS
Load Case 11	HP+HW+HI
Load Case 12	HP+HW+HE
Load Case 13	IP+WE+EW
Load Case 14	IP+WF+CW
Load Case 15	IP+VO+OW
Load Case 16	IP+VE+EW
Load Case 17	NP+VO+OW
Load Case 18	FS+BS+IP+OW
Load Case 19	FS+BS+EP+OW
Load Case 20	BL+IP+OW

Wind Design Code	ASCE/SEI 7-16
Wind Load Reduction Scale Factor	1.100
Basic Wind Speed [V]	97.2 Km/hr
Surface Roughness Category	C: Open Terrain
Type of Surface	Moderately Smooth
Base Elevation	300 mm
Percent Wind for Hydrotest	50.0
Using User defined Wind Press. Vs Elev.	N
Height of Hill or Escarpment H or Hh	0 mm
Distance Upwind of Crest Lh	0 mm
Distance from Crest to the Vessel x	0 mm
Type of Terrain (Hill, Escarpment)	Flat
Damping Factor (Beta) for Wind (Ope)	0.0100
Damping Factor (Beta) for Wind (Empty)	0.0000
Damping Factor (Beta) for Wind (Filled)	0.0000
Site Elevation above Sea Level	0 mm

Seismic Design Code	ASCE/SEI 7-16
Seismic Load Reduction Scale Factor	1.000
Importance Factor	1.250
Table Value Fa	1.600
Table Value Fv	2.400
Max. Mapped Res. Acceleration [Ss]	0.147
Max. Eff. Ground Acceleration [S]	0.065
Force Modification Factor R	3.000
Site Class	D
Component Elevation Ratio z/h	0.000
Amplification Factor Ap	1.000
Force Factor	0.000
Consider Vertical Acceleration	Yes
Minimum Acceleration Multiplier	0.000
User Value of Sds (used if > 0)	0.000
User Value of Sd1 (used if > 0)	0.000
Moment Reduction Factor Tau	1.000

Design Pressure + Static Head	Y
Consider MAP New and Cold in Noz. Design	N
Consider External Loads for Nozzle Des.	Y



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Input Echo:

Step: 1 9:41am Sep 9, 2026

Use ASME VIII-1 Appendix 1-9 N

Perform Blast Load Analysis No

Material Database Year Current w/Addenda or Code Year

Configuration Directives:

g0 is taken as min(g0, attached shell thk) Yes

Do not use Nozzle MDMT Interpretation VIII-1 01-37 No

Use Table G instead of exact equation for "A" Yes

Shell Head Joints are Tapered Yes

Compute "K" in corroded condition Yes

Use Code Case 2286 No

Use the MAWP to compute the MDMT Yes

For thickness ratios <= 0.35, MDMT will be -155F (-104C) Yes

For PWHT & P1 Materials the MDMT can be < -55F (-48C) No

Using Metric Material Databases, ASME II D Yes

Calculate B31.3 type stress for Nozzles with Loads Yes

Reduce the MDMT due to lower membrane stress Yes

Consider Longitudinal Stress in MDMT Calculations Yes

Complete Listing of Vessel Elements and Details:

Element#	1/10	
Element From Node	10	
Element To Node	20	
Element Type	Elliptical	
Description	LHS Bonnet	
Distance "FROM" to "TO"	60	mm
Inside Diameter	770	mm
Element Thickness	19.5	mm
Internal Corrosion Allowance	3	mm
Nominal Thickness	24	mm
External Corrosion Allowance	0	mm
Design Internal Pressure	5850	KPa.
Design Temperature Internal Pressure	65	°C
Design External Pressure	103.42	KPa.
Design Temperature External Pressure	65	°C
Effective Diameter Multiplier	1.2	
Material Name	SA-516 70	
Allowable Stress, Ambient	138	N./mm ²
Allowable Stress, Operating	138	N./mm ²
Allowable Stress, Hydrotest	234	N./mm ²
Material Density	0.00775	kg./cm ³
P Number Thickness	32.004	mm
Yield Stress, Operating	246	N./mm ²
UCS-66 Chart Curve Designation	B	
External Pressure Chart Name	CS-2	
UNS Number	K02700	
Product Form	Plate	
Efficiency, Longitudinal Seam	1.0	
Efficiency, Circumferential Seam	1.0	
Elliptical Head Factor	2.0	
Weld is pre-Heated	No	
Element From Node	10	
Detail Type	Liquid	
Detail ID	Liquid: 10	
Dist. from "FROM" Node / Offset dist	0	mm
Height/Length of Liquid	770	mm
Liquid Density	0.0009984	kg./cm ³
Element From Node	10	
Detail Type	Insulation	



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Input Echo:

Step: 1 9:41am Sep 9,2026

Detail ID	Insulation
Dist. from "FROM" Node / Offset dist	0 mm
Height/Length of Insulation	60 mm
Thickness of Insulation	70 mm
Density	0.00032 kg./cm ³

Element#	2/10
Element From Node	20
Element To Node	30
Element Type	Cylinder
Description	LHS ch shell
Distance "FROM" to "TO"	1000 mm
Inside Diameter	770 mm
Element Thickness	22 mm
Internal Corrosion Allowance	3 mm
Nominal Thickness	22 mm
External Corrosion Allowance	0 mm
Design Internal Pressure	5850 KPa.
Design Temperature Internal Pressure	65 °C
Design External Pressure	103.42 KPa.
Design Temperature External Pressure	65 °C
Effective Diameter Multiplier	1.2
Material Name	SA-516 70
Efficiency, Longitudinal Seam	1.0
Efficiency, Circumferential Seam	1.0
Weld is pre-Heated	No

Element From Node	20
Detail Type	Liquid
Detail ID	Liquid: 20
Dist. from "FROM" Node / Offset dist	0 mm
Height/Length of Liquid	770 mm
Liquid Density	0.0009984 kg./cm ³

Element From Node	20
Detail Type	Insulation
Detail ID	Insulation
Dist. from "FROM" Node / Offset dist	0 mm
Height/Length of Insulation	1000 mm
Thickness of Insulation	70 mm
Density	0.00032 kg./cm ³

Element From Node	20
Detail Type	Nozzle
Detail ID	N1-Tube Inlet
Dist. from "FROM" Node / Offset dist	500 mm
Nozzle Diameter	243 mm
Nozzle Schedule	80
Nozzle Class	600
Layout Angle	0.0
Blind Flange (Y/N)	N
Weight of Nozzle (Used if > 0)	134.48 Kg
Grade of Attached Flange	GR 1.1
Nozzle Matl	SA-350 LF2 +IT

Element From Node	20
Detail Type	Nozzle
Detail ID	N2-Tube outlet
Dist. from "FROM" Node / Offset dist	500 mm
Nozzle Diameter	243 mm
Nozzle Schedule	80
Nozzle Class	600
Layout Angle	180.0



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Input Echo:

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Blind Flange (Y/N)	N
Weight of Nozzle (Used if > 0)	134.48 Kg
Grade of Attached Flange	GR 1.1
Nozzle Matl	SA-350 LF2 +IT

Element#	3/10
Element From Node	30
Element To Node	40
Element Type	Flange
Description	LHS ch.flg
Distance "FROM" to "TO"	189 mm
Flange Inside Diameter	770 mm
Element Thickness	145 mm
Internal Corrosion Allowance	3 mm
Nominal Thickness	153 mm
External Corrosion Allowance	0 mm
Design Internal Pressure	5850 KPa.
Design Temperature Internal Pressure	65 °C
Design External Pressure	103.42 KPa.
Design Temperature External Pressure	65 °C
Effective Diameter Multiplier	1.2
Material Name	SA-350 LF2 +IT
Allowable Stress, Ambient	138 N./mm ²
Allowable Stress, Operating	138 N./mm ²
Allowable Stress, Hydrotest	225 N./mm ²
Material Density	0.00775 kg./cm ³
P Number Thickness	32.004 mm
Yield Stress, Operating	233 N./mm ²
UCS-66 Chart Curve Designation	Impact Tested
External Pressure Chart Name	CS-2
UNS Number	K03011
Class / Thickness / Grade	2 ::
Product Form	Forgings
Perform Flange Stress Calculation (Y/N)	Y
Weld is pre-Heated	No

Element From Node	30
Detail Type	Liquid
Detail ID	Liquid: 30
Dist. from "FROM" Node / Offset dist	0 mm
Height/Length of Liquid	770 mm
Liquid Density	0.0009984 kg./cm ³

Element From Node	30
Detail Type	Insulation
Detail ID	Insulation
Dist. from "FROM" Node / Offset dist	0 mm
Height/Length of Insulation	155 mm
Thickness of Insulation	70 mm
Density	0.00032 kg./cm ³

Element#	4/10
Element From Node	40
Element To Node	50
Element Type	Cylinder
Description	main shell-1
Distance "FROM" to "TO"	2420 mm
Inside Diameter	770 mm
Element Thickness	14 mm
Internal Corrosion Allowance	3 mm
Nominal Thickness	14 mm



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Input Echo:

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External Corrosion Allowance	0	mm
Design Internal Pressure	2250	KPa.
Design Temperature Internal Pressure	65	°C
Design External Pressure	103.42	KPa.
Design Temperature External Pressure	65	°C
Effective Diameter Multiplier	1.2	
Material Name	SA-516 70	
Allowable Stress, Ambient	138	N./mm ²
Allowable Stress, Operating	138	N./mm ²
Allowable Stress, Hydrotest	234	N./mm ²
Material Density	0.00775	kg./cm ³
P Number Thickness	32.004	mm
Yield Stress, Operating	246	N./mm ²
UCS-66 Chart Curve Designation	B	
External Pressure Chart Name	CS-2	
UNS Number	K02700	
Product Form	Plate	
Efficiency, Longitudinal Seam	1.0	
Efficiency, Circumferential Seam	1.0	
Weld is pre-Heated	No	

Element From Node	40	
Detail Type	Saddle	
Detail ID	Left Saddle flo	
Dist. from "FROM" Node / Offset dist	1940	mm
Width of Saddle	281	mm
Height of Saddle at Bottom	915	mm
Saddle Contact Angle	120.0	
Height of Composite Ring Stiffener	0	mm
Width of Wear Plate	425	mm
Thickness of Wear Plate	10	mm
Contact Angle, Wear Plate (degrees)	132.0	
Friction coefficient	0.30000001	
Moment Factor	3.0	
Dimension E at base (optional)	0	mm
Circumferential Eff. over Saddle	1.0	
Circumferential Eff. at Midspan	1.0	
Tangent to Tangent dist. (optional)	0	mm

Element From Node	40	
Detail Type	Liquid	
Detail ID	Liquid: 40	
Dist. from "FROM" Node / Offset dist	0	mm
Height/Length of Liquid	770	mm
Liquid Density	0.0009984	kg./cm ³

Element From Node	40	
Detail Type	Insulation	
Detail ID	Insulation	
Dist. from "FROM" Node / Offset dist	0	mm
Height/Length of Insulation	2420	mm
Thickness of Insulation	100	mm
Density	0.000352	kg./cm ³

Element From Node	40	
Detail Type	Nozzle	
Detail ID	N3-Shell inlet	
Dist. from "FROM" Node / Offset dist	1200	mm
Nozzle Diameter	243	mm
Nozzle Schedule	80	
Nozzle Class	300	
Layout Angle	0.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	149.58	Kgf
Grade of Attached Flange	GR 1.1	



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Input Echo:

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Nozzle Matl SA-350 LF2 +IT

Element From Node	40
Detail Type	Weight
Detail ID	Fireproofing we
Dist. from "FROM" Node / Offset dist	1940 mm
Miscellaneous Weight	250 Kg
Offset from Element Centerline	914.4 mm

Element#	5/10
Element From Node	50
Element To Node	60
Element Type	Cylinder
Description	main shell-2
Distance "FROM" to "TO"	2420 mm
Inside Diameter	770 mm
Element Thickness	14 mm
Internal Corrosion Allowance	3 mm
Nominal Thickness	14 mm
External Corrosion Allowance	0 mm
Design Internal Pressure	2250 KPa.
Design Temperature Internal Pressure	65 °C
Design External Pressure	103.42 KPa.
Design Temperature External Pressure	65 °C
Effective Diameter Multiplier	1.2
Material Name	SA-516 70
Efficiency, Longitudinal Seam	1.0
Efficiency, Circumferential Seam	1.0
Weld is pre-Heated	No

Element From Node	50
Detail Type	Liquid
Detail ID	Liquid: 50
Dist. from "FROM" Node / Offset dist	0 mm
Height/Length of Liquid	770 mm
Liquid Density	0.0009984 kg./cm³

Element From Node	50
Detail Type	Insulation
Detail ID	Insulation
Dist. from "FROM" Node / Offset dist	0 mm
Height/Length of Insulation	2420 mm
Thickness of Insulation	100 mm
Density	0.000352 kg./cm³

Element#	6/10
Element From Node	60
Element To Node	70
Element Type	Cylinder
Description	main shell-3
Distance "FROM" to "TO"	2420 mm
Inside Diameter	770 mm
Element Thickness	14 mm
Internal Corrosion Allowance	3 mm
Nominal Thickness	14 mm
External Corrosion Allowance	0 mm
Design Internal Pressure	2250 KPa.
Design Temperature Internal Pressure	65 °C
Design External Pressure	103.42 KPa.
Design Temperature External Pressure	65 °C
Effective Diameter Multiplier	1.2



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Input Echo:

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Material Name	SA-516 70
Efficiency, Longitudinal Seam	1.0
Efficiency, Circumferential Seam	1.0
Weld is pre-Heated	No

Element From Node	60
Detail Type	Liquid
Detail ID	Liquid: 60
Dist. from "FROM" Node / Offset dist	0 mm
Height/Length of Liquid	770 mm
Liquid Density	0.0009984 kg./cm ³

Element From Node	60
Detail Type	Insulation
Detail ID	Insulation
Dist. from "FROM" Node / Offset dist	0 mm
Height/Length of Insulation	2420 mm
Thickness of Insulation	100 mm
Density	0.000352 kg./cm ³

Element#	7/10
Element From Node	70
Element To Node	80
Element Type	Cylinder
Description	main shell-4
Distance "FROM" to "TO"	2420 mm
Inside Diameter	770 mm
Element Thickness	14 mm
Internal Corrosion Allowance	3 mm
Nominal Thickness	14 mm
External Corrosion Allowance	0 mm
Design Internal Pressure	2250 KPa.
Design Temperature Internal Pressure	65 °C
Design External Pressure	103.42 KPa.
Design Temperature External Pressure	65 °C
Effective Diameter Multiplier	1.2
Material Name	SA-516 70
Efficiency, Longitudinal Seam	1.0
Efficiency, Circumferential Seam	1.0
Weld is pre-Heated	No

Element From Node	70
Detail Type	Saddle
Detail ID	Right Saddle fi
Dist. from "FROM" Node / Offset dist	480 mm
Width of Saddle	281 mm
Height of Saddle at Bottom	915 mm
Saddle Contact Angle	120.0
Height of Composite Ring Stiffener	0 mm
Width of Wear Plate	425 mm
Thickness of Wear Plate	10 mm
Contact Angle, Wear Plate (degrees)	132.0
Friction coefficient	0.0
Moment Factor	3.0
Dimension E at base (optional)	0 mm
Circumferential Eff. over Saddle	1.0
Circumferential Eff. at Midspan	1.0
Tangent to Tangent dist. (optional)	0 mm

Element From Node	70
Detail Type	Liquid
Detail ID	Liquid: 40
Dist. from "FROM" Node / Offset dist	0 mm



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Input Echo:

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Height/Length of Liquid	770	mm
Liquid Density	0.0009984	kg./cm ³

Element From Node	70	
Detail Type	Insulation	
Detail ID	Insulation	
Dist. from "FROM" Node / Offset dist	0	mm
Height/Length of Insulation	2420	mm
Thickness of Insulation	100	mm
Density	0.000352	kg./cm ³

Element From Node	70	
Detail Type	Nozzle	
Detail ID	N4-Shell outlet	
Dist. from "FROM" Node / Offset dist	1200	mm
Nozzle Diameter	243	mm
Nozzle Schedule	80	
Nozzle Class	300	
Layout Angle	180.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	141.39	Kgf
Grade of Attached Flange	GR 1.1	
Nozzle Matl	SA-350 LF2 +IT	

Element From Node	70	
Detail Type	Weight	
Detail ID	Wght:[2 of 2]	
Dist. from "FROM" Node / Offset dist	480	mm
Miscellaneous Weight	250	Kgf
Offset from Element Centerline	914.4	mm

Element#	8/10	
Element From Node	80	
Element To Node	90	
Element Type	Flange	
Description	RHS ch Flg	
Distance "FROM" to "TO"	189	mm
Flange Inside Diameter	770	mm
Element Thickness	145	mm
Internal Corrosion Allowance	3	mm
Nominal Thickness	153	mm
External Corrosion Allowance	0	mm
Design Internal Pressure	5850	KPa.
Design Temperature Internal Pressure	65	°C
Design External Pressure	103.42	KPa.
Design Temperature External Pressure	65	°C
Effective Diameter Multiplier	1.2	
Material Name	SA-350 LF2 +IT	
Allowable Stress, Ambient	138	N./mm ²
Allowable Stress, Operating	138	N./mm ²
Allowable Stress, Hydrotest	225	N./mm ²
Material Density	0.00775	kg./cm ³
P Number Thickness	32.004	mm
Yield Stress, Operating	233	N./mm ²
UCS-66 Chart Curve Designation	Impact Tested	
External Pressure Chart Name	CS-2	
UNS Number	K03011	
Class / Thickness / Grade	2 ::	
Product Form	Forgings	
Perform Flange Stress Calculation (Y/N)	Y	
Weld is pre-Heated	No	
Element From Node	80	



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Input Echo:

Step: 1 9:41am Sep 9,2026

Detail Type	Liquid
Detail ID	Liquid: 50
Dist. from "FROM" Node / Offset dist	0 mm
Height/Length of Liquid	780 mm
Liquid Density	0.0009984 kg./cm ³

Element From Node	80
Detail Type	Insulation
Detail ID	Insulation
Dist. from "FROM" Node / Offset dist	0 mm
Height/Length of Insulation	155 mm
Thickness of Insulation	70 mm
Density	0.00032 kg./cm ³

Element#	9/10
Element From Node	90
Element To Node	100
Element Type	Cylinder
Description	RHS Ch.Shell
Distance "FROM" to "TO"	1000 mm
Inside Diameter	770 mm
Element Thickness	22 mm
Internal Corrosion Allowance	3 mm
Nominal Thickness	22 mm
External Corrosion Allowance	0 mm
Design Internal Pressure	5850 KPa.
Design Temperature Internal Pressure	65 °C
Design External Pressure	103.42 KPa.
Design Temperature External Pressure	65 °C
Effective Diameter Multiplier	1.2
Material Name	SA-516 70
Allowable Stress, Ambient	138 N./mm ²
Allowable Stress, Operating	138 N./mm ²
Allowable Stress, Hydrotest	234 N./mm ²
Material Density	0.00775 kg./cm ³
P Number Thickness	32.004 mm
Yield Stress, Operating	246 N./mm ²
UCS-66 Chart Curve Designation	B
External Pressure Chart Name	CS-2
UNS Number	K02700
Product Form	Plate
Efficiency, Longitudinal Seam	1.0
Efficiency, Circumferential Seam	1.0
Weld is pre-Heated	No

Element From Node	90
Detail Type	Liquid
Detail ID	Liquid: 60
Dist. from "FROM" Node / Offset dist	0 mm
Height/Length of Liquid	770 mm
Liquid Density	0.0009984 kg./cm ³

Element From Node	90
Detail Type	Insulation
Detail ID	Insulation
Dist. from "FROM" Node / Offset dist	0 mm
Height/Length of Insulation	1000 mm
Thickness of Insulation	70 mm
Density	0.00032 kg./cm ³



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Input Echo:

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Element From Node	100
Element To Node	110
Element Type	Elliptical
Description	RHS Bonnet
Distance "FROM" to "TO"	60 mm
Inside Diameter	770 mm
Element Thickness	19.5 mm
Internal Corrosion Allowance	3 mm
Nominal Thickness	24 mm
External Corrosion Allowance	0 mm
Design Internal Pressure	5850 KPa.
Design Temperature Internal Pressure	65 °C
Design External Pressure	103.42 KPa.
Design Temperature External Pressure	65 °C
Effective Diameter Multiplier	1.2
Material Name	SA-516 70
Efficiency, Longitudinal Seam	1.0
Efficiency, Circumferential Seam	1.0
Elliptical Head Factor	2.0
Weld is pre-Heated	No
Element From Node	100
Detail Type	Liquid
Detail ID	Liquid: 70
Dist. from "FROM" Node / Offset dist	0 mm
Height/Length of Liquid	770 mm
Liquid Density	0.0009984 kg./cm ³
Element From Node	100
Detail Type	Insulation
Detail ID	Insulation
Dist. from "FROM" Node / Offset dist	0 mm
Height/Length of Insulation	60 mm
Thickness of Insulation	70 mm
Density	0.00032 kg./cm ³

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XY Coordinate Calculations:

Step: 2 9:41am

Sep 9,2026

XY Coordinate Calculations:

From	To	X (Horiz.) mm	Y (Vert.) mm	DX (Horiz.) mm	DY (Vert.) mm
LHS Bonnet		60	...	60	...
LHS ch shell		1060	...	1000	...
LHS ch.flg		1252.17	...	189	...
main shell-1		3832.18	...	2420	...
main shell-2		6252.17	...	2420	...
main shell-3		8672.17	...	2420	...
main shell-4		11092.2	...	2420	...
RHS ch Flg		11444.3	...	189	...
RHS Ch.Shell		12444.3	...	1000	...
RHS Bonnet		12504.3	...	60	...

The calculated dimensions are based on the given element lengths. Due to variability in manufacturing (weld gaps etc.), the Tangent to Tangent length may not be exact. The length includes gasket thicknesses and raised face heights.

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Flg Calc [Int P]: LHS ch.flg

Flng: 5 9:41am Sep 9,2026

Flange Input Data Values

Description: LHS ch.flg :

LHS ch.flg

Flange Type	Integral Weld Neck, Integral type		
Design Pressure	P	5857.55	KPa.
Design Temperature		65	°C
Internal Corrosion Allowance	ci	3.0000	mm
External Corrosion Allowance	ce	0.0000	mm
Use Corrosion Allowance in Thickness Calcs.		Yes	
Flange Inside Diameter	B	770.000	mm
Flange Outside Diameter	A	1040.000	mm
Flange Thickness	t	145.0000	mm
Thickness of Hub at Small End	go	22.0000	mm
Thickness of Hub at Large End	gl	33.0000	mm
Length of Hub	h	44.0000	mm
Flange Material		SA-350 LF2 +IT	
Flange Material UNS number		K03011	
Flange Allowable Stress At Temperature	Sfo	138.00	N./mm ²
Flange Allowable Stress At Ambient	Sfa	138.00	N./mm ²
Bolt Material		SA-193 B7	
Bolt Allowable Stress At Temperature	Sb	172.00	N./mm ²
Bolt Allowable Stress At Ambient	Sa	172.00	N./mm ²
Diameter of Bolt Circle	C	950.000	mm
Nominal Bolt Diameter	a	38.1000	mm
Type of Threads	TEMA Thread Series		
Number of Bolts		36	
Flange Face Outside Diameter	Fod	839.000	mm
Flange Face Inside Diameter	Fid	770.000	mm
Flange Facing Sketch		Sketch 1a	
Gasket Outside Diameter	Go	828.000	mm
Gasket Inside Diameter	Gi	802.000	mm
Gasket Factor	m	3.0000	
Gasket Design Seating Stress	y	68.95	N./mm ²
Gasket Thickness	tg	3.1750	mm
Raised face depth	Fd	0.0000	mm
Length of Partition Gasket	lp	804.0000	mm
Width of Partition Gasket	tp	10.0000	mm
Partition Gasket Factor	mPart	3.0000	
Partition Gasket Design Seating Stress	yPart	68.95	N./mm ²



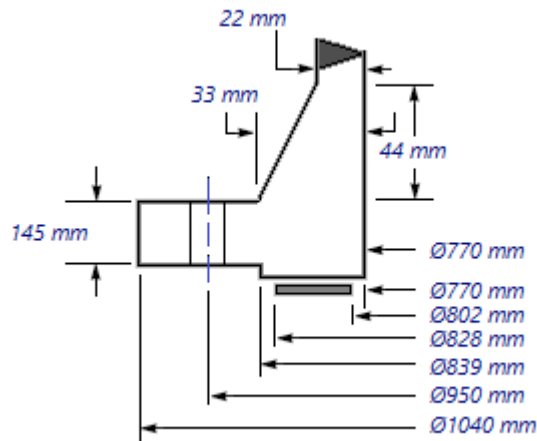
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Flg Calc [Int P]: LHS ch.flg

Flng: 5 9:41am Sep 9, 2026



ASME Code, Section VIII Division 1, 2025

Hub Small End Required Thickness due to Internal Pressure:

$$= (P \cdot (D/2 + Ca)) / (S \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)}$$

$$= (5857.5 \cdot (770/2 + 3)) / (138 \cdot 1 - 0.6 \cdot 5857.5) + Ca$$

$$= 19.9008 \text{ mm}$$

Small End Hub MAWP:

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (138 \cdot 1 \cdot 19) / (388 + 0.6 \cdot 19)$$

$$= 6564.347 \text{ KPa.}$$

Hub Small End Hub MAPnc:

$$= (Sa \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (138 \cdot 1 \cdot 22) / (385 + 0.6 \cdot 22)$$

$$= 7623.729 \text{ KPa.}$$

Corroded Flange Thickness, $t_c = T - c_i$	142.000 mm
Corroded Flange ID, $B_{cor} = B + 2 \cdot c_i$	776.000 mm
Corroded Large Hub, $g1_{cor} = g1 - c_i$	30.000 mm
Corroded Small Hub, $g0_{cor} = g0 - c_i$	19.000 mm
Code R Dimension, $R = ((C - B_{cor})/2) - g1_{cor}$	57.000 mm

Gasket Contact Width, $N = (G_o - G_i)/2$	13.000 mm
Basic Gasket Width, $bo = N/2$	6.500 mm
Effective Gasket Width, $b = C_b \cdot \sqrt{bo}$	6.425 mm
Gasket Reaction Diameter, $G = G_o - 2 \cdot b$	815.151 mm

Basic Flange and Bolt Loads:

Hydrostatic End Load due to Pressure [H or Wo,1]:

$$= \pi/4 \cdot G^2 \cdot P$$

$$= \pi/4 \cdot 815.15^2 \cdot 5857.5$$

$$= 311729.750 \text{ Kg}$$

Contact Load on Gasket Surfaces [Hp or Wo,2]:

$$= 2 \cdot b \cdot \pi \cdot G \cdot m \cdot P + 2 \cdot l_p \cdot b_{Part} \cdot m_{Part} \cdot P$$

$$= 2 \cdot 6.4246 \cdot \pi \cdot 815.15 \cdot 3 \cdot 5857.5$$

$$+ 2 \cdot 804 \cdot 5 \cdot 3 \cdot 5857.5$$

$$= 73372.664 \text{ Kg}$$

Hydrostatic End Load at Flange ID [Hd]:

$$= \pi/4 \cdot B_{cor}^2 \cdot P$$

$$= \pi/4 \cdot 776^2 \cdot 5857.5$$

$$= 282504.719 \text{ Kg}$$



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 FileName : 170-C-AK-Final Page 25 of 199
 Flg Calc [Int P]: LHS ch.flg Flng: 5 9:41am Sep 9,2026

Pressure Force on Flange Face [Ht]:

$$= H - H_d$$

$$= 311730 - 282505$$

$$= 29225.051 \text{ Kgf}$$

Design Bolt Load for the Operating Condition [Wo]:

$$= \max(W_{o,1} + W_{o,2} + H'p, 0)$$

$$= \max(311730 + 73373 + 0, 0)$$

$$= 385102.438 \text{ Kgf}$$

Design Bolt Load for the Gasket Seating Condition [Wg]:

$$= y \cdot b \cdot \pi \cdot G + y_{Part} \cdot b_{Part} \cdot l_p$$

$$= 68.95 \cdot 6.4246 \cdot \pi \cdot 815.15 + 68.95 \cdot 5 \cdot 804$$

$$= 143938.234 \text{ Kgf}$$

Required Bolt Area [Am]:

$$= \max((W_o + F_a + 4Me/G) / S_b, W_g / S_a)$$

$$= \max((385102 + 0 + 4 \cdot 0 / 815.15) / 172, 143938 / 172)$$

$$= 219.576 \text{ cm}^2$$

ASME Maximum Circumferential Spacing between Bolts, Table 4.16.11 [Bsmx]:

$$= 2a + 6t / (m + 0.5)$$

$$= 2 \cdot 38.1 + 6 \cdot 142 / (3 + 0.5)$$

$$= 319.629 \text{ mm}$$

Actual Circumferential Bolt Spacing [Bs]:

$$= C \cdot \sin(\pi / n)$$

$$= 950 \cdot \sin(\pi / 36)$$

$$= 82.798 \text{ mm}$$

ASME Moment Multiplier for Bolt Spacing per Table 4.16.11 [Bsc]:

$$= \max(\sqrt{ Bs / (2a + t) }, 1)$$

$$= \max(\sqrt{ 82.798 / (2 \cdot 38.1 + 142) }, 1)$$

$$= 1.0000$$

Bolting Information for TEMA Imperial Thread Series (Non Mandatory):

	Minimum	Actual	Maximum
Bolt Area:	219.576	326.322	
Radial Distance between Hub and Bolts:	50.800	57.000	
Radial Distance between Bolts and Edge:	38.100	45.000	
Circ. Spacing between the Bolts:	82.550	82.798	319.629

Min. Gasket Contact Width (Brownell Young) [Not an ASME Calc] [Nmin]:

$$= A_b \cdot S_a / (y \cdot \pi (G_o + G_i))$$

$$= 326.32 \cdot 172 / (68.95 \cdot \pi (828 + 802))$$

$$= 15.896 \text{ mm} [\text{Note: Exceeds actual gasket width, 13.000 }]$$

Recommended Minimum Gasket Contact Width per Table 4.16.2

Sheet Gaskets including laminated sheet gaskets with or without a metal core: 16 mm
 Preformed comp. gaskets incl. spiral wound, jacketed, & solid flat metal gaskets: 16 mm

Flange Design Bolt Load, Gasket Seating [W or Wgs]:

$$= S_a \cdot A_b$$

$$= 172 \cdot 326.32$$

$$= 572319.00 \text{ Kgf}$$

Gasket Load for the Operating Condition [HG]:

$$= W_o - H$$

$$= 385102 - 311730$$

$$= 73372.66 \text{ Kgf}$$



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Moment Arm Calculations:

Distance to Gasket Load Reaction [hg]:

$$= (C - G) / 2$$

$$= (950 - 815.15) / 2$$

$$= 67.4246 \text{ mm}$$

Distance to Face Pressure Reaction [ht]:

$$= 0.5 ((C - B) / 2 + hg)$$

$$= 0.5 ((950 - 776) / 2 + 67.425)$$

$$= 77.2123 \text{ mm}$$

Distance to End Pressure Reaction [hd]:

$$= (C - B - g1) / 2$$

$$= (950 - 776 - 30) / 2$$

$$= 72.0000 \text{ mm}$$

Summary of Moments for Internal Pressure: (Kg-m.)

Loading	Force	Distance	Bolt Corr	Moment
End Pressure, Md	282505.	72.0000	1.0000	20341.
Face Pressure, Mt	29225.	77.2123	1.0000	2257.
Gasket Load, Mg	73373.	67.4246	1.0000	4947.
Gasket Seating, Matm	572319.	67.4246	1.0000	38589.

Total Moment for Operation, Mo 27545. Kg-m.

Total Moment for Gasket seating, Mg 38589. Kg-m.

Effective Hub Length, ho = sqrt(Bcor*goCor) 121.425 mm

Hub Ratio, h/h0 = HL / H0 0.362

Thickness Ratio, g1/g0 = (g1Cor/goCor) 1.579

Flange Factors for Integral Flange:

Factor F per 4.16.5 0.866

Factor V per 4.16.5 0.334

Factor f per 4.16.5 1.097

Stress factor from Table 4.16.4 [K]:

$$= A / B$$

$$= 1040 / 776$$

$$= 1.340$$

Stress factor from Table 4.16.4 [Y]:

$$= (0.66845 + (5.7169 (K^2 \cdot \log_{10}(K))) / (K^2 - 1)) / (K - 1)$$

$$= (0.66845 + (5.7169 (1.3402^2 \cdot \log_{10}(1.3402))) / (1.3402^2 - 1)) / (1.3402 - 1)$$

$$= 6.786$$

Stress factor from Table 4.16.4 [T]:

$$= (K^2 (1 + 8.55246 \cdot \log_{10}(K)) - 1) / ((1.0472 + 1.9448 \cdot K^2) \cdot (K - 1))$$

$$= (1.3402^2 (1 + 8.55246 \cdot \log_{10}(1.3402)) - 1) / ((1.0472 + 1.9448 \cdot 1.3402^2) \cdot (1.3402 - 1))$$

$$= 1.780$$

Stress factor from Table 4.16.4 [U]:

$$= (K^2 (1 + 8.55246 \cdot \log_{10}(K)) - 1) / (1.36136 (K^2 - 1) \cdot (K - 1))$$

$$= (1.3402^2 (1 + 8.55246 \cdot \log_{10}(1.3402)) - 1) / (1.36136 (1.3402^2 - 1) \cdot (1.3402 - 1))$$

$$= 7.457$$

Stress factor from Table 4.16.4 [Z]:

$$= (K^2 + 1) / (K^2 - 1)$$

$$= (1.3402^2 + 1) / (1.3402^2 - 1)$$

$$= 3.512$$

Stress factor from Table 4.16.4 for integral flanges [d]:

$$= U \cdot g0^2 \cdot h0 / V$$



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$$= 7.4572 \cdot 19^2 \cdot 121.42 / 0.3339$$

$$= 979121.875 \text{ mm}^3$$

Stress factor from Table 4.16.4 for integral flanges [e]:

$$= F / h_0$$

$$= 0.8664 / 121.42$$

$$= 0.007 \text{ 1/mm}$$

Stress factor from Table 4.16.4 [L]:

$$= (t \cdot e + 1) / T + t^3/d$$

$$= (142 \cdot 0.007135 + 1) / 1.7801 + 142^3 / 979122$$

$$= 4.055$$

Flange Stresses

Longitudinal Hub Stress, Operating [SHo]:

$$= (f \cdot Mop / Bcor) / (L \cdot g1^2)$$

$$= (1.0971 \cdot 27545 / 776) / (4.0553 \cdot 30^2)$$

$$= 104.64 \text{ N./mm}^2$$

Longitudinal Hub Stress, Seating [SHa]:

$$= (f \cdot Matm / Bcor) / (L \cdot g1^2)$$

$$= (1.0971 \cdot 38589 / 776) / (4.0553 \cdot 30^2)$$

$$= 146.59 \text{ N./mm}^2$$

Radial Flange Stress, Operating [SRo]:

$$= (\beta \cdot Mop / Bcor) / (L \cdot t^2)$$

$$= (2.351 \cdot 27545 / 776) / (4.0553 \cdot 142^2)$$

$$= 10.01 \text{ N./mm}^2$$

Radial Flange Stress, Seating [SRa]:

$$= (\beta \cdot Matm / Bcor) / (L \cdot t^2)$$

$$= (2.351 \cdot 38589 / 776) / (4.0553 \cdot 142^2)$$

$$= 14.02 \text{ N./mm}^2$$

Tangential Flange Stress, Operating [STo]:

$$= (Y \cdot Mo / (t^2 \cdot Bcor)) - Z \cdot SRo$$

$$= (6.786 \cdot 27545 / (142^2 \cdot 776)) - 3.5121 \cdot 10.008$$

$$= 82.00 \text{ N./mm}^2$$

Tangential Flange Stress, Seating [STa]:

$$= (y \cdot Matm / (t^2 \cdot Bcor)) - Z \cdot SRa$$

$$= (6.786 \cdot 38589 / (142^2 \cdot 776)) - 3.5121 \cdot 14.021$$

$$= 114.88 \text{ N./mm}^2$$

Average Flange Stress, Operating [SAo]:

$$= (SHo + \max(SRo, STo)) / 2$$

$$= (104.64 + \max(10.008, 81.999)) / 2$$

$$= 93.32 \text{ N./mm}^2$$

Average Flange Stress, Seating [SAa]:

$$= (SHa + \max(SRa, STa)) / 2$$

$$= (146.59 + \max(14.021, 114.88)) / 2$$

$$= 130.74 \text{ N./mm}^2$$

Bolt Stress, Operating [BSO]:

$$= Wm1 / Ab$$

$$= 385102 / 326.32$$

$$= 115.73 \text{ N./mm}^2$$

Bolt Stress, Seating [BSa]:

$$= (Wm2 / Ab)$$

$$= (143938 / 326.32)$$

$$= 43.26 \text{ N./mm}^2$$



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Flange Stress Analysis Results: N./mm²

	Actual	Operating Allowed	Gasket Actual	Seating Allowed
Longitudinal Hub	104.64	207.00	146.59	207.00
Radial Flange	10.01	138.00	14.02	138.00
Tangential Flange	82.00	138.00	114.88	138.00
Maximum Average	93.32	138.00	130.74	138.00
Bolting	115.73	172.00	43.26	172.00

Minimum Required Flange Thickness 140.868 mm

Estimated M.A.W.P. (Operating) 6564.3 KPa.

Estimated Finished Weight of Flange at given Thk. 454.8 kg.

Estimated Unfinished Weight of Forging at given Thk 562.2 kg.

Flange Stress results per WRC-538 at computed S_{fmax} (N./mm²):

		Actual	Operating Allowed	stress ratio	Gasket Actual	Seating Allowed	stress ratio
Longitudinal Hub	Sh	443.40	499.99	0.887	443.40	499.99	0.887
Tangential Flange	St	346.72	374.99	0.925	346.72	374.99	0.925
Tangential Flange	Sto	151.32	250.00	0.605	151.32	250.00	0.605
Combined	Sh/f + St	749.99	749.99	1.000	749.99	749.99	1.000
Combined	Sr + St	389.03	749.99	0.519	389.03	749.99	0.519
Combined	Sh + Sto	594.72	749.99	0.793	594.72	749.99	0.793
Radial Flange	Sr	42.32	499.99	0.085	42.32	499.99	0.085
Maximum Stress Ratio				1.000			1.000



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Coefficients for f and $f2$

	f	$f2$	$f3$
a	-0.71375912	-0.01447638	-0.06456312
b	-0.12846279	-0.27814745	-0.08232239
c	1.08037907	0.01035395	0.13691677
d	0.99766848	1.37984158	-0.77879888
e	1.21823466	-0.34346764	0.14909019
f	0.01024612	0.02536192	0.00660062
g	0.4262313	-0.0014508	0.0616252
h	1.4204976	2.25543162	0.74639834
i	-0.70278181	-0.60889597	-0.079595
j	-0.02483937	-0.15735462	0.09851673
k	-1.59460436	1.22516062	-0.1121436

$$f_n = \frac{\left[a + c \left(\frac{g1}{g0} \right) + e \left(\frac{h}{h0} \right) + g \left(\frac{g1}{g0} \right)^2 + i \left(\frac{h}{h0} \right)^2 + k \left(\frac{g1}{g0} \right) \left(\frac{h}{h0} \right) \right]}{\left[1 + b \left(\frac{g1}{g0} \right) + d \left(\frac{h}{h0} \right) + f \left(\frac{g1}{g0} \right)^2 + h \left(\frac{h}{h0} \right)^2 + j \left(\frac{g1}{g0} \right) \left(\frac{h}{h0} \right) \right]}$$

Hub Stress Correction Factor [f]:

= max(1; F1num / F1den)

= max(1; 1.4915/1.3565)

= 1.0995

Hub Stress Correction Factor [f2]:

= max(0.25; fn2) if $h/h0 < 0.35$, else max (0.25; fn3)

= max(0.25; 0.4948/1.3302) if $h/h0 < 0.35$, else max (0.25; 0.2847/0.7586)

= 0.3752

In this case, the hub ratio $h/h0$, 0.3624 was ≥ 0.35 .

PV Elite iterates on the assembly bolt load stress, Sfmax, until it exceeds the limits provided by WRC 538.

Bolting and Gasket Calculations per ASME PCC-1 Appendix O:

Input Data:

Maximum permissible bolt stress	Sbmax	507.5 N./mm ²
Minimum permissible bolt stress	Sbmin	290.0 N./mm ²
Max. permissible bolt stress prior to flange damage	Sfmax	519.1 N./mm ²
Target assembly gasket stress	SgT	208.0 N./mm ²
Maximum permissible gasket stress	Sgmax	275.0 N./mm ²
Minimum gasket operating stress	Sgmin-O	97.0 N./mm ²
Minimum gasket seating stress	Sgmin-S	140.0 N./mm ²
Nut Factor	K	0.200
Fraction of Gasket Load remaining after relaxation	Phig	0.700



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Root Area of a Single Bolt	Abs	9.0645 cm ²
Max. permissible single flange rotation for gasket	Thetagmax	1.000 deg

Calculated Results:

Flange Material Ambient Yield Stress	Sya	250.0 N./mm ²
Flange Material Operating Yield Stress	Syo	233.0 N./mm ²
Single flange rotation at Sfmax	Thetafmax	0.387 deg

Gasket Contact Area [Ag]:

$$= \pi/4 (GO.D.^2 - GI.D.^2) + (PartFactor \cdot Ip \cdot bPart)$$

$$= \pi/4 (828^2 - 802^2) + (0.5 \cdot 804 \cdot 5)$$

$$= 353.0 \text{ cm}^2$$

Target Bolt Stress [Sbsel]: (Eq. O-1)

$$= SgT \cdot Ag / (nb \cdot Ab)$$

$$= 208 \cdot 352.95 / (36 \cdot 9.0645)$$

$$= 225.0 \text{ N./mm}^2$$

Upper Bolt Stress Limit [Sbsel]: (Eq. O-4)

$$= \min(Sbsel; Sbmax)$$

$$= \min(224.97; 507.5)$$

$$= 225.0 \text{ N./mm}^2$$

Lower Bolt Stress Limit [Sbsel]: (Eq. O-5)

$$= \max(Sbsel; Sbmin)$$

$$= \max(224.97; 290)$$

$$= 290.0 \text{ N./mm}^2$$

Upper Bolt Stress Limit including the Flange [Sbsel]: (Eq. O-6)

$$= \min(Sbsel; Sfmax)$$

$$= \min(290; 519.11)$$

$$= 290.0 \text{ N./mm}^2$$

Check if Gasket Assembly Seating Stress is achieved. (Eq. O-7)

$$:: Sbsel \geq Sgmin-S(Ag / (nb \cdot Abs))$$

$$:: 290 \geq 140 (352.95 / (36 \cdot 9.0645))$$

$$:: 290.0 \geq 151.4 \text{ N./mm}^2 \text{ Check Passes}$$

Check if Gasket Operating Stress is maintained. (Eq. O-8)

$$:: Sbsel \geq (Sgmin-O \cdot Ag + \pi/4 \cdot Pmax \cdot GI.D.^2) / (\phi G \cdot Abs \cdot nb)$$

$$:: 290 \geq (97 \cdot 352.95 + \pi/4 \cdot 5857.5 \cdot 802^2) / (0.7 \cdot 9.0645 \cdot 36)$$

$$:: 290.0 \geq 279.4 \text{ N./mm}^2 \text{ Check Passes}$$

Check if Gasket Maximum Stress is achieved. (Eq. O-9)

$$:: Sbsel \leq Sgmax(Ag / (nb \cdot Abs))$$

$$:: 290 \leq 275 (352.95 / (36 \cdot 9.0645))$$

$$:: 290.0 \leq 297.4 \text{ N./mm}^2 \text{ Check Passes}$$

Check if Flange Rotation Limit is exceeded. (Eq. O-10)

$$:: Sbsel \leq Sfmax(thetagmax / thetafmax)$$

$$:: 290 \leq 519.11 (1/0.3865)$$

$$:: 290.0 \leq 1342.9 \text{ N./mm}^2 \text{ Check Passes}$$

Assembly Bolt Torque [Tb]: (Eq. O-2)

$$= Sbsel \cdot K \cdot Abs \cdot \phi b$$

$$= 290 \cdot 0.2 \cdot 9.0645 \cdot 38.1$$

$$= 204.3 \text{ Kg-m.}$$

Bolt Allowable Stress for: SA-193 B8 Cl. 2 SbB8 122 N./mm²

Bolt Allowable Stress for: SA-193 B7 Sb 171 N./mm²

Note: Since Sb >= SbB8, bolt allowable stress comparison check Passes (see Div 1 UG-44(b)).

Flange rigidity based on required thickness [ASME]:



Flange Rigidity Criterion, Seating per Table 4.16.10 [Jg]:
 $= 52.14 \cdot Ma / Bsc \cdot Cnvfac \cdot V / (\lambda \cdot Eamb \cdot go^2 \cdot ho \cdot Ki)$
 $= 52.14 \cdot 38589 / 1 \cdot 9806.6 \cdot 0.3339 / (3.7907 \cdot 201334 \cdot 19^2 \cdot 121.42 \cdot 0.3)$
 $= 0.656$ (should be ≤ 1)

Flange Rigidity Criterion, Operating per Table 4.16.10 [Jo]:
 $= 52.14 \cdot Mo / Bsc \cdot Cnvfac \cdot V / (\lambda \cdot Eop \cdot goc^2 \cdot ho \cdot Ki)$
 $= 52.14 \cdot 27545 / 1 \cdot 9806.6 \cdot 0.3339 / (3.7907 \cdot 198820 \cdot 19^2 \cdot 121.42 \cdot 0.3)$
 $= 0.474$ (should be ≤ 1)

Flange rigidity based on given thickness [ASME]:

Flange Rigidity Criterion, Seating per Table 4.16.10 [Jg]:
 $= 52.14 \cdot Ma / Bsc \cdot Cnvfac \cdot V / (\lambda \cdot Eamb \cdot go^2 \cdot ho \cdot Ki)$
 $= 52.14 \cdot 38589 / 1 \cdot 9806.6 \cdot 0.3339 / (4.0553 \cdot 201334 \cdot 19^2 \cdot 121.42 \cdot 0.3)$
 $= 0.613$ (should be ≤ 1)

Flange Rigidity Criterion, Operating per Table 4.16.10 [Jo]:
 $= 52.14 \cdot Mo / Bsc \cdot Cnvfac \cdot V / (\lambda \cdot Eop \cdot goc^2 \cdot ho \cdot Ki)$
 $= 52.14 \cdot 27545 / 1 \cdot 9806.6 \cdot 0.3339 / (4.0553 \cdot 198820 \cdot 19^2 \cdot 121.42 \cdot 0.3)$
 $= 0.443$ (should be ≤ 1)

Minimum Design Metal Temperature Results:

Note:
This Material was specified as being an Impact Tested (Low Temperature) Material.

Impact Test Temperature provided per Specification -18 °C

Note: UCS-66(b)(-c) was considered in the flange MDMT calculation.

Per ASME VIII-2, 4.16.2.3, Perform Calculations in the New Condition:

Flange Stress Analysis Results: N./mm²

	Actual	Operating Allowed	Gasket Actual	Seating Allowed
Longitudinal Hub	99.29	207.00	136.67	207.00
Radial Flange	11.18	138.00	15.39	138.00
Tangential Flange	74.05	138.00	101.93	138.00
Maximum Average	86.67	138.00	119.30	138.00
Bolting	115.73	172.00	43.26	172.00

Minimum Req'd Thickness, ambient, no Corrosion: 133.121 mm

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Flg Calc [Int P]: New Flange

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Flange Input Data Values

Description: New Flange :

RHS ch Flg

Flange Type	Integral Weld Neck		
Design Pressure	P	5857.65	KPa.
Design Temperature		65	°C
Internal Corrosion Allowance	ci	3.0000	mm
External Corrosion Allowance	ce	0.0000	mm
Use Corrosion Allowance in Thickness Calcs.		Yes	
Flange Inside Diameter	B	770.000	mm
Flange Outside Diameter	A	1040.000	mm
Flange Thickness	t	145.0000	mm
Thickness of Hub at Small End	go	22.0000	mm
Thickness of Hub at Large End	gl	33.0000	mm
Length of Hub	h	44.0000	mm
Flange Material	SA-350 LF2 +IT		
Flange Material UNS number	K03011		
Flange Allowable Stress At Temperature	Sfo	138.00	N./mm ²
Flange Allowable Stress At Ambient	Sfa	138.00	N./mm ²
Bolt Material	SA-193 B7		
Bolt Allowable Stress At Temperature	Sb	172.00	N./mm ²
Bolt Allowable Stress At Ambient	Sa	172.00	N./mm ²
Diameter of Bolt Circle	C	950.000	mm
Nominal Bolt Diameter	a	38.1000	mm
Type of Threads	TEMA Thread Series		
Number of Bolts	36		
Flange Face Outside Diameter	Fod	839.000	mm
Flange Face Inside Diameter	Fid	770.000	mm
Flange Facing Sketch	Sketch 1a		
Gasket Outside Diameter	Go	828.000	mm
Gasket Inside Diameter	Gi	802.000	mm
Gasket Factor	m	3.0000	
Gasket Design Seating Stress	y	68.95	N./mm ²
Gasket Thickness	tg	3.1750	mm
Raised face depth	Fd	0.0000	mm
Length of Partition Gasket	lp	804.0000	mm
Width of Partition Gasket	tp	10.0000	mm
Partition Gasket Factor	mPart	3.0000	
Partition Gasket Design Seating Stress	yPart	68.95	N./mm ²



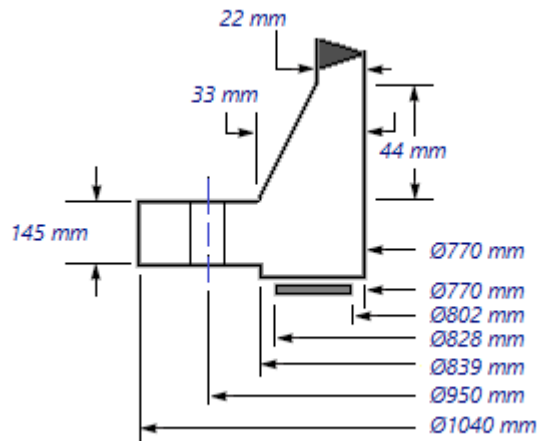
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Flg Calc [Int P]: New Flange

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Hub Small End Required Thickness due to Internal Pressure:

$$= (P \cdot (D/2 + Ca)) / (S \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)}$$

$$= (5857.6 \cdot (770/2 + 3)) / (138 \cdot 1 - 0.6 \cdot 5857.6) + Ca$$

$$= 19.9010 \text{ mm}$$

Small End Hub MAWP:

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (138 \cdot 1 \cdot 19) / (388 + 0.6 \cdot 19)$$

$$= 6564.347 \text{ KPa.}$$

Hub Small End Hub MAPnc:

$$= (Sa \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (138 \cdot 1 \cdot 22) / (385 + 0.6 \cdot 22)$$

$$= 7623.729 \text{ KPa.}$$

Corroded Flange Thickness, $t_c = T - c_i$	142.000 mm
Corroded Flange ID, $B_{cor} = B + 2 \cdot c_i$	776.000 mm
Corroded Large Hub, $g1_{cor} = g1 - c_i$	30.000 mm
Corroded Small Hub, $g0_{cor} = g0 - c_i$	19.000 mm
Code R Dimension, $R = ((C - B_{cor})/2) - g1_{cor}$	57.000 mm

Gasket Contact Width, $N = (G_o - G_i)/2$	13.000 mm
Basic Gasket Width, $bo = N/2$	6.500 mm
Effective Gasket Width, $b = C_b \cdot \sqrt{bo}$	6.425 mm
Gasket Reaction Diameter, $G = G_o - 2 \cdot b$	815.151 mm

Basic Flange and Bolt Loads:

Hydrostatic End Load due to Pressure [H or Wo,1]:

$$= \pi/4 \cdot G^2 \cdot P$$

$$= \pi/4 \cdot 815.15^2 \cdot 5857.6$$

$$= 311734.969 \text{ Kgf}$$

Contact Load on Gasket Surfaces [Hp or Wo,2]:

$$= 2 \cdot b \cdot \pi \cdot G \cdot m \cdot P + 2 \cdot l_p \cdot b_{Part} \cdot m_{Part} \cdot P$$

$$= 2 \cdot 6.4246 \cdot \pi \cdot 815.15 \cdot 3 \cdot 5857.6$$

$$+ 2 \cdot 804 \cdot 5 \cdot 3 \cdot 5857.6$$

$$= 73373.891 \text{ Kgf}$$

Hydrostatic End Load at Flange ID [Hd]:

$$= \pi/4 \cdot B_{cor}^2 \cdot P$$

$$= \pi/4 \cdot 776^2 \cdot 5857.6$$

$$= 282509.438 \text{ Kgf}$$



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Flg Calc [Int P]: New Flange

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Pressure Force on Flange Face [Ht]:

$$\begin{aligned}
 &= H - H_d \\
 &= 311735 - 282509 \\
 &= 29225.533 \text{ Kgf}
 \end{aligned}$$

Design Bolt Load for the Operating Condition [Wo]:

$$\begin{aligned}
 &= \max(W_{o,1} + W_{o,2} + H'p, 0) \\
 &= \max(311735 + 73374 + 0, 0) \\
 &= 385108.844 \text{ Kgf}
 \end{aligned}$$

Design Bolt Load for the Gasket Seating Condition [Wg]:

$$\begin{aligned}
 &= y \cdot b \cdot \pi \cdot G + y_{Part} \cdot b_{Part} \cdot l_p \\
 &= 68.95 \cdot 6.4246 \cdot \pi \cdot 815.15 + 68.95 \cdot 5 \cdot 804 \\
 &= 143938.234 \text{ Kgf}
 \end{aligned}$$

Required Bolt Area [Am]:

$$\begin{aligned}
 &= \max((W_o + F_a + 4Me/G) / S_b, W_g / S_a) \\
 &= \max((385109 + 0 + 4 \cdot 0 / 815.15) / 172, 143938 / 172) \\
 &= 219.579 \text{ cm}^2
 \end{aligned}$$

ASME Maximum Circumferential Spacing between Bolts, Table 4.16.11 [Bsmx]:

$$\begin{aligned}
 &= 2a + 6t / (m + 0.5) \\
 &= 2 \cdot 38.1 + 6 \cdot 142 / (3 + 0.5) \\
 &= 319.629 \text{ mm}
 \end{aligned}$$

Actual Circumferential Bolt Spacing [Bs]:

$$\begin{aligned}
 &= C \cdot \sin(\pi / n) \\
 &= 950 \cdot \sin(\pi / 36) \\
 &= 82.798 \text{ mm}
 \end{aligned}$$

ASME Moment Multiplier for Bolt Spacing per Table 4.16.11 [Bsc]:

$$\begin{aligned}
 &= \max(\sqrt{ Bs / (2a + t) }, 1) \\
 &= \max(\sqrt{ 82.798 / (2 \cdot 38.1 + 142) }, 1) \\
 &= 1.0000
 \end{aligned}$$

Bolting Information for TEMA Imperial Thread Series (Non Mandatory):

	Minimum	Actual	Maximum
Bolt Area:	219.579	326.322	
Radial Distance between Hub and Bolts:	50.800	57.000	
Radial Distance between Bolts and Edge:	38.100	45.000	
Circ. Spacing between the Bolts:	82.550	82.798	319.629

Min. Gasket Contact Width (Brownell Young) [Not an ASME Calc] [Nmin]:

$$\begin{aligned}
 &= A_b \cdot S_a / (y \cdot \pi (G_o + G_i)) \\
 &= 326.32 \cdot 172 / (68.95 \cdot \pi (828 + 802)) \\
 &= 15.896 \text{ mm} [\text{Note: Exceeds actual gasket width, 13.000}]
 \end{aligned}$$

Recommended Minimum Gasket Contact Width per Table 4.16.2

Sheet Gaskets including laminated sheet gaskets with or without a metal core: 16 mm
 Preformed comp. gaskets incl. spiral wound, jacketed, & solid flat metal gaskets: 16 mm

Flange Design Bolt Load, Gasket Seating [W or Wgs]:

$$\begin{aligned}
 &= S_a \cdot A_b \\
 &= 172 \cdot 326.32 \\
 &= 572319.00 \text{ Kgf}
 \end{aligned}$$

Gasket Load for the Operating Condition [HG]:

$$\begin{aligned}
 &= W_o - H \\
 &= 385109 - 311735 \\
 &= 73373.88 \text{ Kgf}
 \end{aligned}$$



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Flg Calc [Int P]: New Flange

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Moment Arm Calculations:

Distance to Gasket Load Reaction [hg]:

$$= (C - G) / 2$$

$$= (950 - 815.15) / 2$$

$$= 67.4246 \text{ mm}$$

Distance to Face Pressure Reaction [ht]:

$$= 0.5 ((C - B) / 2 + hg)$$

$$= 0.5 ((950 - 776) / 2 + 67.425)$$

$$= 77.2123 \text{ mm}$$

Distance to End Pressure Reaction [hd]:

$$= (C - B - g1) / 2$$

$$= (950 - 776 - 30) / 2$$

$$= 72.0000 \text{ mm}$$

Summary of Moments for Internal Pressure: (Kg-m.)

Loading	Force	Distance	Bolt Corr	Moment
End Pressure, Md	282509.	72.0000	1.0000	20341.
Face Pressure, Mt	29226.	77.2123	1.0000	2257.
Gasket Load, Mg	73374.	67.4246	1.0000	4947.
Gasket Seating, Matm	572319.	67.4246	1.0000	38589.

Total Moment for Operation, Mo 27545. Kg-m.
Total Moment for Gasket seating, Mg 38589. Kg-m.

Effective Hub Length, ho = sqrt(Bcor*goCor) 121.425 mm
Hub Ratio, h/h0 = HL / H0 0.362
Thickness Ratio, g1/g0 = (g1Cor/goCor) 1.579

Flange Factors for Integral Flange:

Factor F per 4.16.5 0.866
Factor V per 4.16.5 0.334
Factor f per 4.16.5 1.097

Stress factor from Table 4.16.4 [K]:

$$= A / B$$

$$= 1040 / 776$$

$$= 1.340$$

Stress factor from Table 4.16.4 [Y]:

$$= (0.66845 + (5.7169 (K^2 \cdot \log_{10}(K))) / (K^2 - 1)) / (K - 1)$$

$$= (0.66845 + (5.7169 (1.3402^2 \cdot \log_{10}(1.3402))) / (1.3402^2 - 1)) / (1.3402 - 1)$$

$$= 6.786$$

Stress factor from Table 4.16.4 [T]:

$$= (K^2 (1 + 8.55246 \cdot \log_{10}(K)) - 1) / ((1.0472 + 1.9448 \cdot K^2) \cdot (K - 1))$$

$$= (1.3402^2 (1 + 8.55246 \cdot \log_{10}(1.3402)) - 1) / ((1.0472 + 1.9448 \cdot 1.3402^2) \cdot (1.3402 - 1))$$

$$= 1.780$$

Stress factor from Table 4.16.4 [U]:

$$= (K^2 (1 + 8.55246 \cdot \log_{10}(K)) - 1) / (1.36136 (K^2 - 1) \cdot (K - 1))$$

$$= (1.3402^2 (1 + 8.55246 \cdot \log_{10}(1.3402)) - 1) / (1.36136 (1.3402^2 - 1) \cdot (1.3402 - 1))$$

$$= 7.457$$

Stress factor from Table 4.16.4 [Z]:

$$= (K^2 + 1) / (K^2 - 1)$$

$$= (1.3402^2 + 1) / (1.3402^2 - 1)$$

$$= 3.512$$

Stress factor from Table 4.16.4 for integral flanges [d]:

$$= U \cdot g0^2 \cdot h0 / V$$



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Flg Calc [Int P]: New Flange

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$$= 7.4572 \cdot 19^2 \cdot 121.42 / 0.3339$$

$$= 979121.875 \text{ mm}^3$$

Stress factor from Table 4.16.4 for integral flanges [e]:

$$= F / h_0$$

$$= 0.8664 / 121.42$$

$$= 0.007 \text{ 1/mm}$$

Stress factor from Table 4.16.4 [L]:

$$= (t \cdot e + 1) / T + t^3/d$$

$$= (142 \cdot 0.007135 + 1) / 1.7801 + 142^3 / 979122$$

$$= 4.055$$

Flange Stresses

Longitudinal Hub Stress, Operating [SHo]:

$$= (f \cdot Mop / Bcor) / (L \cdot g1^2)$$

$$= (1.0971 \cdot 27545 / 776) / (4.0553 \cdot 30^2)$$

$$= 104.64 \text{ N./mm}^2$$

Longitudinal Hub Stress, Seating [SHa]:

$$= (f \cdot Matm / Bcor) / (L \cdot g1^2)$$

$$= (1.0971 \cdot 38589 / 776) / (4.0553 \cdot 30^2)$$

$$= 146.59 \text{ N./mm}^2$$

Radial Flange Stress, Operating [SRo]:

$$= (\beta \cdot Mop / Bcor) / (L \cdot t^2)$$

$$= (2.351 \cdot 27545 / 776) / (4.0553 \cdot 142^2)$$

$$= 10.01 \text{ N./mm}^2$$

Radial Flange Stress, Seating [SRa]:

$$= (\beta \cdot Matm / Bcor) / (L \cdot t^2)$$

$$= (2.351 \cdot 38589 / 776) / (4.0553 \cdot 142^2)$$

$$= 14.02 \text{ N./mm}^2$$

Tangential Flange Stress, Operating [STo]:

$$= (Y \cdot Mo / (t^2 \cdot Bcor)) - Z \cdot SRo$$

$$= (6.786 \cdot 27545 / (142^2 \cdot 776)) - 3.5121 \cdot 10.008$$

$$= 82.00 \text{ N./mm}^2$$

Tangential Flange Stress, Seating [STa]:

$$= (y \cdot Matm / (t^2 \cdot Bcor)) - Z \cdot SRa$$

$$= (6.786 \cdot 38589 / (142^2 \cdot 776)) - 3.5121 \cdot 14.021$$

$$= 114.88 \text{ N./mm}^2$$

Average Flange Stress, Operating [SAo]:

$$= (SHo + \max(SRo, STo)) / 2$$

$$= (104.64 + \max(10.008, 82.001)) / 2$$

$$= 93.32 \text{ N./mm}^2$$

Average Flange Stress, Seating [SAa]:

$$= (SHa + \max(SRa, STa)) / 2$$

$$= (146.59 + \max(14.021, 114.88)) / 2$$

$$= 130.74 \text{ N./mm}^2$$

Bolt Stress, Operating [BSo]:

$$= Wm1 / Ab$$

$$= 385109 / 326.32$$

$$= 115.74 \text{ N./mm}^2$$

Bolt Stress, Seating [BSa]:

$$= (Wm2 / Ab)$$

$$= (143938 / 326.32)$$

$$= 43.26 \text{ N./mm}^2$$



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Flg Calc [Int P]: New Flange

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Flange Stress Analysis Results: N./mm²

	Actual	Operating Allowed	Gasket Actual	Seating Allowed
Longitudinal Hub	104.64	207.00	146.59	207.00
Radial Flange	10.01	138.00	14.02	138.00
Tangential Flange	82.00	138.00	114.88	138.00
Maximum Average	93.32	138.00	130.74	138.00
Bolting	115.74	172.00	43.26	172.00

Minimum Required Flange Thickness 140.868 mm

Estimated M.A.W.P. (Operating) 6564.3 KPa.

Estimated Finished Weight of Flange at given Thk. 454.8 kg.

Estimated Unfinished Weight of Forging at given Thk 562.2 kg.

Flange Stress results per WRC-538 at computed S_{fmax} (N./mm²):

		Actual	Operating Allowed	stress ratio	Gasket Actual	Seating Allowed	stress ratio
Longitudinal Hub	Sh	443.40	499.99	0.887	443.40	499.99	0.887
Tangential Flange	St	346.72	374.99	0.925	346.72	374.99	0.925
Tangential Flange	Sto	151.32	250.00	0.605	151.32	250.00	0.605
Combined	Sh/f + St	749.99	749.99	1.000	749.99	749.99	1.000
Combined	Sr + St	389.03	749.99	0.519	389.03	749.99	0.519
Combined	Sh + Sto	594.72	749.99	0.793	594.72	749.99	0.793
Radial Flange	Sr	42.32	499.99	0.085	42.32	499.99	0.085
Maximum Stress Ratio				1.000			1.000



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Flg Calc [Int P]: New Flange

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Coefficients for f and $f2$

	f	$f2$	$f3$
a	-0.71375912	-0.01447638	-0.06456312
b	-0.12846279	-0.27814745	-0.08232239
c	1.08037907	0.01035395	0.13691677
d	0.99766848	1.37984158	-0.77879888
e	1.21823466	-0.34346764	0.14909019
f	0.01024612	0.02536192	0.00660062
g	0.4262313	-0.0014508	0.0616252
h	1.4204976	2.25543162	0.74639834
i	-0.70278181	-0.60889597	-0.079595
j	-0.02483937	-0.15735462	0.09851673
k	-1.59460436	1.22516062	-0.1121436

$$f_n = \frac{\left[a + c \left(\frac{g1}{g0} \right) + e \left(\frac{h}{h0} \right) + g \left(\frac{g1}{g0} \right)^2 + i \left(\frac{h}{h0} \right)^2 + k \left(\frac{g1}{g0} \right) \left(\frac{h}{h0} \right) \right]}{\left[1 + b \left(\frac{g1}{g0} \right) + d \left(\frac{h}{h0} \right) + f \left(\frac{g1}{g0} \right)^2 + h \left(\frac{h}{h0} \right)^2 + j \left(\frac{g1}{g0} \right) \left(\frac{h}{h0} \right) \right]}$$

Hub Stress Correction Factor [f]:

= max(1; F1num / F1den)

= max(1; 1.4915/1.3565)

= 1.0995

Hub Stress Correction Factor [f2]:

= max(0.25; fn2) if $h/h0 < 0.35$, else max (0.25; fn3)

= max(0.25; 0.4948/1.3302) if $h/h0 < 0.35$, else max (0.25; 0.2847/0.7586)

= 0.3752

In this case, the hub ratio $h/h0$, 0.3624 was ≥ 0.35 .

PV Elite iterates on the assembly bolt load stress, Sfmax, until it exceeds the limits provided by WRC 538.

Bolting and Gasket Calculations per ASME PCC-1 Appendix O:

Input Data:

Maximum permissible bolt stress	Sbmax	507.5 N./mm ²
Minimum permissible bolt stress	Sbmin	290.0 N./mm ²
Max. permissible bolt stress prior to flange damage	Sfmax	519.1 N./mm ²
Target assembly gasket stress	SgT	208.0 N./mm ²
Maximum permissible gasket stress	Sgmax	275.0 N./mm ²
Minimum gasket operating stress	Sgmin-O	97.0 N./mm ²
Minimum gasket seating stress	Sgmin-S	140.0 N./mm ²
Nut Factor	K	0.200
Fraction of Gasket Load remaining after relaxation	Phig	0.700



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Flg Calc [Int P]: New Flange

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Root Area of a Single Bolt	Abs	9.0645 cm ²
Max. permissible single flange rotation for gasket	Thetagmax	1.000 deg

Calculated Results:

Flange Material Ambient Yield Stress	Sya	250.0 N./mm ²
Flange Material Operating Yield Stress	Syo	233.0 N./mm ²
Single flange rotation at Sfmax	Thetafmax	0.387 deg

Gasket Contact Area [Ag]:

$$= \pi/4 (GO.D.^2 - GI.D.^2) + (PartFactor \cdot Ip \cdot bPart)$$

$$= \pi/4 (828^2 - 802^2) + (0.5 \cdot 804 \cdot 5)$$

$$= 353.0 \text{ cm}^2$$

Target Bolt Stress [Sbsel]: (Eq. O-1)

$$= SgT \cdot Ag / (nb \cdot Ab)$$

$$= 208 \cdot 352.95 / (36 \cdot 9.0645)$$

$$= 225.0 \text{ N./mm}^2$$

Upper Bolt Stress Limit [Sbsel]: (Eq. O-4)

$$= \min(Sbsel; Sbmax)$$

$$= \min(224.97; 507.5)$$

$$= 225.0 \text{ N./mm}^2$$

Lower Bolt Stress Limit [Sbsel]: (Eq. O-5)

$$= \max(Sbsel; Sbmin)$$

$$= \max(224.97; 290)$$

$$= 290.0 \text{ N./mm}^2$$

Upper Bolt Stress Limit including the Flange [Sbsel]: (Eq. O-6)

$$= \min(Sbsel; Sfmax)$$

$$= \min(290; 519.11)$$

$$= 290.0 \text{ N./mm}^2$$

Check if Gasket Assembly Seating Stress is achieved. (Eq. O-7)

$$:: Sbsel \geq Sgmin-S(Ag / (nb \cdot Abs))$$

$$:: 290 \geq 140 (352.95 / (36 \cdot 9.0645))$$

$$:: 290.0 \geq 151.4 \text{ N./mm}^2 \text{ Check Passes}$$

Check if Gasket Operating Stress is maintained. (Eq. O-8)

$$:: Sbsel \geq (Sgmin-O \cdot Ag + \pi/4 \cdot Pmax \cdot GI.D.^2) / (\phi G \cdot Abs \cdot nb)$$

$$:: 290 \geq (97 \cdot 352.95 + \pi/4 \cdot 5857.6 \cdot 802^2) / (0.7 \cdot 9.0645 \cdot 36)$$

$$:: 290.0 \geq 279.4 \text{ N./mm}^2 \text{ Check Passes}$$

Check if Gasket Maximum Stress is achieved. (Eq. O-9)

$$:: Sbsel \leq Sgmax(Ag / (nb \cdot Abs))$$

$$:: 290 \leq 275 (352.95 / (36 \cdot 9.0645))$$

$$:: 290.0 \leq 297.4 \text{ N./mm}^2 \text{ Check Passes}$$

Check if Flange Rotation Limit is exceeded. (Eq. O-10)

$$:: Sbsel \leq Sfmax(thetagmax / thetafmax)$$

$$:: 290 \leq 519.11 (1/0.3865)$$

$$:: 290.0 \leq 1342.9 \text{ N./mm}^2 \text{ Check Passes}$$

Assembly Bolt Torque [Tb]: (Eq. O-2)

$$= Sbsel \cdot K \cdot Abs \cdot \phi b$$

$$= 290 \cdot 0.2 \cdot 9.0645 \cdot 38.1$$

$$= 204.3 \text{ Kg-m.}$$

Bolt Allowable Stress for: SA-193 B8 Cl. 2 SbB8 122 N./mm²

Bolt Allowable Stress for: SA-193 B7 Sb 171 N./mm²

Note: Since Sb >= SbB8, bolt allowable stress comparison check Passes (see Div 1 UG-44(b)).

Flange rigidity based on required thickness [ASME]:



Flange Rigidity Criterion, Seating per Table 4.16.10 [Jg]:
 $= 52.14 \cdot Ma / Bsc \cdot Cnvfac \cdot V / (\lambda \cdot Eamb \cdot go^2 \cdot ho \cdot Ki)$
 $= 52.14 \cdot 38589 / 1 \cdot 9806.6 \cdot 0.3339 / (3.7907 \cdot 201334 \cdot 19^2 \cdot 121.42 \cdot 0.3)$
 $= 0.656$ (should be ≤ 1)

Flange Rigidity Criterion, Operating per Table 4.16.10 [Jo]:
 $= 52.14 \cdot Mo / Bsc \cdot Cnvfac \cdot V / (\lambda \cdot Eop \cdot goc^2 \cdot ho \cdot Ki)$
 $= 52.14 \cdot 27545 / 1 \cdot 9806.6 \cdot 0.3339 / (3.7907 \cdot 198820 \cdot 19^2 \cdot 121.42 \cdot 0.3)$
 $= 0.474$ (should be ≤ 1)

Flange rigidity based on given thickness [ASME]:

Flange Rigidity Criterion, Seating per Table 4.16.10 [Jg]:
 $= 52.14 \cdot Ma / Bsc \cdot Cnvfac \cdot V / (\lambda \cdot Eamb \cdot go^2 \cdot ho \cdot Ki)$
 $= 52.14 \cdot 38589 / 1 \cdot 9806.6 \cdot 0.3339 / (4.0553 \cdot 201334 \cdot 19^2 \cdot 121.42 \cdot 0.3)$
 $= 0.613$ (should be ≤ 1)

Flange Rigidity Criterion, Operating per Table 4.16.10 [Jo]:
 $= 52.14 \cdot Mo / Bsc \cdot Cnvfac \cdot V / (\lambda \cdot Eop \cdot goc^2 \cdot ho \cdot Ki)$
 $= 52.14 \cdot 27545 / 1 \cdot 9806.6 \cdot 0.3339 / (4.0553 \cdot 198820 \cdot 19^2 \cdot 121.42 \cdot 0.3)$
 $= 0.443$ (should be ≤ 1)

Minimum Design Metal Temperature Results:

Note:
This Material was specified as being an Impact Tested (Low Temperature) Material.

Impact Test Temperature provided per Specification -18 °C

Note: UCS-66(b)(-c) was considered in the flange MDMT calculation.

Per ASME VIII-2, 4.16.2.3, Perform Calculations in the New Condition:

Flange Stress Analysis Results: N./mm²

	Actual	Operating Allowed	Gasket Actual	Seating Allowed
Longitudinal Hub	99.29	207.00	136.67	207.00
Radial Flange	11.18	138.00	15.39	138.00
Tangential Flange	74.05	138.00	101.93	138.00
Maximum Average	86.67	138.00	119.30	138.00
Bolting	115.74	172.00	43.26	172.00

Minimum Req'd Thickness, ambient, no Corrosion: 133.121 mm

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Internal Pressure Calculations: Step: 5 9:41am Sep 9,2026

Internal Pressure Results Summary:

Element Thickness, Pressure, Diameter and Allowable Stress :

From	To	Int. Press + Liq. Hd KPa.	Nominal Thickness mm	Total Corr Allowance mm	Element Diameter mm	Allowable Stress(SE) N./mm ²	Element#
LHS Bonnet		5857.6	24	3	770	138	1
LHS ch shell		5857.6	22	3	770	138	2
LHS ch.flg		5857.5	153	3	770	138	3
main shell-1		2257.6	14	3	770	138	4
main shell-2		2257.6	14	3	770	138	5
main shell-3		2257.6	14	3	770	138	6
main shell-4		2257.6	14	3	770	138	7
RHS ch Flg		5857.6	153	3	770	138	8
RHS Ch.Shell		5857.6	22	3	770	138	9
RHS Bonnet		5857.6	24	3	770	138	10

Element Required Thickness and MAWP :

From	To	Design Pressure KPa.	M.A.W.P. Corroded KPa.	M.A.P. New & Cold KPa.	Minimum Thickness mm	Required Thickness mm	Element#
LHS Bonnet		5850	5895.6	6953.9	19.5	19.372	1
LHS ch shell		5850	6556.8	7623.7	22	19.9008	2
LHS ch.flg		5850	6556.8	7623.7	145	140.868	3
main shell-1		2250	3839.1	4910.7	14	9.4108	4
main shell-2		2250	3839.1	4910.7	14	9.4108	5
main shell-3		2250	3839.1	4910.7	14	9.4108	6
main shell-4		2250	3839.1	4910.7	14	9.4108	7
RHS ch Flg		5850	6556.7	7623.7	145	140.868	8
RHS Ch.Shell		5850	6556.8	7623.7	22	19.9008	9
RHS Bonnet		5850	5895.6	6953.9	19.5	19.372	10

Summary of Heat Exchanger Maximum Allowable Working Pressures :

Note:

For Exchanger designs, the following values include MAWPs that consider the tubesheet, tubes, tube/tubesheet joint etc. These values were determined by iteration. Review the tubesheet analysis report for more information.

Shell Side MAWP = 3839.1 KPa.
 Shell Side MAPnc = 4910.7 KPa.
 Channel Side MAWP = 5895.6 KPa.
 Channel Side MAPnc = 6953.9 KPa.

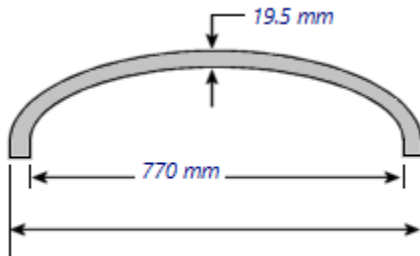
Elements Suitable for Design Internal Pressure.

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Internal Pressure Calculation Results:

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Internal Pressure Results for: LHS Bonnet

Elliptical Head Design Information:			
Design Pressure P	5857.577 KPa.	Int. Design Temperature	65 °C
Material	SA-516 70	Ext. Design Temperature	65 °C
Length	60.000 mm	External Pressure Chart	CS-2
		UNS Number	K02700
Int. Corr. All. c	3.0000 mm	Allowable Stress (ope) S	138.0 N./mm ²
Ext. Corr. All. ce	0.0000 mm	Allowable Stress (amb) Sa	138.0 N./mm ²
Inside Diameter Di	770.000 mm	Actual Stress	136.9 N./mm ²
Spec. Min. Thk t,e, or sa	19.5000 mm	Spec. Nominal Thk.	24.000 mm
Surface Area	8795.1 cm ²	SG of Contents	1.000
Element Volume	87715.5 cm ³	Weight of Contents	87.7 Kg
Empty Weight	168.2 Kg	Operating Weight	255.9 Kg
		Comp. All. Ope. Stress	143.5 N./mm ²
		Aspect Ratio	2.00

Radiography Information:			
User Defined		User Defined	
Circ. Joint Efficiency	100 %	Long. Joint Eff. E or v	100 %

Tolerance for Formed Heads per UG-81(a):

Head inner surface maximum deviation outside the specified shape, 1.25% of D: 9.625 mm

Head inner surface maximum deviation inside the specified shape, 0.625% of D: 4.812 mm

Required Thickness due to Internal Pressure [tr]:

$$= (P \cdot D \cdot K_{cor}) / (2 \cdot S \cdot E - 0.2 \cdot P) \text{ Appendix 1-4(c)}$$

$$= (5857.6 \cdot 776 \cdot 0.9898) / (2 \cdot 138 \cdot 1 - 0.2 \cdot 5857.6)$$

$$= 16.3720 + 3.0000 = 19.3720 \text{ mm}$$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

Less Operating Hydrostatic Head Pressure of 7.577 KPa.

$$= (2 \cdot S \cdot E \cdot t) / (K_{cor} \cdot D + 0.2 \cdot t) \text{ per Appendix 1-4 (c)}$$

$$= (2 \cdot 138 \cdot 1 \cdot 16.5) / (0.9898 \cdot 776 + 0.2 \cdot 16.5)$$

$$= 5903.2 - 7.577 = 5895.6 \text{ KPa.}$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$= (2 \cdot S \cdot E \cdot t) / (K \cdot D + 0.2 \cdot t) \text{ per Appendix 1-4 (c)}$$

$$= (2 \cdot 138 \cdot 1 \cdot 19.5) / (1 \cdot 770 + 0.2 \cdot 19.5)$$

$$= 6953.9 \text{ KPa.}$$



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Element: LHS Bonnet

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Actual stress at given pressure and thickness, corroded [Sact]:

$$= (P \cdot (K_{cor} \cdot D + 0.2 \cdot t)) / (2 \cdot E \cdot t)$$

$$= (5857.6 \cdot (0.9898 \cdot 776 + 0.2 \cdot 16.5)) / (2 \cdot 1 \cdot 16.5)$$

$$= 136.932 \text{ N./mm}^2$$

Straight Flange Required Thickness:

$$= (P \cdot R) / (S \cdot E - 0.6 \cdot P) + c \text{ per UG-27 (c)(1)}$$

$$= (5857.6 \cdot 388) / (138 \cdot 1 - 0.6 \cdot 5857.6) + 3$$

$$= 19.901 \text{ mm}$$

Straight Flange Maximum Allowable Working Pressure:

[Less Operating Hydrostatic Head Pressure of 7.577 KPa.](#)

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (138 \cdot 1 \cdot 21) / (388 + 0.6 \cdot 21)$$

$$= 7233.6 - 7.577 = 7226 \text{ KPa.}$$

Factor K, corroded condition [Kcor]:

$$= (2 + (\text{Inside Diameter} / (2 \cdot \text{Inside Head Depth})))^2 / 6$$

$$= (2 + (776 / (2 \cdot 195.5)))^2 / 6$$

$$= 0.989809$$

Percent Elong. per UCS-79, VIII-1-01-57 $(75 \cdot t_{nom} / R_f) \cdot (1 - R_f / R_o)$ 12.596 %

[Note: Please Check Requirements of UCS-79 as Elongation is > 5%.](#)

MDMT Calculations in the Knuckle Portion:

Govrn. thk, tg = 19.5, tr = 16.479, c = 3 mm, E* = 1

Thickness Ratio = $tr \cdot E* / (tg - c) = 0.9987$, Temp. Reduction = 0 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-8 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-8 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

MDMT Calculations in the Head Straight Flange:

Govrn. thk, tg = 24, tr = 17.013, c = 3 mm, E* = 1

Thickness Ratio = $tr \cdot E* / (tg - c) = 0.8102$, Temp. Reduction = 11 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-2 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-13 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

[Note:](#)

[Post Weld Heat Treatment is required for this Element/Joint and it was specified as being heat treated.](#)

MDMT Design Information:			
MDMT Curve	B	Press. at MDMT	5895.587 KPa.
UCS-66(b) red.	Yes	Min. Des. Mtl. Temp.	-33 °C
Governing Thk.	24.000 mm	Basic MDMT	-2 °C
Stress/Thk. Ratio	0.810	Temperature Difference	11 °C
UCS-66(c) red.	No	Computed Min. Temp.	-13 °C
		MDMT per UG-20(f)	-29 °C
		Joint Efficiency E*	100 %

[Note: Temperature values displayed above are for the location with the highest basic MDMT](#)

Tolerance for Formed Heads per UG-81(a):

Head inner surface maximum deviation outside the specified shape, 1.25% of D: 9.625 mm

Head inner surface maximum deviation inside the specified shape, 0.625% of D: 4.812 mm



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Element: LHS Bonnet

Step: 6

9:41am

Sep 9,2026

EXTERNAL PRESSURE RESULTS for: LHS Bonnet**Elliptical Head From 10 to 20 Ext. Chart: CS-2 at 65 °C**

Elastic Modulus from Chart: CS-2 at 65 °C: 199943392 KPa.

Results for Maximum Allowable Ext. Pressure				MAEP
Tca	Outer Dia	Do/t	Factor A	Factor B
16.500	809.00	49.03	0.0028327	110.87

$$MAEP = B / (K0 * Do / t) = 110.87 / (0.9 * 49.03) = 2512.4 \text{ KPa.}$$

Results for Required Thickness				Tca
Tca	Outer Dia	Do/t	Factor A	Factor B
2.095	809.00	386.22	0.0003596	35.95

$$MAEP = B / (K0 * Do / t) = 35.953 / (0.9 * 386.22) = 103.43 \text{ KPa.}$$

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FileName : 170-C-AK-Final

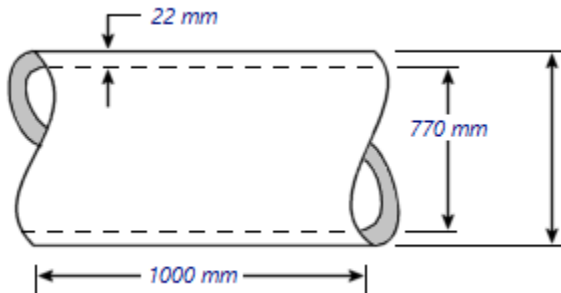
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Element: LHS ch shell

Step: 7 9:41am Sep 9,2026

Internal Pressure Calculation Results:

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Internal Pressure Results for: LHS ch shell

Cylindrical Shell Design Information:			
Design Pressure P	5857.577 KPa.	Int. Design Temperature	65 °C
Material	SA-516 70	Ext. Design Temperature	65 °C
Length	1000.000 mm	External Pressure Chart	CS-2
		UNS Number	K02700
Int. Corr. All. c	3.0000 mm	Allowable Stress (ope) S	138.0 N./mm ²
Ext. Corr. All. ce	0.0000 mm	Allowable Stress (amb) Sa	138.0 N./mm ²
Inside Diameter Di	770.000 mm	Actual Stress	123.1 N./mm ²
Spec. Min. Thk t,e, or sa	22.0000 mm	Spec. Nominal Thk.	22.000 mm
Surface Area	25572.6 cm ²	SG of Contents	1.000
Element Volume	465746.0 cm ³	Weight of Contents	465.5 Kg
Empty Weight	424.3 Kg	Operating Weight	889.7 Kg
		Comp. All. Ope. Stress	145.3 N./mm ²

Radiography Information:			
User Defined		User Defined	
Circ. Joint Efficiency	100 %	Long. Joint Eff.	E or v 100 %

Required Thickness due to Internal Pressure [tr]:

$$\begin{aligned}
 &= (P \cdot R) / (S \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)} \\
 &= (5857.6 \cdot 388) / (138 \cdot 1 - 0.6 \cdot 5857.6) \\
 &= 16.9008 + 3.0000 = 19.9008 \text{ mm}
 \end{aligned}$$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

$$\begin{aligned}
 &\text{Less Operating Hydrostatic Head Pressure of 7.577 KPa.} \\
 &= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)} \\
 &= (138 \cdot 1 \cdot 19) / (388 + 0.6 \cdot 19) \\
 &= 6564.3 - 7.577 = 6556.8 \text{ KPa.}
 \end{aligned}$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$\begin{aligned}
 &= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)} \\
 &= (138 \cdot 1 \cdot 22) / (385 + 0.6 \cdot 22) \\
 &= 7623.7 \text{ KPa.}
 \end{aligned}$$

Actual stress at given pressure and thickness, corroded [Sact]:

$$\begin{aligned}
 &= (P \cdot (R + 0.6 \cdot t)) / (E \cdot t) \\
 &= (5857.6 \cdot (388 + 0.6 \cdot 19)) / (1 \cdot 19) \\
 &= 123.140 \text{ N./mm}^2
 \end{aligned}$$



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 Element: LHS ch shell Step: 7 9:41am Sep 9,2026

% Elongation per Table UG-79-1 (50*tnom/Rf*(1-Rf/Ro)) 2.778 %

Minimum Design Metal Temperature Results:

Govrn. thk, tg = 22, tr = 17.013, c = 3 mm, E* = 1
 Thickness Ratio = tr * E*/(tg - c) = 0.8954, Temp. Reduction = 6 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-5 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-10 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Note:
 Post Weld Heat Treatment is required for this Element/Joint and it was specified as being heat treated.

MDMT Design Information:			
MDMT Curve	B	Press. at MDMT	5895.587 KPa.
UCS-66(b) red.	Yes	Min. Des. Mtl. Temp.	-33 °C
Governing Thk.	22.000 mm	Basic MDMT	-5 °C
Stress/Thk. Ratio	0.895	Temperature Difference	6 °C
UCS-66(c) red.	No	Computed Min. Temp.	-10 °C
		MDMT per UG-20(f)	-29 °C
		Joint Efficiency E*	100 %

EXTERNAL PRESSURE RESULTS for: LHS ch shell

Cylindrical Shell From 20 to 30 Ext. Chart: CS-2 at 65 °C

Elastic Modulus from Chart: CS-2 at 65 °C: 199943392 KPa.

Results for Maximum Allowable Ext. Pressure				MAEP		
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
19.000	814.00	1313.17	42.84	1.6132	0.0029052	111.33

$$MAEP = (4*B)/(3*(Do/t)) = (4*111.33)/(3*42.842) = 3464.7 \text{ KPa.}$$

Results for Required Thickness				Tca		
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
3.189	814.00	1313.17	255.27	1.6132	0.0001981	19.80

$$MAEP = (4*B)/(3*(Do/t)) = (4*19.802)/(3*255.27) = 103.42 \text{ KPa.}$$

Results for Maximum Stiffened Length				Slen		
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
19.000	814.00	33216.03	42.84	40.8059	0.0006086	60.84

$$MAEP = (4*B)/(3*(Do/t)) = (4*60.843)/(3*42.842) = 1893.4 \text{ KPa.}$$



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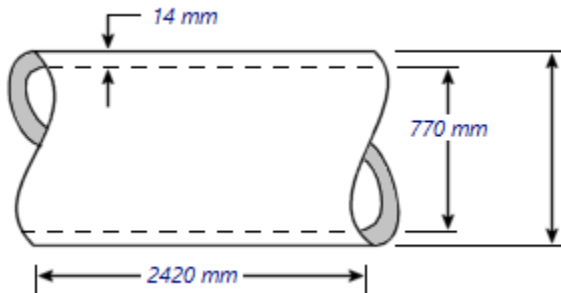
Element: main shell-1

Step: 8 9:41am

Sep 9,2026

Internal Pressure Calculation Results:

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Internal Pressure Results for: main shell-1

Cylindrical Shell Design Information:			
Design Pressure P	2257.577 KPa.	Int. Design Temperature	65 °C
Material	SA-516 70	Ext. Design Temperature	65 °C
Length	2420.000 mm	External Pressure Chart	CS-2
		UNS Number	K02700
Int. Corr. All. c	3.0000 mm	Allowable Stress (ope) S	138.0 N./mm ²
Ext. Corr. All. ce	0.0000 mm	Allowable Stress (amb) Sa	138.0 N./mm ²
Inside Diameter Di	770.000 mm	Actual Stress	81.0 N./mm ²
Spec. Min. Thk t,e, or sa	14.0000 mm	Spec. Nominal Thk.	14.000 mm
Surface Area	60669.2 cm ²	SG of Contents	1.000
Element Volume	781245.9 cm ³	Weight of Contents	780.8 Kg
Empty Weight	646.8 Kg	Operating Weight	1427.5 Kg
		Comp. All. Ope. Stress	137.1 N./mm ²

Radiography Information:			
User Defined		User Defined	
Circ. Joint Efficiency	100 %	Long. Joint Eff. E or v	100 %

Required Thickness due to Internal Pressure [tr]:

$$\begin{aligned}
 &= (P \cdot R) / (S \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)} \\
 &= (2257.6 \cdot 388) / (138 \cdot 1 - 0.6 \cdot 2257.6) \\
 &= 6.4108 + 3.0000 = 9.4108 \text{ mm}
 \end{aligned}$$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

$$\begin{aligned}
 &\text{Less Operating Hydrostatic Head Pressure of 7.577 KPa.} \\
 &= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)} \\
 &= (138 \cdot 1 \cdot 11) / (388 + 0.6 \cdot 11) \\
 &= 3846.6 - 7.577 = 3839.1 \text{ KPa.}
 \end{aligned}$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$\begin{aligned}
 &= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)} \\
 &= (138 \cdot 1 \cdot 14) / (385 + 0.6 \cdot 14) \\
 &= 4910.7 \text{ KPa.}
 \end{aligned}$$

Actual stress at given pressure and thickness, corroded [Sact]:

$$\begin{aligned}
 &= (P \cdot (R + 0.6 \cdot t)) / (E \cdot t) \\
 &= (2257.6 \cdot (388 + 0.6 \cdot 11)) / (1 \cdot 11) \\
 &= 80.990 \text{ N./mm}^2
 \end{aligned}$$



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Element: main shell-1

Step: 8

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% Elongation per Table UG-79-1 ($50 \cdot t_{nom} / R_f \cdot (1 - R_f / R_o)$) 1.786 %

Minimum Design Metal Temperature Results:

Govrn. thk, $t_g = 14$, $t_r = 10.978$, $c = 3$ mm, $E^* = 1$

Thickness Ratio = $t_r \cdot E^* / (t_g - c) = 0.998$, Temp. Reduction = 0 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -18 °C

Min Metal Temp. at Required thickness (UCS 66.1) -18 °C

Min Metal Temp. w/o impact per UG-20(f) -29 °C

Note:

Post Weld Heat Treatment is required for this Element/Joint and it was specified as being heat treated.

MDMT Design Information:			
MDMT Curve	B	Press. at MDMT	3839.064 KPa.
UCS-66(b) red.	Yes	Min. Des. Mtl. Temp.	-33 °C
Governing Thk.	14.000 mm	Basic MDMT	-18 °C
Stress/Thk. Ratio	0.998	Temperature Difference	0 °C
UCS-66(c) red.	No	Computed Min. Temp.	-18 °C
		MDMT per UG-20(f)	-29 °C
		Joint Efficiency E^*	100 %

EXTERNAL PRESSURE RESULTS for: main shell-1

Cylindrical Shell From 40 to 50 Ext. Chart: CS-2 at 65 °C

Elastic Modulus from Chart: CS-2 at 65 °C: 199943392 KPa.

Results for Maximum Allowable Ext. Pressure				MAEP		
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
11.000	798.00	9680.00	72.55	12.1303	0.0002242	22.41

$$MAEP = (4 \cdot B) / (3 \cdot (Do/t)) = (4 \cdot 22.413) / (3 \cdot 72.545) = 411.9 \text{ KPa.}$$

Results for Required Thickness				Tca		
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
6.874	798.00	9680.00	116.08	12.1303	0.0000901	9.00

$$MAEP = (4 \cdot B) / (3 \cdot (Do/t)) = (4 \cdot 9.0045) / (3 \cdot 116.08) = 103.42 \text{ KPa.}$$

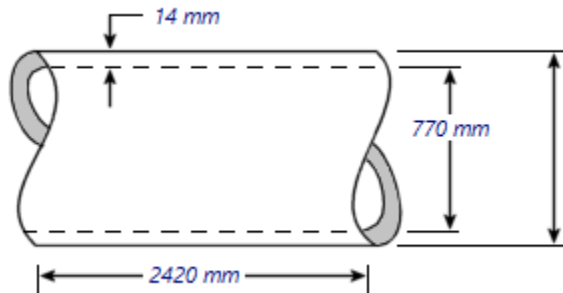
Results for Maximum Stiffened Length				Slen		
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
11.000	798.00	284305.03	72.55	50.0000	0.0002156	21.55

$$MAEP = (4 \cdot B) / (3 \cdot (Do/t)) = (4 \cdot 21.553) / (3 \cdot 72.545) = 396.11 \text{ KPa.}$$



Internal Pressure Calculation Results:

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Internal Pressure Results for: main shell-2

Cylindrical Shell Design Information:			
Design Pressure P	2257.577 KPa.	Int. Design Temperature	65 °C
Material	SA-516 70	Ext. Design Temperature	65 °C
Length	2420.000 mm	External Pressure Chart	CS-2
		UNS Number	K02700
Int. Corr. All. c	3.0000 mm	Allowable Stress (ope) S	138.0 N./mm ²
Ext. Corr. All. ce	0.0000 mm	Allowable Stress (amb) Sa	138.0 N./mm ²
Inside Diameter Di	770.000 mm	Actual Stress	81.0 N./mm ²
Spec. Min. Thk t,e, or sa	14.0000 mm	Spec. Nominal Thk.	14.000 mm
Surface Area	60669.2 cm ²	SG of Contents	1.000
Element Volume	781246.0 cm ³	Weight of Contents	780.8 Kg
Empty Weight	646.8 Kg	Operating Weight	1427.5 Kg
		Comp. All. Ope. Stress	137.1 N./mm ²
Radiography Information:			
User Defined		User Defined	
Circ. Joint Efficiency	100 %	Long. Joint Eff. E or v	100 %

Required Thickness due to Internal Pressure [tr]:

$$\begin{aligned}
 &= (P \cdot R) / (S \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)} \\
 &= (2257.6 \cdot 388) / (138 \cdot 1 - 0.6 \cdot 2257.6) \\
 &= 6.4108 + 3.0000 = 9.4108 \text{ mm}
 \end{aligned}$$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

Less Operating Hydrostatic Head Pressure of 7.577 KPa.

$$\begin{aligned}
 &= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)} \\
 &= (138 \cdot 1 \cdot 11) / (388 + 0.6 \cdot 11) \\
 &= 3846.6 - 7.577 = 3839.1 \text{ KPa.}
 \end{aligned}$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$\begin{aligned}
 &= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)} \\
 &= (138 \cdot 1 \cdot 14) / (385 + 0.6 \cdot 14) \\
 &= 4910.7 \text{ KPa.}
 \end{aligned}$$

Actual stress at given pressure and thickness, corroded [Sact]:

$$\begin{aligned}
 &= (P \cdot (R + 0.6 \cdot t)) / (E \cdot t) \\
 &= (2257.6 \cdot (388 + 0.6 \cdot 11)) / (1 \cdot 11) \\
 &= 80.990 \text{ N./mm}^2
 \end{aligned}$$



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Element: main shell-2

Step: 9

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% Elongation per Table UG-79-1 ($50 \cdot t_{nom} / R_f \cdot (1 - R_f / R_o)$) 1.786 %

Minimum Design Metal Temperature Results:

Govrn. thk, $t_g = 14$, $t_r = 10.978$, $c = 3$ mm, $E^* = 1$

Thickness Ratio = $t_r \cdot E^* / (t_g - c) = 0.998$, Temp. Reduction = 0 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -18 °C

Min Metal Temp. at Required thickness (UCS 66.1) -18 °C

Min Metal Temp. w/o impact per UG-20(f) -29 °C

Note:

Post Weld Heat Treatment is required for this Element/Joint and it was specified as being heat treated.

MDMT Design Information:			
MDMT Curve	B	Press. at MDMT	3839.064 KPa.
UCS-66(b) red.	Yes	Min. Des. Mtl. Temp.	-33 °C
Governing Thk.	14.000 mm	Basic MDMT	-18 °C
Stress/Thk. Ratio	0.998	Temperature Difference	0 °C
UCS-66(c) red.	No	Computed Min. Temp.	-18 °C
		MDMT per UG-20(f)	-29 °C
		Joint Efficiency E^*	100 %

EXTERNAL PRESSURE RESULTS for: main shell-2

Cylindrical Shell From 50 to 60 Ext. Chart: CS-2 at 65 °C

Elastic Modulus from Chart: CS-2 at 65 °C: 199943392 KPa.

Results for Maximum Allowable Ext. Pressure				MAEP		
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
11.000	798.00	9680.00	72.55	12.1303	0.0002242	22.41

$$MAEP = (4 \cdot B) / (3 \cdot (Do/t)) = (4 \cdot 22.413) / (3 \cdot 72.545) = 411.9 \text{ KPa.}$$

Results for Required Thickness				Tca		
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
6.874	798.00	9680.00	116.08	12.1303	0.0000901	9.00

$$MAEP = (4 \cdot B) / (3 \cdot (Do/t)) = (4 \cdot 9.0045) / (3 \cdot 116.08) = 103.42 \text{ KPa.}$$

Results for Maximum Stiffened Length				Slen		
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
11.000	798.00	284305.03	72.55	50.0000	0.0002156	21.55

$$MAEP = (4 \cdot B) / (3 \cdot (Do/t)) = (4 \cdot 21.553) / (3 \cdot 72.545) = 396.11 \text{ KPa.}$$



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Element: main shell-3

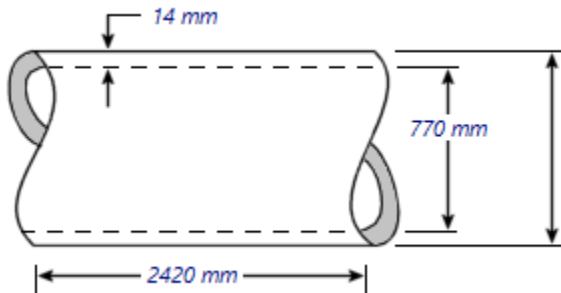
Step: 10

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Internal Pressure Calculation Results:

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Internal Pressure Results for: main shell-3

Cylindrical Shell Design Information:			
Design Pressure P	2257.577 KPa.	Int. Design Temperature	65 °C
Material	SA-516 70	Ext. Design Temperature	65 °C
Length	2420.000 mm	External Pressure Chart	CS-2
		UNS Number	K02700
Int. Corr. All. c	3.0000 mm	Allowable Stress (ope) S	138.0 N./mm ²
Ext. Corr. All. ce	0.0000 mm	Allowable Stress (amb) Sa	138.0 N./mm ²
Inside Diameter Di	770.000 mm	Actual Stress	81.0 N./mm ²
Spec. Min. Thk t,e, or sa	14.0000 mm	Spec. Nominal Thk.	14.000 mm
Surface Area	60669.2 cm ²	SG of Contents	1.000
Element Volume	781246.0 cm ³	Weight of Contents	780.8 Kg
Empty Weight	646.8 Kg	Operating Weight	1427.5 Kg
		Comp. All. Ope. Stress	137.1 N./mm ²

Radiography Information:			
User Defined		User Defined	
Circ. Joint Efficiency	100 %	Long. Joint Eff.	E or v 100 %

Required Thickness due to Internal Pressure [tr]:

$$\begin{aligned}
 &= (P \cdot R) / (S \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)} \\
 &= (2257.6 \cdot 388) / (138 \cdot 1 - 0.6 \cdot 2257.6) \\
 &= 6.4108 + 3.0000 = 9.4108 \text{ mm}
 \end{aligned}$$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

Less Operating Hydrostatic Head Pressure of 7.577 KPa.

$$\begin{aligned}
 &= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)} \\
 &= (138 \cdot 1 \cdot 11) / (388 + 0.6 \cdot 11) \\
 &= 3846.6 - 7.577 = 3839.1 \text{ KPa.}
 \end{aligned}$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$\begin{aligned}
 &= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)} \\
 &= (138 \cdot 1 \cdot 14) / (385 + 0.6 \cdot 14) \\
 &= 4910.7 \text{ KPa.}
 \end{aligned}$$

Actual stress at given pressure and thickness, corroded [Sact]:

$$\begin{aligned}
 &= (P \cdot (R + 0.6 \cdot t)) / (E \cdot t) \\
 &= (2257.6 \cdot (388 + 0.6 \cdot 11)) / (1 \cdot 11) \\
 &= 80.990 \text{ N./mm}^2
 \end{aligned}$$



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Element: main shell-3

Step: 10

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% Elongation per Table UG-79-1 ($50 \cdot t_{nom} / R_f \cdot (1 - R_f / R_o)$) 1.786 %

Minimum Design Metal Temperature Results:

Govrn. thk, $t_g = 14$, $t_r = 10.978$, $c = 3$ mm, $E^* = 1$

Thickness Ratio = $t_r \cdot E^* / (t_g - c) = 0.998$, Temp. Reduction = 0 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -18 °C

Min Metal Temp. at Required thickness (UCS 66.1) -18 °C

Min Metal Temp. w/o impact per UG-20(f) -29 °C

Note:

Post Weld Heat Treatment is required for this Element/Joint and it was specified as being heat treated.

MDMT Design Information:			
MDMT Curve	B	Press. at MDMT	3839.064 KPa.
UCS-66(b) red.	Yes	Min. Des. Mtl. Temp.	-33 °C
Governing Thk.	14.000 mm	Basic MDMT	-18 °C
Stress/Thk. Ratio	0.998	Temperature Difference	0 °C
UCS-66(c) red.	No	Computed Min. Temp.	-18 °C
		MDMT per UG-20(f)	-29 °C
		Joint Efficiency E^*	100 %

EXTERNAL PRESSURE RESULTS for: main shell-3

Cylindrical Shell From 60 to 70 Ext. Chart: CS-2 at 65 °C

Elastic Modulus from Chart: CS-2 at 65 °C: 199943392 KPa.

Results for Maximum Allowable Ext. Pressure				MAEP		
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
11.000	798.00	9680.00	72.55	12.1303	0.0002242	22.41

$$MAEP = (4 \cdot B) / (3 \cdot (Do/t)) = (4 \cdot 22.413) / (3 \cdot 72.545) = 411.9 \text{ KPa.}$$

Results for Required Thickness				Tca		
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
6.874	798.00	9680.00	116.08	12.1303	0.0000901	9.00

$$MAEP = (4 \cdot B) / (3 \cdot (Do/t)) = (4 \cdot 9.0045) / (3 \cdot 116.08) = 103.42 \text{ KPa.}$$

Results for Maximum Stiffened Length				Slen		
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
11.000	798.00	284305.03	72.55	50.0000	0.0002156	21.55

$$MAEP = (4 \cdot B) / (3 \cdot (Do/t)) = (4 \cdot 21.553) / (3 \cdot 72.545) = 396.11 \text{ KPa.}$$



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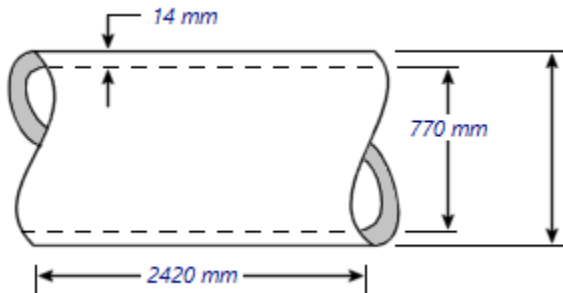
Element: main shell-4

Step: 11 9:41am

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Internal Pressure Calculation Results:

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Internal Pressure Results for: main shell-4

Cylindrical Shell Design Information:			
Design Pressure P	2257.577 KPa.	Int. Design Temperature	65 °C
Material	SA-516 70	Ext. Design Temperature	65 °C
Length	2420.000 mm	External Pressure Chart	CS-2
		UNS Number	K02700
Int. Corr. All. c	3.0000 mm	Allowable Stress (ope) S	138.0 N./mm ²
Ext. Corr. All. ce	0.0000 mm	Allowable Stress (amb) Sa	138.0 N./mm ²
Inside Diameter Di	770.000 mm	Actual Stress	81.0 N./mm ²
Spec. Min. Thk t,e, or sa	14.0000 mm	Spec. Nominal Thk.	14.000 mm
Surface Area	60669.2 cm ²	SG of Contents	1.000
Element Volume	781246.0 cm ³	Weight of Contents	780.8 Kg
Empty Weight	646.8 Kg	Operating Weight	1427.5 Kg
		Comp. All. Ope. Stress	137.1 N./mm ²

Radiography Information:			
User Defined		User Defined	
Circ. Joint Efficiency	100 %	Long. Joint Eff.	E or v 100 %

Required Thickness due to Internal Pressure [tr]:

$$\begin{aligned}
 &= (P \cdot R) / (S \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)} \\
 &= (2257.6 \cdot 388) / (138 \cdot 1 - 0.6 \cdot 2257.6) \\
 &= 6.4108 + 3.0000 = 9.4108 \text{ mm}
 \end{aligned}$$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

Less Operating Hydrostatic Head Pressure of 7.577 KPa.

$$\begin{aligned}
 &= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)} \\
 &= (138 \cdot 1 \cdot 11) / (388 + 0.6 \cdot 11) \\
 &= 3846.6 - 7.577 = 3839.1 \text{ KPa.}
 \end{aligned}$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$\begin{aligned}
 &= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)} \\
 &= (138 \cdot 1 \cdot 14) / (385 + 0.6 \cdot 14) \\
 &= 4910.7 \text{ KPa.}
 \end{aligned}$$

Actual stress at given pressure and thickness, corroded [Sact]:

$$\begin{aligned}
 &= (P \cdot (R + 0.6 \cdot t)) / (E \cdot t) \\
 &= (2257.6 \cdot (388 + 0.6 \cdot 11)) / (1 \cdot 11) \\
 &= 80.990 \text{ N./mm}^2
 \end{aligned}$$



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Element: main shell-4

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% Elongation per Table UG-79-1 ($50 \cdot t_{nom} / R_f \cdot (1 - R_f / R_o)$) 1.786 %

Minimum Design Metal Temperature Results:

Govrn. thk, $t_g = 14$, $t_r = 10.978$, $c = 3$ mm, $E^* = 1$

Thickness Ratio = $t_r \cdot E^* / (t_g - c) = 0.998$, Temp. Reduction = 0 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -18 °C

Min Metal Temp. at Required thickness (UCS 66.1) -18 °C

Min Metal Temp. w/o impact per UG-20(f) -29 °C

Note:

Post Weld Heat Treatment is required for this Element/Joint and it was specified as being heat treated.

MDMT Design Information:			
MDMT Curve	B	Press. at MDMT	3839.064 KPa.
UCS-66(b) red.	Yes	Min. Des. Mtl. Temp.	-33 °C
Governing Thk.	14.000 mm	Basic MDMT	-18 °C
Stress/Thk. Ratio	0.998	Temperature Difference	0 °C
UCS-66(c) red.	No	Computed Min. Temp.	-18 °C
		MDMT per UG-20(f)	-29 °C
		Joint Efficiency E^*	100 %

EXTERNAL PRESSURE RESULTS for: main shell-4

Cylindrical Shell From 70 to 80 Ext. Chart: CS-2 at 65 °C

Elastic Modulus from Chart: CS-2 at 65 °C: 199943392 KPa.

Results for Maximum Allowable Ext. Pressure				MAEP		
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
11.000	798.00	9680.00	72.55	12.1303	0.0002242	22.41

$$MAEP = (4 \cdot B) / (3 \cdot (Do/t)) = (4 \cdot 22.413) / (3 \cdot 72.545) = 411.9 \text{ KPa.}$$

Results for Required Thickness				Tca		
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
6.874	798.00	9680.00	116.08	12.1303	0.0000901	9.00

$$MAEP = (4 \cdot B) / (3 \cdot (Do/t)) = (4 \cdot 9.0045) / (3 \cdot 116.08) = 103.42 \text{ KPa.}$$

Results for Maximum Stiffened Length				Slen		
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
11.000	798.00	284305.03	72.55	50.0000	0.0002156	21.55

$$MAEP = (4 \cdot B) / (3 \cdot (Do/t)) = (4 \cdot 21.553) / (3 \cdot 72.545) = 396.11 \text{ KPa.}$$



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Element: RHS Ch.Shell

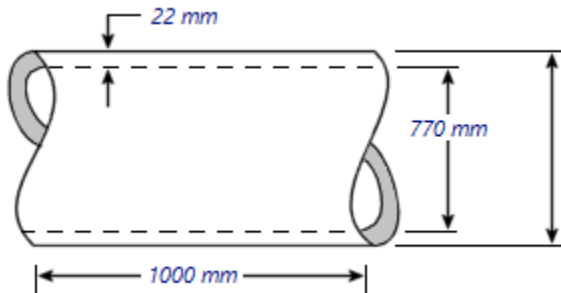
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Internal Pressure Calculation Results:

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Internal Pressure Results for: RHS Ch.Shell

Cylindrical Shell Design Information:			
Design Pressure P	5857.577 KPa.	Int. Design Temperature	65 °C
Material	SA-516 70	Ext. Design Temperature	65 °C
Length	1000.000 mm	External Pressure Chart	CS-2
		UNS Number	K02700
Int. Corr. All. c	3.0000 mm	Allowable Stress (ope) S	138.0 N./mm ²
Ext. Corr. All. ce	0.0000 mm	Allowable Stress (amb) Sa	138.0 N./mm ²
Inside Diameter Di	770.000 mm	Actual Stress	123.1 N./mm ²
Spec. Min. Thk t,e, or sa	22.0000 mm	Spec. Nominal Thk.	22.000 mm
Surface Area	25572.6 cm ²	SG of Contents	1.000
Element Volume	465746.0 cm ³	Weight of Contents	465.5 Kg
Empty Weight	424.3 Kg	Operating Weight	889.7 Kg
		Comp. All. Ope. Stress	145.3 N./mm ²

Radiography Information:			
User Defined		User Defined	
Circ. Joint Efficiency	100 %	Long. Joint Eff.	E or v 100 %

Required Thickness due to Internal Pressure [tr]:

$$\begin{aligned}
 &= (P \cdot R) / (S \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)} \\
 &= (5857.6 \cdot 388) / (138 \cdot 1 - 0.6 \cdot 5857.6) \\
 &= 16.9008 + 3.0000 = 19.9008 \text{ mm}
 \end{aligned}$$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

Less Operating Hydrostatic Head Pressure of 7.577 KPa.

$$\begin{aligned}
 &= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)} \\
 &= (138 \cdot 1 \cdot 19) / (388 + 0.6 \cdot 19) \\
 &= 6564.3 - 7.577 = 6556.8 \text{ KPa.}
 \end{aligned}$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$\begin{aligned}
 &= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)} \\
 &= (138 \cdot 1 \cdot 22) / (385 + 0.6 \cdot 22) \\
 &= 7623.7 \text{ KPa.}
 \end{aligned}$$

Actual stress at given pressure and thickness, corroded [Sact]:

$$\begin{aligned}
 &= (P \cdot (R + 0.6 \cdot t)) / (E \cdot t) \\
 &= (5857.6 \cdot (388 + 0.6 \cdot 19)) / (1 \cdot 19) \\
 &= 123.140 \text{ N./mm}^2
 \end{aligned}$$



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Element: RHS Ch.Shell

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% Elongation per Table UG-79-1 ($50 \cdot t_{nom} / R_f \cdot (1 - R_f / R_o)$) 2.778 %

Minimum Design Metal Temperature Results:

Govrn. thk, $t_g = 22$, $t_r = 17.013$, $c = 3$ mm, $E^* = 1$

Thickness Ratio = $t_r \cdot E^* / (t_g - c) = 0.8954$, Temp. Reduction = 6 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -5 °C

Min Metal Temp. at Required thickness (UCS 66.1) -10 °C

Min Metal Temp. w/o impact per UG-20(f) -29 °C

Note:

Post Weld Heat Treatment is required for this Element/Joint and it was specified as being heat treated.

MDMT Design Information:			
MDMT Curve	B	Press. at MDMT	5895.587 KPa.
UCS-66(b) red.	Yes	Min. Des. Mtl. Temp.	-33 °C
Governing Thk.	22.000 mm	Basic MDMT	-5 °C
Stress/Thk. Ratio	0.895	Temperature Difference	6 °C
UCS-66(c) red.	No	Computed Min. Temp.	-10 °C
		MDMT per UG-20(f)	-29 °C
		Joint Efficiency E^*	100 %

EXTERNAL PRESSURE RESULTS for: RHS Ch.Shell

Cylindrical Shell From 90 to 100 Ext. Chart: CS-2 at 65 °C

Elastic Modulus from Chart: CS-2 at 65 °C: 199943392 KPa.

Results for Maximum Allowable Ext. Pressure				MAEP		
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
19.000	814.00	1313.17	42.84	1.6132	0.0029052	111.33

$$MAEP = (4 \cdot B) / (3 \cdot (Do/t)) = (4 \cdot 111.33) / (3 \cdot 42.842) = 3464.7 \text{ KPa.}$$

Results for Required Thickness				Tca		
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
3.189	814.00	1313.17	255.27	1.6132	0.0001981	19.80

$$MAEP = (4 \cdot B) / (3 \cdot (Do/t)) = (4 \cdot 19.802) / (3 \cdot 255.27) = 103.42 \text{ KPa.}$$

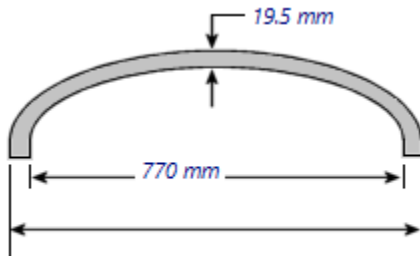
Results for Maximum Stiffened Length				Slen		
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
19.000	814.00	33216.03	42.84	40.8059	0.0006086	60.84

$$MAEP = (4 \cdot B) / (3 \cdot (Do/t)) = (4 \cdot 60.843) / (3 \cdot 42.842) = 1893.4 \text{ KPa.}$$



Internal Pressure Calculation Results:

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Internal Pressure Results for: RHS Bonnet

Elliptical Head Design Information:			
Design Pressure P	5857.577 KPa.	Int. Design Temperature	65 °C
Material	SA-516 70	Ext. Design Temperature	65 °C
Length	60.000 mm	External Pressure Chart	CS-2
		UNS Number	K02700
Int. Corr. All. c	3.0000 mm	Allowable Stress (ope) S	138.0 N./mm ²
Ext. Corr. All. ce	0.0000 mm	Allowable Stress (amb) Sa	138.0 N./mm ²
Inside Diameter Di	770.000 mm	Actual Stress	136.9 N./mm ²
Spec. Min. Thk t,e, or sa	19.5000 mm	Spec. Nominal Thk.	24.000 mm
Surface Area	8795.1 cm ²	SG of Contents	1.000
Element Volume	87715.5 cm ³	Weight of Contents	87.7 Kg
Empty Weight	168.2 Kg	Operating Weight	255.9 Kg
		Comp. All. Ope. Stress	145.3 N./mm ²
		Aspect Ratio	2.00

Radiography Information:			
User Defined		User Defined	
Circ. Joint Efficiency	100 %	Long. Joint Eff. E or v	100 %

Tolerance for Formed Heads per UG-81(a):

Head inner surface maximum deviation outside the specified shape, 1.25% of D: 9.625 mm

Head inner surface maximum deviation inside the specified shape, 0.625% of D: 4.812 mm

Required Thickness due to Internal Pressure [tr]:

$$= (P \cdot D \cdot K_{cor}) / (2 \cdot S \cdot E - 0.2 \cdot P) \text{ Appendix 1-4(c)}$$

$$= (5857.6 \cdot 776 \cdot 0.9898) / (2 \cdot 138 \cdot 1 - 0.2 \cdot 5857.6)$$

$$= 16.3720 + 3.0000 = 19.3720 \text{ mm}$$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

Less Operating Hydrostatic Head Pressure of 7.577 KPa.

$$= (2 \cdot S \cdot E \cdot t) / (K_{cor} \cdot D + 0.2 \cdot t) \text{ per Appendix 1-4 (c)}$$

$$= (2 \cdot 138 \cdot 1 \cdot 16.5) / (0.9898 \cdot 776 + 0.2 \cdot 16.5)$$

$$= 5903.2 - 7.577 = 5895.6 \text{ KPa.}$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$= (2 \cdot S \cdot E \cdot t) / (K \cdot D + 0.2 \cdot t) \text{ per Appendix 1-4 (c)}$$

$$= (2 \cdot 138 \cdot 1 \cdot 19.5) / (1 \cdot 770 + 0.2 \cdot 19.5)$$

$$= 6953.9 \text{ KPa.}$$



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Element: RHS Bonnet

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Actual stress at given pressure and thickness, corroded [Sact]:

$$= (P \cdot (K_{cor} \cdot D + 0.2 \cdot t)) / (2 \cdot E \cdot t)$$

$$= (5857.6 \cdot (0.9898 \cdot 776 + 0.2 \cdot 16.5)) / (2 \cdot 1 \cdot 16.5)$$

$$= 136.932 \text{ N./mm}^2$$

Straight Flange Required Thickness:

$$= (P \cdot R) / (S \cdot E - 0.6 \cdot P) + c \text{ per UG-27 (c)(1)}$$

$$= (5857.6 \cdot 388) / (138 \cdot 1 - 0.6 \cdot 5857.6) + 3$$

$$= 19.901 \text{ mm}$$

Straight Flange Maximum Allowable Working Pressure:

[Less Operating Hydrostatic Head Pressure of 7.577 KPa.](#)

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (138 \cdot 1 \cdot 21) / (388 + 0.6 \cdot 21)$$

$$= 7233.6 - 7.577 = 7226 \text{ KPa.}$$

Factor K, corroded condition [Kcor]:

$$= (2 + (\text{Inside Diameter} / (2 \cdot \text{Inside Head Depth})))^2 / 6$$

$$= (2 + (776 / (2 \cdot 195.5)))^2 / 6$$

$$= 0.989809$$

Percent Elong. per UCS-79, VIII-1-01-57 $(75 \cdot t_{nom} / R_f) \cdot (1 - R_f / R_o)$ 12.596 %

[Note: Please Check Requirements of UCS-79 as Elongation is > 5%.](#)

MDMT Calculations in the Knuckle Portion:

Govrn. thk, tg = 19.5, tr = 16.479, c = 3 mm, E* = 1

Thickness Ratio = $tr \cdot E^* / (tg - c) = 0.9987$, Temp. Reduction = 0 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-8 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-8 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

MDMT Calculations in the Head Straight Flange:

Govrn. thk, tg = 24, tr = 17.013, c = 3 mm, E* = 1

Thickness Ratio = $tr \cdot E^* / (tg - c) = 0.8102$, Temp. Reduction = 11 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-2 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-13 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

[Note:](#)

[Post Weld Heat Treatment is required for this Element/Joint and it was specified as being heat treated.](#)

MDMT Design Information:			
MDMT Curve	B	Press. at MDMT	5895.587 KPa.
UCS-66(b) red.	Yes	Min. Des. Mtl. Temp.	-33 °C
Governing Thk.	24.000 mm	Basic MDMT	-2 °C
Stress/Thk. Ratio	0.810	Temperature Difference	11 °C
UCS-66(c) red.	No	Computed Min. Temp.	-13 °C
		MDMT per UG-20(f)	-29 °C
		Joint Efficiency E*	100 %

[Note: Temperature values displayed above are for the location with the highest basic MDMT](#)

Tolerance for Formed Heads per UG-81(a):

Head inner surface maximum deviation outside the specified shape, 1.25% of D: 9.625 mm

Head inner surface maximum deviation inside the specified shape, 0.625% of D: 4.812 mm



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Element: RHS Bonnet

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EXTERNAL PRESSURE RESULTS for: RHS Bonnet

Elliptical Head From 100 to 110 Ext. Chart: CS-2 at 65 °C

Elastic Modulus from Chart: CS-2 at 65 °C: 199943392 KPa.

Results for Maximum Allowable Ext. Pressure				MAEP
Tca	Outer Dia	Do/t	Factor A	Factor B
16.500	809.00	49.03	0.0028327	110.87

$$MAEP = B / (K0 * Do / t) = 110.87 / (0.9 * 49.03) = 2512.4 \text{ KPa.}$$

Results for Required Thickness				Tca
Tca	Outer Dia	Do/t	Factor A	Factor B
2.095	809.00	386.22	0.0003596	35.95

$$MAEP = B / (K0 * Do / t) = 35.953 / (0.9 * 386.22) = 103.43 \text{ KPa.}$$

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Shop Pressure Test Results:

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Hydrostatic Test Pressure Results:

Exchanger Shell Side Hydrostatic Test Pressures:

UG99(b)	= 1.30 * M.A.W.P. * Sa/S	4990.783 KPa.
UG99b(b)(35)	= 1.30 * Design Pres * Sa/S	2925.000 KPa.
UG99(c)	= 1.30 * M.A.P. - Head(Hyd)	6376.309 KPa.
UG100	= 1.10 * M.A.W.P. * Sa/S	4222.970 KPa.
PED	= max(1.43*DP, 1.25*DP*ratio)	3217.500 KPa.
PED&UG99(b)(35)	= max(1.43*DP, 1.3*DP*ratio)	3217.500 KPa.
App 27-4	= M.A.W.P.	3839.064 KPa.

Exchanger Channel Side Hydrostatic Test Pressures:

UG99(b)	= 1.30 * M.A.W.P. * Sa/S	7664.264 KPa.
UG99b(b)(35)	= 1.30 * Design Pres * Sa/S	7605.000 KPa.
UG99(c)	= 1.30 * M.A.P. - Head(Hyd)	9032.466 KPa.
UG100	= 1.10 * M.A.W.P. * Sa/S	6485.146 KPa.
PED	= max(1.43*DP, 1.25*DP*ratio)	8365.500 KPa.
PED&UG99(b)(35)	= max(1.43*DP, 1.3*DP*ratio)	8365.500 KPa.
App 27-4	= M.A.W.P.	5895.587 KPa.

Horizontal Test performed per: UG-99c

Please note that Nozzle, Shell, Head, Flange, etc MAWPs are all considered when determining the hydrotest pressure for those test types that are based on the MAWP of the vessel.

Stresses on Elements due to Test Pressure (N./mm² & KPa.):

From To	Stress	Allowable	Ratio	Pressure
LHS Bonnet	211.3	234.0	0.903	9040.01
LHS ch shell	190.0	234.0	0.812	9040.01
main shell-1	229.0	234.0	0.979	6383.86
main shell-2	229.0	234.0	0.979	6383.86
main shell-3	229.0	234.0	0.979	6383.86
main shell-4	229.0	234.0	0.979	6383.86
RHS Ch.Shell	190.0	234.0	0.812	9040.01
RHS Bonnet	211.3	234.0	0.903	9040.01

Stress ratios for Nozzle and Pad Materials (N./mm²):

Description	Pad/Nozzle	Ambient	Operating	Ratio
N1-Tube Inlet	Nozzle	138.00	138.00	1.000
N2-Tube outlet	Nozzle	138.00	138.00	1.000
N3-Shell inlet	Nozzle	138.00	138.00	1.000
N4-Shell outlet	Nozzle	138.00	138.00	1.000

Minimum

1.000

Stress ratios for Pressurized Vessel Elements (N./mm²):

Description	Ambient	Operating	Ratio
LHS Bonnet	138.00	138.00	1.000
LHS ch shell	138.00	138.00	1.000
LHS ch.flg	138.00	138.00	1.000
main shell-1	138.00	138.00	1.000
main shell-2	138.00	138.00	1.000
main shell-3	138.00	138.00	1.000
main shell-4	138.00	138.00	1.000



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Shop Pressure Test Results:

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RHS ch Flg	138.00	138.00	1.000
RHS Ch.Shell	138.00	138.00	1.000
RHS Bonnet	138.00	138.00	1.000

Minimum

1.000

Stress ratios for Exchanger Materials (N./mm²):

Description	Ambient	Operating	Ratio
Tube Material	118.00	118.00	1.000
Tubesheet Material	138.00	138.00	1.000

Minimum

1.000

Hoop Stress in Nozzle Wall during Pressure Test (N./mm²):

Description	Ambient	Operating	Ratio
N1-Tube Inlet	99.22	225.00	0.441
N2-Tube outlet	99.22	225.00	0.441
N3-Shell inlet	70.07	225.00	0.311
N4-Shell outlet	70.07	225.00	0.311

Elements Suitable for Test Pressure.

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Field Pressure Test Results: Step: 15 9:41am Sep 9, 2026

Hydrostatic Test Pressure Results:

Exchanger Shell Side Hydrostatic Test Pressures:

UG99(b)	= 1.30 * M.A.W.P. * Sa/S	4990.783 KPa.
UG99b(b)(35)	= 1.30 * Design Pres * Sa/S	2925.000 KPa.
UG99(c)	= 1.30 * M.A.P. - Head(Hyd)	6376.309 KPa.
UG100	= 1.10 * M.A.W.P. * Sa/S	4222.970 KPa.
PED	= max(1.43*DP, 1.25*DP*ratio)	3217.500 KPa.
PED&UG99(b)(35)	= max(1.43*DP, 1.3*DP*ratio)	3217.500 KPa.
App 27-4	= M.A.W.P.	3839.064 KPa.

Exchanger Channel Side Hydrostatic Test Pressures:

UG99(b)	= 1.30 * M.A.W.P. * Sa/S	7664.264 KPa.
UG99b(b)(35)	= 1.30 * Design Pres * Sa/S	7605.000 KPa.
UG99(c)	= 1.30 * M.A.P. - Head(Hyd)	9032.466 KPa.
UG100	= 1.10 * M.A.W.P. * Sa/S	6485.146 KPa.
PED	= max(1.43*DP, 1.25*DP*ratio)	8365.500 KPa.
PED&UG99(b)(35)	= max(1.43*DP, 1.3*DP*ratio)	8365.500 KPa.
App 27-4	= M.A.W.P.	5895.587 KPa.

UG-99(b)(35), Test Pressure Calculation [Shell Side]:

= Test Factor • Design Pressure • Stress Ratio

= 1.3 • 2250 • 1

= 2925.000 KPa.

UG-99(b) Note 35, Test Pressure Calculation [Channel Side]:

= Test Factor • Design Pressure • Stress Ratio

= 1.3 • 5850 • 1

= 7605.000 KPa.

Horizontal Test performed per: UG-99b (Note 35)

Please note that Nozzle, Shell, Head, Flange, etc MAWPs are all considered when determining the hydrotest pressure for those test types that are based on the MAWP of the vessel.

Stresses on Elements due to Test Pressure (N./mm² & KPa.):

From To	Stress	Allowable	Ratio	Pressure
LHS Bonnet	178.0	234.0	0.761	7612.55
LHS ch shell	160.0	234.0	0.684	7612.55
main shell-1	105.2	234.0	0.450	2932.55
main shell-2	105.2	234.0	0.450	2932.55
main shell-3	105.2	234.0	0.450	2932.55
main shell-4	105.2	234.0	0.450	2932.55
RHS Ch.Shell	160.0	234.0	0.684	7612.55
RHS Bonnet	178.0	234.0	0.761	7612.55

Stress ratios for Nozzle and Pad Materials (N./mm²):

Description	Pad/Nozzle	Ambient	Operating	Ratio
N1-Tube Inlet	Nozzle	138.00	138.00	1.000
N2-Tube outlet	Nozzle	138.00	138.00	1.000
N3-Shell inlet	Nozzle	138.00	138.00	1.000
N4-Shell outlet	Nozzle	138.00	138.00	1.000

Minimum

1.000

Stress ratios for Pressurized Vessel Elements (N./mm²):



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Field Pressure Test Results:

Step: 15

9:41am

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Description	Ambient	Operating	Ratio
LHS Bonnet	138.00	138.00	1.000
LHS ch shell	138.00	138.00	1.000
LHS ch.flg	138.00	138.00	1.000
main shell-1	138.00	138.00	1.000
main shell-2	138.00	138.00	1.000
main shell-3	138.00	138.00	1.000
main shell-4	138.00	138.00	1.000
RHS ch Flg	138.00	138.00	1.000
RHS Ch.Shell	138.00	138.00	1.000
RHS Bonnet	138.00	138.00	1.000

Minimum

1.000

Stress ratios for Exchanger Materials (N./mm²):

Description	Ambient	Operating	Ratio
Tube Material	118.00	118.00	1.000
Tubesheet Material	138.00	138.00	1.000

Minimum

1.000

Hoop Stress in Nozzle Wall during Pressure Test (N./mm²):

Description	Ambient	Operating	Ratio
N1-Tube Inlet	83.55	225.00	0.371
N2-Tube outlet	83.55	225.00	0.371
N3-Shell inlet	32.19	225.00	0.143
N4-Shell outlet	32.19	225.00	0.143

Elements Suitable for Test Pressure.

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ASME TS Calc: Pressure Test

Case: 1 9:41a Sep 9,2026

Input Echo, Tubesheet Number 1, Description: Fixed Tubesheet

Shell Data:

Main Shell Description: main shell-1

Shell Test Pressure		2925.00 KPa.
Shell Thickness	ts	14.0000 mm
Shell Internal Corrosion Allowance	cas	0.0000 mm
Shell External Corrosion Allowance	casext	0.0000 mm
Inside Diameter of Shell	Ds	770.000 mm
Shell Circumferential Joint Efficiency	Esw	1.000
Shell Temperature for Internal Pressure	Ts	21.11 °C
Shell Material		SA-516 70

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Shell Material UNS Number		K02700
Shell Allowable Stress at Temperature	Ss	138.00 N./mm ²
Shell Allowable Stress at Ambient		138.00 N./mm ²

Channel Data:

Channel Description: LHS ch shell

Channel Type:		Cylinder
Channel Test Pressure		7605.00 KPa.
Channel Thickness	tc	22.0000 mm
Channel Internal Corrosion Allowance	cac	0.0000 mm
Channel External Corrosion Allowance	cacext	0.0000 mm
Inside Diameter of Channel	Dc	770.000 mm
Channel Design Temperature	TEMPC	21.11 °C
Channel Flange Matl		SA-516 70

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Channel Material UNS Number		K02700
Channel Allowable Stress at Temperature	Sc	138.00 N./mm ²
Channel Allowable Stress at Ambient		138.00 N./mm ²

Tube Data:

Number of Tube Holes	Nt	282
Tube Wall Thickness	et	2.1100 mm
Tube Outside Diameter	D	25.4000 mm
Total Straight Tube Length	Lt	10000.00 mm
Straight Tube Length (bet. inner tubsht faces) L		9680.00 mm
Design Temperature of the Tubes		21.11 °C
Tube Material		SA-334 6
Tube Material UNS Number		K03006
Is this a Welded Tube		No
Tube Material Specification used	Smls. & wld. tube	
Tube Allowable Stress at Temperature		118.00 N./mm ²
Tube Allowable Stress At Ambient		118.00 N./mm ²
Tube Yield Stress At design Temperature	Syt	240.00 N./mm ²
Tube Pitch (Center to Center Spacing)	P	33.8670 mm
Tube Layout Pattern		Triangular

Fillet Weld Leg	af	2.1100 mm
Groove Weld Leg	ag	2.1100 mm
Tube-Tubesheet Joint Weld Type		Full Strength
Method for Tube-Tubesheet Jt. Allowable		UW-20.3
Is the Tube Inset		No

Radius to Outermost Tube Hole Center	ro	362.479 mm
Largest Center-to-Center Tube Distance	Ul	0.0000 mm
Length of Expanded Portion of Tube	ltx	0.0000 mm
Tube-side pass partition groove depth	hg	6.0000 mm
Area of Tube Layout	Ap	2578.1 cm ²



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ASME TS Calc: Pressure Test

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Perimeter of Tube Layout Cp 2112.8 mm

Tubesheet Data:

Tubesheet TYPE: Fixed Tubesheet Exchanger, Conf B

Tubesheet Design Metal Temperature T 21.11 °C
Tubesheet Material SA-350 LF2

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Tubesheet Material UNS Number		K03011
Tubesheet Allowable Stress at Temperature	S	138.00 N./mm ²
Tubesheet Allowable Stress at Ambient	Tt	138.00 N./mm ²
Thickness of Tubesheet	h	160.0000 mm
Tubesheet Corr. Allowance (Shell side)	Cats	0.0000 mm
Tubesheet Corr. Allowance (Channel side)	Catc	0.0000 mm
Tubesheet Outside Diameter	A	1170.000 mm

Additional Data for Stepped Tubesheets:

Is the Tubesheet Stepped?		Yes
Is the Tubesheet Flat on Tubeside?		No
Step 1 Diameter on the Tubeside	dt1	846.00 mm
Step 1 Depth on the Tubeside	ht1	9.00 mm
Step 2 Diameter on the Tubeside	dt2	777.00 mm
Step 2 Depth on the Tubeside	ht2	7.00 mm
Is the Tubesheet Flat on Shellside?		Yes
Calculated Tubesheet Diameter per Div 2, 4.18.4		1170.00 mm

Note: Tubesheet diameter is now: 1170.000 mm Div 2, 4.18.4.

Area of the Untubed Lanes AL 0.0 cm²

Additional Data for Fixed/Floating Tubesheet Exchangers:

Unsupported Tube Span under consideration	l	220.000 mm
Tube End condition corresponding to Span (l)	k	1.00
Ignore Radial Thermal Exp. effects (4.18.8/4.18.9)		Yes
Perform Differential Pressure Design		No
Run Multiple Load Cases		Yes
Shell Side Min. Design Pressure	Psd,min	-103.4190 KPa.
Mean Shell Metal Temp. along Shell len.	Tsm	21.11 °C
Mean Tube Metal Temp. along Tube length	Ttm	21.11 °C
Junction Stress Reduction option		Perform Plastic Calculation

Additional Data for Gasketed Tubesheets:

Tubesheet Gasket on which Side		Channel
Flange Outside Diameter	A	1170.000 mm
Flange Inside Diameter	B	770.000 mm
Flange Face Outside Diameter	Fod	839.000 mm
Flange Face Inside Diameter	Fid	770.000 mm
Gasket Outside Diameter	Go	828.000 mm
Gasket Inside Diameter	Gi	802.000 mm
Small end Hub thk.	g0	22.0000 mm
Large end Hub thk.	g1	33.0000 mm
Gasket Factor,	m	3.00
Gasket Design Seating Stress	y	68.95 N./mm ²
Flange Facing Sketch		Code Sketch 1a
Gasket Thickness	tg	3.1750 mm
Full face Gasket Flange Option		Program Selects
Length of Partition Gasket	lp	804.000 mm
Width of Partition Gasket	wp	10.0000 mm
Partition Gasket Factor,	mPart	3.0000
Partition Gasket Design Seating Stress	yPart	68.95 N./mm ²



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ASME TS Calc: Pressure Test

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Partition Gasket Facing Sketch Code Sketch 1a
Partition Gasket Column for Gasket Seating Code Column II

Bolting Information:

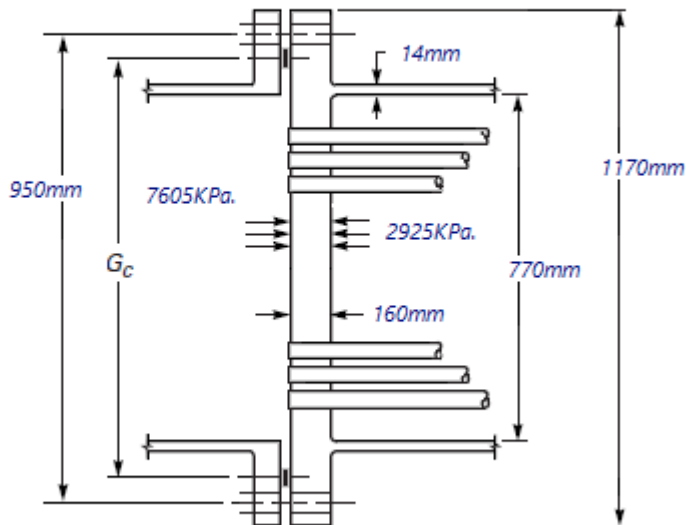
Diameter of Bolt Circle	C	950.000 mm
Nominal Bolt Diameter	dB	38.1000 mm
Thread Series	TEMA Thread Series	
Number of Bolts	n	36
Bolt Material		SA-193 B7
Bolt Allowable Stress At Temperature	Sb	172.00 N./mm ²
Bolt Allowable Stress At Ambient	Sa	172.00 N./mm ²
Weld between Flange and Shell/Channel		0.0000 mm

Note: The full bolt load $Sa \cdot Ab$, was used for all load cases.

Tubesheet Integral with	Shell
Tubesheet Extended as Flange	Yes
Thickness of Extended Portion of Tubesheet	Tf 128.0000 mm
Is Bolt Load Transferred to the Tubesheet	Yes

Is Exchanger in Creep range (skip EP, Use 3S for Sps) No

Tubesheet Integral w/Shell, Gasketed w/Channel, Extended as a Flange



Configuration b:

ASME TubeSheet Results per Part UHX/4.18, 2025

Elasticity/Expansion Material Properties:

Shell - TE-1 Carbon & Low Alloy Steels, Group 1
Shell - TM-1 Carbon Steels with $C \leq 0.3\%$

Th. Exp. Coeff. Metal Temp. along Len	21.1 °C	0.0000115191 /°C
Elastic Mod. at Design Temperature	21.1 °C	0.20270E+09 KPa.
Th. Exp. Coeff. Metal Temp. at Tubsht	21.1 °C	0.0000115191 /°C
Elastic Mod. at Metal Temp. along Len	21.1 °C	0.20270E+09 KPa.
Elastic Mod. at Ambient Temperature	21.1 °C	0.20270E+09 KPa.

Channel - TE-1 Carbon & Low Alloy Steels, Group 1
Channel - TM-1 Carbon Steels with $C \leq 0.3\%$



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Th. Exp. Coeff. Metal Temp. at Tubsht	21.1 °C	0.0000115191 /°C
Elastic Mod. at Design Temperature	21.1 °C	0.20270E+09 KPa.
Elastic Mod. at Ambient Temperature	21.1 °C	0.20270E+09 KPa.

Tubes - TE-1 Carbon & Low Alloy Steels, Group 1
Tubes - TM-1 Carbon Steels with C<= 0.3%

Th. Exp. Coeff. Metal Temp. along Len	21.1 °C	0.0000115191 /°C
Elastic Mod. at Design Temperature	21.1 °C	0.20270E+09 KPa.
Elastic Mod. at Metal Temp. along Len	21.1 °C	0.20270E+09 KPa.
Elastic Mod. at Tubsht. Design Temp.	21.1 °C	0.20270E+09 KPa.
Elastic Mod. at Ambient Temperature	21.1 °C	0.20270E+09 KPa.

TubeSheet - TE-1 Carbon & Low Alloy Steels, Group 1
TubeSheet - TM-1 Carbon Steels with C> 0.3%

Elastic Mod. at Design Temperature	21.1 °C	0.20132E+09 KPa.
Elastic Mod. at Ambient Temperature	21.1 °C	0.20132E+09 KPa.

Note:

The Elasticity and Alpha values are taken from Tables in ASME II D.
Please insure these properties are consistent with the
type of Material for the tubes, shell, channel etc.

Tube Required Thickness under Internal Pressure (Tubeside pressure):

Thickness Due to Internal Pressure:

$$= (P \cdot (D/2 - CAE)) / (S \cdot E + 0.4 \cdot P) \text{ per Appendix 1-1 (a)(1)}$$

$$= (7708.4 \cdot (25.4/2 - 0)) / (118 \cdot 1 + 0.4 \cdot 7708.4)$$

$$= 0.8086 + 0.0000 = 0.8086 \text{ mm}$$

Tube Required Thickness under External Pressure (Shellside pressure) :

External Pressure Chart CS-2 at 21.11 °C
Elastic Modulus for Material 199943392.00 KPa.

Results for Max. Allowable External Pressure (Emawp):

TCA	ODCA	SLEN	D/T	L/D	Factor A	B
2.1100	25.40	9680.00	12.04	50.0000	0.0075908	122.73

$$EAWP = (4 \cdot B) / (3 \cdot (D/T)) = (4 \cdot 122.73) / (3 \cdot 12.038) = 13593 \text{ KPa.}$$

Results for Req'd Thickness for Ext. Pressure (Tca):

TCA	ODCA	SLEN	D/T	L/D	Factor A	B
0.7002	25.40	9680.00	36.28	50.0000	0.0008358	79.60

$$EAWP = (4 \cdot B) / (3 \cdot (D/T)) = (4 \cdot 79.596) / (3 \cdot 36.277) = 2925.3 \text{ KPa.}$$

Summary of Tube Required Thickness Results:

Total Required Thickness including Corrosion all.	0.8086 mm
Allowable Internal Pressure at Corroded thickness	20998.74 KPa.
Required Internal Design Pressure	7708.42 KPa.
Allowable External Pressure at Corroded thickness	13593.04 KPa.
Required External Design Pressure	2925.00 KPa.
Required Thickness due to Shell Side pressure	0.7002 mm

Detailed Results for load Case D3 un-corr. (Psd,max + Ptd,max)

Intermediate Calculations For Tubesheets Extended As Flanges:

ASME Code, Section VIII Division 1, 2025

Gasket Contact Width,	$N = (G_{oc} - G_{ic}) / 2$	13.000 mm
Basic Gasket Width,	$b_0 = N / 2$	6.500 mm



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Effective Gasket Width, $b = \sqrt{b_0} * 2.5$ 6.422 mm
Gasket Reaction Diameter, $G = G_0 - 2.0 * b$ 815.156 mm

Results for ASME Fixed Tubesheet Calculations for Configuration b:

Results for Tubesheet Calculations Original Thickness:

ASME Section VIII-2 Part 4.18.8.4 Step 1:

Tube expansion depth ratio $[\rho]$:

= l_{tx} / h (modified for corrosion if present)

= 0/160

= 0.0000 (must be $0 \leq \rho \leq 1$)

Effective tube hole diameter $[d^*]$:

= $\max(dt - 2 * tt(E_t/E) * (S_t/S) * (\rho); dt - 2 * tt)$

= $\max(25.4 - 2 * 2.11 (202701232/201322320) * (118/138) * (0); 25.4 - 2 * 2.11)$

= 25.4000 mm

Equivalent outer tube limit circle diameter $[Do]$:

= $2 * r_o + dt$

= $2 * 362.48 + 25.4$

= 750.358 mm

Determine the basic ligament efficiency for shear $[\mu]$:

= $(p - dt) / p$

= $(33.867 - 25.4) / 33.867$

= 0.250

Equivalent outer tube limit radius $[ao]$:

= $Do / 2$

= 750.36/2

= 375.1790 mm

Effective tube pitch $[p^*]$:

= $p / \sqrt{ 1 - 4 * \min(AL * CNVfactor; 4 * Do * p) / (\pi * Do^2) }$

= $33.867 / \sqrt{ 1 - 4 * \min(0 * 100 ; 4 * 750.36 * 33.867) / (\pi * 750.36^2) }$

= 33.8670 mm

Effective ligament efficiency for bending $[\mu^*]$:

= $(p^* - d^*) / p^*$

= $(33.867 - 25.4) / 33.867$

= 0.2500

Ratio $[ps]$:

= as / ao

= 385/375.18

= 1.0262

Ratio $[pc]$:

= ac / ao

= 407.58/375.18

= 1.0864

Compute parameter $[xt]$:

= $1 - Nt((dt - 2 * tt) / (2 * ao))^2$

= $1 - 282 ((25.4 - 2 * 2.11) / (2 * 375.18))^2$

= 0.7753

Determine parameter $[xs]$:

= $1 - Nt(dt / (2 * ao))^2$

= $1 - 282 (25.4 / (2 * 375.18))^2$

= 0.6769



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Determine the value [h'g]:

$$= \max((h_g - CATC); 0) \text{ (For pressure only cases)}$$

$$= \max((6 - 0); 0)$$

$$= 6.000 \text{ mm}$$

ASME Section VIII-2 Part 4.18.8.4 Step 2:

Axial Shell Stiffness [Ks]:

$$= \pi \cdot t_s (D_s + t_s) E_s / L$$

$$= \pi \cdot 14 (770 + 14) 202701232 / 9680$$

$$= 73632.8438 \text{ Kgf/mm}$$

Axial Tube Stiffness [Kt]:

$$= \pi \cdot t_t (D_t - t_t) E_t / L$$

$$= \pi \cdot 2.11 (25.4 - 2.11) 202701232 / 9680$$

$$= 329.6700 \text{ Kgf/mm}$$

Stiffness Factor [Ks,t]:

$$= K_s / (N_t \cdot K_t)$$

$$= 73633 / (282 \cdot 329.67)$$

$$= 0.79203$$

Rigidity Ratio [J]:

$$= 1 / (1 + K_s / K_j)$$

$$= 1 / (1 + 73633 / 0)$$

$$= 1.0000 (= 1 \text{ if No Exp. Jt.})$$

Shell coefficient [βs]:

$$= ((12(1 - v_s^2))^{1/4}) / ((D_s + t_s) t_s)$$

$$= ((12(1 - 0.3^2))^{1/4}) / ((770 + 14) 14)$$

$$= 0.0174 \text{ 1/mm}$$

Shell coefficient [ks]:

$$= \beta_s \cdot E_s \cdot t_s^3 / (6(1 - v_s^2))$$

$$= 0.01735 \cdot 202701232 \cdot 14^3 / (6(1 - 0.3^2))$$

$$= 1767590144.0000 \text{ KPa} \cdot \text{mm}^2$$

Shell coefficient [λs]:

$$= 6(D_s + t_s) \cdot k_s / h^3 \cdot (1 + h \cdot \beta_s + (h \cdot \beta_s)^2 / 2)$$

$$= 6(770 + 14) \cdot 1767590144 / 160^3 \cdot (1 + 160 \cdot 0.01735 + (160 \cdot 0.01735)^2 / 2)$$

$$= 15488467.0000 \text{ KPa}$$

Shell coefficient [δs]:

$$= D_s^2 / (4 \cdot E_s \cdot t_s) \cdot (1 - D_s \cdot v_s / (2(D_s + t_s)))$$

$$= 770^2 / (4 \cdot 202701232 \cdot 14) \cdot (1 - 770 \cdot 0.3 / (2(770 + 14)))$$

$$= 0.44535\text{E-}01 \text{ mm/N} \cdot \text{mm}^2$$

Intermediate parameters for Tubesheet Gasketed on the Channel Side:

βC, kc, λC, δC are all = 0

ASME Section VIII-2 Part 4.18.8.4 Step 3:

h/p = 4.724363 ; μ* = 0.250007 ; E* = 43610424 KPa.

E*/E = 0.216620 ; ν* = 0.396449

Note: As h/p (4.724) is > 2, data values for h/p = 2 were used.

Tube Bundle Stiffness Factor [Xa]:

$$= ((24(1 - \nu^{*2}) N_t \cdot E_t \cdot t_t (d_t - t_t) a_o^2) / (E^* \cdot L \cdot H^3))^{1/4}$$

$$= ((24(1 - 0.3964^2) 282 \cdot 202701232 \cdot 2.11$$

$$(25.4 - 2.11) 375.18^2) / (43610424 \cdot 9680 \cdot 160^3))^{1/4}$$

$$= 1.4665$$

Values from Table 4.18.3



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Zd = 0.484881 ; Zv = 0.171120 ; Zm = 0.692654
Za = 0.542097E+00 ; Zw = 0.171120

ASME Section VIII-2 Part 4.18.8.4 Step 4:

Diameter ratio [K]:

= A / Do
= 1170/750.36
= 1.5593

Compute coefficient [F]:

= (1 - v*) / E* * (λs + λc + E * ln(K))
= (1 - 0.3964) / 43610424 * (15488467 + 0 +
201322320 * ln(1.5593))
= 1.4520

Compute parameter [φ]:

= (1 + v*) F
= (1 + 0.3964) 1.452
= 2.0277

Compute parameter [Q1]:

= (ps - 1 - φ * Zv) / (1 + φ * Zm)
= (1.0262 - 1 - 2.0277 * 0.1711) / (1 + 2.0277 * 0.6927)
= -0.133417

Compute Parameter [Qz1]:

= (Zd + Q1 * Zw) / 2 * Xa⁴
= (0.4849 + -0.1334 * 0.1711) / 2 * 1.4665⁴
= 1.0686

Compute parameter [Qz2]:

= (Zv + Q1 * Zm) / 2 * Xa⁴
= (0.1711 + -0.1334 * 0.6927) / 2 * 1.4665⁴
= 0.1820

Compute parameter [U]:

= (Zw + (ps - 1) Zm) Xa⁴ / (1 + φ * Zm)
= (0.1711 + (1.0262 - 1) 0.6927) 1.4665⁴ / (1 + 2.0277 * 0.6927)
= 0.3641

ASME Section VIII-2 Part 4.18.8.4 Step 5:

Determine factor [yb]:

= (Gc - C) / Do (config b)
= (815.16 - 950) / 750.36 = -0.1797

Compute Parameter [γ]:

= 0.000 mm (For Pressure only cases)

Calculate parameter [ws]:

= ps * ks * βs * δS(1 + h * βs)
= 1.0262 * 1767590144 * 0.01735 * 0.044534557 (1 + 160 * 0.01735)
= 5293.1807 mm²

Calculate parameter [ω*s]:

= ao²(ps² - 1) * (ps - 1) / 4 - ωs
= 375.18²(1.0262² - 1) * (1.0262 - 1) / 4 - 5293.2
= -5244.3237 mm²

Parameter [ωc]:

= pc * kc[δC(CmTheta * βC + CqΔ * h * βC²) -
ζc(CmTheta + h * CqTheta * h * βC / 2))
= 1.0864 * 0 [0.000 (0 + 0 * 0 * 160) -
0.000 (0/160 + 0 * 0/2))



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$$= 0.0000 \text{ mm}^2$$

Parameter [$\omega \cdot c$]:

$$\begin{aligned} &= a_o^2 [(p_c^2 + 1) \cdot (p_c - 1)/4 - (p_s - 1)/2] - \omega c \\ &= 375.18^2 [(1.0864^2 + 1) \cdot (1.0864 - 1)/4 - (1.0262 - 1)/2] - 0 \\ &= 4782.8521 \text{ mm}^2 \end{aligned}$$

Pressure [$P \cdot S$]:

$$= 0 \text{ For Pressure only cases or Configurations d,e,f,A,B,C,D}$$

Pressure [$P \cdot C$]:

$$= 0 \text{ For Pressure only cases or Configurations b,c,d,B,C,D}$$

ASME Section VIII-2 Part 4.18.8.4 Step 6:

Pressure [$P's$]:

$$\begin{aligned} &= P_s \{ x_s + 2(1 - x_s) v_t + [2/Kst \cdot (D_s/D_o)^2] v_s - \\ &\quad [(p_s^2 - 1) / (J \cdot Kst)] - [(1 - J) / (2J \cdot Kst)] [(D_j^2 - (D_s)^2) / D_o^2] \} \\ &= 2925 \{ 0.6769 + 2(1 - 0.6769) 0.3 + [2/0.792 \cdot \\ &\quad (770/750.36)^2] 0.3 - [(1.0262^2 - 1) / (1 \cdot 0.792)] - \\ &\quad [(1 - 1) / (2 \cdot 1 \cdot 0.792)] [(0^2 - (770)^2) / 750.36^2] \} \\ &= 4684.4033 \text{ KPa.} \end{aligned}$$

Pressure [$P't$]:

$$\begin{aligned} &= [x_t + 2(1 - x_t) v_t + 1/(J \cdot Kst)] P_t \\ &= [0.7753 + 2(1 - 0.7753) 0.3 + 1/(1 \cdot 0.792)] 7605 \\ &= 16523.4004 \text{ KPa.} \end{aligned}$$

Pressure [P_y]:

$$\begin{aligned} &= N_t \cdot K_t \cdot \gamma / (\pi \cdot a_o^2) \\ &= 282 \cdot 329.67 \cdot 0 / (3.1416 \cdot 375.18^2) \\ &= 0.0000 \text{ KPa.} \end{aligned}$$

Pressure [P_w]:

$$\begin{aligned} &= -\gamma b \cdot U \cdot W \cdot / (2 \cdot \pi \cdot a_o^2) \\ &= -0.1797 \cdot 0.3641 \cdot 572319 / (2 \cdot \pi \cdot 375.18^2) \\ &= 415.1726 \text{ KPa.} \end{aligned}$$

Calculate the pressure [$Prim$]:

$$\begin{aligned} &= - (U/a_o^2) \cdot (\omega \cdot s \cdot P_s - \omega \cdot c \cdot P_t) \\ &= - (0.3641/375.18^2) \cdot (-5244.3 \cdot 2925 - 4782.9 \cdot 7605) \\ &= 133.7531 \text{ KPa.} \end{aligned}$$

Calculate the pressure [P_ω]:

$$\begin{aligned} &= (U/a_o^2) \cdot (\omega \cdot s \cdot P \cdot s - \omega \cdot c \cdot P \cdot c) \\ &= (0.3641/375.18^2) \cdot (5293.2 \cdot 0 - 0 \cdot 0) \\ &= 0.0000 \text{ KPa.} \end{aligned}$$

Determine the effective pressure [Pe]:

$$\begin{aligned} &= J \cdot Kst / (1 + J \cdot Kst(Qz1 + (p_s - 1) Qz2)) \cdot \\ &\quad (P's - P't + P_y + P_\omega + P_w + Prim) \\ &= 0.1000E+01 \cdot 0.792 / (1 + 1 \cdot 0.792 (1.0686 + (1.0262 - 1) 0.182)) \cdot (4684.4 - 16523 + 0 + 0 + 415.17 + 133.75) \\ &= -4833.1885 \text{ KPa.} \end{aligned}$$

ASME Section VIII-2 Part 4.18.8.4 Step 7:

Determine factor [$Q2$]:

$$\begin{aligned} &= [((\omega \cdot s \cdot P_s - \omega \cdot c \cdot P_t) - (\omega \cdot s \cdot P \cdot s - \omega \cdot c \cdot P \cdot c)) + \\ &\quad W \cdot \gamma b / (2 \cdot \pi)] / (1 + \phi \cdot Z_m) \\ &= [((-5244.3 \cdot 2925 - 4782.9 \cdot 7605) - \\ &\quad (5293.2 \cdot 0 - 0 \cdot 0)) \cdot 0.000102 + \\ &\quad 572319 \cdot -0.1797 / (2 \cdot \pi)] / (1 + 2.0277 \cdot 0.6927) \\ &= -9000.9463 \text{ Kg} \end{aligned}$$



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ASME TS Calc: Pressure Test

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Calculate factor [Q3]:

$$= Q1 + 2 \cdot Q2 / (Pe \cdot ao^2)$$

$$= -0.1334 + 2 \cdot -9000.9 / (-4833.2 \cdot 375.18^2)$$

$$= 0.12607$$

Fm Value from Table 4.18.3 = 0.256825

Tubesheet Bending Stress - original thickness [σ]:

$$= (1.5 \cdot Fm / \mu \cdot) \cdot (2 \cdot ao / (H - h'g))^2 \cdot Pe$$

$$= (1.5 \cdot 0.2568 / 0.25) \cdot (2 \cdot 375.18 / (160 - 6))^2 \cdot -4833.2$$

$$= -176.8198 \text{ N./mm}^2$$

Allowable Tubesheet Bending Stress [σ allowed]:

$$= 1.5 \cdot \beta T \cdot Sy = 1.5 \cdot 0.95 \cdot 250 = 356.24 \text{ N./mm}^2, 4.18.233$$

Tubesheet Bending Stress - final thickness [σ_f]:

$$= (1.5 \cdot Fm / \mu \cdot) \cdot (2 \cdot ao / (h - h'g))^2 \cdot Pe$$

$$= (1.5 \cdot 0.2169 / 0.25) \cdot (2 \cdot 375.18 / (94.571 - 6))^2 \cdot -3813.9$$

$$= 356.2436 \text{ N./mm}^2$$

Reqd Tubesheet Thickness, for Bending Stress (including CA) [HReqB]:

$$= h + Cats + Catc$$

$$= 94.571 + 0 + 0$$

$$= 94.5709 \text{ mm}$$

ASME Section VIII-2 Part 4.18.8.4 Step 8:

Shear Stress check [τ limit]:

$$= (2 \cdot \mu \cdot h / ao) \cdot (0.8 \cdot \beta T \cdot Sy)$$

$$= (2 \cdot 0.25 \cdot 160 / 375.18) \cdot 190$$

$$= 40.5145 \text{ N./mm}^2$$

Shear Stress is not required to be computed; [Pe] <= τ limit

Tubesheet Shear Stress is probably low, use the following req. thk:

$$\text{Tubesheet thickness (Incl. Corr.)} = 3.8100 \text{ mm}$$

$$\text{Tubesheet Shear Stress} = 40.9505 \text{ N./mm}^2$$

Reqd Tubesheet Thickness for Given Loadings (including CA) [Hreqd] :

$$= \max(HreqB; HreqS)$$

$$= \max(94.571; 3.81)$$

$$= 94.5709 \text{ mm}$$

ASME Section VIII-2 Part 4.18.8.4 Step 9:

The Ftmin and Ftmax Coefficients from Table 4.18.4:

$$Ftmin = 0.8076, Ftmax = 1.1713$$

First Extreme Tube Axial Stress from among all the tubes [σ_1]:

$$= ((Ps \cdot xs - Pt \cdot xt) - Pe \cdot Ftmin) / (Xt - Xs)$$

$$= ((2925 \cdot 0.6769 - 7605 \cdot 0.7753) - (-4833.2) \cdot 0.8076) /$$

$$(0.7753 - 0.6769)$$

$$= -0.1326 \text{ N./mm}^2$$

Second Extreme value of Tube Axial Stress from among all the tubes [σ_2]:

$$= ((Ps \cdot xs - Pt \cdot xt) - Pe \cdot Ftmax) / (Xt - Xs)$$

$$= ((2925 \cdot 0.6769 - 7605 \cdot 0.7753) - (-4833.2) \cdot 1.1713) /$$

$$(0.7753 - 0.6769)$$

$$= 17.7218 \text{ N./mm}^2$$

Maximum Tube Axial Stress [σ ,max]:

$$= \max(|\sigma_1|; |\sigma_2|) = 17.722 \text{ N./mm}^2$$

Allowable Tube Stress [σ ,A]:



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$$= \beta T \cdot S_{y,t} = 227.9959 \text{ N./mm}^2$$

Check for Buckling as some of the tubes are in compression.

Determine the Factor of Safety [Fs]:

$$= \max((3.25 - 0.25(Z_d + Q_3 \cdot Z_w)X_{a^4}); 1.25)$$

$$= \max((3.25 - 0.25(0.4849 + 0.1261 \cdot 0.1711)1.4665^4); 1.25)$$

$$= 2.0000 \text{ (Should be } \leq 2 \text{)}$$

Determine the factor [rt]:

$$= (\sqrt{ (d_t^2 + (d_t - 2 \cdot t_t)^2) }) / 4$$

$$= (\sqrt{ (25.4^2 + (25.4 - 2 \cdot 2.11)^2) }) / 4$$

$$= 8.2680 \text{ mm}$$

Determine the factor [Ct]:

$$= \sqrt{ (2 \cdot \pi^2 \cdot E_t / S_{y,t})}$$

$$= \sqrt{ (2 \cdot \pi^2 \cdot 202701232 / 240)}$$

$$= 133.8137$$

Determine the factor [Ft]:

$$= k \cdot L / r$$

$$= 1 \cdot 220 / 8.268$$

$$= 26.6087$$

Buckling Allowable Stress [Stb]:

$$= (S_{y,t} / F_s) \cdot (1 - F_t / (2 \cdot C_t))$$

$$= (240 / 2) \cdot (1 - 26.609 / (2 \cdot 133.81))$$

$$= 227.996 \text{ N./mm}^2 \text{ (Never greater than } S_{ot} \text{)}$$

Note: The Axial Compressive stress in tubes is within limits.

Largest tube-to-tubesheet Joint Load [Wt]:

$$= \sigma_{t,max} \cdot \text{Tube Area}$$

$$= 17.722 \cdot 1.5438$$

$$= 278.99 \text{ Kgf}$$

Tube Metal Area [At]:

$$= \pi (d_o - t) t$$

$$= \pi (25.4 - 2.11) 2.11$$

$$= 1.544 \text{ cm}^2$$

Tube Weld Size Results per UW-20.3:

Tube Strength [Ft]:

$$= \pi \cdot t (d_o - t) S$$

$$= \pi \cdot 2.11 (25.4 - 2.11) 228$$

$$= 3589.221 \text{ Kgf}$$

Groove Weld Strength [Fg]:

$$= 0.85 \cdot \pi \cdot a_g (d_o + 0.67 \cdot a_g) S_w \text{ (but not } > F_t \text{)}$$

$$= 0.85 \cdot \pi \cdot 2.11 (25.4 + 0.67 \cdot 2.11) 228$$

$$= 3512.4197 \text{ Kgf}$$

Weld Strength Factor [fw]:

$$= S / S_w \text{ where } S_w = \min(S \text{ and } S_t)$$

$$= 228 / 228$$

$$= 1.000$$

Design Strength Ratio [fd]:

$$= 1 \text{ for full strength welds}$$

Fillet Weld Strength [Ff]:

$$= 0.55 \cdot \pi \cdot a_f (d_o + 0.67 \cdot a_f) S_w \text{ (but not } > F_t \text{)}$$

$$= 0.55 \cdot \pi \cdot 2.11 (25.4 + 0.67 \cdot 2.11) 228$$

$$= 2272.7422 \text{ Kgf}$$



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Minimum Length of Weld [ar]:

$$= 2 \left(\sqrt{(0.75 \cdot do)^2 + 1.07 \cdot t \cdot (do - t) \cdot fw \cdot fd} - 0.75 \cdot do \right)$$

$$= 2 \left(\sqrt{(0.75 \cdot 25.4)^2 + 1.07 \cdot 2.11 \cdot (25.4 - 2.11) \cdot 1 \cdot 1} - 0.75 \cdot 25.4 \right)$$

$$= 2.667 \text{ mm}$$

Minimum Required Fillet Weld Leg [afr]: 1.3334 mm

Minimum Required Groove Weld Leg [agr]: 1.3334 mm

Tube-Tubesheet Jt allowable, 3589.2 is >= tube strength 3589.2 Kgf

Note: This tube-tubesheet joint is a Full Strength joint

ASME Section VIII-2 Part 4.18.8.4 Step 10:

Shell Axial Membrane Allowable Stress:

$$= \beta T \cdot S_{y,s} = 0.95 \cdot 260 = 370.49 \text{ N./mm}^2$$

Axial Membrane Stress in Shell [σ_s, m]:

$$= a \sigma^2 / ((D_s + t_s) t_s) \cdot [P_e + (p_s^2 - 1)(P_s - P_t)] + a s^2 \cdot P_t / ((D_s + t_s) t_s)$$

$$= 375.18^2 / ((770 + 14) \cdot 14) \cdot [-4833.2 + (1.0262^2 - 1) \cdot (2925 - 7605)] + 385^2 \cdot 7605 / ((770 + 14) \cdot 14)$$

$$= 37.5382 \text{ N./mm}^2$$

ASME Section VIII-2 Part 4.18.8.4 Step 11:

Note:

For a given Shell thickness of 14 mm, the minimum Shell length adjacent to the tubesheet should be 186.888 mm

Computations completed for ASME tubesheet configuration: b

Stress/Force Summary for Loadcase D3 un-corr. (Psd,max + Ptd,max):

Stress Description	Actual	Allowable	Pass/Fail
Tubesheet Bend. Stress	176.8 <=	356.2 N./mm ²	Pass
Tubesheet Shear Stress	14.7 <=	190.0 N./mm ²	Pass
Maximum Tube Stress	17.7 <=	228.0 N./mm ²	Pass
Minimum Tube Stress (Buckling)	-0.1 <=	-228.0 N./mm ²	Pass
Maximum Force on any one Tube	279.0 <=	3589.2 Kgf	Pass
Axial Membrane Stress in Shell	37.5 <=	370.5 N./mm ²	Pass

Thickness Results for Loadcase D3 un-corr. (Psd,max + Ptd,max):

Thickness (mm)	Required	Actual	P/F
Tubesheet Thickness :	94.571	160.000	Ok
Tube-Tubesheet Fillet Weld Leg :	1.333	2.110	Ok
Tube-Tubesheet Groove Weld Leg :	1.333	2.110	Ok

Fixed Tubesheet results per ASME Div. 2 Part 4.18.8 2025

Results for 3 Load Cases:

Case#	--Reqd. Thk. + CA Tbsht Extnsn	----	Tubesheet Stresses Bend Allwd Shear Allwd	Case Type	Pass/Fail
D1uc	114.400	60.942	212 356 21 190	Ps+Pt-Th	D1 Ok
D2uc	33.650	...	34 356 6 190	Ps+Pt-Th	D2 Ok
D3uc	94.571	...	177 356 15 190	Ps+Pt-Th	D3 Ok
Max:	114.3999	60.942 mm	0.594	0.111 (Str. Ratio)	



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Load Case Definitions:

[Ps & Pt]:

Shell-side and Tube-side Design or Operating Pressures

derived from Psd,min Ptd,max, Psox,min, Ptox,max etc. per the Load Case Tables

[(+)-Th]:

With or Without Thermal Expansion, Tt,mx & Ts,mx

[c]:

With or Without Corrosion Allowance

[D1, D2, D3]:

Design Load Cases using the Maximum and Minimum Design Pressures

[D4]:

Design Load Case using the Minimum (Vacuum) Pressures (if specified)

[O1, O2, O3, O4]:

Operating Load Cases using the Maximum and Minimum Operating Pressures and Operating Temperatures

Shell Axial Membrane Stress Summary:

---- Shell Stresses ----					:	---- Shell Band Stress ----					:	Pass
Case#	Ten	Allwd	Cmp	Allwd	:	Ten	Allwd	Cmp	Allwd	:	Fail	
D1uc	9	370	...	...		...	...	...	...			Ok
D2uc	28	370	...	...		...	...	...	...			Ok
D3uc	38	370	...	...		...	...	...	...			Ok
Max RATIO	0.101			...			...		...			

Tube, Shell and Channel Stress Summary:

---- Tube Stresses ----					Tube Loads		Shell Stress		Channel Stress		Pass
Case#	Ten	Allwd	Cmp	Allwd	Ld	Allwd	Stress	Allwd	Stress	Allwd	Fail
D1uc	19	227	-1	-227	303	3589	...	370	...	...	Ok
D2uc	1	227	-1	-227	23	3589	...	370	...	...	Ok
D3uc	18	227	...	-227	279	3589	...	370	...	...	Ok
Max RATIO	0.084			0.006		0.084		...		...	

Summary of Thickness Comparisons for 3 Load Cases:

Thickness (mm)	Required	Actual	P/F
Tubesheet thickness :	114.400	160.000	Ok
Tubesheet thickness flanged extension :	60.942	128.000	Ok
Tube thickness :	0.809	2.110	Ok
Tube-tubesheet fillet weld leg :	1.333	2.110	Ok
Tube-tubesheet groove weld leg :	1.333	2.110	Ok

Min Shell length of thk, (14.000) adj. to tubesheet: 186.888 mm

Note: This is a full strength Tube to Tubesheet Joint.

Summary of Axial Differential Expansion between Shell and Tubes :

Due to Thermal Expansion Shell Extends by : +.000 mm
 Due to Pressure Shell Extends by : +.235 mm
 Due to Pressure + Thermal Shell Extends by : +.235 mm



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Summary of Exchanger Pressure and Thermal Displacements: (mm)

Load Case#	Due to Pressure		Due to Thermal Exp.		Computed Delta Ms	
	UnCorroded	Corroded	UnCorroded	Corroded	UnCorroded	Corroded
1	0.2348	...	...	...	...	...
2	-0.2062	...	...	...	...	...
3	0.0213	...	...	...	...	...
4	...	...	...	...	...	...
5	...	...	...	...	...	...
6	...	...	...	...	...	...
7	...	...	...	...	...	...
8	...	...	...	...	...	...

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External Pressure Calculations: Step: 16 9:41am Sep 9,2026

External Pressure Calculation Results Summary:

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External Pressure Calculations:

From	To	Section Length mm	Outside Diameter mm	Corroded Thickness mm	Factor A	Factor B N./mm ²
10	20	No Calc	809	16.5	0.0028327	110.871
20	30	1313.17	814	19	0.0029052	111.332
30	40	No Calc	...	142	No Calc	No Calc
40	50	9680	798	11	0.0022418	22.4126
50	60	9680	798	11	0.0022418	22.4126
60	70	9680	798	11	0.0022418	22.4126
70	80	9680	798	11	0.0022418	22.4126
80	90	No Calc	...	142	No Calc	No Calc
90	100	1313.17	814	19	0.0029052	111.332
100	110	No Calc	809	16.5	0.0028327	110.871

External Pressure Calculations:

From	To	External Actual T. mm	External Required T. mm	External Design Pressure KPa.	External M.A.W.P. KPa.
10	20	19.5	5.09469	103.419	2512.4
20	30	22	6.18877	103.419	3464.7
30	40	145	140.868	103.419	No Calc
40	50	14	9.87445	103.419	411.9
50	60	14	9.87445	103.419	411.9
60	70	14	9.87445	103.419	411.9
70	80	14	9.87445	103.419	411.9
80	90	145	140.868	103.419	No Calc
90	100	22	6.18877	103.419	3464.7
100	110	19.5	5.09469	103.419	2512.4

Minimum

411.9

External Pressure Calculations:

From	To	Actual Length Bet. Stiffeners mm	Allowable Length Bet. Stiffeners mm	Ring Inertia Required cm ⁴	Ring Inertia Available cm ⁴
10	20	No Calc	No Calc	No Calc	No Calc
20	30	1313.17	33216	No Calc	No Calc
30	40	No Calc	No Calc	No Calc	No Calc
40	50	9680	284305	No Calc	No Calc
50	60	9680	284305	No Calc	No Calc
60	70	9680	284305	No Calc	No Calc
70	80	9680	284305	No Calc	No Calc
80	90	No Calc	No Calc	No Calc	No Calc
90	100	1313.17	33216	No Calc	No Calc
100	110	No Calc	No Calc	No Calc	No Calc

Elements Suitable for External Pressure.

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Element and Detail Weights:

Step: 17

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Element and Detail Weights:

From	To	Element Metal Wgt. kg.	Element ID Volume cm ³	Corroded Metal Wgt. kg.	Corroded ID Volume cm ³	Extra due Misc % kg.
10	20	168.193	87715.5	147.169	89560.9	16.8193
20	30	424.256	465746	367.791	473033	42.4256
30	40	478.645	92866.8	467.521	93187.4	47.8645
40	50	646.754	781246	510.108	798880	64.6754
50	60	646.754	781246	510.108	798880	64.6754
60	70	646.754	781246	510.108	798880	64.6754
70	80	646.754	781246	510.108	798880	64.6754
80	90	478.645	92866.8	467.521	93187.4	47.8645
90	100	424.256	465746	367.791	473033	42.4256
100	110	168.193	87715.5	147.169	89560.9	16.8193
Total		4729	4417640.50	4005	4507081.50	472



For elements specified as shell side elements, the volume(s) shown above for those elements, reflects the displacement of the tubes.

Weight of Details:

From	Type	Weight of Detail kg.	X Offset, Dtl. Cent. mm	Y Offset, Dtl. Cent. mm	Z Offset, Dtl. Cent. mm	Description
10	Liqd	87.662	-64.1667	...	...	Liquid: 10
10	Insl	27.7867	30	...	...	Insulation
20	Liqd	465.462	500	...	...	Liquid: 20
20	Insl	62.2814	500	...	...	Insulation
20	Nozl	147.933	500	506.5	...	N1-Tube Inlet
20	Nozl	147.933	500	-506.5	...	N2-Tube outlet
30	Liqd	92.8102	94.5	...	...	Liquid: 30
30	Insl	12.9754	77.5	...	...	Insulation
40	Sadl	151.501	1940	638	...	Left Saddle flo
40	Liqd	780.769	1210	...	...	Liquid: 40
40	Insl	240.598	1210	...	...	Insulation
40	Nozl	164.534	1200	506.5	...	N3-Shell inlet
40	Wght	250	1940	-914.4	...	Fireproofing we
50	Liqd	780.769	1210	...	...	Liquid: 50
50	Insl	240.598	1210	...	...	Insulation
60	Liqd	780.769	1210	...	...	Liquid: 60
60	Insl	240.598	1210	...	...	Insulation
70	Sadl	151.501	480	638	...	Right Saddle fi
70	Liqd	780.769	1210	...	...	Liquid: 40
70	Insl	240.598	1210	...	...	Insulation
70	Nozl	155.526	1200	-506.5	...	N4-Shell outlet
70	Wght	250	480	-914.4	...	Wght:[2 of 2]
80	Liqd	92.8102	94.5	-5.00002	...	Liquid: 50
80	Insl	12.9754	77.5	...	...	Insulation
90	Liqd	465.462	500	...	...	Liquid: 60
90	Insl	62.2814	500	...	...	Insulation
100	Liqd	87.662	124.167	...	...	Liquid: 70
100	Insl	27.7867	30	...	...	Insulation
30	FTsh	1271.66	275.35	...	...	Fixed Tubesheet
30	Tube	3711.71	5195.35	...	...	Tubes
30	RTsh	1271.66	10115.3	...	...	Tubesheet
40	Bafl	17.8593	220	...	...	Baffles
40	Bafl	17.8593	440	...	...	Baffles
40	Bafl	17.8593	660	...	...	Baffles



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Element and Detail Weights:

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40	Baf1	17.8593	880	...	...	Baffles
40	Baf1	17.8593	1100	...	...	Baffles
40	Baf1	17.8593	1320	...	...	Baffles
40	Baf1	17.8593	1540	...	...	Baffles
40	Baf1	17.8593	1760	...	...	Baffles
40	Baf1	17.8593	1980	...	...	Baffles
40	Baf1	17.8593	2200	...	...	Baffles
40	Baf1	17.8593	2420	...	...	Baffles
40	Baf1	17.8593	2640	...	...	Baffles
40	Baf1	17.8593	2860	...	...	Baffles
40	Baf1	17.8593	3080	...	...	Baffles
40	Baf1	17.8593	3300	...	...	Baffles
40	Baf1	17.8593	3520	...	...	Baffles
40	Baf1	17.8593	3740	...	...	Baffles
40	Baf1	17.8593	3960	...	...	Baffles
40	Baf1	17.8593	4180	...	...	Baffles
40	Baf1	17.8593	4400	...	...	Baffles
40	Baf1	17.8593	4620	...	...	Baffles
40	Baf1	17.8593	4840	...	...	Baffles
40	Baf1	17.8593	5060	...	...	Baffles
40	Baf1	17.8593	5280	...	...	Baffles
40	Baf1	17.8593	5500	...	...	Baffles
40	Baf1	17.8593	5720	...	...	Baffles
40	Baf1	17.8593	5940	...	...	Baffles
40	Baf1	17.8593	6160	...	...	Baffles
40	Baf1	17.8593	6380	...	...	Baffles
40	Baf1	17.8593	6600	...	...	Baffles
40	Baf1	17.8593	6820	...	...	Baffles
40	Baf1	17.8593	7040	...	...	Baffles
40	Baf1	17.8593	7260	...	...	Baffles
40	Baf1	17.8593	7480	...	...	Baffles
40	Baf1	17.8593	7700	...	...	Baffles
40	Baf1	17.8593	7920	...	...	Baffles
40	Baf1	17.8593	8140	...	...	Baffles
40	Baf1	17.8593	8360	...	...	Baffles
40	Baf1	17.8593	8580	...	...	Baffles
40	Baf1	17.8593	8800	...	...	Baffles
40	Baf1	17.8593	9020	...	...	Baffles
40	Baf1	17.8593	9240	...	...	Baffles
40	Baf1	17.8593	9460	...	...	Baffles
40	Baf1	17.8593	9680	...	...	Baffles
40	Baf1	17.9546	9900	...	...	Baffles

Total Weight of Each Detail Type:

Saddles	303.0
Liquid	4414.9
Insulation	1168.5
Nozzles	615.9
Weights	500.0
Exchanger Components	7058.8
Liquid in Tubes	993.1

Sum of the Detail Weights 15054.3 kg.

Weight Summation Results: (kg.)

	Fabricated	Shop Test	Shipping	Erected	Empty	Operating
Main Elements	5202.1	5202.1	5202.1	5202.1	5202.1	5202.1
Saddles	303.0	303.0	303.0	303.0	303.0	303.0
Nozzles	615.9	615.9	615.9	615.9	615.9	615.9
Exchanger	7058.8	7058.8	7058.8	7058.8	7058.8	7058.8
Insulation	...	...	...	...	1168.5	1168.5 *



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Element and Detail Weights:

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Empty Weights	...	...	...	500.0	500.0	...
Test Weights	...	500.0	...	...	...	...
Ope Weights	...	...	...	...	...	500.0
Ope. Liquid	...	...	...	...	...	4414.9
Tube Ope Lqd	...	...	...	...	...	993.1
Test Liquid	...	4414.9	...	...	...	...
Tube Tst Lqd	...	993.1	...	...	...	...
Totals	13179.8	19087.9	13179.8	13679.8	14848.3	20256.4

Field Installation Options:

* Insulation installed after lifting.

Miscellaneous Weight Percent: 10.0 %

Note that the above value for the miscellaneous weight percent has been applied to the shells/heads/flange/tubesheets/tubes etc. in the weight calculations for metallic components.

Weight Summary:

Fabricated Wt.	- Bare weight without removable internals	13179.8 kg.
Shop Test Wt.	- Fabricated weight + water (full)	19087.9 kg.
Shipping Wt.	- Fab. weight + removable intls.+ shipping app.	13179.8 kg.
Erected Wt.	- Fab. wt + or - loose items (trays,platforms etc.)	13679.8 kg.
Ope. Wt. no Liq	- Fab. weight + internals. + details + weights	14848.3 kg.
Operating Wt.	- Empty weight + operating liq. uncorroded	20256.4 kg.
Oper. Wt. + CA	- Corr wt. + operating liquid	19460.2 kg.
Field Test Wt.	- Empty weight + test medium (full)	19549.6 kg.

Exchanger Tube Data

Volume of Exchanger tubes :	993730.7 cm ³
Weight of Ope Liq in tubes :	993.1 kg.
Weight of Water in tubes :	993.1 kg.

Field Test weight is computed in the corroded condition.

Surface Areas of Elements:

From	To	Outside Surface Area cm ²	Inside Surface Area cm ²
10	20	8795.1	7878.36
20	30	25572.6	24190.3
30	40	9992.75	4765.48
40	50	60669.2	58540.4
50	60	60669.2	58540.4
60	70	60669.2	58540.4
70	80	60669.2	58540.4
80	90	9992.75	4765.48
90	100	25572.6	24190.3
100	110	8795.1	7878.36

Total	331397.6	307830.0 cm ²
Total	356.7	331.3 ft ²

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Nozzle Flange MAWP:

Step: 18 9:41am Sep 9,2026

Nozzle Flange MAWP Results: (KPa. & °C)

Nozzle Description	Flange Rating Ope.	Flange Rating Ambient	Des. Temp	Class	Grade/Group	Equiv. Press	- - - UG-44	Max Pressure 50%	DNV
N1-Tube Inlet	9774.1	10200.0	65	600	GR 1.1	3795.8	9774.1	9774.1	9774.1
N2-Tube outlet	9774.1	10200.0	65	600	GR 1.1	3468.9	9774.1	9774.1	9774.1
N3-Shell inlet	4887.1	5100.0	65	300	GR 1.1	2539.5	4791.1	4887.1	4791.1
N4-Shell outle	4887.1	5100.0	65	300	GR 1.1	2539.5	4791.1	4887.1	4791.1

Shellside Flange Rating

Lowest Flange Pressure Rating was (Ope)[ShellSide]: 4791.058 KPa.

Lowest Flange Pressure Rating was (Amb)[ShellSide]: 5100.005 KPa.

Channelside Flange Rating

Lowest Flange Pressure Rating was (Ope)[TubeSide]: 9774.126 KPa.

Lowest Flange Pressure Rating was (Amb)[TubeSide]: 10200.003 KPa.

Selected Method for Derating ASME B16.5 Flange MAWP: ASME UG-44(b)

The UG-44(b) Method is based on the paper PVP 2013-97814. PV Elite uses the maximum loads from each load category to determine ME and FE. In many cases, the computed maximum allowable pressure will be greater than the flange rating. In these cases, the minimum of the rating from the table and the UG-44(b) method will be used. SA-193 B8 Cl. 2 bolts or ones with higher allowable stresses at the specified bolt size shall be used.

The axial force FE must put the nozzle in tension (-P for 107/297; +Fr for PD 5500 AnnexG) to be considered.

Nozzle Description	Gasket Dimension G mm	Fm Value	Resolved Moment ME Kg-m.	Axial Force FE Kgf	MAWP PD KPa.	Maximum Pressure Pmax KPa.	Pass Fail
N1-Tube Inlet	296.037	0.50	1802	2295	5895.6	9774.1	Pass
N2-Tube outlet	296.037	0.50	1802	0	5895.6	9774.1	Pass
N3-Shell inlet	302.387	0.50	1261	1912	3839.1	4791.1	Pass
N4-Shell outlet	302.387	0.50	1261	1912	3839.1	4791.1	Pass

Evaluating Flange on Nozzle [N1-Tube Inlet]:

$$= 16 \cdot ME + 4 \cdot FE \cdot G \leq \pi \cdot G^3 [(PR - PD) + FM \cdot PR]$$

$$= 16(1802) + 4 \cdot \text{abs}(2295) 296.037 \leq \pi \cdot 296.037^3 [(9774.126 - 5895.587) + .500 \cdot 9774.126]$$

$$= 0.31550E+05 \leq 0.72858E+05 \text{ Check Passes}$$

Maximum Pressure for Flange on Nozzle N1-Tube Inlet [Pmax]:

$$= \min(PR, FM \cdot PR + PR - (16 \cdot ME + 4 \cdot FE \cdot G) / (\pi \cdot G^3))$$

$$= \min(9774.126, 0.500 \cdot 9774.126 + 9774.126 - 31549.8 / (\pi \cdot 296.037^3))$$

$$= \min(9774.126, 10865.393)$$

$$= 9774.126 \text{ KPa.}$$

Evaluating Flange on Nozzle [N2-Tube outlet]:

$$= 16 \cdot ME + 4 \cdot FE \cdot G \leq \pi \cdot G^3 [(PR - PD) + FM \cdot PR]$$

$$= 16(1802) + 4 \cdot \text{abs}(0) 296.037 \leq \pi \cdot 296.037^3 [(9774.126 - 5895.587) + .500 \cdot 9774.126]$$

$$= 0.28833E+05 \leq 0.72858E+05 \text{ Check Passes}$$

Maximum Pressure for Flange on Nozzle N2-Tube outlet [Pmax]:

$$= \min(PR, FM \cdot PR + PR - (16 \cdot ME + 4 \cdot FE \cdot G) / (\pi \cdot G^3))$$

$$= \min(9774.126, 0.500 \cdot 9774.126 + 9774.126 - 28832.7 / (\pi \cdot 296.037^3))$$

$$= \min(9774.126, 11192.291)$$

$$= 9774.126 \text{ KPa.}$$

Evaluating Flange on Nozzle [N3-Shell inlet]:



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Nozzle Flange MAWP:

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$$\begin{aligned} &= 16 \cdot ME + 4 \cdot FE \cdot G \leq \pi \cdot G^3 [(PR - PD) + FM \cdot PR] \\ &= 16(1261) + 4 \cdot \text{abs}(1912) 302.387 \leq \pi \cdot 302.387^3 [(4887.066 - 3839.064) + .500 \cdot 4887.066] \\ &= 0.22496E+05 \leq 0.30929E+05 \text{ Check Passes} \end{aligned}$$

Maximum Pressure for Flange on Nozzle N3-Shell inlet [Pmax]:

$$\begin{aligned} &= \min(PR, FM \cdot PR + PR - (16 \cdot ME + 4 \cdot FE \cdot G) / (\pi \cdot G^3)) \\ &= \min(4887.066, 0.500 \cdot 4887.066 + 4887.066 - 22495.7 / (\pi \cdot 302.387^3)) \\ &= \min(4887.066, 4791.058) \\ &= 4791.058 \text{ KPa.} \end{aligned}$$

Evaluating Flange on Nozzle [N4-Shell outlet]:

$$\begin{aligned} &= 16 \cdot ME + 4 \cdot FE \cdot G \leq \pi \cdot G^3 [(PR - PD) + FM \cdot PR] \\ &= 16(1261) + 4 \cdot \text{abs}(1912) 302.387 \leq \pi \cdot 302.387^3 [(4887.066 - 3839.064) + .500 \cdot 4887.066] \\ &= 0.22496E+05 \leq 0.30929E+05 \text{ Check Passes} \end{aligned}$$

Maximum Pressure for Flange on Nozzle N4-Shell outlet [Pmax]:

$$\begin{aligned} &= \min(PR, FM \cdot PR + PR - (16 \cdot ME + 4 \cdot FE \cdot G) / (\pi \cdot G^3)) \\ &= \min(4887.066, 0.500 \cdot 4887.066 + 4887.066 - 22495.7 / (\pi \cdot 302.387^3)) \\ &= \min(4887.066, 4791.058) \\ &= 4791.058 \text{ KPa.} \end{aligned}$$

Flange Pressure Ratings are per ASME B16.5 2025 Metric Edition

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Wind Load Calculation:

Step: 19 9:41am Sep 9,2026

Wind Load Calculation Results:

Input Values:

Wind Design Code	ASCE/SEI 7-16
Wind Load Reduction Scale Factor	1.100
Basic Wind Speed [V]	97.2 Km/hr
Surface Roughness Category	C: Open Terrain
Type of Surface	Moderately Smooth
Base Elevation	300 mm
Percent Wind for Hydrotest	50.0
Using User defined Wind Press. Vs Elev.	N
Height of Hill or Escarpment H or Hh	0 mm
Distance Upwind of Crest Lh	0 mm
Distance from Crest to the Vessel x	0 mm
Type of Terrain (Hill, Escarpment)	Flat
Damping Factor (Beta) for Wind (Ope)	0.0100
Damping Factor (Beta) for Wind (Empty)	0.0000
Damping Factor (Beta) for Wind (Filled)	0.0000
Site Elevation above Sea Level	0 mm

Wind Analysis Results

The importance factor is not used in the calculation of wind pressure vs. elevation for the selected wind code.

Static Gust-Effect Factor, Operating Case [G]:

$$= \min(0.85, 0.925((1 + 1.7 \cdot gQ \cdot I_{zbar} \cdot Q) / (1 + 1.7 \cdot gV \cdot I_{zbar})))$$

$$= \min(0.85, 0.925((1 + 1.7 \cdot 3.4 \cdot 0.2281 \cdot 0.9841) / (1 + 1.7 \cdot 3.4 \cdot 0.2281)))$$

$$= \min(0.85, 0.9166)$$

$$= 0.850$$

Natural Frequency of Vessel (Operating)	33.000 Hz
Natural Frequency of Vessel (Empty)	33.000 Hz
Natural Frequency of Vessel (Test)	33.000 Hz

User Entered Importance Factor is	1.000
Force Coefficient [Cf]	0.635
Structure Height to Diameter ratio	13.371

This is classified as a rigid structure. Static analysis performed.

Sample Calculation for the First Element:

The ASCE code performs calculations in Imperial Units. The wind pressure is therefore computed in these units.

Value of $[\alpha]$ and $[Zg]$:

Exposure Category: C from Table 26.11-1
 $\alpha = 9.5 : Zg = 274320 \text{ mm}$

Effective Height [z]:

$$= \text{Centroid Height} + \text{Vessel Base Elevation}$$

$$= 915 + 300 = 1215 \text{ mm}$$

$$= 3.9862 \text{ ft. Imperial Units}$$

Velocity Pressure coefficient evaluated at height z [Kz]:

$$\text{Because } z (3.9862 \text{ ft.}) < 15 \text{ ft.}$$

$$= 2.01(15 / Zg)^{2/\alpha}$$

$$= 2.01(15/900)^{2/9.5}$$

$$= 0.8489$$

Type of Hill: No Hill



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Wind Load Calculation:

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Wind Directionality Factor [Kd]:

= 1, per Table 26.6-1

As there is No Hill Present: [Kzt]:

K1 = 0, K2 = 0, K3 = 0

Topographical Factor [Kzt]:

= (1 + K1 • K2 • K3)²

= (1 + 0 • 0 • 0)²

= 1

Velocity Pressure evaluated at height z, Imperial Units [qz]:

= max(16, 0.00256 • Kz • Kzt • Kd • Ke • V(mph)²)

= max(16, 0.00256 • 0.8489 • 1 • 1 • 1 • 60.399²)

= 16 psf [78.12] Kgs/m²

Force on the first element [F]:

= qz • G • Cf • WindArea

= 16 • 0.85 • 0.6354 • 2.9979

= 25.906 lbs. [11.751] Kgf

Element	Hgt (z) mm	K1	K2	K3	Kz	Kzt	qz Kgs/m ²
LHS Bonnet	1215.0	0.000	0.000	0.000	0.849	1.000	78.120
LHS ch shell	1215.0	0.000	0.000	0.000	0.849	1.000	78.120
LHS ch.flg	1215.0	0.000	0.000	0.000	0.849	1.000	78.120
main shell-1	1215.0	0.000	0.000	0.000	0.849	1.000	78.120
main shell-2	1215.0	0.000	0.000	0.000	0.849	1.000	78.120
main shell-3	1215.0	0.000	0.000	0.000	0.849	1.000	78.120
main shell-4	1215.0	0.000	0.000	0.000	0.849	1.000	78.120
RHS ch Flg	1215.0	0.000	0.000	0.000	0.849	1.000	78.120
RHS Ch.Shell	1215.0	0.000	0.000	0.000	0.849	1.000	78.120
RHS Bonnet	1215.0	0.000	0.000	0.000	0.849	1.000	78.120

Wind Loads on Masses/Equipment/Piping

ID	Wind Area cm ²	Elevation mm	Pressure Kgs/m ²	Force Kgf
Fireproofing	0.00	300.00	78.12	0.00
Wght:[2 of 2]	0.00	300.00	78.12	0.00

Wind Load Calculation:

From	To	Wind Height mm	Wind Diameter mm	Wind Area cm ²	Wind Pressure Kgs/m ²	Element Wind Load Kgf
10	20	1215	1138.8	2785.15	78.12	12.9261
20	30	1215	1144.8	11448	78.12	53.131
30	40	1215	1061.78	2006.76	78.12	9.31352
40	50	1215	1197.6	28981.9	78.12	134.507
50	60	1215	1197.6	28981.9	78.12	134.507
60	70	1215	1197.6	28981.9	78.12	134.507
70	80	1215	1197.6	28981.9	78.12	134.507
80	90	1215	1061.78	2006.76	78.12	9.31352
90	100	1215	1144.8	11448	78.12	53.131
100	110	1215	1138.8	2785.15	78.12	12.9261

The Wind Loads calculated and printed in the Wind Load calculation report have been factored by the input scalar/load reduction factor of: 1.100.



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Wind Load Calculation:

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Be sure the wind speed is in accordance with the specified wind design code.

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Earthquake Load Calculation:

Step: 20

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Earthquake Load Calculation Results:

Earthquake Load Calculation:

Input Values:

Seismic Design Code		ASCE/SEI 7-16
Seismic Load Reduction Scale Factor		1.000
Importance Factor		1.250
Table Value Fa		1.600
Table Value Fv		2.400
Max. Mapped Res. Acceleration	[Ss]	0.147
Max. Eff. Ground Acceleration	[S]	0.065
Force Modification Factor R		3.000
Site Class		D
Component Elevation Ratio	z/h	0.000
Amplification Factor	Ap	1.000
Force Factor		0.000
Consider Vertical Acceleration		Yes
Minimum Acceleration Multiplier		0.000
User Value of Sds (used if > 0)		0.000
User Value of Sd1 (used if > 0)		0.000
Moment Reduction Factor Tau		1.000

Seismic Analysis Results:

$$\begin{aligned} SMS &= Fa * Ss = 1.6 * 0.147 = 0.2352 \\ SM1 &= Fv * S1 = 2.4 * 0.065 = 0.156 \\ SDS &= 2/3 * Sms = 2/3 * 0.2352 = 0.1568 \\ SD1 &= 2/3 * Sm1 = 2/3 * 0.156 = 0.104 \end{aligned}$$

Check Approximate Fundamental Period from 12.8-7 [Ta]:
 $= Ct \cdot h_n^x$ where $Ct = 0.020$, $x = 0.75$ and h_n = Structural Height (ft.)
 $= 0.020(4.265^{0.75})$
 $= 0.059$ seconds

The Coefficient C_u from Table 12.8-1 is : 1.700

Fundamental Period (1/Frequency) [T]:
 $= (1/\text{Natural Frequency}) = (1/33)$
 $= 0.030$

Check the Value of T which is the smaller of $C_u \cdot T_a$ and T:
 $= \text{Minimum Value of } (1.7 \cdot 0.05936, 0.0303)$ per 12.8.2
 $= 0.030$
 As the time period is < 0.06 second, use section 15.4.2.

Base Shear per equation 15.4-5, [V]:
 $= 0.3 \cdot Sds \cdot W \cdot I$
 $= 0.3 \cdot 0.1568 \cdot 19953 \cdot 1.25$
 $= 1173.259$ Kgf

Vertical load per 12.4-4, [Ev]:
 $= 0.2 \cdot Sds \cdot W$
 $= 0.2 \cdot 0.1568 \cdot 19953$
 $= 625.7$ Kgf

Earthquake Load Calculation:

From	To	Earthquake Height mm	Earthquake Weight Kgf	Element Ope Load Kgf
10	20	385	1662.78	97.7716



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Earthquake Load Calculation:

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20	30	385	1662.78	97.7716
30	40	385	1662.78	97.7716
40	50	385	1662.78	97.7716
40	50	385	1662.78	97.7716
50	60	385	1662.78	97.7716
60	70	385	1662.78	97.7716
70	0	385	1662.78	97.7716
70	80	385	1662.78	97.7716
80	90	385	1662.78	97.7716
90	100	385	1662.78	97.7716
100	110	385	1662.78	97.7716

The Earthquake Weight does not include the weight of the saddles.

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Center of Gravity Calculation: Step: 21 9:41am Sep 9,2026

Shop/Field Installation Options :

Insulation is installed in the Field after being lifted.

Note : The CG is computed from the first Element From Node

Center of Gravity of the Saddles	6252.174 mm
Center of Gravity of the Liquid	6252.241 mm
Center of Gravity of the Insulation	6251.832 mm
Center of Gravity of the Nozzles	3459.602 mm
Center of Gravity of the Added Weights (Operating)	6252.175 mm
Center of Gravity of the Added Weights (Empty)	6252.175 mm
Center of Gravity of the Tubesheet(s)	6255.349 mm
Center of Gravity of the Tubes	6255.350 mm
Center of Gravity of the Baffles	6472.752 mm
Center of Gravity of Bare Shell New and Cold	6252.175 mm
Center of Gravity of Bare Shell Corroded	6252.174 mm
Vessel CG in the Operating Condition	6174.076 mm
Vessel CG in the Fabricated (Shop/Empty) Condition	6140.853 mm
Vessel CG in the Test Condition	6172.558 mm

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Saddle Calcs: Road Transport:

Step: 22

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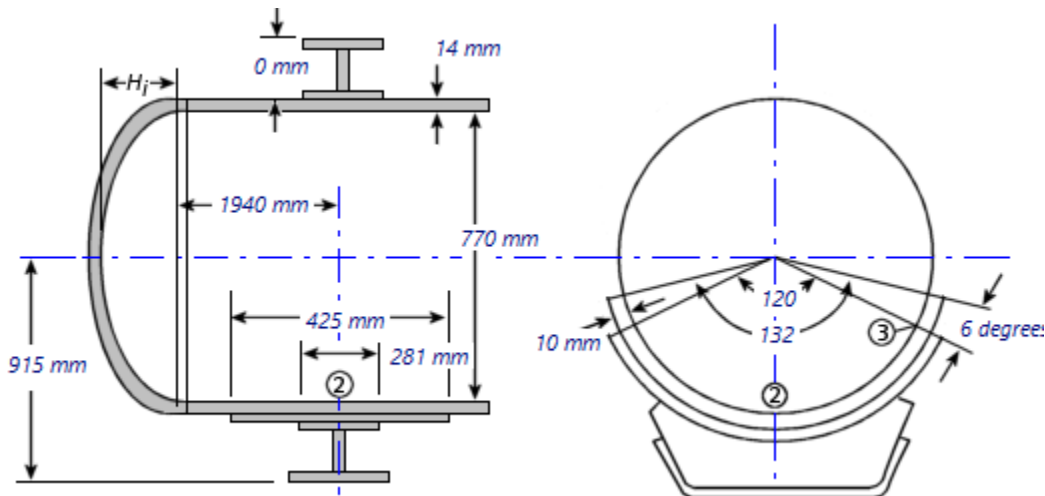
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ASME VIII Division 2 Horizontal Vessel Analysis, Left Saddle:

Horizontal Vessel Stress Calculations : Road Transport

Input and Calculated Values:

Vessel Mean Radius	Rm	392.00	mm
Shell Thickness used in this Case	t	14.000	mm
Stiffened Vessel Length per 4.15.6	L	9680.00	mm
Distance from Saddle to Vessel tangent	a1 or a	1940.00	mm
Saddle Width	b1 or b	281.00	mm
Saddle Bearing Angle	delta or theta	120.00	degrees
Wear Plate Width	b2 or b1	425.00	mm
Wear Plate Bearing Angle	delta2 or theta1	132.00	degrees
Wear Plate Thickness	e2 or tr	10.0	mm
Wear Plate Allowable Stress	fw or Sr	138.00	N./mm ²
Shell Allowable Stress used in Calculation		138.00	N./mm ²
Head Allowable Stress used in Calculation		138.00	N./mm ²
Circumferential Efficiency in Plane of Saddle		1.00	
Circumferential Efficiency at Mid-Span		1.00	
Distance from Saddle Base to Centerline	B	915.00	mm
Coefficient of Friction	mu	0.30	



Results for Horizontal Vessel Analysis:

Saddle Force Q for this Transportation Case		8527.63	Kgf
Shipping Longitudinal Acceleration	gx	0.10	g
Shipping Transverse Acceleration	gz	0.10	g
Shipping Vertical Acceleration	gy	0.05	g
Vertical Acceleration Acting with Transverse	gw	1.00	g
Internal Pressure during Transport		0.00	KPa.
Transportation Wind Load Scalar Factor	wlsf	1.00	
Transportation Wind Velocity	wv	92.7	Km/hr



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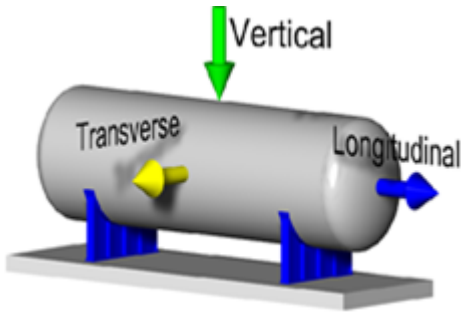
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Saddle Calcs: Road Transport:

Step: 22

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Horizontal Vessel Analysis Results:	Actual N./mm ²	Allowable N./mm ²
Long. Stress at Top of Midspan	-6.04	-234.00
Long. Stress at Top of Midspan	6.04	138.00
Long. Stress at Bottom of Midspan	6.04	138.00
Long. Stress at Top of Saddles	44.21	138.00
Long. Stress at Bottom of Saddles	-24.50	-234.00
Long. Stress at Bottom of Saddles	24.50	138.00
Tangential Shear in Shell	10.69	110.40
Circ. Stress at Horn of Saddle	13.71	207.00
Circ. Compressive Stress in Shell	0.67	138.00

Intermediate Results: Saddle Reaction Q due to Road Transport Load:

Transverse Force due to Wind during Transport [F_{tw}]:

$$\begin{aligned}
 &= \text{Transverse Area} \cdot 0.00256 \cdot wv^2 \cdot wlsf \\
 &= 159.74 \cdot 0.00256 \cdot 57.603^2 \cdot 1 \\
 &= 1356.9 \text{ lbf} \\
 &= 615.495 \text{ Kg}
 \end{aligned}$$

Longitudinal Force due to Wind during Transport [F_{lw}]:

$$\begin{aligned}
 &= \pi \cdot D^2/4 \cdot 0.00256 \cdot wv^2 \cdot wlsf \cdot \text{Wind Dia. Mult} \\
 &= \pi \cdot 3.0774^2/4 \cdot 0.00256 \cdot 57.603^2 \cdot 1 \cdot 1.2 \\
 &= 75.8 \text{ lbf} \\
 &= 34.4 \text{ Kg}
 \end{aligned}$$

Transverse Force due to Lateral Acceleration [F_t]:

$$\begin{aligned}
 &= g_z \cdot \text{Shipping Weight} \cdot g_w + F_{tw} \\
 &= 0.1 \cdot 12877 \cdot 1 + 615.49 \\
 &= 1903.2 \text{ Kg}
 \end{aligned}$$

Longitudinal Force due to Axial Acceleration [F_l]:

$$\begin{aligned}
 &= g_x \cdot \text{Shipping Weight} \cdot g_w + F_{lw} \\
 &= 0.1 \cdot 12877 \cdot 1 + 34.391 \\
 &= 1322.1 \text{ Kg}
 \end{aligned}$$

Saddle Reaction Force due to Transverse Load [F_{wt}]:

$$\begin{aligned}
 &= F_t / \text{Num of Saddles} \cdot B / E \\
 &= 1903.2/2 \cdot 915/708.41 \\
 &= 1229.1 \text{ Kg}
 \end{aligned}$$

Saddle Reaction Force due to Longitudinal Load [F_{wl}]:

$$\begin{aligned}
 &= F_l \cdot B / L_s \\
 &= 1322.1 \cdot 915/5800 \\
 &= 208.6 \text{ Kg}
 \end{aligned}$$

Saddle Reaction Force due to Vertical Transportation Acceleration [F_{rv}]:

$$\begin{aligned}
 &= \text{Saddle Load} \cdot g_y \\
 &= 6951 \cdot 0.05
 \end{aligned}$$



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Saddle Calcs: Road Transport: Step: 22 9:41am Sep 9, 2026

= 347.549 Kgf

Load Combination Results due to Transport Loading [Q]:

= Saddle Load + Frv + max(Fwl, Fwt)
 = 6951 + 347.55 + max(208.57, 1229.1)
 = 8527.6 Kgf

Summary of Loads at the Base of this Saddle:

Vertical Load (including saddle weight)	8679.13	Kgf
Transverse Shear Load	951.59	Kgf
Longitudinal Shear Load	1322.07	Kgf

Note: Wear Plate is Welded to the Shell, k = 0.1

Saddle Dimension [E]:

= min(2(ShellID/2 + t + WearPadThickness)sin(θ/2), 2•Rm)
 = min(2(770/2 + 14 + 10)sin(120/2), 2•392)
 = 708.409 mm

The Computed K values from Table 4.15.1:

K1 = 0.1066	K2 = 1.1707	K3 = 0.8799	K4 = 0.4011
K5 = 0.7603	K6 = 0.0529	K7 = 0.0529	K8 = 0.3405
K9 = 0.2711	K10 = 0.0581	K1* = 0.1923	K6p = 0.0434
K7p = 0.0434			

The suffix 'p' denotes the values for a wear plate if it exists.

Note: Dimension a is greater than or equal to Rm/2.

Moment per Equation 4.15.1 [M1]:

= -Q•a [1 - (1 - a/L + (Rm²-hm²)/(2a•L))/(1+(4hm)/3L)]
 = -8527.6•1940[1-(1-1940/9680+(392²-0²)/
 (2•1940•9680))/(1+(4•0)/(3•9680))]
 = -3247.9 Kg-m.

Moment per Equation 4.15.2 [M2]:

= Q•L/4((1+2(Rm²-hm²)/(L²))/(1+(4hm²)/(3L))-4a/L)
 = 8527.6•9680/4((1+2(392²-0²)/(9680²))/(1+(4•0)/
 (3•9680))-4•1940/9680)
 = 4161.0 Kg-m.

Longitudinal Stress at Top of Shell (4.15.4) [σ1]:

= P • Rm/(2t) - M2/(π•Rm²t)
 = 0 • 392/(2•14) - 4161/(π•392²•14)
 = -6.04 N./mm²

Longitudinal Stress at Bottom of Shell (4.15.5) [σ2]:

= P • Rm/(2t) + M2/(π • Rm² • t)
 = 0 • 392/(2 • 14) + 4161/(π • 392² • 14)
 = 6.04 N./mm²

Longitudinal Stress at Top of Shell at Support (4.15.8) [σ*3]:

= P • Rm/(2t) - M1/(K1•π•Rm²t)
 = 0•392/(2•14)-3247.9/(0.1066•π•392²•14)
 = 44.21 N./mm²

Longitudinal Stress at Bottom of Shell at Support (4.15.9) [σ*4]:

= P • Rm/(2t) + M1/(K1* • π • Rm² • t)
 = 0•392/(2•14)+3247.9/(0.1923•π•392²•14)
 = -24.50 N./mm²

Maximum Shear Force in the Saddle (4.15.3) [T]:

= Q(L-2a)/(L+(4•hm/3))
 = 8527.6 (9680 - 2 • 1940)/(9680 + (4 • 0/3))
 = 5109.5 Kgf



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Shear Stress in the shell no rings, not stiffened (4.15.12) [τ_2]:

$$= K_2 \cdot T / (R_m \cdot t)$$

$$= 1.1707 \cdot 5109.5 / (392 \cdot 14)$$

$$= 10.69 \text{ N./mm}^2$$

Decay Length (4.15.20) [x_1, x_2]:

$$= 0.78 \cdot \sqrt{(R_m \cdot t)}$$

$$= 0.78 \cdot \sqrt{(392 \cdot 14)}$$

$$= 57.783 \text{ mm}$$

Effective reinforcing plate width (4.15.24) [B1]:

$$= \min(b + 1.56 \cdot \sqrt{(R_m \cdot t)}, 2a)$$

$$= \min(281 + 1.56 \cdot \sqrt{(392 \cdot 14)}, 2 \cdot 1940)$$

$$= 396.57 \text{ mm}$$

Wear Plate/Shell Stress ratio (4.15.26) [η]:

$$= 1.0000 \text{ Materials are the same, transport case}$$

Circumferential Stress at Saddle Base with Wear Plate (4.15.25) [σ_6, r]:

$$= -K_5 \cdot Q \cdot k / (B_1(t + \eta \cdot tr))$$

$$= -0.7603 \cdot 8527.6 \cdot 0.1 / (396.57(14 + 1 \cdot 10))$$

$$= -0.67 \text{ N./mm}^2$$

Circ. Comp. Stress at Horn of Saddle, $L \geq 8R_m$ (4.15.28) [σ_7, r]:

$$= -Q / (4(t + \eta \cdot tr)b_1) - 3 \cdot K_7 \cdot Q / (2(t + \eta \cdot tr)^2)$$

$$= -8527.6 / (4(14 + 1 \cdot 10)396.57) -$$

$$3 \cdot 0.05285 \cdot 8527.6 / (2(14 + 1 \cdot 10)^2)$$

$$= -13.71 \text{ N./mm}^2$$

Results for Vessel Ribs, Web and Base:

Baseplate Length	Bplen	780.0000	mm
Baseplate Thickness	Bpthk	16.0000	mm
Baseplate Width	Bpwid	310.0000	mm
Number of Ribs (inc. outside ribs)	Nribs	3	
Rib Thickness	Ribtk	10.0000	mm
Web Thickness	Webtk	10.0000	mm
Web Location	Webloc	Center	
Saddle Yield Stress	Sy	260.0	N./mm ²
Height of Web at Center	Hw, c	525.0	mm
Friction Coefficient	mu	0.300	

Moment of Inertia of Saddle - Transverse Direction (90 degrees to long axis)

Inertia of Shell [ShellInertia]:

$$= ((1.56 \cdot \sqrt{(R \cdot t)} + \text{WearPlateWidth})t^3) / 12$$

$$= ((1.56 \cdot \sqrt{(385 \cdot 14)} + 425)14^3) / 12$$

$$= 12.337 \text{ cm}^4$$

Resolved Inertia for the Shell [I, shell]:

$$= \text{ShellInertia} + A_{\text{Shell}}(C_1 - Y)^2$$

$$= 12.337 + 75.534(186.53 - 7)^2$$

$$= 24357.342 \text{ cm}^4$$

	B mm	D mm	Y mm	A cm ²	AY mm ³	I + AD ⁽²⁾ cm ⁴
Shell	539.5	14.0	7.0	75.5	52873.9	0.244E+05
Wearplate	425.0	10.0	19.0	42.5	80750.0	0.119E+05
Web	10.0	490.0	269.0	49.0	1318100.4	0.131E+05
BasePlate	310.0	16.0	522.0	49.6	2589120.0	0.558E+05
Totals	...	...	...	216.6	4040844.5	0.105E+06

Distance to Centroid [C1]:

$$= AY / A$$



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$$= 4040844/216.63$$

$$= 186.528 \text{ mm}$$

Angle [β]:

$$= 180 - \text{Saddle Angle}/2$$

$$= 180 - 120/2$$

$$= 120.0$$

Saddle Splitting Coefficient [K1]:

$$= (1 + \cos(\beta) - 0.5 \cdot \sin(\beta)^2) / (\pi - \beta + \sin(\beta) \cos(\beta))$$

$$= (1 + \cos(120) - 0.5 \cdot \sin(120)^2) / (\pi - 2.0944 + \sin(120) \cos(120))$$

$$= 0.2035$$

Saddle Splitting Force [Fh]:

$$= K1 \cdot Q$$

$$= 0.2035 \cdot 8527.6$$

$$= 1735.5570 \text{ Kgf}$$

Tension Stress, St = (Fh/As) = 1.2063 N./mm²

Allowed Stress, Sa = 0.6 * Yield Str = 155.9972 N./mm²

Saddle Splitting Dimension [d]:

$$= B - R \cdot \sin(\text{theta}/2) / (\text{theta}/2 \text{ in radians})$$

$$= 915 - 385 \cdot \sin(120/2) / 1.0472$$

$$= 596.608 \text{ mm}$$

Bending Moment, M = Fh * d = 1035.4670 Kg-m.

Bending Stress, Sb = (M * C1 / I) = 1.7995 N./mm²

Allowed Stress, Sa = 2/3 * Yield Str = 173.3302 N./mm²

Minimum Thickness of Baseplate per Moss:

$$= \sqrt{(3(Q + \text{SaddleWt}) \text{BasePlateWidth}) / (4 \cdot \text{BasePlateLength} \cdot \text{AllStress})}$$

$$= \sqrt{(3(8527.6 + 151.5)310 / (4 \cdot 780 \cdot 173.33))}$$

$$= 12.098 \text{ mm}$$

Calculation of Axial Load, Intermediate Values and Compressive Stress:

Web Length Dimension [Web Length]:

$$= 2 \cdot \cos(90 - \text{Saddle Angle}/2) (\text{Inside Radius} + \text{Shell Thk} + \text{Wear Plate Thk})$$

$$= 2 \cdot \cos(90 - 120/2) (385 + 14 + 10)$$

$$= 708.409 \text{ mm}$$

Distance between Ribs [e]:

$$= \text{Web Length} / (\text{Nr ribs} - 1)$$

$$= 708.41 / (3 - 1)$$

$$= 354.204 \text{ mm}$$

Baseplate Pressure Area [Ap]:

$$= e \cdot \text{Bpwid} / 2$$

$$= 354.2 \cdot 310 / 2$$

$$= 549.017 \text{ cm}^2$$

Bearing Pressure [Bp]:

$$= (Q + \text{SaddleWt}) / (\text{BasePlateLength} \cdot \text{BasePlateWidth})$$

$$= 8679.1 / (780 \cdot 310)$$

$$= 3.589 \text{ Kgf/cm}^2$$

Axial Load [P]:

$$= Ap \cdot Bp$$

$$= 549.02 \cdot 3.5894$$

$$= 1970.632 \text{ Kgf}$$

Area of the Rib and Web [Ar]:

$$= \text{Rib Area} + \text{Web Area}$$



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$$= 27.1 + 17.71$$

$$= 44.810 \text{ cm}^2$$

Compressive Stress [Sc]:

$$= P/A_r$$

$$= 1970.6/44.81$$

$$= 4.313 \text{ N./mm}^2$$

Check of Outside Ribs:

Inertia of Saddle, Outer Ribs - Longitudinal Direction

	B	D	Y	A	AY	Io
Rib+Web	10.0	281.0	...	28.1	...	0.185E+04

Rib dimension [D]:

$$= \text{Saddle Width} - \text{Web Thickness}$$

$$= 281 - 10$$

$$= 271.000 \text{ mm}$$

Distance to Centroid from Datum [ytot]:

$$= AY / A$$

$$= 0/44.81$$

$$= 0.000 \text{ mm}$$

Distance to Centroid [C1]:

$$= \text{Saddle Width} / 2$$

$$= 281/2$$

$$= 140.500 \text{ mm}$$

Radius of Gyration [r]:

$$= \sqrt{(\text{Total Inertia} / \text{Total Area})}$$

$$= \sqrt{(1849/44.81)}$$

$$= 64.236 \text{ mm}$$

Length of Outer Rib [L]:

$$= \text{Saddle Height} - \cos(\theta/2) (\text{radius} + \text{shlthk} + \text{wpdthk}) - \text{bpthk}$$

$$= 915 - \cos(120/2) (385 + 14 + 10) - 16$$

$$= 694.500 \text{ mm}$$

Intermediate Term [Cc]:

$$= \sqrt{(2 \cdot \pi^2 \cdot \text{Elastic Modulus} / \text{Yield Stress})}$$

$$= \sqrt{(2 \cdot \pi^2 \cdot 199943088/260)}$$

$$= 123.211$$

Slenderness ratio [KL/r]:

$$= KL/r$$

$$= 1 \cdot 694.5/64.236$$

$$= 10.812$$

Bending Moment [Rm]:

$$= FI / (2 \cdot \text{Bplen}) \cdot e \cdot L / 2$$

$$= 1322.1 / (2 \cdot 780) \cdot 354.2 \cdot 694.5/2$$

$$= 104.240 \text{ Kg-m.}$$

Compressive Allowable, $KL/r < C_c$ ($10.812 < 123.21$) per AISC E2-1 [Sca]:

$$= (1 - (KL/r)^2 / (2 \cdot C_c^2)) F_y / (5/3 + 3 \cdot (KL/r) / (8 \cdot C_c) - (KL/r)^3 / (8 \cdot C_c^3))$$

$$= (1 - (10.812)^2 / (2 \cdot 123.21^2)) 260 /$$

$$(5/3 + 3 \cdot (10.812) / (8 \cdot 123.21) - (10.812)^3 / (8 \cdot 123.21^3))$$

$$= 152.4 \text{ N./mm}^2$$

AISC Unity Check of Outside Ribs (must be ≤ 1)

$$= Sc/Sca + (Rm \cdot C1 / I) / Sba$$

$$= 4.3128/152.4 + (104.24 \cdot 140.5/18490034) / 173.33$$



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= 0.073

Check of Inside Ribs:

Inertia of Saddle, Inner Ribs - Axial Direction

	B	D	Y	A	AY	Io
Rib	10.0	271.0	0.0	27.1	0.0	0.185E+04
Web	354.2	10.0	0.0	35.4	0.0	2.95
Totals	...	...	...	62.5	...	0.185E+04

Distance to Centroid from Datum [ytot]:

= AY / A

= 0/62.52

= 0.000 mm

Distance to Centroid [C1]:

= Saddle Width / 2

= 281/2

= 140.500 mm

Length of Inner Rib [L]:

= Saddle Height - Outside Radius - Bpthk

= 915 - cos(409/2)(16 + 0 + 0) - 0

= 490.000 mm

Radius of Gyration [r]:

= $\sqrt{\text{Total Inertia} / \text{Total Area}}$

= $\sqrt{(1851.9/62.52)}$

= 54.424 mm

Slenderness ratio [KL/r]:

= KL/r

= 1 • 490/54.424

= 9.003

Unit Force [Force,u]:

= FI / (2 • Baseplate Length)

= 1322.1/(2 • 780)

= 0.847 Kg/mm

Moment at base of inner Rib [Mbase,c]:

= Unit Force • e • L

= 0.8475 • 354.2 • 490

= 147.092 Kg-m.

Bending Stress due to Transverse Force and Weight Load [σ_B ,base,c]:

= Bending Moment / Section Modulus

= 147.09/131806

= 10.944 N./mm²

Compressive Allowable, KL/r < Cc (9.0033 < 123.21) per AISC E2-1 [Sca]:

= $(1 - (KL/r)^2 / (2 \cdot Cc^2)) F_y / (5/3 + 3 \cdot (KL/r) / (8 \cdot Cc) - (KL/r)^3 / (8 \cdot Cc^3))$

= $(1 - (9.0033)^2 / (2 \cdot 123.21^2)) 260 /$

$(5/3 + 3 \cdot (9.0033) / (8 \cdot 123.21) - (9.0033^3) / (8 \cdot 123.21^3))$

= 153.1 N./mm²

AISC Unity Check of Inside Ribs (must be ≤ 1)

= Sc/Sca + (Mbase,c • C1/I)/Sba

= 6.1468/153.07 + (147.09 • 140.5/1851.9)/173.33

= 0.103

ASME VIII Division 2 Horizontal Vessel Analysis, Right Saddle:

Input and Calculated Values:



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Vessel Mean Radius	Rm	392.00	mm
Shell Thickness used in this Case	t	14.000	mm
Stiffened Vessel Length per 4.15.6	L	9680.00	mm
Distance from Saddle to Vessel tangent	a1 or a	1940.00	mm
Saddle Width	b1 or b	281.00	mm
Saddle Bearing Angle	delta or theta	120.00	degrees
Wear Plate Width	b2 or b1	425.00	mm
Wear Plate Bearing Angle	delta2 or theta1	132.00	degrees
Wear Plate Thickness	e2 or tr	10.0	mm
Wear Plate Allowable Stress	fw or Sr	138.00	N./mm ²
Shell Allowable Stress used in Calculation		138.00	N./mm ²
Head Allowable Stress used in Calculation		138.00	N./mm ²
Circumferential Efficiency in Plane of Saddle		1.00	
Circumferential Efficiency at Mid-Span		1.00	
Distance from Saddle Base to Centerline	B	915.00	mm
Coefficient of Friction	mu	0.00	

Results for Horizontal Vessel Analysis:

Saddle Force Q for this Transportation Case		7976.24	Kgf
Shipping Longitudinal Acceleration	gx	0.10	g
Shipping Transverse Acceleration	gz	0.10	g
Shipping Vertical Acceleration	gy	0.05	g
Vertical Acceleration Acting with Transverse	gw	1.00	g
Internal Pressure during Transport		0.00	KPa.
Transportation Wind Load Scalar Factor	wlsf	1.00	
Transportation Wind Velocity	wv	92.7	Km/hr

Horizontal Vessel Analysis Results:		Actual N./mm ²	Allowable N./mm ²
Long. Stress at Top of Midspan		-5.65	-234.00
Long. Stress at Top of Midspan		5.65	138.00
Long. Stress at Bottom of Midspan		5.65	138.00
Long. Stress at Top of Saddles		41.35	138.00
Long. Stress at Bottom of Saddles		-22.92	-234.00
Long. Stress at Bottom of Saddles		22.92	138.00
Tangential Shear in Shell		10.00	110.40
Circ. Stress at Horn of Saddle		12.82	207.00
Circ. Compressive Stress in Shell		0.62	138.00

Intermediate Results: Saddle Reaction Q due to Road Transport Load:

Transverse Force due to Wind during Transport [Ftw]:

$$\begin{aligned}
 &= \text{Transverse Area} \cdot 0.00256 \cdot wv^2 \cdot wlsf \\
 &= 159.74 \cdot 0.00256 \cdot 57.603^2 \cdot 1 \\
 &= 1356.9 \text{ lbf} \\
 &= 615.495 \text{ Kgf}
 \end{aligned}$$

Longitudinal Force due to Wind during Transport [Flw]:

$$\begin{aligned}
 &= \pi \cdot D^2/4 \cdot 0.00256 \cdot wv^2 \cdot wlsf \cdot \text{Wind Dia. Mult} \\
 &= \pi \cdot 3.0774^2/4 \cdot 0.00256 \cdot 57.603^2 \cdot 1 \cdot 1.2 \\
 &= 75.8 \text{ lbf} \\
 &= 34.4 \text{ Kgf}
 \end{aligned}$$

Transverse Force due to Lateral Acceleration [Ft]:

$$\begin{aligned}
 &= gz \cdot \text{Shipping Weight} \cdot gw + Ftw \\
 &= 0.1 \cdot 12877 \cdot 1 + 615.49
 \end{aligned}$$



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= 1903.2 Kgf

Longitudinal Force due to Axial Acceleration [FI]:

= $g_x \cdot \text{Shipping Weight} \cdot g_w + F_{lw}$

= $0.1 \cdot 12877 \cdot 1 + 34.391$

= 1322.1 Kgf

Saddle Reaction Force due to Transverse Load [Fwt]:

= $F_t / \text{Num of Saddles} \cdot B / E$

= $1903.2 / 2 \cdot 915 / 708.41$

= 1229.1 Kgf

Saddle Reaction Force due to Longitudinal Load [Fwl]:

= $F_l \cdot B / L_s$

= $1322.1 \cdot 915 / 5800$

= 208.6 Kgf

Saddle Reaction Force due to Vertical Transportation Acceleration [Frv]:

= Saddle Load $\cdot g_y$

= $6425.9 \cdot 0.05$

= 321.293 Kgf

Load Combination Results due to Transport Loading [Q]:

= Saddle Load + Frv + max(Fwl, Fwt)

= $6425.9 + 321.29 + \max(208.57, 1229.1)$

= 7976.2 Kgf

Summary of Loads at the Base of this Saddle:

Vertical Load (including saddle weight)

8127.75 Kgf

Transverse Shear Load

951.59 Kgf

Longitudinal Shear Load

1322.07 Kgf

Note: Wear Plate is Welded to the Shell, $k = 0.1$

Saddle Dimension [E]:

= $\min(2(\text{ShellID}/2 + t + \text{WearPadThickness}) \sin(\theta/2), 2 \cdot R_m)$

= $\min(2(770/2 + 14 + 10) \sin(120/2), 2 \cdot 392)$

= 708.409 mm

The Computed K values from Table 4.15.1:

K1 = 0.1066 K2 = 1.1707 K3 = 0.8799 K4 = 0.4011

K5 = 0.7603 K6 = 0.0529 K7 = 0.0529 K8 = 0.3405

K9 = 0.2711 K10 = 0.0581 K1* = 0.1923 K6p = 0.0434

K7p = 0.0434

The suffix 'p' denotes the values for a wear plate if it exists.

Note: Dimension a is greater than or equal to $R_m/2$.

Moment per Equation 4.15.1 [M1]:

= $-Q \cdot a [1 - (1 - a/L + (R_m^2 - h_m^2)/(2a \cdot L)) / (1 + (4h_m)/3L)]$

= $-7976.2 \cdot 1940 [1 - (1 - 1940/9680 + (392^2 - 0^2)/$

$(2 \cdot 1940 \cdot 9680) / (1 + (4 \cdot 0) / (3 \cdot 9680))]$

= -3037.9 Kg-m.

Moment per Equation 4.15.2 [M2]:

= $Q \cdot L / 4 ((1 + 2(R_m^2 - h_m^2)/(L^2)) / (1 + (4h_m^2)/(3L)) - 4a/L)$

= $7976.2 \cdot 9680 / 4 ((1 + 2(392^2 - 0^2)/(9680^2)) / (1 + (4 \cdot 0) /$

$(3 \cdot 9680)) - 4 \cdot 1940 / 9680)$

= 3892.0 Kg-m.

Longitudinal Stress at Top of Shell (4.15.4) [σ_1]:

= $P \cdot R_m / (2t) - M_2 / (\pi \cdot R_m^3 t)$

= $0 \cdot 392 / (2 \cdot 14) - 3892 / (\pi \cdot 392^3 \cdot 14)$

= -5.65 N./mm²



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Longitudinal Stress at Bottom of Shell (4.15.5) [σ_2]:

$$= P \cdot R_m / (2t) + M_2 / (\pi \cdot R_m^2 \cdot t)$$

$$= 0 \cdot 392 / (2 \cdot 14) + 3892 / (\pi \cdot 392^2 \cdot 14)$$

$$= 5.65 \text{ N./mm}^2$$

Longitudinal Stress at Top of Shell at Support (4.15.8) [σ_{*3}]:

$$= P \cdot R_m / (2t) - M_1 / (K_1 \cdot \pi \cdot R_m^2 t)$$

$$= 0 \cdot 392 / (2 \cdot 14) - 3037.9 / (0.1066 \cdot \pi \cdot 392^2 \cdot 14)$$

$$= 41.35 \text{ N./mm}^2$$

Longitudinal Stress at Bottom of Shell at Support (4.15.9) [σ_{*4}]:

$$= P \cdot R_m / (2t) + M_1 / (K_1 \cdot \pi \cdot R_m^2 \cdot t)$$

$$= 0 \cdot 392 / (2 \cdot 14) + 3037.9 / (0.1923 \cdot \pi \cdot 392^2 \cdot 14)$$

$$= -22.92 \text{ N./mm}^2$$

Maximum Shear Force in the Saddle (4.15.3) [T]:

$$= Q(L - 2a) / (L + (4 \cdot h_m / 3))$$

$$= 7976.2 (9680 - 2 \cdot 1940) / (9680 + (4 \cdot 0 / 3))$$

$$= 4779.2 \text{ Kgf}$$

Shear Stress in the shell no rings, not stiffened (4.15.12) [τ_2]:

$$= K_2 \cdot T / (R_m \cdot t)$$

$$= 1.1707 \cdot 4779.2 / (392 \cdot 14)$$

$$= 10.00 \text{ N./mm}^2$$

Decay Length (4.15.20) [x_1, x_2]:

$$= 0.78 \cdot \sqrt{(R_m \cdot t)}$$

$$= 0.78 \cdot \sqrt{(392 \cdot 14)}$$

$$= 57.783 \text{ mm}$$

Effective reinforcing plate width (4.15.24) [B1]:

$$= \min(b + 1.56 \cdot \sqrt{(R_m \cdot t)}, 2a)$$

$$= \min(281 + 1.56 \cdot \sqrt{(392 \cdot 14)}, 2 \cdot 1940)$$

$$= 396.57 \text{ mm}$$

Wear Plate/Shell Stress ratio (4.15.26) [eta]:

$$= 1.0000 \text{ Materials are the same, transport case}$$

Circumferential Stress at Saddle Base with Wear Plate (4.15.25) [σ_6, r]:

$$= -K_5 \cdot Q \cdot k / (B_1(t + \eta \cdot tr))$$

$$= -0.7603 \cdot 7976.2 \cdot 0.1 / (396.57(14 + 1 \cdot 10))$$

$$= -0.62 \text{ N./mm}^2$$

Circ. Comp. Stress at Horn of Saddle, $L \geq 8R_m$ (4.15.28) [σ_7, r]:

$$= -Q / (4(t + \eta \cdot tr)b_1) - 3 \cdot K_7 \cdot Q / (2(t + \eta \cdot tr)^2)$$

$$= -7976.2 / (4(14 + 1 \cdot 10)396.57) -$$

$$3 \cdot 0.05285 \cdot 7976.2 / (2(14 + 1 \cdot 10)^2)$$

$$= -12.82 \text{ N./mm}^2$$

Results for Vessel Ribs, Web and Base:

Baseplate Length	Bplen	780.0000	mm
Baseplate Thickness	Bpthk	16.0000	mm
Baseplate Width	Bpwid	310.0000	mm
Number of Ribs (inc. outside ribs)	Nribs	3	
Rib Thickness	Ribtk	10.0000	mm
Web Thickness	Webtk	10.0000	mm
Web Location	Webloc	Center	
Saddle Yield Stress	Sy	260.0	N./mm ²
Height of Web at Center	Hw, c	525.0	mm
Friction Coefficient	mu	0.000	

Moment of Inertia of Saddle - Transverse Direction (90 degrees to long axis)



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Inertia of Shell [ShellInertia]:

$$= ((1.56 \cdot \sqrt{(R \cdot t)} + \text{WearPlateWidth})^3) / 12$$

$$= ((1.56 \cdot \sqrt{(385 \cdot 14)} + 425)^3) / 12$$

$$= 12.337 \text{ cm}^4$$

Resolved Inertia for the Shell [I,shell]:

$$= \text{ShellInertia} + A_{\text{Shell}} (C1 - Y)^2$$

$$= 12.337 + 75.534 (186.53 - 7)^2$$

$$= 24357.342 \text{ cm}^4$$

	B mm	D mm	Y mm	A cm ²	AY mm ³	I + AD ² cm ⁴
Shell	539.5	14.0	7.0	75.5	52873.9	0.244E+05
Wearplate	425.0	10.0	19.0	42.5	80750.0	0.119E+05
Web	10.0	490.0	269.0	49.0	1318100.4	0.131E+05
BasePlate	310.0	16.0	522.0	49.6	2589120.0	0.558E+05
Totals	...	...	...	216.6	4040844.5	0.105E+06

Distance to Centroid [C1]:

$$= AY / A$$

$$= 4040844 / 216.63$$

$$= 186.528 \text{ mm}$$

Angle [β]:

$$= 180 - \text{Saddle Angle} / 2$$

$$= 180 - 120 / 2$$

$$= 120.0$$

Saddle Splitting Coefficient [K1]:

$$= (1 + \cos(\beta) - 0.5 \cdot \sin(\beta)^2) / (\pi - \beta + \sin(\beta) \cos(\beta))$$

$$= (1 + \cos(120) - 0.5 \cdot \sin(120)^2) / (\pi - 2.0944 + \sin(120) \cos(120))$$

$$= 0.2035$$

Saddle Splitting Force [Fh]:

$$= K1 \cdot Q$$

$$= 0.2035 \cdot 7976.2$$

$$= 1623.3386 \text{ Kgf}$$

$$\text{Tension Stress, } St = (Fh / As) = 1.1283 \text{ N./mm}^2$$

$$\text{Allowed Stress, } Sa = 0.6 \cdot \text{Yield Str} = 155.9972 \text{ N./mm}^2$$

Saddle Splitting Dimension [d]:

$$= B - R \cdot \sin(\theta/2) / (\theta/2 \text{ in radians})$$

$$= 915 - 385 \cdot \sin(120/2) / 1.0472$$

$$= 596.608 \text{ mm}$$

$$\text{Bending Moment, } M = Fh \cdot d = 968.5153 \text{ Kg-m.}$$

$$\text{Bending Stress, } Sb = (M \cdot C1 / I) = 1.6831 \text{ N./mm}^2$$

$$\text{Allowed Stress, } Sa = 2/3 \cdot \text{Yield Str} = 173.3302 \text{ N./mm}^2$$

Minimum Thickness of Baseplate per Moss:

$$= \sqrt{(3(Q + \text{SaddleWt}) \cdot \text{BasePlateWidth}) / (4 \cdot \text{BasePlateLength} \cdot \text{AllStress})}$$

$$= \sqrt{(3(7976.2 + 151.5) \cdot 310) / (4 \cdot 780 \cdot 173.33)}$$

$$= 11.708 \text{ mm}$$

Calculation of Axial Load, Intermediate Values and Compressive Stress:

Web Length Dimension [Web Length]:

$$= 2 \cdot \cos(90 - \text{Saddle Angle} / 2) (\text{Inside Radius} + \text{Shell Thk} + \text{Wear Plate Thk})$$

$$= 2 \cdot \cos(90 - 120/2) (385 + 14 + 10)$$

$$= 708.409 \text{ mm}$$

Distance between Ribs [e]:



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= Web Length / (Nribs - 1)
 = 708.41 / (3 - 1)
 = 354.204 mm

Baseplate Pressure Area [Ap]:

= e • Bpwid / 2
 = 354.2 • 310 / 2
 = 549.017 cm²

Bearing Pressure [Bp]:

= (Q + SaddleWt) / (BasePlateLength • BasePlateWidth)
 = 8127.7 / (780 • 310)
 = 3.361 Kg/cm²

Axial Load [P]:

= Ap • Bp
 = 549.02 • 3.3614
 = 1845.438 Kgf

Area of the Rib and Web [Ar]:

= Rib Area + Web Area
 = 27.1 + 17.71
 = 44.810 cm²

Compressive Stress [Sc]:

= P / Ar
 = 1845.4 / 44.81
 = 4.039 N./mm²

Check of Outside Ribs:

Inertia of Saddle, Outer Ribs - Longitudinal Direction

	B	D	Y	A	AY	Io
Rib+Web	10.0	281.0	...	28.1	...	0.185E+04

Rib dimension [D]:

= Saddle Width - Web Thickness
 = 281 - 10
 = 271.000 mm

Distance to Centroid from Datum [ytot]:

= AY / A
 = 0 / 44.81
 = 0.000 mm

Distance to Centroid [C1]:

= Saddle Width / 2
 = 281 / 2
 = 140.500 mm

Radius of Gyration [r]:

= $\sqrt{\text{Total Inertia} / \text{Total Area}}$
 = $\sqrt{1849 / 44.81}$
 = 64.236 mm

Length of Outer Rib [L]:

= Saddle Height - cos(theta/2)(radius + shlthk + wpdthk) - bpthk
 = 915 - cos(120/2)(385 + 14 + 10) - 16
 = 694.500 mm

Intermediate Term [Cc]:

= $\sqrt{2 \cdot \pi^2 \cdot \text{Elastic Modulus} / \text{Yield Stress}}$
 = $\sqrt{2 \cdot \pi^2 \cdot 199943008 / 260}$



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$$= 123.211$$

Slenderness ratio [KL/r]:

$$= KL/r$$

$$= 1 \cdot 694.5/64.236$$

$$= 10.812$$

Bending Moment [Rm]:

$$= FI / (2 \cdot B_{plen}) \cdot e \cdot L / 2$$

$$= 1322.1 / (2 \cdot 780) \cdot 354.2 \cdot 694.5/2$$

$$= 104.240 \text{ Kg-m.}$$

Compressive Allowable, $KL/r < C_c$ ($10.812 < 123.21$) per AISC E2-1 [Sca]:

$$= (1 - (KL/r)^2 / (2 \cdot C_c^2)) F_y / (5/3 + 3 \cdot (KL/r) / (8 \cdot C_c) - (KL/r)^3 / (8 \cdot C_c^3))$$

$$= (1 - (10.812)^2 / (2 \cdot 123.21^2)) 260 /$$

$$(5/3 + 3 \cdot (10.812) / (8 \cdot 123.21) - (10.812^3) / (8 \cdot 123.21^3))$$

$$= 152.4 \text{ N./mm}^2$$

AISC Unity Check of Outside Ribs (must be ≤ 1)

$$= S_c / S_{ca} + (R_m \cdot C_1 / I) / S_{ba}$$

$$= 4.0388/152.4 + (104.24 \cdot 140.5/18490034) / 173.33$$

$$= 0.071$$

Check of Inside Ribs:

Inertia of Saddle, Inner Ribs - Axial Direction

	B	D	Y	A	AY	Io
Rib	10.0	271.0	0.0	27.1	0.0	0.185E+04
Web	354.2	10.0	0.0	35.4	0.0	2.95
Totals	...	...	...	62.5	...	0.185E+04

Distance to Centroid from Datum [ytot]:

$$= AY / A$$

$$= 0/62.52$$

$$= 0.000 \text{ mm}$$

Distance to Centroid [C1]:

$$= \text{Saddle Width} / 2$$

$$= 281/2$$

$$= 140.500 \text{ mm}$$

Length of Inner Rib [L]:

$$= \text{Saddle Height} - \text{Outside Radius} - B_{pthk}$$

$$= 915 - \cos(409/2) (16 + 0 + 0) - 0$$

$$= 490.000 \text{ mm}$$

Radius of Gyration [r]:

$$= \sqrt{ (\text{Total Inertia} / \text{Total Area}) }$$

$$= \sqrt{ (1851.9/62.52) }$$

$$= 54.424 \text{ mm}$$

Slenderness ratio [KL/r]:

$$= KL/r$$

$$= 1 \cdot 490/54.424$$

$$= 9.003$$

Unit Force [Force,u]:

$$= FI / (2 \cdot \text{Baseplate Length})$$

$$= 1322.1 / (2 \cdot 780)$$

$$= 0.847 \text{ Kg/mm}$$

Moment at base of inner Rib [Mbase,c]:

$$= \text{Unit Force} \cdot e \cdot L$$

$$= 0.8475 \cdot 354.2 \cdot 490$$



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= 147.092 Kg-m.

Bending Stress due to Transverse Force and Weight Load [σ_B , base, c]:

= Bending Moment / Section Modulus

= 147.09/131806

= 10.944 N./mm²

Compressive Allowable, $KL/r < C_c$ (9.0033 < 123.21) per AISC E2-1 [Sca]:

= ($1 - (KL/r)^2 / (2 \cdot C_c^2)$) $F_y / (5/3 + 3 \cdot (KL/r) / (8 \cdot C_c) - (KL/r)^3 / (8 \cdot C_c^3)$

= ($1 - (9.0033)^2 / (2 \cdot 123.21^2)$) 260 /

($5/3 + 3 \cdot (9.0033) / (8 \cdot 123.21) - (9.0033^3) / (8 \cdot 123.21^3)$)

= 153.1 N./mm²

AISC Unity Check of Inside Ribs (must be ≤ 1)

= $S_c / S_{ca} + (M_{base, c} \cdot C1 / I) / S_{ba}$

= 5.7563/153.07 + (147.09 • 140.5/1851.9)/173.33

= 0.101

Review notes about nozzle loadings and supports in the Warnings report.

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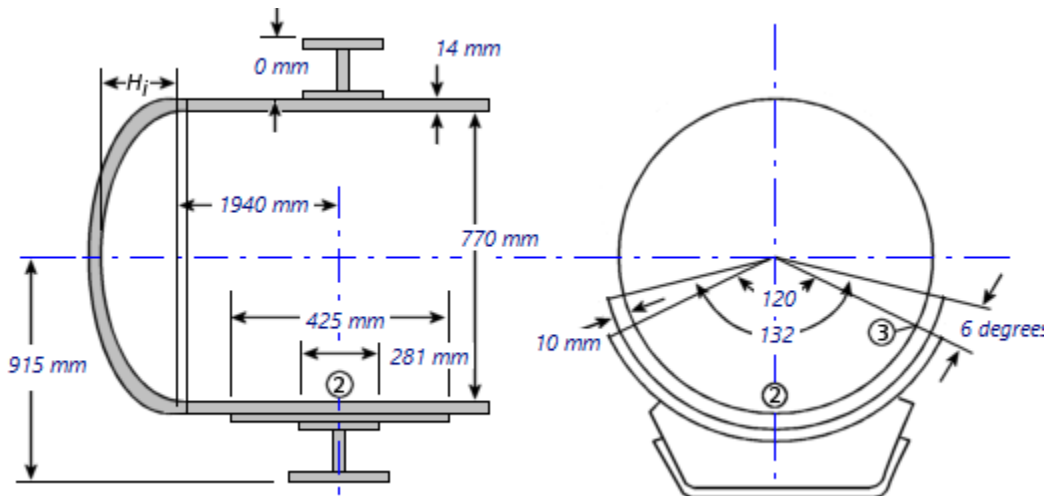
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ASME VIII Division 2 Horizontal Vessel Analysis, Left Saddle:

Horizontal Vessel Stress Calculations : Operating Case

Input and Calculated Values:

Vessel Mean Radius	Rm	393.50	mm
Shell Thickness used in this Case	t	14.000	mm
Stiffened Vessel Length per 4.15.6	L	9680.00	mm
Distance from Saddle to Vessel tangent	a1 or a	1940.00	mm
Saddle Width	b1 or b	281.00	mm
Saddle Bearing Angle	delta or theta	120.00	degrees
Wear Plate Width	b2 or b1	425.00	mm
Wear Plate Bearing Angle	delta2 or theta1	132.00	degrees
Wear Plate Thickness	e2 or tr	10.0	mm
Wear Plate Allowable Stress	fw or Sr	138.00	N./mm ²
Shell Allowable Stress used in Calculation		138.00	N./mm ²
Head Allowable Stress used in Calculation		138.00	N./mm ²
Circumferential Efficiency in Plane of Saddle		1.00	
Circumferential Efficiency at Mid-Span		1.00	
Distance from Saddle Base to Centerline	B	915.00	mm
Coefficient of Friction	mu	0.30	



Results for Horizontal Vessel Analysis:

Saddle Force Q, Operating Case	12824.72	Kgf
Pressure used in Analysis	P 2253.774	KPa.

Horizontal Vessel Analysis Results:	Actual N./mm ²	Allowable N./mm ²
Long. Stress at Top of Midspan	28.84	138.00
Long. Stress at Bottom of Midspan	51.78	138.00
Long. Stress at Top of Saddles	124.27	138.00
Long. Stress at Bottom of Saddles	-6.22	-221.40
Long. Stress at Bottom of Saddles	6.22	138.00
Tangential Shear in Shell	20.38	110.40
Circ. Stress at Horn of Saddle	26.51	172.50



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Circ. Compressive Stress in Shell	1.19	138.00
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Intermediate Results: Saddle Reaction Q due to Wind or Seismic:

Transverse Saddle Reaction Force [Fwt]:
 $= \text{Ftr} (\text{Ft/Num of Saddles} + \text{Z Force Load}) \cdot B / E$
 $= 3 (688.77/2 + 0) \cdot 915/708.41$
 $= 1334.5 \text{ Kgf}$

Longitudinal Saddle Reaction Force [Fwl]:
 $= \max(\text{Fl,wind}; \text{Sum of X Forces}) \cdot B / \text{Ls}$
 $= \max(85.509, 0) \cdot 915/5800$
 $= 13.5 \text{ Kgf}$

Saddle Reaction Force due to Earthquake Fl [Fsl]:
 $= \max(\text{Fl,eq}; \text{Sum of X Forces}) \cdot B / \text{Ls}$
 $= \max(1173.3, 0) \cdot 915/5800$
 $= 185.1 \text{ Kgf}$

Saddle Reaction Force due to Earthquake Ft [Fst]:
 $= \text{Ftr} (\text{Ft/Num of Saddles} + \text{Z Force Load}) \cdot B / E$
 $= 3 (1173.3/2 + 0) \cdot 915/708.41$
 $= 2273.1 \text{ Kgf}$

Load Combination Results for Q + Wind or Seismic [Q]:
 $= \text{Saddle Load} + \max(\text{Fwl}, \text{Fwt}, \text{Fsl}, \text{Fst})$
 $= 10552 + \max(13.49, 1334.5, 185.09, 2273.1)$
 $= 12824.7 \text{ Kgf}$

Longitudinal Wind Force [Fl]:
 $= \text{WindScalar} \cdot \text{WindPress} (\text{Platform Area} + (\pi/4 (\text{OD} \cdot \text{WindDiaMult})^2))$
 $= 1.1 \cdot 766.08 (0 + \pi/4 (0.938 \cdot 1.2)^2)$
 $= 838.548 \text{ N}$

Summary of Loads at the Base of this Saddle:

Vertical Load (including saddle weight)	12976.22	Kgf
Transverse Shear Load	586.63	Kgf
Longitudinal Shear Load	1173.26	Kgf

Note: Wear Plate is Welded to the Shell, $k = 0.1$

Saddle Dimension [E]:
 $= \min(2 (\text{ShellID}/2 + t + \text{WearPadThickness}) \sin(\theta/2), 2 \cdot \text{Rm})$
 $= \min(2 (770/2 + 14 + 10) \sin(120/2), 2 \cdot 393.5)$
 $= 708.409 \text{ mm}$

The Computed K values from Table 4.15.1:

K1 = 0.1066	K2 = 1.1707	K3 = 0.8799	K4 = 0.4011
K5 = 0.7603	K6 = 0.0529	K7 = 0.0529	K8 = 0.3405
K9 = 0.2711	K10 = 0.0581	K1* = 0.1923	K6p = 0.0434
K7p = 0.0434			

The suffix 'p' denotes the values for a wear plate if it exists.

Note: Dimension a is greater than or equal to $\text{Rm}/2$.

Moment per Equation 4.15.1 [M1]:
 $= -Q \cdot a [1 - (1 - a/L + (\text{Rm}^2 - \text{hm}^2) / (2a \cdot L)) / (1 + (4\text{hm}) / (3L))]$
 $= -12825 \cdot 1940 [1 - (1 - 1940/9680 + (393.5^2 - 0^2) / (2 \cdot 1940 \cdot 9680)) / (1 + (4 \cdot 0) / (3 \cdot 9680))]$
 $= -4883.8 \text{ Kg-m.}$

Moment per Equation 4.15.2 [M2]:
 $= Q \cdot L / 4 ((1 + 2 (\text{Rm}^2 - \text{hm}^2) / (L^2)) / (1 + (4\text{hm}^2) / (3L)) - 4a/L)$



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$$= 12825 \cdot 9680 / 4 \cdot ((1 + 2(393.5^2 - 0^2) / (9680^2)) / (1 + (4 \cdot 0) / (3 \cdot 9680))) - 4 \cdot 1940 / 9680$$

$$= 6258.6 \text{ Kg-m.}$$

Longitudinal Stress at Top of Shell (4.15.4) [σ_1]:

$$= P \cdot R_m / (2t) - M_2 / (\pi \cdot R_m^2 \cdot t)$$

$$= 2253.8 \cdot 393.5 / (2 \cdot 11) - 6258.6 / (\pi \cdot 393.5^2 \cdot 11)$$

$$= 28.84 \text{ N./mm}^2$$

Longitudinal Stress at Bottom of Shell (4.15.5) [σ_2]:

$$= P \cdot R_m / (2t) + M_2 / (\pi \cdot R_m^2 \cdot t)$$

$$= 2253.8 \cdot 393.5 / (2 \cdot 11) + 6258.6 / (\pi \cdot 393.5^2 \cdot 11)$$

$$= 51.78 \text{ N./mm}^2$$

Longitudinal Stress at Top of Shell at Support (4.15.8) [σ_{*3}]:

$$= P \cdot R_m / (2t) - M_1 / (K_1 \cdot \pi \cdot R_m^2 \cdot t)$$

$$= 2253.8 \cdot 393.5 / (2 \cdot 11) - 4883.8 / (0.1066 \cdot \pi \cdot 393.5^2 \cdot 11)$$

$$= 124.27 \text{ N./mm}^2$$

Longitudinal Stress at Bottom of Shell at Support (4.15.9) [σ_{*4}]:

$$= P \cdot R_m / (2t) + M_1 / (K_1 \cdot \pi \cdot R_m^2 \cdot t)$$

$$= 2253.8 \cdot 393.5 / (2 \cdot 11) + 4883.8 / (0.1923 \cdot \pi \cdot 393.5^2 \cdot 11)$$

$$= -6.22 \text{ N./mm}^2$$

Maximum Shear Force in the Saddle (4.15.3) [T]:

$$= Q(L - 2a) / (L + (4 \cdot h_m / 3))$$

$$= 12825 \cdot (9680 - 2 \cdot 1940) / (9680 + (4 \cdot 0 / 3))$$

$$= 7684.2 \text{ Kgf}$$

Shear Stress in the shell no rings, not stiffened (4.15.12) [τ_2]:

$$= K_2 \cdot T / (R_m \cdot t)$$

$$= 1.1707 \cdot 7684.2 / (393.5 \cdot 11)$$

$$= 20.38 \text{ N./mm}^2$$

Decay Length (4.15.20) [x_1, x_2]:

$$= 0.78 \cdot \sqrt{(R_m \cdot t)}$$

$$= 0.78 \cdot \sqrt{(393.5 \cdot 11)}$$

$$= 51.317 \text{ mm}$$

Effective reinforcing plate width (4.15.24) [B1]:

$$= \min(b + 1.56 \cdot \sqrt{(R_m \cdot t)}, 2a)$$

$$= \min(281 + 1.56 \cdot \sqrt{(393.5 \cdot 11)}, 2 \cdot 1940)$$

$$= 383.63 \text{ mm}$$

Wear Plate/Shell Stress ratio (4.15.26) [η]:

$$= \min(S_r / S, 1)$$

$$= \min(138 / 138, 1)$$

$$= 1.0000$$

Circumferential Stress at Saddle Base with Wear Plate (4.15.25) [σ_6, r]:

$$= -K_5 \cdot Q \cdot k / (B_1(t + \eta \cdot t_r))$$

$$= -0.7603 \cdot 12825 \cdot 0.1 / (383.63(11 + 1 \cdot 10))$$

$$= -1.19 \text{ N./mm}^2$$

Circ. Comp. Stress at Horn of Saddle, $L \geq 8R_m$ (4.15.28) [σ_7, r]:

$$= -Q / (4(t + \eta \cdot t_r) b_1) - 3 \cdot K_7 \cdot Q / (2(t + \eta \cdot t_r)^2)$$

$$= -12825 / (4(11 + 1 \cdot 10) 383.63) -$$

$$3 \cdot 0.05285 \cdot 12825 / (2(11 + 1 \cdot 10)^2)$$

$$= -26.51 \text{ N./mm}^2$$

Distance between Saddle Supports [Ls]:

$$= 5800.0 \text{ mm}$$

Free Un-Restrained Thermal Expansion between the Saddles [Exp]:

$$= \alpha \cdot L_s (\text{Design Temperature} - \text{Ambient Temperature})$$



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$$= 0.11875E-04 \cdot 5800 (65 - 21.114)$$

$$= 3.023 \text{ mm}$$

Results for Vessel Ribs, Web and Base:

Baseplate Length	Bplen	780.0000	mm
Baseplate Thickness	Bpthk	16.0000	mm
Baseplate Width	Bpwid	310.0000	mm
Number of Ribs (inc. outside ribs)	Nribs	3	
Rib Thickness	Ribtk	10.0000	mm
Web Thickness	Webtk	10.0000	mm
Web Location	Webloc	Center	
Saddle Design Temperature		21.1	°C
Saddle Yield Stress	Sy	246.0	N./mm ²
Height of Web at Center	Hw,c	525.0	mm
Friction Coefficient	mu	0.300	

Moment of Inertia of Saddle - Transverse Direction (90 degrees to long axis)

Inertia of Shell [ShellInertia]:

$$= ((1.56 \cdot \sqrt{(R \cdot t)} + \text{WearPlateWidth})t^3)/12$$

$$= ((1.56 \cdot \sqrt{(388 \cdot 11)} + 425)11^3)/12$$

$$= 5.844 \text{ cm}^4$$

Resolved Inertia for the Shell [I,shell]:

$$= \text{ShellInertia} + A_{\text{Shell}}(C1 - Y)^2$$

$$= 5.8444 + 57.961(199.81 - 5.5)^2$$

$$= 21890.658 \text{ cm}^4$$

	B mm	D mm	Y mm	A cm ²	AY mm ³	I + AD ⁽²⁾ cm ⁴
Shell	526.9	11.0	5.5	58.0	31878.3	0.219E+05
Wearplate	425.0	10.0	16.0	42.5	68000.0	0.144E+05
Web	10.0	490.0	266.0	49.0	1303400.4	0.120E+05
BasePlate	310.0	16.0	519.0	49.6	2574240.0	0.505E+05
Totals	...	...	...	199.1	3977518.8	0.987E+05

Distance to Centroid [C1]:

$$= AY / A$$

$$= 3977518/199.06$$

$$= 199.814 \text{ mm}$$

Angle [β]:

$$= 180 - \text{Saddle Angle}/2$$

$$= 180 - 120/2$$

$$= 120.0$$

Saddle Splitting Coefficient [K1]:

$$= (1 + \cos(\beta) - 0.5 \cdot \sin(\beta)^2) / (\pi - \beta + \sin(\beta)\cos(\beta))$$

$$= (1 + \cos(120) - 0.5 \cdot \sin(120)^2) / (\pi - 2.0944 + \sin(120)\cos(120))$$

$$= 0.2035$$

Saddle Splitting Force [Fh]:

$$= K1 \cdot Q$$

$$= 0.2035 \cdot 12825$$

$$= 2610.1079 \text{ Kgf}$$

$$\text{Tension Stress, } St = (Fh/As) = 1.8141 \text{ N./mm}^2$$

$$\text{Allowed Stress, } Sa = 0.6 \cdot \text{Yield Str} = 147.5999 \text{ N./mm}^2$$

Saddle Splitting Dimension [d]:

$$= B - R \cdot \sin(\theta/2) / (\theta/2 \text{ in radians})$$

$$= 915 - 388 \cdot \sin(120/2) / 1.0472$$

$$= 594.127 \text{ mm}$$



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Bending Moment, $M = F_h \cdot d = 1550.7653 \text{ Kg-m.}$

Bending Stress, $S_b = (M \cdot C_1 / I) = 3.0773 \text{ N./mm}^2$

Allowed Stress, $S_a = 2/3 \cdot \text{Yield Str} = 163.9998 \text{ N./mm}^2$

Minimum Thickness of Baseplate per Moss:

$$= \sqrt{\frac{3(Q + \text{SaddleWt}) \cdot \text{BasePlateWidth}}{4 \cdot \text{BasePlateLength} \cdot \text{AllStress}}}$$

$$= \sqrt{\frac{3(12825 + 151.5) \cdot 310}{4 \cdot 780 \cdot 164}}$$

$$= 15.208 \text{ mm}$$

Calculation of Axial Load, Intermediate Values and Compressive Stress:

Web Length Dimension [Web Length]:

$$= 2 \cdot \cos(90 - \text{Saddle Angle}/2) (\text{Inside Radius} + \text{Shell Thk} + \text{Wear Plate Thk})$$

$$= 2 \cdot \cos(90 - 120/2) (385 + 14 + 10)$$

$$= 708.409 \text{ mm}$$

Distance between Ribs [e]:

$$= \text{Web Length} / (\text{Nr ribs} - 1)$$

$$= 708.41 / (3 - 1)$$

$$= 354.204 \text{ mm}$$

Baseplate Pressure Area [Ap]:

$$= e \cdot \text{Bpwid} / 2$$

$$= 354.2 \cdot 310 / 2$$

$$= 549.017 \text{ cm}^2$$

Bearing Pressure [Bp]:

$$= (Q + \text{SaddleWt}) / (\text{BasePlateLength} \cdot \text{BasePlateWidth})$$

$$= 12976 / (780 \cdot 310)$$

$$= 5.367 \text{ Kg/cm}^2$$

Axial Load [P]:

$$= A_p \cdot B_p$$

$$= 549.02 \cdot 5.3665$$

$$= 2946.303 \text{ Kg}$$

Area of the Rib and Web [Ar]:

$$= \text{Rib Area} + \text{Web Area}$$

$$= 27.1 + 17.71$$

$$= 44.810 \text{ cm}^2$$

Compressive Stress [Sc]:

$$= P / A_r$$

$$= 2946.3 / 44.81$$

$$= 6.448 \text{ N./mm}^2$$

Check of Outside Ribs:

Inertia of Saddle, Outer Ribs - Longitudinal Direction

	B	D	Y	A	AY	Io
Rib+Web	10.0	281.0	...	28.1	...	0.185E+04

Rib dimension [D]:

$$= \text{Saddle Width} - \text{Web Thickness}$$

$$= 281 - 10$$

$$= 271.000 \text{ mm}$$

Distance to Centroid from Datum [ytot]:

$$= AY / A$$

$$= 0 / 44.81$$

$$= 0.000 \text{ mm}$$



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Distance to Centroid [C1]:

$$= \text{Saddle Width} / 2$$

$$= 281/2$$

$$= 140.500 \text{ mm}$$

Radius of Gyration [r]:

$$= \sqrt{\text{Total Inertia} / \text{Total Area}}$$

$$= \sqrt{1849/44.81}$$

$$= 64.236 \text{ mm}$$

Length of Outer Rib [L]:

$$= \text{Saddle Height} - \cos(\theta/2) (\text{radius} + \text{shlthk} + \text{wpdthk}) - \text{bpthk}$$

$$= 915 - \cos(120/2) (385 + 14 + 10) - 16$$

$$= 694.500 \text{ mm}$$

Intermediate Term [Cc]:

$$= \sqrt{\frac{2 \cdot \pi^2 \cdot \text{Elastic Modulus}}{\text{Yield Stress}}}$$

$$= \sqrt{\frac{2 \cdot \pi^2 \cdot 199943088}{246}}$$

$$= 126.667$$

Slenderness ratio [KL/r]:

$$= KL/r$$

$$= 1 \cdot 694.5/64.236$$

$$= 10.812$$

Bending Moment [Rm]:

$$= FI / (2 \cdot B_{\text{plen}}) \cdot e \cdot L / 2$$

$$= 1173.3 / (2 \cdot 780) \cdot 354.2 \cdot 694.5/2$$

$$= 92.507 \text{ Kg-m.}$$

Compressive Allowable, $KL/r < Cc$ ($10.812 < 126.67$) per AISC E2-1 [Sca]:

$$= (1 - (KL/r)^2 / (2 \cdot Cc^2)) F_y / (5/3 + 3 \cdot (KL/r) / (8 \cdot Cc) - (KL/r)^3 / (8 \cdot Cc^3))$$

$$= (1 - (10.812)^2 / (2 \cdot 126.67^2)) 246 / (5/3 + 3 \cdot (10.812) / (8 \cdot 126.67) - (10.812^3) / (8 \cdot 126.67^3))$$

$$= 144.3 \text{ N./mm}^2$$

AISC Unity Check of Outside Ribs (must be ≤ 1)

$$= S_c / S_{ca} + (R_m \cdot C_1 / I) / S_{ba}$$

$$= 6.4481/144.3 + (92.507 \cdot 140.5/18490034) / 164$$

$$= 0.087$$

Check of Inside Ribs:

Inertia of Saddle, Inner Ribs - Axial Direction

	B	D	Y	A	AY	Io
Rib	10.0	271.0	0.0	27.1	0.0	0.185E+04
Web	354.2	10.0	0.0	35.4	0.0	2.95
Totals	...	...	...	62.5	...	0.185E+04

Distance to Centroid from Datum [ytot]:

$$= AY / A$$

$$= 0/62.52$$

$$= 0.000 \text{ mm}$$

Distance to Centroid [C1]:

$$= \text{Saddle Width} / 2$$

$$= 281/2$$

$$= 140.500 \text{ mm}$$

Length of Inner Rib [L]:

$$= \text{Saddle Height} - \text{Outside Radius} - \text{Bpthk}$$

$$= 915 - \cos(409/2) (16 + 0 + 0) - 0$$

$$= 490.000 \text{ mm}$$



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Radius of Gyration [r]:
 $= \sqrt{\text{Total Inertia} / \text{Total Area}}$
 $= \sqrt{1851.9 / 62.52}$
 $= 54.424 \text{ mm}$

Slenderness ratio [KL/r]:
 $= KL/r$
 $= 1 \cdot 490 / 54.424$
 $= 9.003$

Unit Force [Force,u]:
 $= FI / (2 \cdot \text{Baseplate Length})$
 $= 1173.3 / (2 \cdot 780)$
 $= 0.752 \text{ Kg/mm}$

Moment at base of inner Rib [Mbase,c]:
 $= \text{Unit Force} \cdot e \cdot L$
 $= 0.7521 \cdot 354.2 \cdot 490$
 $= 130.535 \text{ Kg-m.}$

Bending Stress due to Transverse Force and Weight Load [σ_B ,base,c]:
 $= \text{Bending Moment} / \text{Section Modulus}$
 $= 130.54 / 131806$
 $= 9.712 \text{ N./mm}^2$

Compressive Allowable, $KL/r < C_c$ ($9.0033 < 126.67$) per AISC E2-1 [Sca]:
 $= (1 - (KL/r)^2 / (2 \cdot C_c^2)) F_y / (5/3 + 3 \cdot (KL/r) / (8 \cdot C_c) - (KL/r)^3 / (8 \cdot C_c^3))$
 $= (1 - (9.0033)^2 / (2 \cdot 126.67^2)) 246 /$
 $(5/3 + 3 \cdot (9.0033) / (8 \cdot 126.67) - (9.0033^3) / (8 \cdot 126.67^3))$
 $= 144.9 \text{ N./mm}^2$

AISC Unity Check of Inside Ribs (must be ≤ 1)
 $= S_c / S_{ca} + (M_{base,c} \cdot C1/I) / S_{ba}$
 $= 9.1901 / 144.91 + (130.54 \cdot 140.5 / 1851.9) / 164$
 $= 0.123$

Input Data for Base Plate Bolting Calculations:

Total Number of Bolts per BasePlate	Nbolts	4
Total Number of Bolts in Tension/Baseplate	Nbt	2
Bolt Design Temperature		21.1 °C
Bolt Material Specification		SA-193 B7
Bolt Allowable Stress	Stba	138.00 N./mm ²
Bolt Corrosion Allowance	Bca	1.5 mm
Distance from Bolts to Edge	Edgedis	50.8 mm
User Defined Area in a Single Bolt	UserBoltArea	2.4000 cm ²
BasePlate Allowable Stress	S	138.00 N./mm ²
Area Available in a Single Bolt	BltArea	1.6469 cm ²
Saddle Load Q0 (Weight)	Q0	10703.1 Kg
Saddle Load QL (Wind/Seismic contribution)	QL	185.1 Kg
Maximum Transverse Force	Ft	586.6 Kg
Maximum Longitudinal Force	F1	1173.3 Kg
Saddle Bolted to Steel Foundation		No

Shear Stress in a Single Bolt, Longitudinal Direction [τ_b ,l]:
 $= FI / (\text{Bolt Area} \cdot \text{Number of Bolts})$
 $= 1173.3 / (1.6469 \cdot 4)$
 $= 17.5 \text{ N./mm}^2$. Must be less than 82.8 N./mm^2 .

Shear Stress in a Single Bolt, Transverse Direction [τ_b ,t]:
 $= Ft / (\text{Bolt Area} \cdot \text{Number of Bolts})$
 $= 586.63 / (1.6469 \cdot 4)$
 $= 8.7 \text{ N./mm}^2$. Must be less than 82.8 N./mm^2 .

Bolt Area Calculation per Dennis R. Moss



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Bolt Area Requirement Due to Longitudinal Load [Bltarear]:
= 0.0 (QO > QL --> No Uplift in Longitudinal direction)

Bolt Area due to Shear Load [Bltarears]:
= $F_l / (BoltShearAllowable \cdot Nbolts)$
= $1173.3 / (82.8 \cdot 4)$
= 0.3474 cm²

Bolt Area due to Transverse Load:

Moment on Baseplate Due to Transverse Load [Rmom]:
= $B \cdot F_t + \text{Sum of X Moments}$
= $915 \cdot 586.63 + 0$
= 536.78 Kg-m.

Eccentricity (e):
= $Rmom / QO$
= $536.78 / 10703$
= 50.15 mm < $B_{plen} / 6$ --> No Uplift in Transverse direction

Bolt Area due to Transverse Load [Bltareart]:
= 0 (No Uplift)

Required Area of a Single Bolt [Bltarear]:
= $\max[Bltarear, Bltarears, Bltareart]$
= $\max[0, 0.3474, 0]$
= 0.3474 cm²

MDMT Calculations for: Left Saddle flo, Saddle to Wearplate Junction, Curve: B

Govrn. thk, $t_g = 10$, $c = 3$ mm, $E^* = 1$
Thickness Ratio = $tr \cdot E^* / (t_g - c) = 0.998$, Temp. Reduction = 0 °C
Attachment governing, Attachment stress = Shell stress per Code

Min Metal Temp. w/o impact per UCS-66, Curve B	-29 °C
Min Metal Temp per UCS-66 and UCS-68(c), PWHT credit	-46 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-46 °C

ASME VIII Division 2 Horizontal Vessel Analysis, Right Saddle:

Input and Calculated Values:

Vessel Mean Radius	Rm	393.50	mm
Shell Thickness used in this Case	t	14.000	mm
Stiffened Vessel Length per 4.15.6	L	9680.00	mm
Distance from Saddle to Vessel tangent	a1 or a	1940.00	mm
Saddle Width	b1 or b	281.00	mm
Saddle Bearing Angle	delta or theta	120.00	degrees
Wear Plate Width	b2 or b1	425.00	mm
Wear Plate Bearing Angle	delta2 or theta1	132.00	degrees
Wear Plate Thickness	e2 or tr	10.0	mm
Wear Plate Allowable Stress	fw or Sr	138.00	N./mm ²
Shell Allowable Stress used in Calculation		138.00	N./mm ²
Head Allowable Stress used in Calculation		138.00	N./mm ²
Circumferential Efficiency in Plane of Saddle		1.00	
Circumferential Efficiency at Mid-Span		1.00	
Distance from Saddle Base to Centerline	B	915.00	mm
Coefficient of Friction	mu	0.00	



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Results for Horizontal Vessel Analysis:

Saddle Force Q, Operating Case 12300.64 Kgf
Pressure used in Analysis P 2253.774 KPa.

Horizontal Vessel Analysis Results:	Actual N./mm ²	Allowable N./mm ²
Long. Stress at Top of Midspan	29.31	138.00
Long. Stress at Bottom of Midspan	51.32	138.00
Long. Stress at Top of Saddles	120.84	138.00
Long. Stress at Bottom of Saddles	-4.32	-221.40
Long. Stress at Bottom of Saddles	4.32	138.00
Tangential Shear in Shell	19.55	110.40
Circ. Stress at Horn of Saddle	25.43	172.50
Circ. Compressive Stress in Shell	1.14	138.00

Intermediate Results: Saddle Reaction Q due to Wind or Seismic:

Transverse Saddle Reaction Force [Fwt]:
= Ftr(Ft/Num of Saddles + Z Force Load) • B / E
= 3 (688.77/2 + 0) • 915/708.41
= 1334.5 Kgf

Longitudinal Saddle Reaction Force [Fwl]:
= max(Fl,wind; Sum of X Forces) • B / Ls
= max(85.509, 0) • 915/5800
= 13.5 Kgf

Saddle Reaction Force due to Earthquake Fl [Fsl]:
= max(Fl,eq; Sum of X Forces) • B / Ls
= max(1173.3, 0) • 915/5800
= 185.1 Kgf

Saddle Reaction Force due to Earthquake Ft [Fst]:
= Ftr(Ft/Num of Saddles + Z Force Load) • B / E
= 3 (1173.3/2 + 0) • 915/708.41
= 2273.1 Kgf

Load Combination Results for Q + Wind or Seismic [Q]:
= Saddle Load + max(Fwl, Fwt, Fsl, Fst)
= 10028 + max(13.49, 1334.5, 185.09, 2273.1)
= 12300.6 Kgf

Longitudinal Wind Force [Fl]:
= WindScalar • WindPress(Platform Area + ($\pi/4$ (OD • WindDiaMult)²))
= 1.1 • 766.08 (0 + $\pi/4$ (0.938 • 1.2)²)
= 838.548 N

Summary of Loads at the Base of this Saddle:

Vertical Load (including saddle weight)	12452.14	Kgf
Transverse Shear Load	586.63	Kgf
Longitudinal Shear Load	1173.26	Kgf

Note: Wear Plate is Welded to the Shell, k = 0.1

Saddle Dimension [E]:
= min(2(ShellID/2 + t + WearPadThickness)sin(θ/2), 2•Rm)
= min(2(770/2 + 14 + 10)sin(120/2), 2•393.5)
= 708.409 mm

The Computed K values from Table 4.15.1:

K1 = 0.1066 K2 = 1.1707 K3 = 0.8799 K4 = 0.4011



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K5 = 0.7603 K6 = 0.0529 K7 = 0.0529 K8 = 0.3405
K9 = 0.2711 K10 = 0.0581 K1* = 0.1923 K6p = 0.0434
K7p = 0.0434

The suffix 'p' denotes the values for a wear plate if it exists.

Note: Dimension a is greater than or equal to Rm/2.

Moment per Equation 4.15.1 [M1]:

$$= -Q \cdot a [1 - (1 - a/L + (Rm^2 - hm^2)/(2a \cdot L))/(1 + (4hm)/3L)]$$

$$= -12301 \cdot 1940 [1 - (1 - 1940/9680 + (393.5^2 - 0^2)/(2 \cdot 1940 \cdot 9680))/(1 + (4 \cdot 0)/(3 \cdot 9680))]$$

$$= -4684.2 \text{ Kg-m.}$$

Moment per Equation 4.15.2 [M2]:

$$= Q \cdot L/4 [(1 + 2(Rm^2 - hm^2)/(L^2))/(1 + (4hm^2)/(3L)) - 4a/L]$$

$$= 12301 \cdot 9680/4 [(1 + 2(393.5^2 - 0^2)/(9680^2))/(1 + (4 \cdot 0)/(3 \cdot 9680)) - 4 \cdot 1940/9680]$$

$$= 6002.8 \text{ Kg-m.}$$

Longitudinal Stress at Top of Shell (4.15.4) [σ_1]:

$$= P \cdot Rm/(2t) - M2/(\pi \cdot Rm^2 t)$$

$$= 2253.8 \cdot 393.5/(2 \cdot 11) - 6002.8/(\pi \cdot 393.5^2 \cdot 11)$$

$$= 29.31 \text{ N./mm}^2$$

Longitudinal Stress at Bottom of Shell (4.15.5) [σ_2]:

$$= P \cdot Rm/(2t) + M2/(\pi \cdot Rm^2 \cdot t)$$

$$= 2253.8 \cdot 393.5/(2 \cdot 11) + 6002.8/(\pi \cdot 393.5^2 \cdot 11)$$

$$= 51.32 \text{ N./mm}^2$$

Longitudinal Stress at Top of Shell at Support (4.15.8) [σ_{*3}]:

$$= P \cdot Rm/(2t) - M1/(K1 \cdot \pi \cdot Rm^2 t)$$

$$= 2253.8 \cdot 393.5/(2 \cdot 11) - 4684.2/(0.1066 \cdot \pi \cdot 393.5^2 \cdot 11)$$

$$= 120.84 \text{ N./mm}^2$$

Longitudinal Stress at Bottom of Shell at Support (4.15.9) [σ_{*4}]:

$$= P \cdot Rm/(2t) + M1/(K1 \cdot \pi \cdot Rm^2 \cdot t)$$

$$= 2253.8 \cdot 393.5/(2 \cdot 11) + 4684.2/(0.1923 \cdot \pi \cdot 393.5^2 \cdot 11)$$

$$= -4.32 \text{ N./mm}^2$$

Maximum Shear Force in the Saddle (4.15.3) [T]:

$$= Q(L - 2a)/(L + (4 \cdot hm/3))$$

$$= 12301 (9680 - 2 \cdot 1940)/(9680 + (4 \cdot 0/3))$$

$$= 7370.2 \text{ Kg}$$

Shear Stress in the shell no rings, not stiffened (4.15.12) [τ_2]:

$$= K2 \cdot T / (Rm \cdot t)$$

$$= 1.1707 \cdot 7370.2/(393.5 \cdot 11)$$

$$= 19.55 \text{ N./mm}^2$$

Decay Length (4.15.20) [x1, x2]:

$$= 0.78 \cdot \sqrt{(Rm \cdot t)}$$

$$= 0.78 \cdot \sqrt{(393.5 \cdot 11)}$$

$$= 51.317 \text{ mm}$$

Effective reinforcing plate width (4.15.24) [B1]:

$$= \min(b + 1.56 \cdot \sqrt{(Rm \cdot t)}, 2a)$$

$$= \min(281 + 1.56 \cdot \sqrt{(393.5 \cdot 11)}, 2 \cdot 1940)$$

$$= 383.63 \text{ mm}$$

Wear Plate/Shell Stress ratio (4.15.26) [eta]:

$$= \min(Sr/S, 1)$$

$$= \min(138/138, 1)$$

$$= 1.0000$$



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Circumferential Stress at Saddle Base with Wear Plate (4.15.25) [σ_6, r]:

$$= -K5 \cdot Q \cdot k / (B1(t + \eta \cdot tr))$$

$$= -0.7603 \cdot 12301 \cdot 0.1 / (383.63(11 + 1 \cdot 10))$$

$$= -1.14 \text{ N./mm}^2$$

Circ. Comp. Stress at Horn of Saddle, $L \geq 8R_m$ (4.15.28) [σ_7, r]:

$$= -Q / (4(t + \eta \cdot tr)b1) - 3 \cdot K7 \cdot Q / (2(t + \eta \cdot tr)^2)$$

$$= -12301 / (4(11 + 1 \cdot 10)383.63) - 3 \cdot 0.05285 \cdot 12301 / (2(11 + 1 \cdot 10)^2)$$

$$= -25.43 \text{ N./mm}^2$$

Results for Vessel Ribs, Web and Base:

Baseplate Length	Bplen	780.0000	mm
Baseplate Thickness	Bpthk	16.0000	mm
Baseplate Width	Bpwid	310.0000	mm
Number of Ribs (inc. outside ribs)	Nribs	3	
Rib Thickness	Ribtk	10.0000	mm
Web Thickness	Webtk	10.0000	mm
Web Location	Webloc	Center	
Saddle Design Temperature		21.1	°C
Saddle Yield Stress	Sy	246.0	N./mm ²
Height of Web at Center	Hw,c	525.0	mm
Friction Coefficient	mu	0.000	

Moment of Inertia of Saddle - Transverse Direction (90 degrees to long axis)

Inertia of Shell [ShellInertia]:

$$= ((1.56 \cdot \sqrt{(R \cdot t) + \text{WearPlateWidth}})t^3) / 12$$

$$= ((1.56 \cdot \sqrt{(388 \cdot 11) + 425})11^3) / 12$$

$$= 5.844 \text{ cm}^4$$

Resolved Inertia for the Shell [I,shell]:

$$= \text{ShellInertia} + A_{\text{Shell}}(C1 - Y)^2$$

$$= 5.8444 + 57.961(199.81 - 5.5)^2$$

$$= 21890.658 \text{ cm}^4$$

	B mm	D mm	Y mm	A cm ²	AY mm ³	I + AD ² cm ⁴
Shell	526.9	11.0	5.5	58.0	31878.3	0.219E+05
Wearplate	425.0	10.0	16.0	42.5	68000.0	0.144E+05
Web	10.0	490.0	266.0	49.0	1303400.4	0.120E+05
BasePlate	310.0	16.0	519.0	49.6	2574240.0	0.505E+05
Totals	...	...	...	199.1	3977518.8	0.987E+05

Distance to Centroid [C1]:

$$= AY / A$$

$$= 3977518 / 199.06$$

$$= 199.814 \text{ mm}$$

Angle [β]:

$$= 180 - \text{Saddle Angle} / 2$$

$$= 180 - 120 / 2$$

$$= 120.0$$

Saddle Splitting Coefficient [K1]:

$$= (1 + \cos(\beta) - 0.5 \cdot \sin(\beta)^2) / (\pi - \beta + \sin(\beta) \cos(\beta))$$

$$= (1 + \cos(120) - 0.5 \cdot \sin(120)^2) / (\pi - 2.0944 + \sin(120) \cos(120))$$

$$= 0.2035$$

Saddle Splitting Force [Fh]:

$$= K1 \cdot Q$$

$$= 0.2035 \cdot 12301$$

$$= 2503.4470 \text{ Kgf}$$



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Tension Stress, $St = (Fh/As) = 1.7400 \text{ N./mm}^2$
 Allowed Stress, $Sa = 0.6 * \text{Yield Str} = 147.5999 \text{ N./mm}^2$

Saddle Splitting Dimension [d]:
 $= B - R \cdot \sin(\theta/2) / (\theta/2 \text{ in radians})$
 $= 915 - 388 \cdot \sin(120/2) / 1.0472$
 $= 594.127 \text{ mm}$

Bending Moment, $M = Fh * d = 1487.3939 \text{ Kg-m.}$

Bending Stress, $Sb = (M * C1 / I) = 2.9515 \text{ N./mm}^2$
 Allowed Stress, $Sa = 2/3 * \text{Yield Str} = 163.9998 \text{ N./mm}^2$

Minimum Thickness of Baseplate per Moss:

$= \sqrt{(3(Q + \text{SaddleWt}) \text{BasePlateWidth} / (4 * \text{BasePlateLength} * \text{AllStress}))}$
 $= \sqrt{(3(12301 + 151.5)310 / (4 * 780 * 164))}$
 $= 14.898 \text{ mm}$

Calculation of Axial Load, Intermediate Values and Compressive Stress:

Web Length Dimension [Web Length]:
 $= 2 * \cos(90 - \text{Saddle Angle}/2) (\text{Inside Radius} + \text{Shell Thk} + \text{Wear Plate Thk})$
 $= 2 * \cos(90 - 120/2) (385 + 14 + 10)$
 $= 708.409 \text{ mm}$

Distance between Ribs [e]:
 $= \text{Web Length} / (N_{\text{ribs}} - 1)$
 $= 708.41 / (3 - 1)$
 $= 354.204 \text{ mm}$

Baseplate Pressure Area [Ap]:
 $= e * B_{\text{pwid}} / 2$
 $= 354.2 * 310 / 2$
 $= 549.017 \text{ cm}^2$

Bearing Pressure [Bp]:
 $= (Q + \text{SaddleWt}) / (\text{BasePlateLength} * \text{BasePlateWidth})$
 $= 12452 / (780 * 310)$
 $= 5.150 \text{ Kg/cm}^2$

Axial Load [P]:
 $= Ap * Bp$
 $= 549.02 * 5.1498$
 $= 2827.310 \text{ Kg}$

Area of the Rib and Web [Ar]:
 $= \text{Rib Area} + \text{Web Area}$
 $= 27.1 + 17.71$
 $= 44.810 \text{ cm}^2$

Compressive Stress [Sc]:
 $= P/Ar$
 $= 2827.3/44.81$
 $= 6.188 \text{ N./mm}^2$

Check of Outside Ribs:

Inertia of Saddle, Outer Ribs - Longitudinal Direction

	B	D	Y	A	AY	Io
Rib+Web	10.0	281.0	...	28.1	...	0.185E+04

Rib dimension [D]:



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= Saddle Width - Web Thickness

= 281 - 10

= 271.000 mm

Distance to Centroid from Datum [ytot]:

= AY / A

= 0/44.81

= 0.000 mm

Distance to Centroid [C1]:

= Saddle Width / 2

= 281/2

= 140.500 mm

Radius of Gyration [r]:

= $\sqrt{\text{Total Inertia} / \text{Total Area}}$

= $\sqrt{1849/44.81}$

= 64.236 mm

Length of Outer Rib [L]:

= Saddle Height - cos(theta/2)(radius + shlthk + wpdthk) - bpthk

= 915 - cos(120/2)(385 + 14 + 10) - 16

= 694.500 mm

Intermediate Term [Cc]:

= $\sqrt{2 \cdot \pi^2 \cdot \text{Elastic Modulus} / \text{Yield Stress}}$

= $\sqrt{2 \cdot \pi^2 \cdot 199943008/246}$

= 126.667

Slenderness ratio [KL/r]:

= KL/r

= 1 • 694.5/64.236

= 10.812

Bending Moment [Rm]:

= FI / (2 • Bplen) • e • L / 2

= 1173.3/ (2 • 780) • 354.2 • 694.5/2

= 92.507 Kg-m.

Compressive Allowable, KL/r < Cc (10.812 < 126.67) per AISC E2-1 [Sca]:

= $(1 - (KL/r)^2 / (2 \cdot Cc^2)) Fy / (5/3 + 3 \cdot (KL/r) / (8 \cdot Cc) - (KL/r)^3 / (8 \cdot Cc^3))$

= $(1 - (10.812)^2 / (2 \cdot 126.67^2)) 246 /$

$(5/3 + 3 \cdot (10.812) / (8 \cdot 126.67) - (10.812^3) / (8 \cdot 126.67^3))$

= 144.3 N./mm²

AISC Unity Check of Outside Ribs (must be ≤ 1)

= Sc/Sca + (Rm • C1 / I)/Sba

= 6.1876/144.3 + (92.507 • 140.5/18490034)/164

= 0.085

Check of Inside Ribs:

Inertia of Saddle, Inner Ribs - Axial Direction

	B	D	Y	A	AY	Io
Rib	10.0	271.0	0.0	27.1	0.0	0.185E+04
Web	354.2	10.0	0.0	35.4	0.0	2.95
Totals	...	...	...	62.5	...	0.185E+04

Distance to Centroid from Datum [ytot]:

= AY / A

= 0/62.52

= 0.000 mm

Distance to Centroid [C1]:



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= Saddle Width / 2
= 281/2
= 140.500 mm

Length of Inner Rib [L]:
= Saddle Height - Outside Radius - Bpthk
= 915 - cos(409/2)(16 + 0 + 0) - 0
= 490.000 mm

Radius of Gyration [r]:
= $\sqrt{\text{Total Inertia} / \text{Total Area}}$
= $\sqrt{1851.9/62.52}$
= 54.424 mm

Slenderness ratio [KL/r]:
= KL/r
= 1 • 490/54.424
= 9.003

Unit Force [Force,u]:
= FI / (2 • Baseplate Length)
= 1173.3/(2 • 780)
= 0.752 Kg/mm

Moment at base of inner Rib [Mbase,c]:
= Unit Force • e • L
= 0.7521 • 354.2 • 490
= 130.535 Kg-m.

Bending Stress due to Transverse Force and Weight Load [σ_B ,base,c]:
= Bending Moment / Section Modulus
= 130.54/131806
= 9.712 N./mm²

Compressive Allowable, KL/r < Cc (9.0033 < 126.67) per AISC E2-1 [Sca]:
= $(1 - (KL/r)^2 / (2 \cdot Cc^2)) F_y / (5/3 + 3 \cdot (KL/r) / (8 \cdot Cc) - (KL/r)^3 / (8 \cdot Cc^3))$
= $(1 - (9.0033)^2 / (2 \cdot 126.67^2)) 246 / (5/3 + 3 \cdot (9.0033) / (8 \cdot 126.67) - (9.0033^3) / (8 \cdot 126.67^3))$
= 144.9 N./mm²

AISC Unity Check of Inside Ribs (must be ≤ 1)
= Sc/Sca + (Mbase,c • C1/I)/Sba
= 8.8189/144.91 + (130.54 • 140.5/1851.9)/164
= 0.120

Input Data for Base Plate Bolting Calculations:

Total Number of Bolts per BasePlate	Nbolts	4
Total Number of Bolts in Tension/Baseplate	Nbt	2
Bolt Design Temperature		21.1 °C
Bolt Material Specification		SA-193 B7
Bolt Allowable Stress	Stba	138.00 N./mm ²
Bolt Corrosion Allowance	Bca	1.5 mm
Distance from Bolts to Edge	Edgedis	50.8 mm
User Defined Area in a Single Bolt	UserBoltArea	2.4000 cm ²
BasePlate Allowable Stress	S	138.00 N./mm ²
Area Available in a Single Bolt	BltArea	1.6469 cm ²
Saddle Load Q0 (Weight)	Q0	10179.0 Kg
Saddle Load QL (Wind/Seismic contribution)	QL	185.1 Kg
Maximum Transverse Force	Ft	586.6 Kg
Maximum Longitudinal Force	F1	1173.3 Kg
Saddle Bolted to Steel Foundation		No

Friction force due to thermal growth, [Ff]:
= 3210.9 Kg



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Saddle Calcs: Operating Case: Step: 23 9:41am Sep 9, 2026

Shear Stress in a Single Bolt, Longitudinal Direction [τ_b, l]:

= $\max(F_f, F_l) / (\text{Bolt Area} \cdot \text{Number of Bolts})$

= $3210.9 / (1.6469 \cdot 4)$

= 47.8 N./mm². Must be less than 82.8 N./mm².

Shear Stress in a Single Bolt, Transverse Direction [τ_b, t]:

= $F_t / (\text{Bolt Area} \cdot \text{Number of Bolts})$

= $586.63 / (1.6469 \cdot 4)$

= 8.7 N./mm². Must be less than 82.8 N./mm².

Bolt Area Calculation per Dennis R. Moss

Bolt Area Requirement Due to Longitudinal Load [Bltarearl]:

= 0.0 (QO > QL --> No Uplift in Longitudinal direction)

Bolt Area due to Shear Load [Bltarears]:

= $F_l / (\text{BoltShearAllowable} \cdot \text{Nbols})$

= $1173.3 / (82.8 \cdot 4)$

= 0.3474 cm²

Bolt Area due to Transverse Load:

Moment on Baseplate Due to Transverse Load [Rmom]:

= $B \cdot F_t + \text{Sum of X Moments}$

= $915 \cdot 586.63 + 0$

= 536.78 Kg-m.

Eccentricity (e):

= R_{mom} / QO

= $536.78 / 10179$

= 52.73 mm < Bplen/6 --> No Uplift in Transverse direction

Bolt Area due to Transverse Load [Bltareart]:

= 0 (No Uplift)

Required Area of a Single Bolt [Bltarear]:

= $\max[Bltarearl, Bltarears, Bltareart]$

= $\max[0, 0.3474, 0]$

= 0.3474 cm²

MDMT Calculations for: Right Saddle fi, Saddle to Wearplate Junction, Curve: B

Govrn. thk, tg = 10, c = 3 mm, E* = 1

Thickness Ratio = $tr \cdot E^* / (tg - c) = 0.998$, Temp. Reduction = 0 °C

Attachment governing, Attachment stress = Shell stress per Code

Min Metal Temp. w/o impact per UCS-66, Curve B -29 °C

Min Metal Temp per UCS-66 and UCS-68(c), PWHT credit -46 °C

Min Metal Temp. at Required thickness (UCS 66.1) -46 °C

Review notes about nozzle loadings and supports in the Warnings report.

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Saddle Calcs: Test Case:

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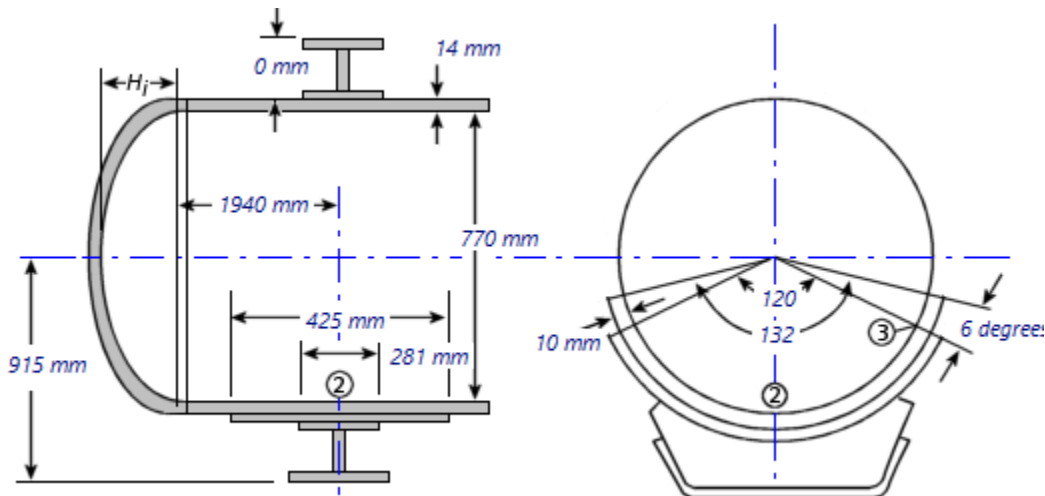
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ASME VIII Division 2 Horizontal Vessel Analysis, Left Saddle:

Horizontal Vessel Stress Calculations : Test Case

Input and Calculated Values:

Vessel Mean Radius	Rm	393.50	mm
Shell Thickness used in this Case	t	14.000	mm
Stiffened Vessel Length per 4.15.6	L	9680.00	mm
Distance from Saddle to Vessel tangent	a1 or a	1940.00	mm
Saddle Width	b1 or b	281.00	mm
Saddle Bearing Angle	delta or theta	120.00	degrees
Wear Plate Width	b2 or b1	425.00	mm
Wear Plate Bearing Angle	delta2 or theta1	132.00	degrees
Wear Plate Thickness	e2 or tr	10.0	mm
Wear Plate Allowable Stress	fw or Sr	138.00	N./mm ²
Shell Allowable Stress used in Calculation		234.00	N./mm ²
Head Allowable Stress used in Calculation		234.00	N./mm ²
Circumferential Efficiency in Plane of Saddle		1.00	
Circumferential Efficiency at Mid-Span		1.00	
Distance from Saddle Base to Centerline	B	915.00	mm
Coefficient of Friction	mu	0.30	



Results for Horizontal Vessel Analysis:

Saddle Force Q, Test Case, no Ext. Forces	10321.70	Kgf
Pressure used in Analysis	P	2928.857 KPa.

Horizontal Vessel Analysis Results:	Actual N./mm ²	Allowable N./mm ²
Long. Stress at Top of Midspan	43.16	234.00
Long. Stress at Bottom of Midspan	61.62	234.00
Long. Stress at Top of Saddles	119.96	234.00
Long. Stress at Bottom of Saddles	14.94	234.00
Tangential Shear in Shell	16.40	138.58
Circ. Stress at Horn of Saddle	21.34	350.99
Circ. Compressive Stress in Shell	0.96	234.00



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Saddle Calcs: Test Case:

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Intermediate Results: Saddle Reaction Q due to Wind or Seismic:

Transverse Saddle Reaction Force [Fwt]:

$$= \text{Ftr}(\text{Ft/Num of Saddles} + \text{Z Force Load}) \cdot B / E$$

$$= 3 (344.39/2 + 0) \cdot 915/708.41$$

$$= 667.2 \text{ Kgf}$$

Longitudinal Saddle Reaction Force [Fwl]:

$$= \max(\text{Fl,wind}; \text{Sum of X Forces}) \cdot B / \text{Ls}$$

$$= \max(42.755, 0) \cdot 915/5800$$

$$= 6.7 \text{ Kgf}$$

Load Combination Results for Q + Wind or Seismic [Q]:

$$= \text{Saddle Load} + \max(\text{Fwl}, \text{Fwt}, \text{Fsl}, \text{Fst})$$

$$= 9654.5 + \max(6.7449, 667.23, 0, 0)$$

$$= 10321.7 \text{ Kgf}$$

Longitudinal Wind Force [Fl]:

$$= \text{WindScalar} \cdot \text{WindPress}(\text{Platform Area} + (\pi/4(\text{OD} \cdot \text{WindDiaMult})^2))$$

$$= 1.1 \cdot 766.08 (0 + \pi/4(0.938 \cdot 1.2)^2)$$

$$= 838.548 \text{ N}$$

Summary of Loads at the Base of this Saddle:

Vertical Load (including saddle weight)	10473.20	Kgf
Transverse Shear Load	172.19	Kgf
Longitudinal Shear Load	42.75	Kgf

Note: Wear Plate is Welded to the Shell, $k = 0.1$

Saddle Dimension [E]:

$$= \min(2(\text{ShellID}/2 + t + \text{WearPadThickness})\sin(\theta/2), 2 \cdot \text{Rm})$$

$$= \min(2(770/2 + 14 + 10)\sin(120/2), 2 \cdot 393.5)$$

$$= 708.409 \text{ mm}$$

The Computed K values from Table 4.15.1:

K1 = 0.1066	K2 = 1.1707	K3 = 0.8799	K4 = 0.4011
K5 = 0.7603	K6 = 0.0529	K7 = 0.0529	K8 = 0.3405
K9 = 0.2711	K10 = 0.0581	K1* = 0.1923	K6p = 0.0434
K7p = 0.0434			

The suffix 'p' denotes the values for a wear plate if it exists.

Note: Dimension a is greater than or equal to $\text{Rm}/2$.

Moment per Equation 4.15.1 [M1]:

$$= -Q \cdot a [1 - (1 - a/L + (\text{Rm}^2 - \text{hm}^2)/(\text{L}^2)) / (1 + (4\text{hm})/3\text{L})]$$

$$= -10322 \cdot 1940 [1 - (1 - 1940/9680 + (393.5^2 - 0^2)/(2 \cdot 1940 \cdot 9680)) / (1 + (4 \cdot 0)/(3 \cdot 9680))]$$

$$= -3930.6 \text{ Kg-m.}$$

Moment per Equation 4.15.2 [M2]:

$$= Q \cdot L/4 ((1 + 2(\text{Rm}^2 - \text{hm}^2)/(\text{L}^2)) / (1 + (4\text{hm}^2)/(3\text{L})) - 4a/L)$$

$$= 10322 \cdot 9680/4 ((1 + 2(393.5^2 - 0^2)/(9680^2)) / (1 + (4 \cdot 0)/(3 \cdot 9680)) - 4 \cdot 1940/9680)$$

$$= 5037.1 \text{ Kg-m.}$$

Longitudinal Stress at Top of Shell (4.15.4) [σ_1]:

$$= P \cdot \text{Rm}/(2t) - M2/(\pi \cdot \text{Rm}^2 t)$$

$$= 2928.9 \cdot 393.5/(2 \cdot 11) - 5037.1/(\pi \cdot 393.5^2 \cdot 11)$$

$$= 43.16 \text{ N/mm}^2$$

Longitudinal Stress at Bottom of Shell (4.15.5) [σ_2]:

$$= P \cdot \text{Rm}/(2t) + M2/(\pi \cdot \text{Rm}^2 \cdot t)$$



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$$= 2928.9 \cdot 393.5 / (2 \cdot 11) + 5037.1 / (\pi \cdot 393.5^2 \cdot 11)$$

$$= 61.62 \text{ N./mm}^2$$

Longitudinal Stress at Top of Shell at Support (4.15.8) [σ_3]:

$$= P \cdot R_m / (2t) - M1 / (K1 \cdot \pi \cdot R_m^2 t)$$

$$= 2928.9 \cdot 393.5 / (2 \cdot 11) - 3930.6 / (0.1066 \cdot \pi \cdot 393.5^2 \cdot 11)$$

$$= 119.96 \text{ N./mm}^2$$

Longitudinal Stress at Bottom of Shell at Support (4.15.9) [σ_4]:

$$= P \cdot R_m / (2t) + M1 / (K1 \cdot \pi \cdot R_m^2 \cdot t)$$

$$= 2928.9 \cdot 393.5 / (2 \cdot 11) + 3930.6 / (0.1923 \cdot \pi \cdot 393.5^2 \cdot 11)$$

$$= 14.94 \text{ N./mm}^2$$

Maximum Shear Force in the Saddle (4.15.3) [T]:

$$= Q(L - 2a) / (L + (4 \cdot h_m / 3))$$

$$= 10322 (9680 - 2 \cdot 1940) / (9680 + (4 \cdot 0 / 3))$$

$$= 6184.5 \text{ Kg}$$

Shear Stress in the shell no rings, not stiffened (4.15.12) [τ_2]:

$$= K2 \cdot T / (R_m \cdot t)$$

$$= 1.1707 \cdot 6184.5 / (393.5 \cdot 11)$$

$$= 16.40 \text{ N./mm}^2$$

Decay Length (4.15.20) [x_1, x_2]:

$$= 0.78 \cdot \sqrt{(R_m \cdot t)}$$

$$= 0.78 \cdot \sqrt{(393.5 \cdot 11)}$$

$$= 51.317 \text{ mm}$$

Effective reinforcing plate width (4.15.24) [B1]:

$$= \min(b + 1.56 \cdot \sqrt{(R_m \cdot t)}, 2a)$$

$$= \min(281 + 1.56 \cdot \sqrt{(393.5 \cdot 11)}, 2 \cdot 1940)$$

$$= 383.63 \text{ mm}$$

Wear Plate/Shell Stress ratio (4.15.26) [η]:

$$= 1.0000 \text{ Materials are the same, test case}$$

Circumferential Stress at Saddle Base with Wear Plate (4.15.25) [σ_6, r]:

$$= -K5 \cdot Q \cdot k / (B1(t + \eta \cdot tr))$$

$$= -0.7603 \cdot 10322 \cdot 0.1 / (383.63(11 + 1 \cdot 10))$$

$$= -0.96 \text{ N./mm}^2$$

Circ. Comp. Stress at Horn of Saddle, $L \geq 8R_m$ (4.15.28) [σ_7, r]:

$$= -Q / (4(t + \eta \cdot tr)b1) - 3 \cdot K7 \cdot Q / (2(t + \eta \cdot tr)^2)$$

$$= -10322 / (4(11 + 1 \cdot 10)383.63) -$$

$$3 \cdot 0.05285 \cdot 10322 / (2(11 + 1 \cdot 10)^2)$$

$$= -21.34 \text{ N./mm}^2$$

Results for Vessel Ribs, Web and Base:

Baseplate Length	Bplen	780.0000	mm
Baseplate Thickness	Bpthk	16.0000	mm
Baseplate Width	Bpwid	310.0000	mm
Number of Ribs (inc. outside ribs)	Nribs	3	
Rib Thickness	Ribtk	10.0000	mm
Web Thickness	Webtk	10.0000	mm
Web Location	Webloc	Center	
Saddle Yield Stress	Sy	260.0	N./mm ²
Height of Web at Center	Hw,c	525.0	mm
Friction Coefficient	mu	0.300	

Moment of Inertia of Saddle - Transverse Direction (90 degrees to long axis)

Inertia of Shell [ShellInertia]:

$$= ((1.56 \cdot \sqrt{(R \cdot t)} + \text{WearPlateWidth})^3) / 12$$

$$= ((1.56 \cdot \sqrt{(388 \cdot 11)} + 425)^3) / 12$$

$$= 5.844 \text{ cm}^4$$



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Resolved Inertia for the Shell [I,shell]:

$$= \text{ShellInertia} + A_{\text{Shell}}(C1 - Y)^2$$

$$= 5.8444 + 57.961 (199.81 - 5.5)^2$$

$$= 21890.658 \text{ cm}^4$$

	B mm	D mm	Y mm	A cm ²	AY mm ³	I + AD ² cm ⁴
Shell	526.9	11.0	5.5	58.0	31878.3	0.219E+05
Wearplate	425.0	10.0	16.0	42.5	68000.0	0.144E+05
Web	10.0	490.0	266.0	49.0	1303400.4	0.120E+05
BasePlate	310.0	16.0	519.0	49.6	2574240.0	0.505E+05
Totals	...	...	...	199.1	3977518.8	0.987E+05

Distance to Centroid [C1]:

$$= AY / A$$

$$= 3977518/199.06$$

$$= 199.814 \text{ mm}$$

Angle [β]:

$$= 180 - \text{Saddle Angle}/2$$

$$= 180 - 120/2$$

$$= 120.0$$

Saddle Splitting Coefficient [K1]:

$$= (1 + \cos(\beta) - 0.5 \cdot \sin(\beta)^2) / (\pi - \beta + \sin(\beta) \cos(\beta))$$

$$= (1 + \cos(120) - 0.5 \cdot \sin(120)^2) / (\pi - 2.0944 + \sin(120) \cos(120))$$

$$= 0.2035$$

Saddle Splitting Force [Fh]:

$$= K1 \cdot Q$$

$$= 0.2035 \cdot 10322$$

$$= 2100.6892 \text{ Kg}$$

$$\text{Tension Stress, } St = (Fh / As) = 1.4600 \text{ N./mm}^2$$

$$\text{Allowed Stress, } Sa = 0.6 \cdot \text{Yield Str} = 155.9972 \text{ N./mm}^2$$

Saddle Splitting Dimension [d]:

$$= B - R \cdot \sin(\theta/2) / (\theta/2 \text{ in radians})$$

$$= 915 - 388 \cdot \sin(120/2) / 1.0472$$

$$= 594.127 \text{ mm}$$

$$\text{Bending Moment, } M = Fh \cdot d = 1248.1000 \text{ Kg-m.}$$

$$\text{Bending Stress, } Sb = (M \cdot C1 / I) = 2.4767 \text{ N./mm}^2$$

$$\text{Allowed Stress, } Sa = 2/3 \cdot \text{Yield Str} = 173.3302 \text{ N./mm}^2$$

Minimum Thickness of Baseplate per Moss:

$$= \sqrt[3]{(3(Q + \text{SaddleWt}) \cdot \text{BasePlateWidth}) / (4 \cdot \text{BasePlateLength} \cdot \text{AllStress})}$$

$$= \sqrt[3]{(3(10322 + 151.5) \cdot 310) / (4 \cdot 780 \cdot 173.33)}$$

$$= 13.290 \text{ mm}$$

Calculation of Axial Load, Intermediate Values and Compressive Stress:

Web Length Dimension [Web Length]:

$$= 2 \cdot \cos(90 - \text{Saddle Angle}/2) (\text{Inside Radius} + \text{Shell Thk} + \text{Wear Plate Thk})$$

$$= 2 \cdot \cos(90 - 120/2) (385 + 14 + 10)$$

$$= 708.409 \text{ mm}$$

Distance between Ribs [e]:

$$= \text{Web Length} / (N_{\text{ribs}} - 1)$$

$$= 708.41 / (3 - 1)$$

$$= 354.204 \text{ mm}$$



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Baseplate Pressure Area [Ap]:

$$= e \cdot Bpwid / 2$$

$$= 354.2 \cdot 310 / 2$$

$$= 549.017 \text{ cm}^2$$

Bearing Pressure [Bp]:

$$= (Q + SaddleWt) / (BasePlateLength \cdot BasePlateWidth)$$

$$= 10473 / (780 \cdot 310)$$

$$= 4.331 \text{ Kg/cm}^2$$

Axial Load [P]:

$$= Ap \cdot Bp$$

$$= 549.02 \cdot 4.3313$$

$$= 2377.982 \text{ Kg}$$

Area of the Rib and Web [Ar]:

$$= Rib \text{ Area} + Web \text{ Area}$$

$$= 27.1 + 17.71$$

$$= 44.810 \text{ cm}^2$$

Compressive Stress [Sc]:

$$= P / Ar$$

$$= 2378 / 44.81$$

$$= 5.204 \text{ N./mm}^2$$

Check of Outside Ribs:

Inertia of Saddle, Outer Ribs - Longitudinal Direction

	B	D	Y	A	AY	Io
Rib+Web	10.0	281.0	...	28.1	...	0.185E+04

Rib dimension [D]:

$$= Saddle \text{ Width} - Web \text{ Thickness}$$

$$= 281 - 10$$

$$= 271.000 \text{ mm}$$

Distance to Centroid from Datum [ytot]:

$$= AY / A$$

$$= 0 / 44.81$$

$$= 0.000 \text{ mm}$$

Distance to Centroid [C1]:

$$= Saddle \text{ Width} / 2$$

$$= 281 / 2$$

$$= 140.500 \text{ mm}$$

Radius of Gyration [r]:

$$= \sqrt{ (Total \text{ Inertia} / Total \text{ Area}) }$$

$$= \sqrt{ (1849 / 44.81) }$$

$$= 64.236 \text{ mm}$$

Length of Outer Rib [L]:

$$= Saddle \text{ Height} - \cos(\theta / 2) (radius + shlthk + wpdthk) - bpthk$$

$$= 915 - \cos(120 / 2) (385 + 14 + 10) - 16$$

$$= 694.500 \text{ mm}$$

Intermediate Term [Cc]:

$$= \sqrt{ (2 \cdot \pi^2 \cdot Elastic \text{ Modulus} / Yield \text{ Stress}) }$$

$$= \sqrt{ (2 \cdot \pi^2 \cdot 199943088 / 260) }$$

$$= 123.211$$

Slenderness ratio [KL/r]:

$$= KL / r$$



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$$= 1 \cdot 694.5/64.236$$

$$= 10.812$$

Bending Moment [Rm]:

$$= FI / (2 \cdot Bplen) \cdot e \cdot L / 2$$

$$= 42.755 / (2 \cdot 780) \cdot 354.2 \cdot 694.5/2$$

$$= 3.371 \text{ Kg-m.}$$

Compressive Allowable, $KL/r < Cc$ ($10.812 < 123.21$) per AISC E2-1 [Sca]:

$$= (1 - (KL/r)^2 / (2 \cdot Cc^2)) Fy / (5/3 + 3 \cdot (KL/r) / (8 \cdot Cc) - (KL/r)^3 / (8 \cdot Cc^3))$$

$$= (1 - (10.812)^2 / (2 \cdot 123.21^2)) 260 /$$

$$(5/3 + 3 \cdot (10.812) / (8 \cdot 123.21) - (10.812^3) / (8 \cdot 123.21^3))$$

$$= 152.4 \text{ N/mm}^2$$

AISC Unity Check of Outside Ribs (must be ≤ 1)

$$= Sc/Sca + (Rm \cdot C1 / I) / Sba$$

$$= 5.2043/152.4 + (3.371 \cdot 140.5/18490034) / 173.33$$

$$= 0.036$$

Check of Inside Ribs:

Inertia of Saddle, Inner Ribs - Axial Direction

	B	D	Y	A	AY	Io
Rib	10.0	271.0	0.0	27.1	0.0	0.185E+04
Web	354.2	10.0	0.0	35.4	0.0	2.95
Totals	...	...	...	62.5	...	0.185E+04

Distance to Centroid from Datum [ytot]:

$$= AY / A$$

$$= 0/62.52$$

$$= 0.000 \text{ mm}$$

Distance to Centroid [C1]:

$$= \text{Saddle Width} / 2$$

$$= 281/2$$

$$= 140.500 \text{ mm}$$

Length of Inner Rib [L]:

$$= \text{Saddle Height} - \text{Outside Radius} - Bpthk$$

$$= 915 - \cos(409/2) (16 + 0 + 0) - 0$$

$$= 490.000 \text{ mm}$$

Radius of Gyration [r]:

$$= \sqrt{ (\text{Total Inertia} / \text{Total Area}) }$$

$$= \sqrt{ (1851.9/62.52) }$$

$$= 54.424 \text{ mm}$$

Slenderness ratio [KL/r]:

$$= KL/r$$

$$= 1 \cdot 490/54.424$$

$$= 9.003$$

Unit Force [Force,u]:

$$= FI / (2 \cdot \text{Baseplate Length})$$

$$= 42.755 / (2 \cdot 780)$$

$$= 0.027 \text{ Kg/mm}$$

Moment at base of inner Rib [Mbase,c]:

$$= \text{Unit Force} \cdot e \cdot L$$

$$= 0.02741 \cdot 354.2 \cdot 490$$

$$= 4.757 \text{ Kg-m.}$$

Bending Stress due to Transverse Force and Weight Load [$\sigma B, \text{base}, c$]:

$$= \text{Bending Moment} / \text{Section Modulus}$$



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= 4.7568/131806
= 0.354 N./mm²

Compressive Allowable, $KL/r < C_c$ (9.0033 < 123.21) per AISC E2-1 [Sca]:

= $(1 - (KL/r)^2 / (2 \cdot C_c^2)) F_y / (5/3 + 3 \cdot (KL/r) / (8 \cdot C_c) - (KL/r)^3 / (8 \cdot C_c^3))$
= $(1 - (9.0033)^2 / (2 \cdot 123.21^2)) 260 /$
 $(5/3 + 3 \cdot (9.0033) / (8 \cdot 123.21) - (9.0033^3) / (8 \cdot 123.21^3))$
= 153.1 N./mm²

AISC Unity Check of Inside Ribs (must be ≤ 1)

= $S_c / S_{ca} + (M_{base,c} \cdot C1/I) / S_{ba}$
= $7.4174 / 153.07 + (4.7568 \cdot 140.5 / 1851.9) / 173.33$
= 0.050

Input Data for Base Plate Bolting Calculations:

Total Number of Bolts per BasePlate	Nbolts	4
Total Number of Bolts in Tension/Baseplate	Nbt	2
Bolt Material Specification		SA-193 B7
Bolt Allowable Stress	Stba	172.00 N./mm ²
Bolt Corrosion Allowance	Bca	1.5 mm
Distance from Bolts to Edge	Edgedis	50.8 mm
User Defined Area in a Single Bolt	UserBoltArea	2.4000 cm ²
BasePlate Allowable Stress	S	138.00 N./mm ²
Area Available in a Single Bolt	BltArea	1.6469 cm ²
Saddle Load QO (Weight)	QO	9806.0 Kgf
Saddle Load QL (Wind/Seismic contribution)	QL	6.7 Kgf
Maximum Transverse Force	Ft	172.2 Kgf
Maximum Longitudinal Force	F1	42.8 Kgf
Saddle Bolted to Steel Foundation		No

Shear Stress in a Single Bolt, Longitudinal Direction [tb,l]:

= $F1 / (Bolt\ Area \cdot Number\ of\ Bolts)$
= $42.755 / (1.6469 \cdot 4)$
= 0.6 N./mm². Must be less than 103.2 N./mm².

Shear Stress in a Single Bolt, Transverse Direction [tb,t]:

= $Ft / (Bolt\ Area \cdot Number\ of\ Bolts)$
= $172.19 / (1.6469 \cdot 4)$
= 2.6 N./mm². Must be less than 103.2 N./mm².

Bolt Area Calculation per Dennis R. Moss

Bolt Area Requirement Due to Longitudinal Load [Bltarearl]:

= 0.0 (QO > QL --> No Uplift in Longitudinal direction)

Bolt Area due to Shear Load [Bltarears]:

= $F1 / (BoltShearAllowable \cdot Nbolts)$
= $42.755 / (103.2 \cdot 4)$
= 0.0102 cm²

Bolt Area due to Transverse Load:

Moment on Baseplate Due to Transverse Load [Rmom]:

= $B \cdot Ft + Sum\ of\ X\ Moments$
= $915 \cdot 172.19 + 0$
= 157.56 Kg-m.

Eccentricity (e):

= $Rmom / QO$
= $157.56 / 9806$
= 16.07 mm < Bplen/6 --> No Uplift in Transverse direction

Bolt Area due to Transverse Load [Bltareart]:

= 0 (No Uplift)



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Required Area of a Single Bolt [Bltarear]:

= max[Bltarearl, Bltarears, Bltareart]

= max[0, 0.01016, 0]

= 0.0102 cm²

ASME VIII Division 2 Horizontal Vessel Analysis, Right Saddle:

Input and Calculated Values:

Vessel Mean Radius	Rm	393.50	mm
Shell Thickness used in this Case	t	14.000	mm
Stiffened Vessel Length per 4.15.6	L	9680.00	mm
Distance from Saddle to Vessel tangent	a1 or a	1940.00	mm
Saddle Width	b1 or b	281.00	mm
Saddle Bearing Angle	delta or theta	120.00	degrees
Wear Plate Width	b2 or b1	425.00	mm
Wear Plate Bearing Angle	delta2 or theta1	132.00	degrees
Wear Plate Thickness	e2 or tr	10.0	mm
Wear Plate Allowable Stress	fw or Sr	138.00	N./mm²
Shell Allowable Stress used in Calculation		234.00	N./mm²
Head Allowable Stress used in Calculation		234.00	N./mm²
Circumferential Efficiency in Plane of Saddle		1.00	
Circumferential Efficiency at Mid-Span		1.00	
Distance from Saddle Base to Centerline	B	915.00	mm
Coefficient of Friction	mu	0.00	

Results for Horizontal Vessel Analysis:

Saddle Force Q, Test Case, no Ext. Forces		9797.66	Kgf
Pressure used in Analysis	P	2928.857	KPa.

Horizontal Vessel Analysis Results:	Actual N./mm²	Allowable N./mm²
Long. Stress at Top of Midspan	43.63	234.00
Long. Stress at Bottom of Midspan	61.15	234.00
Long. Stress at Top of Saddles	116.53	234.00
Long. Stress at Bottom of Saddles	16.84	234.00
Tangential Shear in Shell	15.57	138.58
Circ. Stress at Horn of Saddle	20.25	350.99
Circ. Compressive Stress in Shell	0.91	234.00

Intermediate Results: Saddle Reaction Q due to Wind or Seismic:

Transverse Saddle Reaction Force [Fwt]:

= Ftr(Ft/Num of Saddles + Z Force Load) • B / E

= 3 (344.39/2 + 0) • 915/708.41

= 667.2 Kgf

Longitudinal Saddle Reaction Force [Fwl]:

= max(Fl,wind; Sum of X Forces) • B / Ls

= max(42.755, 0) • 915/5800

= 6.7 Kgf

Load Combination Results for Q + Wind or Seismic [Q]:

= Saddle Load + max(Fwl, Fwt, Fsl, Fst)

= 9130.4 + max(6.7449, 667.23, 0, 0)



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Saddle Calcs: Test Case:

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= 9797.7 Kgf

Longitudinal Wind Force [FI]:

= WindScalar • WindPress (Platform Area + ($\pi/4$ (OD • WindDiaMult)²))

= 1.1 • 766.08 (0 + $\pi/4$ (0.938 • 1.2)²)

= 838.548 N

Summary of Loads at the Base of this Saddle:

Vertical Load (including saddle weight) 9949.16 Kgf

Transverse Shear Load 172.19 Kgf

Longitudinal Shear Load 42.75 Kgf

Note: Wear Plate is Welded to the Shell, k = 0.1

Saddle Dimension [E]:

= min(2(ShellID/2 + t + WearPadThickness)sin($\theta/2$), 2•Rm)

= min(2(770/2 + 14 + 10)sin(120/2), 2•393.5)

= 708.409 mm

The Computed K values from Table 4.15.1:

K1 = 0.1066 K2 = 1.1707 K3 = 0.8799 K4 = 0.4011

K5 = 0.7603 K6 = 0.0529 K7 = 0.0529 K8 = 0.3405

K9 = 0.2711 K10 = 0.0581 K1* = 0.1923 K6p = 0.0434

K7p = 0.0434

The suffix 'p' denotes the values for a wear plate if it exists.

Note: Dimension a is greater than or equal to Rm/2.

Moment per Equation 4.15.1 [M1]:

= -Q•a [1 - (1- a/L + (Rm²-hm²)/(2a•L))/(1+(4hm)/3L)]

= -9797.7•1940[1 - (1-1940/9680 + (393.5²-0²)/

(2•1940•9680))/(1+(4•0)/(3•9680))]

= -3731.1 Kg-m.

Moment per Equation 4.15.2 [M2]:

= Q•L/4((1+2(Rm²-hm²)/(L²))/(1+(4hm²)/(3L))-4a/L)

= 9797.7•9680/4((1+2(393.5²-0²)/(9680²))/(1+(4•0)/

(3•9680))-4•1940/9680)

= 4781.3 Kg-m.

Longitudinal Stress at Top of Shell (4.15.4) [σ_1]:

= P • Rm/(2t) - M2/(π •Rm²t)

= 2928.9 • 393.5/(2•11) - 4781.3/(π •393.5²•11)

= 43.63 N./mm²

Longitudinal Stress at Bottom of Shell (4.15.5) [σ_2]:

= P • Rm/(2t) + M2/(π • Rm² • t)

= 2928.9 • 393.5/(2 • 11) + 4781.3/(π • 393.5² • 11)

= 61.15 N./mm²

Longitudinal Stress at Top of Shell at Support (4.15.8) [σ_{*3}]:

= P • Rm/(2t) - M1/(K1• π •Rm²t)

= 2928.9•393.5/(2•11)--3731.1/(0.1066• π •393.5²•11)

= 116.53 N./mm²

Longitudinal Stress at Bottom of Shell at Support (4.15.9) [σ_{*4}]:

= P • Rm/(2t) + M1/(K1* • π • Rm² • t)

= 2928.9•393.5/(2•11)+-3731.1/(0.1923• π •393.5²•11)

= 16.84 N./mm²

Maximum Shear Force in the Saddle (4.15.3) [T]:

= Q(L-2a)/(L+(4•hm/3))

= 9797.7 (9680 - 2 • 1940)/(9680 + (4 • 0/3))

= 5870.5 Kgf



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Shear Stress in the shell no rings, not stiffened (4.15.12) [τ_2]:

$$= K_2 \cdot T / (R_m \cdot t)$$

$$= 1.1707 \cdot 5870.5 / (393.5 \cdot 11)$$

$$= 15.57 \text{ N./mm}^2$$

Decay Length (4.15.20) [x_1, x_2]:

$$= 0.78 \cdot \sqrt{(R_m \cdot t)}$$

$$= 0.78 \cdot \sqrt{(393.5 \cdot 11)}$$

$$= 51.317 \text{ mm}$$

Effective reinforcing plate width (4.15.24) [B1]:

$$= \min(b + 1.56 \cdot \sqrt{(R_m \cdot t)}, 2a)$$

$$= \min(281 + 1.56 \cdot \sqrt{(393.5 \cdot 11)}, 2 \cdot 1940)$$

$$= 383.63 \text{ mm}$$

Wear Plate/Shell Stress ratio (4.15.26) [η]:

$$= 1.0000 \text{ Materials are the same, test case}$$

Circumferential Stress at Saddle Base with Wear Plate (4.15.25) [σ_6, r]:

$$= -K_5 \cdot Q \cdot k / (B_1(t + \eta \cdot tr))$$

$$= -0.7603 \cdot 9797.7 \cdot 0.1 / (383.63(11 + 1 \cdot 10))$$

$$= -0.91 \text{ N./mm}^2$$

Circ. Comp. Stress at Horn of Saddle, $L \geq 8R_m$ (4.15.28) [σ_7, r]:

$$= -Q / (4(t + \eta \cdot tr)b_1) - 3 \cdot K_7 \cdot Q / (2(t + \eta \cdot tr)^2)$$

$$= -9797.7 / (4(11 + 1 \cdot 10)383.63) -$$

$$3 \cdot 0.05285 \cdot 9797.7 / (2(11 + 1 \cdot 10)^2)$$

$$= -20.25 \text{ N./mm}^2$$

Results for Vessel Ribs, Web and Base:

Baseplate Length	Bplen	780.0000	mm
Baseplate Thickness	Bpthk	16.0000	mm
Baseplate Width	Bpwid	310.0000	mm
Number of Ribs (inc. outside ribs)	Nribs	3	
Rib Thickness	Ribtk	10.0000	mm
Web Thickness	Webtk	10.0000	mm
Web Location	Webloc	Center	
Saddle Yield Stress	Sy	260.0	N./mm ²
Height of Web at Center	Hw,c	525.0	mm
Friction Coefficient	mu	0.000	

Moment of Inertia of Saddle - Transverse Direction (90 degrees to long axis)

Inertia of Shell [ShellInertia]:

$$= ((1.56 \cdot \sqrt{(R \cdot t)} + \text{WearPlateWidth})t^3) / 12$$

$$= ((1.56 \cdot \sqrt{(388 \cdot 11)} + 425)11^3) / 12$$

$$= 5.844 \text{ cm}^4$$

Resolved Inertia for the Shell [I,shell]:

$$= \text{ShellInertia} + A_{\text{Shell}}(C_1 - Y)^2$$

$$= 5.8444 + 57.961(199.81 - 5.5)^2$$

$$= 21890.658 \text{ cm}^4$$

	B mm	D mm	Y mm	A cm ²	AY mm ³	I + AD ² (2) cm ⁴
Shell	526.9	11.0	5.5	58.0	31878.3	0.219E+05
Wearplate	425.0	10.0	16.0	42.5	68000.0	0.144E+05
Web	10.0	490.0	266.0	49.0	1303400.4	0.120E+05
BasePlate	310.0	16.0	519.0	49.6	2574240.0	0.505E+05
Totals	...	...	...	199.1	3977518.8	0.987E+05

Distance to Centroid [C1]:



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$$\begin{aligned}
 &= AY / A \\
 &= 3977518 / 199.06 \\
 &= 199.814 \text{ mm}
 \end{aligned}$$

$$\begin{aligned}
 \text{Angle } [\beta]: \\
 &= 180 - \text{Saddle Angle} / 2 \\
 &= 180 - 120 / 2 \\
 &= 120.0
 \end{aligned}$$

$$\begin{aligned}
 \text{Saddle Splitting Coefficient [K1]:} \\
 &= (1 + \cos(\beta) - 0.5 \cdot \sin(\beta)^2) / (\pi - \beta + \sin(\beta) \cos(\beta)) \\
 &= (1 + \cos(120) - 0.5 \cdot \sin(120)^2) / (\pi - 2.0944 + \sin(120) \cos(120)) \\
 &= 0.2035
 \end{aligned}$$

$$\begin{aligned}
 \text{Saddle Splitting Force [Fh]:} \\
 &= K1 \cdot Q \\
 &= 0.2035 \cdot 9797.7 \\
 &= 1994.0355 \text{ Kgf}
 \end{aligned}$$

$$\begin{aligned}
 \text{Tension Stress, } St &= (Fh / As) = 1.3859 \text{ N./mm}^2 \\
 \text{Allowed Stress, } Sa &= 0.6 \cdot \text{Yield Str} = 155.9972 \text{ N./mm}^2
 \end{aligned}$$

$$\begin{aligned}
 \text{Saddle Splitting Dimension [d]:} \\
 &= B - R \cdot \sin(\theta/2) / (\theta/2 \text{ in radians}) \\
 &= 915 - 388 \cdot \sin(120/2) / 1.0472 \\
 &= 594.127 \text{ mm}
 \end{aligned}$$

$$\text{Bending Moment, } M = Fh \cdot d = 1184.7330 \text{ Kg-m.}$$

$$\begin{aligned}
 \text{Bending Stress, } Sb &= (M \cdot C1 / I) = 2.3509 \text{ N./mm}^2 \\
 \text{Allowed Stress, } Sa &= 2/3 \cdot \text{Yield Str} = 173.3302 \text{ N./mm}^2
 \end{aligned}$$

Minimum Thickness of Baseplate per Moss:

$$\begin{aligned}
 &= \sqrt{ 3(Q + \text{SaddleWt}) \text{BasePlateWidth} / (4 \cdot \text{BasePlateLength} \cdot \text{AllStress}) } \\
 &= \sqrt{ 3(9797.7 + 151.5) 310 / (4 \cdot 780 \cdot 173.33) } \\
 &= 12.953 \text{ mm}
 \end{aligned}$$

Calculation of Axial Load, Intermediate Values and Compressive Stress:

$$\begin{aligned}
 \text{Web Length Dimension [Web Length]:} \\
 &= 2 \cdot \cos(90 - \text{Saddle Angle} / 2) (\text{Inside Radius} + \text{Shell Thk} + \text{Wear Plate Thk}) \\
 &= 2 \cdot \cos(90 - 120 / 2) (385 + 14 + 10) \\
 &= 708.409 \text{ mm}
 \end{aligned}$$

$$\begin{aligned}
 \text{Distance between Ribs [e]:} \\
 &= \text{Web Length} / (\text{Nr ribs} - 1) \\
 &= 708.41 / (3 - 1) \\
 &= 354.204 \text{ mm}
 \end{aligned}$$

$$\begin{aligned}
 \text{Baseplate Pressure Area [Ap]:} \\
 &= e \cdot \text{Bpwid} / 2 \\
 &= 354.2 \cdot 310 / 2 \\
 &= 549.017 \text{ cm}^2
 \end{aligned}$$

$$\begin{aligned}
 \text{Bearing Pressure [Bp]:} \\
 &= (Q + \text{SaddleWt}) / (\text{BasePlateLength} \cdot \text{BasePlateWidth}) \\
 &= 9949.2 / (780 \cdot 310) \\
 &= 4.115 \text{ Kgf/cm}^2
 \end{aligned}$$

$$\begin{aligned}
 \text{Axial Load [P]:} \\
 &= Ap \cdot Bp \\
 &= 549.02 \cdot 4.1146 \\
 &= 2258.997 \text{ Kgf}
 \end{aligned}$$

$$\text{Area of the Rib and Web [Ar]:}$$



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Saddle Calcs: Test Case:

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= Rib Area + Web Area

= 27.1 + 17.71

= 44.810 cm²

Compressive Stress [Sc]:

= P/Ar

= 2259/44.81

= 4.944 N./mm²

Check of Outside Ribs:

Inertia of Saddle, Outer Ribs - Longitudinal Direction

	B	D	Y	A	AY	Io
Rib+Web	10.0	281.0	...	28.1	...	0.185E+04

Rib dimension [D]:

= Saddle Width - Web Thickness

= 281 - 10

= 271.000 mm

Distance to Centroid from Datum [ytot]:

= AY / A

= 0/44.81

= 0.000 mm

Distance to Centroid [C1]:

= Saddle Width / 2

= 281/2

= 140.500 mm

Radius of Gyration [r]:

= $\sqrt{\text{Total Inertia} / \text{Total Area}}$

= $\sqrt{(1849/44.81)}$

= 64.236 mm

Length of Outer Rib [L]:

= Saddle Height - cos(theta/2)(radius + shlthk + wpdthk) - bpthk

= 915 - cos(120/2)(385 + 14 + 10) - 16

= 694.500 mm

Intermediate Term [Cc]:

= $\sqrt{(2 \cdot \pi^2 \cdot \text{Elastic Modulus} / \text{Yield Stress})}$

= $\sqrt{(2 \cdot \pi^2 \cdot 199943008/260)}$

= 123.211

Slenderness ratio [KL/r]:

= KL/r

= 1 • 694.5/64.236

= 10.812

Bending Moment [Rm]:

= FI / (2 • Bplen) • e • L / 2

= 42.755 / (2 • 780) • 354.2 • 694.5/2

= 3.371 Kg-m.

Compressive Allowable, KL/r < Cc (10.812 < 123.21) per AISC E2-1 [Sca]:

= $(1 - (KL/r)^2 / (2 \cdot Cc^2)) Fy / (5/3 + 3 \cdot (KL/r) / (8 \cdot Cc) - (KL/r)^3 / (8 \cdot Cc^3))$

= $(1 - (10.812)^2 / (2 \cdot 123.21^2)) 260 /$

$(5/3 + 3 \cdot (10.812) / (8 \cdot 123.21) - (10.812^3) / (8 \cdot 123.21^3))$

= 152.4 N./mm²

AISC Unity Check of Outside Ribs (must be ≤ 1)

= Sc/Sca + (Rm • C1 / I) / Sba



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Saddle Calcs: Test Case:

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$$= 4.9439/152.4 + (3.371 \cdot 140.5/18490034)/173.33$$

$$= 0.034$$

Check of Inside Ribs:

Inertia of Saddle, Inner Ribs - Axial Direction

	B	D	Y	A	AY	Io
Rib	10.0	271.0	0.0	27.1	0.0	0.185E+04
Web	354.2	10.0	0.0	35.4	0.0	2.95
Totals	...	...	...	62.5	...	0.185E+04

Distance to Centroid from Datum [ytot]:

$$= AY / A$$

$$= 0/62.52$$

$$= 0.000 \text{ mm}$$

Distance to Centroid [C1]:

$$= \text{Saddle Width} / 2$$

$$= 281/2$$

$$= 140.500 \text{ mm}$$

Length of Inner Rib [L]:

$$= \text{Saddle Height} - \text{Outside Radius} - \text{Bpthk}$$

$$= 915 - \cos(409/2)(16 + 0 + 0) - 0$$

$$= 490.000 \text{ mm}$$

Radius of Gyration [r]:

$$= \sqrt{(\text{Total Inertia} / \text{Total Area})}$$

$$= \sqrt{(1851.9/62.52)}$$

$$= 54.424 \text{ mm}$$

Slenderness ratio [KL/r]:

$$= KL/r$$

$$= 1 \cdot 490/54.424$$

$$= 9.003$$

Unit Force [Force,u]:

$$= FI / (2 \cdot \text{Baseplate Length})$$

$$= 42.755/(2 \cdot 780)$$

$$= 0.027 \text{ Kg/mm}$$

Moment at base of inner Rib [Mbase,c]:

$$= \text{Unit Force} \cdot e \cdot L$$

$$= 0.02741 \cdot 354.2 \cdot 490$$

$$= 4.757 \text{ Kg-m.}$$

Bending Stress due to Transverse Force and Weight Load [σ_B ,base,c]:

$$= \text{Bending Moment} / \text{Section Modulus}$$

$$= 4.7568/131806$$

$$= 0.354 \text{ N./mm}^2$$

Compressive Allowable, $KL/r < Cc$ ($9.0033 < 123.21$) per AISC E2-1 [Sca]:

$$= (1 - (KL/r)^2 / (2 \cdot Cc^2)) F_y / (5/3 + 3 \cdot (KL/r) / (8 \cdot Cc) - (KL/r)^3 / (8 \cdot Cc^3))$$

$$= (1 - (9.0033)^2 / (2 \cdot 123.21^2)) 260 /$$

$$(5/3 + 3 \cdot (9.0033) / (8 \cdot 123.21) - (9.0033^3) / (8 \cdot 123.21^3))$$

$$= 153.1 \text{ N./mm}^2$$

AISC Unity Check of Inside Ribs (must be ≤ 1)

$$= Sc/Sca + (M_{base,c} \cdot C1/I) / S_{ba}$$

$$= 7.0463/153.07 + (4.7568 \cdot 140.5/1851.9) / 173.33$$

$$= 0.048$$

Input Data for Base Plate Bolting Calculations:



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Saddle Calcs: Test Case:

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Total Number of Bolts per BasePlate	Nbolts	4
Total Number of Bolts in Tension/Baseplate	Nbt	2
Bolt Material Specification		SA-193 B7
Bolt Allowable Stress	Stba	172.00 N./mm ²
Bolt Corrosion Allowance	Bca	1.5 mm
Distance from Bolts to Edge	Edgedis	50.8 mm
User Defined Area in a Single Bolt	UserBoltArea	2.4000 cm ²
BasePlate Allowable Stress	S	138.00 N./mm ²
Area Available in a Single Bolt	BltArea	1.6469 cm ²
Saddle Load Q0 (Weight)	Q0	9281.9 Kgf
Saddle Load QL (Wind/Seismic contribution)	QL	6.7 Kgf
Maximum Transverse Force	Ft	172.2 Kgf
Maximum Longitudinal Force	F1	42.8 Kgf
Saddle Bolted to Steel Foundation		No

Shear Stress in a Single Bolt, Longitudinal Direction [τ_{b,l}]:

= $F1 / (\text{Bolt Area} \cdot \text{Number of Bolts})$

= $42.755 / (1.6469 \cdot 4)$

= 0.6 N./mm². Must be less than 103.2 N./mm².

Shear Stress in a Single Bolt, Transverse Direction [τ_{b,t}]:

= $Ft / (\text{Bolt Area} \cdot \text{Number of Bolts})$

= $172.19 / (1.6469 \cdot 4)$

= 2.6 N./mm². Must be less than 103.2 N./mm².

Bolt Area Calculation per Dennis R. Moss

Bolt Area Requirement Due to Longitudinal Load [Bltarearl]:

= 0.0 (Q0 > QL --> No Uplift in Longitudinal direction)

Bolt Area due to Shear Load [Bltarears]:

= $F1 / (\text{BoltShearAllowable} \cdot \text{Nbolts})$

= $42.755 / (103.2 \cdot 4)$

= 0.0102 cm²

Bolt Area due to Transverse Load:

Moment on Baseplate Due to Transverse Load [Rmom]:

= $B \cdot Ft + \text{Sum of X Moments}$

= $915 \cdot 172.19 + 0$

= 157.56 Kg-m.

Eccentricity (e):

= $Rmom / Q0$

= $157.56 / 9281.9$

= 16.97 mm < Bplen/6 --> No Uplift in Transverse direction

Bolt Area due to Transverse Load [Bltareart]:

= 0 (No Uplift)

Required Area of a Single Bolt [Bltarear]:

= $\max[\text{Bltarearl}, \text{Bltarears}, \text{Bltareart}]$

= $\max[0, 0.01016, 0]$

= 0.0102 cm²

Review notes about nozzle loadings and supports in the Warnings report.

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Saddle Calcs: Ext. Pressure:

Step: 25

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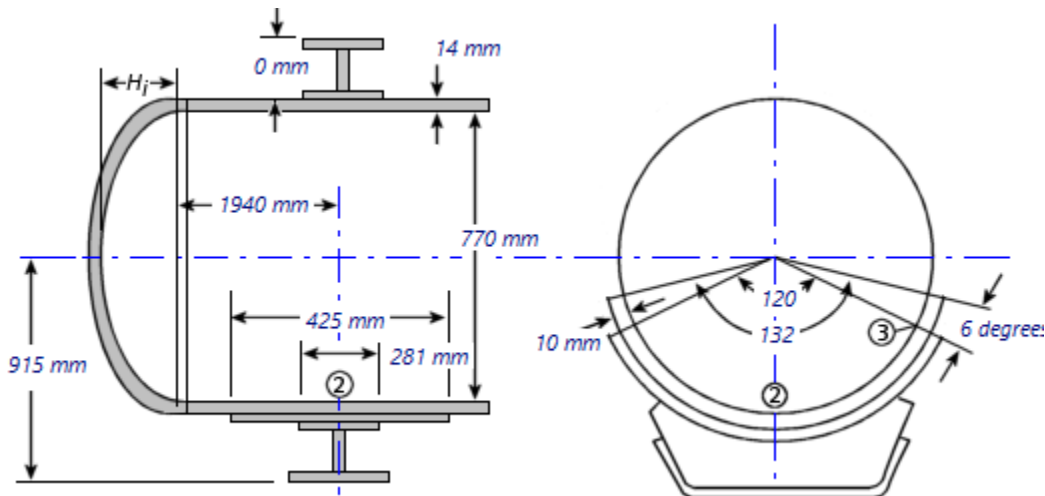
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ASME VIII Division 2 Horizontal Vessel Analysis, Left Saddle:

Horizontal Vessel Stress Calculations : External Pressure

Input and Calculated Values:

Vessel Mean Radius	Rm	393.50	mm
Shell Thickness used in this Case	t	14.000	mm
Stiffened Vessel Length per 4.15.6	L	9680.00	mm
Distance from Saddle to Vessel tangent	a1 or a	1940.00	mm
Saddle Width	b1 or b	281.00	mm
Saddle Bearing Angle	delta or theta	120.00	degrees
Wear Plate Width	b2 or b1	425.00	mm
Wear Plate Bearing Angle	delta2 or theta1	132.00	degrees
Wear Plate Thickness	e2 or tr	10.0	mm
Wear Plate Allowable Stress	fw or Sr	138.00	N./mm ²
Shell Allowable Stress used in Calculation		138.00	N./mm ²
Head Allowable Stress used in Calculation		138.00	N./mm ²
Circumferential Efficiency in Plane of Saddle		1.00	
Circumferential Efficiency at Mid-Span		1.00	
Distance from Saddle Base to Centerline	B	915.00	mm
Coefficient of Friction	mu	0.30	



Results for Horizontal Vessel Analysis:

Saddle Force Q, External Pressure Case	12824.72	Kgf
Pressure used in Analysis	P	-103.419 KPa.

Horizontal Vessel Analysis Results:		Actual N./mm ²	Allowable N./mm ²
Long. Stress at Top of Midspan	-13.32	-221.40	
Long. Stress at Top of Midspan	13.32	138.00	
Long. Stress at Bottom of Midspan	9.62	138.00	
Long. Stress at Top of Saddles	82.10	138.00	
Long. Stress at Bottom of Saddles	-48.38	-221.40	
Long. Stress at Bottom of Saddles	48.38	138.00	
Tangential Shear in Shell	20.38	110.40	



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Saddle Calcs: Ext. Pressure: Step: 25 9:41am Sep 9, 2026

Circ. Stress at Horn of Saddle	26.51	172.50
Circ. Compressive Stress in Shell	1.19	138.00

Intermediate Results: Saddle Reaction Q due to Wind or Seismic:

Transverse Saddle Reaction Force [Fwt]:

$$= \text{Ftr}(\text{Ft/Num of Saddles} + \text{Z Force Load}) \cdot B / E$$

$$= 3 (688.77/2 + 0) \cdot 915/708.41$$

$$= 1334.5 \text{ Kg}$$

Longitudinal Saddle Reaction Force [Fwl]:

$$= \max(\text{Fl,wind}; \text{Sum of X Forces}) \cdot B / \text{Ls}$$

$$= \max(85.509, 0) \cdot 915/5800$$

$$= 13.5 \text{ Kg}$$

Saddle Reaction Force due to Earthquake Fl [Fsl]:

$$= \max(\text{Fl,eq}; \text{Sum of X Forces}) \cdot B / \text{Ls}$$

$$= \max(1173.3, 0) \cdot 915/5800$$

$$= 185.1 \text{ Kg}$$

Saddle Reaction Force due to Earthquake Ft [Fst]:

$$= \text{Ftr}(\text{Ft/Num of Saddles} + \text{Z Force Load}) \cdot B / E$$

$$= 3 (1173.3/2 + 0) \cdot 915/708.41$$

$$= 2273.1 \text{ Kg}$$

Load Combination Results for Q + Wind or Seismic [Q]:

$$= \text{Saddle Load} + \max(\text{Fwl}, \text{Fwt}, \text{Fsl}, \text{Fst})$$

$$= 10552 + \max(13.49, 1334.5, 185.09, 2273.1)$$

$$= 12824.7 \text{ Kg}$$

Longitudinal Wind Force [Fl]:

$$= \text{WindScalar} \cdot \text{WindPress}(\text{Platform Area} + (\pi/4(\text{OD} \cdot \text{WindDiaMult})^2))$$

$$= 1.1 \cdot 766.08 (0 + \pi/4(0.938 \cdot 1.2)^2)$$

$$= 838.548 \text{ N}$$

Summary of Loads at the Base of this Saddle:

Vertical Load (including saddle weight)	12976.22	Kgf
Transverse Shear Load	586.63	Kgf
Longitudinal Shear Load	1173.26	Kgf

Note: Wear Plate is Welded to the Shell, k = 0.1

Saddle Dimension [E]:

$$= \min(2(\text{ShellID}/2 + t + \text{WearPadThickness})\sin(\theta/2), 2 \cdot \text{Rm})$$

$$= \min(2(770/2 + 14 + 10)\sin(120/2), 2 \cdot 393.5)$$

$$= 708.409 \text{ mm}$$

The Computed K values from Table 4.15.1:

K1 = 0.1066	K2 = 1.1707	K3 = 0.8799	K4 = 0.4011
K5 = 0.7603	K6 = 0.0529	K7 = 0.0529	K8 = 0.3405
K9 = 0.2711	K10 = 0.0581	K1* = 0.1923	K6p = 0.0434
K7p = 0.0434			

The suffix 'p' denotes the values for a wear plate if it exists.

Note: Dimension a is greater than or equal to Rm/2.

Moment per Equation 4.15.1 [M1]:

$$= -Q \cdot a [1 - (1 - a/L + (Rm^2 - hm^2)/(2a \cdot L))/(1 + (4hm)/3L)]$$

$$= -12825 \cdot 1940 [1 - (1 - 1940/9680 + (393.5^2 - 0^2)/(2 \cdot 1940 \cdot 9680))/(1 + (4 \cdot 0)/(3 \cdot 9680))]$$

$$= -4883.8 \text{ Kg-m.}$$

Moment per Equation 4.15.2 [M2]:



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$$\begin{aligned}
 &= Q \cdot L / 4 \cdot ((1 + 2(Rm^2 - hm^2) / (L^2)) / (1 + (4hm^2) / (3L)) - 4a / L) \\
 &= 12825 \cdot 9680 / 4 \cdot ((1 + 2(393.5^2 - 0^2) / (9680^2)) / (1 + (4 \cdot 0) / (3 \cdot 9680)) - 4 \cdot 1940 / 9680) \\
 &= 6258.6 \text{ Kg-m.}
 \end{aligned}$$

Longitudinal Stress at Top of Shell (4.15.4) [σ_1]:

$$\begin{aligned}
 &= P \cdot Rm / (2t) - M2 / (\pi \cdot Rm^2 \cdot t) \\
 &= -103.42 \cdot 393.5 / (2 \cdot 11) - 6258.6 / (\pi \cdot 393.5^2 \cdot 11) \\
 &= -13.32 \text{ N./mm}^2
 \end{aligned}$$

Longitudinal Stress at Bottom of Shell (4.15.5) [σ_2]:

$$\begin{aligned}
 &= P \cdot Rm / (2t) + M2 / (\pi \cdot Rm^2 \cdot t) \\
 &= -103.42 \cdot 393.5 / (2 \cdot 11) + 6258.6 / (\pi \cdot 393.5^2 \cdot 11) \\
 &= 9.62 \text{ N./mm}^2
 \end{aligned}$$

Longitudinal Stress at Top of Shell at Support (4.15.8) [σ_{*3}]:

$$\begin{aligned}
 &= P \cdot Rm / (2t) - M1 / (K1 \cdot \pi \cdot Rm^2 \cdot t) \\
 &= -103.42 \cdot 393.5 / (2 \cdot 11) - 4883.8 / (0.1066 \cdot \pi \cdot 393.5^2 \cdot 11) \\
 &= 82.10 \text{ N./mm}^2
 \end{aligned}$$

Longitudinal Stress at Bottom of Shell at Support (4.15.9) [σ_{*4}]:

$$\begin{aligned}
 &= P \cdot Rm / (2t) + M1 / (K1 \cdot \pi \cdot Rm^2 \cdot t) \\
 &= -103.42 \cdot 393.5 / (2 \cdot 11) + 4883.8 / (0.1923 \cdot \pi \cdot 393.5^2 \cdot 11) \\
 &= -48.38 \text{ N./mm}^2
 \end{aligned}$$

Maximum Shear Force in the Saddle (4.15.3) [T]:

$$\begin{aligned}
 &= Q(L - 2a) / (L + (4 \cdot hm / 3)) \\
 &= 12825 \cdot (9680 - 2 \cdot 1940) / (9680 + (4 \cdot 0 / 3)) \\
 &= 7684.2 \text{ Kgf}
 \end{aligned}$$

Shear Stress in the shell no rings, not stiffened (4.15.12) [τ_2]:

$$\begin{aligned}
 &= K2 \cdot T / (Rm \cdot t) \\
 &= 1.1707 \cdot 7684.2 / (393.5 \cdot 11) \\
 &= 20.38 \text{ N./mm}^2
 \end{aligned}$$

Decay Length (4.15.20) [x_1, x_2]:

$$\begin{aligned}
 &= 0.78 \cdot \sqrt{(Rm \cdot t)} \\
 &= 0.78 \cdot \sqrt{(393.5 \cdot 11)} \\
 &= 51.317 \text{ mm}
 \end{aligned}$$

Effective reinforcing plate width (4.15.24) [B1]:

$$\begin{aligned}
 &= \min(b + 1.56 \cdot \sqrt{(Rm \cdot t)}, 2a) \\
 &= \min(281 + 1.56 \cdot \sqrt{(393.5 \cdot 11)}, 2 \cdot 1940) \\
 &= 383.63 \text{ mm}
 \end{aligned}$$

Wear Plate/Shell Stress ratio (4.15.26) [η]:

$$\begin{aligned}
 &= \min(Sr / S, 1) \\
 &= \min(138 / 138, 1) \\
 &= 1.0000
 \end{aligned}$$

Circumferential Stress at Saddle Base with Wear Plate (4.15.25) [σ_6, r]:

$$\begin{aligned}
 &= -K5 \cdot Q \cdot k / (B1(t + \eta \cdot tr)) \\
 &= -0.7603 \cdot 12825 \cdot 0.1 / (383.63(11 + 1 \cdot 10)) \\
 &= -1.19 \text{ N./mm}^2
 \end{aligned}$$

Circ. Comp. Stress at Horn of Saddle, $L \geq 8Rm$ (4.15.28) [σ_7, r]:

$$\begin{aligned}
 &= -Q / (4(t + \eta \cdot tr)b1) - 3 \cdot K7 \cdot Q / (2(t + \eta \cdot tr)^2) \\
 &= -12825 / (4(11 + 1 \cdot 10)383.63) - 3 \cdot 0.05285 \cdot 12825 / (2(11 + 1 \cdot 10)^2) \\
 &= -26.51 \text{ N./mm}^2
 \end{aligned}$$

Results for Vessel Ribs, Web and Base:

Baseplate Length	Bplen	780.0000 mm
Baseplate Thickness	Bpthk	16.0000 mm
Baseplate Width	Bpwid	310.0000 mm



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Number of Ribs (inc. outside ribs)	Nribs	3	
Rib Thickness	Ribtk	10.0000	mm
Web Thickness	Webtk	10.0000	mm
Web Location	Webloc	Center	
Saddle Design Temperature		21.1	°C
Saddle Yield Stress	Sy	246.0	N./mm ²
Height of Web at Center	Hw,c	525.0	mm
Friction Coefficient	mu	0.300	

Moment of Inertia of Saddle - Transverse Direction (90 degrees to long axis)

Inertia of Shell [ShellInertia]:

$$= ((1.56 \cdot \sqrt{(R \cdot t)} + \text{WearPlateWidth})^3) / 12$$

$$= ((1.56 \cdot \sqrt{(388 \cdot 11)} + 425)^3) / 12$$

$$= 5.844 \text{ cm}^4$$

Resolved Inertia for the Shell [I,shell]:

$$= \text{ShellInertia} + A_{\text{Shell}} (C1 - Y)^2$$

$$= 5.8444 + 57.961 (199.81 - 5.5)^2$$

$$= 21890.658 \text{ cm}^4$$

	B mm	D mm	Y mm	A cm ²	AY mm ³	I + AD ⁽²⁾ cm ⁴
Shell	526.9	11.0	5.5	58.0	31878.3	0.219E+05
Wearplate	425.0	10.0	16.0	42.5	68000.0	0.144E+05
Web	10.0	490.0	266.0	49.0	1303400.4	0.120E+05
BasePlate	310.0	16.0	519.0	49.6	2574240.0	0.505E+05
Totals	...	...	...	199.1	3977518.8	0.987E+05

Distance to Centroid [C1]:

$$= AY / A$$

$$= 3977518 / 199.06$$

$$= 199.814 \text{ mm}$$

Angle [β]:

$$= 180 - \text{Saddle Angle} / 2$$

$$= 180 - 120 / 2$$

$$= 120.0$$

Saddle Splitting Coefficient [K1]:

$$= (1 + \cos(\beta) - 0.5 \cdot \sin(\beta)^2) / (\pi - \beta + \sin(\beta) \cos(\beta))$$

$$= (1 + \cos(120) - 0.5 \cdot \sin(120)^2) / (\pi - 2.0944 + \sin(120) \cos(120))$$

$$= 0.2035$$

Saddle Splitting Force [Fh]:

$$= K1 \cdot Q$$

$$= 0.2035 \cdot 12825$$

$$= 2610.1079 \text{ Kgf}$$

$$\text{Tension Stress, } St = (Fh / As) = 1.8141 \text{ N./mm}^2$$

$$\text{Allowed Stress, } Sa = 0.6 \cdot \text{Yield Str} = 147.5999 \text{ N./mm}^2$$

Saddle Splitting Dimension [d]:

$$= B - R \cdot \sin(\text{theta} / 2) / (\text{theta} / 2 \text{ in radians})$$

$$= 915 - 388 \cdot \sin(120 / 2) / 1.0472$$

$$= 594.127 \text{ mm}$$

$$\text{Bending Moment, } M = Fh \cdot d = 1550.7653 \text{ Kg-m.}$$

$$\text{Bending Stress, } Sb = (M \cdot C1 / I) = 3.0773 \text{ N./mm}^2$$

$$\text{Allowed Stress, } Sa = 2/3 \cdot \text{Yield Str} = 163.9998 \text{ N./mm}^2$$

Minimum Thickness of Baseplate per Moss:

$$= \sqrt{3(Q + \text{SaddleWt}) \text{BasePlateWidth} / (4 \cdot \text{BasePlateLength} \cdot \text{AllStress})}$$



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$$= \sqrt{3(12825 + 151.5)310 / (4 \cdot 780 \cdot 164)}$$

$$= 15.208 \text{ mm}$$

Calculation of Axial Load, Intermediate Values and Compressive Stress:

Web Length Dimension [Web Length]:

$$= 2 \cdot \cos(90 - \text{Saddle Angle}/2) (\text{Inside Radius} + \text{Shell Thk} + \text{Wear Plate Thk})$$

$$= 2 \cdot \cos(90 - 120/2) (385 + 14 + 10)$$

$$= 708.409 \text{ mm}$$

Distance between Ribs [e]:

$$= \text{Web Length} / (\text{Nr ribs} - 1)$$

$$= 708.41 / (3 - 1)$$

$$= 354.204 \text{ mm}$$

Baseplate Pressure Area [Ap]:

$$= e \cdot \text{Bpwid} / 2$$

$$= 354.2 \cdot 310 / 2$$

$$= 549.017 \text{ cm}^2$$

Bearing Pressure [Bp]:

$$= (Q + \text{SaddleWt}) / (\text{BasePlateLength} \cdot \text{BasePlateWidth})$$

$$= 12976 / (780 \cdot 310)$$

$$= 5.367 \text{ Kg/cm}^2$$

Axial Load [P]:

$$= \text{Ap} \cdot \text{Bp}$$

$$= 549.02 \cdot 5.3665$$

$$= 2946.303 \text{ Kg}$$

Area of the Rib and Web [Ar]:

$$= \text{Rib Area} + \text{Web Area}$$

$$= 27.1 + 17.71$$

$$= 44.810 \text{ cm}^2$$

Compressive Stress [Sc]:

$$= P / \text{Ar}$$

$$= 2946.3 / 44.81$$

$$= 6.448 \text{ N/mm}^2$$

Check of Outside Ribs:

Inertia of Saddle, Outer Ribs - Longitudinal Direction

	B	D	Y	A	AY	Io
Rib+Web	10.0	281.0	...	28.1	...	0.185E+04

Rib dimension [D]:

$$= \text{Saddle Width} - \text{Web Thickness}$$

$$= 281 - 10$$

$$= 271.000 \text{ mm}$$

Distance to Centroid from Datum [ytot]:

$$= \text{AY} / \text{A}$$

$$= 0 / 44.81$$

$$= 0.000 \text{ mm}$$

Distance to Centroid [C1]:

$$= \text{Saddle Width} / 2$$

$$= 281 / 2$$

$$= 140.500 \text{ mm}$$

Radius of Gyration [r]:

$$= \sqrt{(\text{Total Inertia} / \text{Total Area})}$$



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$$= \sqrt{1849/44.81}$$

$$= 64.236 \text{ mm}$$

Length of Outer Rib [L]:

$$= \text{Saddle Height} - \cos(\theta/2) (\text{radius} + \text{shlthk} + \text{wpdthk}) - \text{bpthk}$$

$$= 915 - \cos(120/2) (385 + 14 + 10) - 16$$

$$= 694.500 \text{ mm}$$

Intermediate Term [Cc]:

$$= \sqrt{2 \cdot \pi^2 \cdot \text{Elastic Modulus} / \text{Yield Stress}}$$

$$= \sqrt{2 \cdot \pi^2 \cdot 199943088 / 246}$$

$$= 126.667$$

Slenderness ratio [KL/r]:

$$= KL/r$$

$$= 1 \cdot 694.5 / 64.236$$

$$= 10.812$$

Bending Moment [Rm]:

$$= FI / (2 \cdot B_{plen}) \cdot e \cdot L / 2$$

$$= 1173.3 / (2 \cdot 780) \cdot 354.2 \cdot 694.5 / 2$$

$$= 92.507 \text{ Kg-m.}$$

Compressive Allowable, $KL/r < Cc$ ($10.812 < 126.67$) per AISC E2-1 [Sca]:

$$= (1 - (KL/r)^2 / (2 \cdot Cc^2)) F_y / (5/3 + 3 \cdot (KL/r) / (8 \cdot Cc) - (KL/r)^3 / (8 \cdot Cc^3))$$

$$= (1 - (10.812)^2 / (2 \cdot 126.67^2)) 246 / (5/3 + 3 \cdot (10.812) / (8 \cdot 126.67) - (10.812^3) / (8 \cdot 126.67^3))$$

$$= 144.3 \text{ N./mm}^2$$

AISC Unity Check of Outside Ribs (must be ≤ 1)

$$= S_c / S_{ca} + (R_m \cdot C_1 / I) / S_{ba}$$

$$= 6.4481 / 144.3 + (92.507 \cdot 140.5 / 18490034) / 164$$

$$= 0.087$$

Check of Inside Ribs:

Inertia of Saddle, Inner Ribs - Axial Direction

	B	D	Y	A	AY	Io
Rib	10.0	271.0	0.0	27.1	0.0	0.185E+04
Web	354.2	10.0	0.0	35.4	0.0	2.95
Totals	...	...	...	62.5	...	0.185E+04

Distance to Centroid from Datum [ytot]:

$$= AY / A$$

$$= 0 / 62.52$$

$$= 0.000 \text{ mm}$$

Distance to Centroid [C1]:

$$= \text{Saddle Width} / 2$$

$$= 281 / 2$$

$$= 140.500 \text{ mm}$$

Length of Inner Rib [L]:

$$= \text{Saddle Height} - \text{Outside Radius} - \text{Bpthk}$$

$$= 915 - \cos(409/2) (16 + 0 + 0) - 0$$

$$= 490.000 \text{ mm}$$

Radius of Gyration [r]:

$$= \sqrt{\text{Total Inertia} / \text{Total Area}}$$

$$= \sqrt{1851.9 / 62.52}$$

$$= 54.424 \text{ mm}$$

Slenderness ratio [KL/r]:

$$= KL/r$$



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$$= 1 \cdot 490/54.424$$

$$= 9.003$$

Unit Force [Force,u]:

$$= F / (2 \cdot \text{Baseplate Length})$$

$$= 1173.3 / (2 \cdot 780)$$

$$= 0.752 \text{ Kg/mm}$$

Moment at base of inner Rib [Mbase,c]:

$$= \text{Unit Force} \cdot e \cdot L$$

$$= 0.7521 \cdot 354.2 \cdot 490$$

$$= 130.535 \text{ Kg-m.}$$

Bending Stress due to Transverse Force and Weight Load [σ_B ,base,c]:

$$= \text{Bending Moment} / \text{Section Modulus}$$

$$= 130.54/131806$$

$$= 9.712 \text{ N./mm}^2$$

Compressive Allowable, $KL/r < C_c$ ($9.0033 < 126.67$) per AISC E2-1 [Sca]:

$$= (1 - (KL/r)^2 / (2 \cdot C_c^2)) F_y / (5/3 + 3 \cdot (KL/r) / (8 \cdot C_c) - (KL/r)^3 / (8 \cdot C_c^3))$$

$$= (1 - (9.0033)^2 / (2 \cdot 126.67^2)) 246 / (5/3 + 3 \cdot (9.0033) / (8 \cdot 126.67) - (9.0033^3) / (8 \cdot 126.67^3))$$

$$= 144.9 \text{ N./mm}^2$$

AISC Unity Check of Inside Ribs (must be ≤ 1)

$$= S_c / S_{ca} + (M_{base,c} \cdot C_1 / I) / S_{ba}$$

$$= 9.1901/144.91 + (130.54 \cdot 140.5 / 1851.9) / 164$$

$$= 0.123$$

ASME VIII Division 2 Horizontal Vessel Analysis, Right Saddle:

Input and Calculated Values:

Vessel Mean Radius	Rm	393.50	mm
Shell Thickness used in this Case	t	14.000	mm
Stiffened Vessel Length per 4.15.6	L	9680.00	mm
Distance from Saddle to Vessel tangent	a1 or a	1940.00	mm
Saddle Width	b1 or b	281.00	mm
Saddle Bearing Angle	delta or theta	120.00	degrees
Wear Plate Width	b2 or b1	425.00	mm
Wear Plate Bearing Angle	delta2 or theta1	132.00	degrees
Wear Plate Thickness	e2 or tr	10.0	mm
Wear Plate Allowable Stress	fw or Sr	138.00	N./mm ²
Shell Allowable Stress used in Calculation		138.00	N./mm ²
Head Allowable Stress used in Calculation		138.00	N./mm ²
Circumferential Efficiency in Plane of Saddle		1.00	
Circumferential Efficiency at Mid-Span		1.00	
Distance from Saddle Base to Centerline	B	915.00	mm
Coefficient of Friction	mu	0.00	

Results for Horizontal Vessel Analysis:

Saddle Force Q, External Pressure Case		12300.64	Kgf
Pressure used in Analysis	P	-103.419	KPa.

Horizontal Vessel Analysis Results:		Actual	Allowable
		N./mm ²	N./mm ²
Long. Stress at Top	of Midspan	-12.85	-221.40
Long. Stress at Top	of Midspan	12.85	138.00
Long. Stress at Bottom	of Midspan	9.15	138.00



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Long. Stress at Top of Saddles	78.67	138.00
Long. Stress at Bottom of Saddles	-46.48	-221.40
Long. Stress at Bottom of Saddles	46.48	138.00
Tangential Shear in Shell	19.55	110.40
Circ. Stress at Horn of Saddle	25.43	172.50
Circ. Compressive Stress in Shell	1.14	138.00

Intermediate Results: Saddle Reaction Q due to Wind or Seismic:

Transverse Saddle Reaction Force [Fwt]:

$$= Ftr(Ft/Num of Saddles + Z Force Load) \cdot B / E$$

$$= 3 (688.77/2 + 0) \cdot 915/708.41$$

$$= 1334.5 \text{ Kgf}$$

Longitudinal Saddle Reaction Force [Fwl]:

$$= \max(Fl,wind; \text{Sum of X Forces}) \cdot B / Ls$$

$$= \max(85.509, 0) \cdot 915/5800$$

$$= 13.5 \text{ Kgf}$$

Saddle Reaction Force due to Earthquake Fl [Fsl]:

$$= \max(Fl,eq; \text{Sum of X Forces}) \cdot B / Ls$$

$$= \max(1173.3, 0) \cdot 915/5800$$

$$= 185.1 \text{ Kgf}$$

Saddle Reaction Force due to Earthquake Ft [Fst]:

$$= Ftr(Ft/Num of Saddles + Z Force Load) \cdot B / E$$

$$= 3 (1173.3/2 + 0) \cdot 915/708.41$$

$$= 2273.1 \text{ Kgf}$$

Load Combination Results for Q + Wind or Seismic [Q]:

$$= \text{Saddle Load} + \max(Fwl, Fwt, Fsl, Fst)$$

$$= 10028 + \max(13.49, 1334.5, 185.09, 2273.1)$$

$$= 12300.6 \text{ Kgf}$$

Longitudinal Wind Force [Fl]:

$$= \text{WindScalar} \cdot \text{WindPress}(\text{Platform Area} + (\pi/4(OD \cdot \text{WindDiaMult})^2))$$

$$= 1.1 \cdot 766.08 (0 + \pi/4(0.938 \cdot 1.2)^2)$$

$$= 838.548 \text{ N}$$

Summary of Loads at the Base of this Saddle:

Vertical Load (including saddle weight) 12452.14 Kgf

Transverse Shear Load 586.63 Kgf

Longitudinal Shear Load 1173.26 Kgf

Note: Wear Plate is Welded to the Shell, k = 0.1

Saddle Dimension [E]:

$$= \min(2(ShellID/2 + t + \text{WearPadThickness}) \sin(\theta/2), 2 \cdot Rm)$$

$$= \min(2(770/2 + 14 + 10) \sin(120/2), 2 \cdot 393.5)$$

$$= 708.409 \text{ mm}$$

The Computed K values from Table 4.15.1:

K1 = 0.1066 K2 = 1.1707 K3 = 0.8799 K4 = 0.4011

K5 = 0.7603 K6 = 0.0529 K7 = 0.0529 K8 = 0.3405

K9 = 0.2711 K10 = 0.0581 K1* = 0.1923 K6p = 0.0434

K7p = 0.0434

The suffix 'p' denotes the values for a wear plate if it exists.

Note: Dimension a is greater than or equal to Rm/2.

Moment per Equation 4.15.1 [M1]:

$$= -Q \cdot a [1 - (1 - a/L + (Rm^2 - hm^2) / (2a \cdot L)) / (1 + (4hm) / 3L)]$$



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$$= -12301 \cdot 1940 [1 - (1 - 1940/9680 + (393.5^2 - 0^2) / (2 \cdot 1940 \cdot 9680)) / (1 + (4 \cdot 0) / (3 \cdot 9680))] \\ = -4684.2 \text{ Kg-m.}$$

Moment per Equation 4.15.2 [M2]:

$$= Q \cdot L / 4 ((1 + 2(Rm^2 - hm^2) / (L^2)) / (1 + (4hm^2) / (3L)) - 4a/L) \\ = 12301 \cdot 9680 / 4 ((1 + 2(393.5^2 - 0^2) / (9680^2)) / (1 + (4 \cdot 0) / (3 \cdot 9680)) - 4 \cdot 1940 / 9680) \\ = 6002.8 \text{ Kg-m.}$$

Longitudinal Stress at Top of Shell (4.15.4) [σ_1]:

$$= P \cdot Rm / (2t) - M2 / (\pi \cdot Rm^2 \cdot t) \\ = -103.42 \cdot 393.5 / (2 \cdot 11) - 6002.8 / (\pi \cdot 393.5^2 \cdot 11) \\ = -12.85 \text{ N./mm}^2$$

Longitudinal Stress at Bottom of Shell (4.15.5) [σ_2]:

$$= P \cdot Rm / (2t) + M2 / (\pi \cdot Rm^2 \cdot t) \\ = -103.42 \cdot 393.5 / (2 \cdot 11) + 6002.8 / (\pi \cdot 393.5^2 \cdot 11) \\ = 9.15 \text{ N./mm}^2$$

Longitudinal Stress at Top of Shell at Support (4.15.8) [σ^*3]:

$$= P \cdot Rm / (2t) - M1 / (K1 \cdot \pi \cdot Rm^2 \cdot t) \\ = -103.42 \cdot 393.5 / (2 \cdot 11) - 4684.2 / (0.1066 \cdot \pi \cdot 393.5^2 \cdot 11) \\ = 78.67 \text{ N./mm}^2$$

Longitudinal Stress at Bottom of Shell at Support (4.15.9) [σ^*4]:

$$= P \cdot Rm / (2t) + M1 / (K1 \cdot \pi \cdot Rm^2 \cdot t) \\ = -103.42 \cdot 393.5 / (2 \cdot 11) + 4684.2 / (0.1923 \cdot \pi \cdot 393.5^2 \cdot 11) \\ = -46.48 \text{ N./mm}^2$$

Maximum Shear Force in the Saddle (4.15.3) [T]:

$$= Q(L - 2a) / (L + (4 \cdot hm / 3)) \\ = 12301 (9680 - 2 \cdot 1940) / (9680 + (4 \cdot 0 / 3)) \\ = 7370.2 \text{ Kgf}$$

Shear Stress in the shell no rings, not stiffened (4.15.12) [τ_2]:

$$= K2 \cdot T / (Rm \cdot t) \\ = 1.1707 \cdot 7370.2 / (393.5 \cdot 11) \\ = 19.55 \text{ N./mm}^2$$

Decay Length (4.15.20) [x_1, x_2]:

$$= 0.78 \cdot \sqrt{(Rm \cdot t)} \\ = 0.78 \cdot \sqrt{(393.5 \cdot 11)} \\ = 51.317 \text{ mm}$$

Effective reinforcing plate width (4.15.24) [B1]:

$$= \min(b + 1.56 \cdot \sqrt{(Rm \cdot t)}, 2a) \\ = \min(281 + 1.56 \cdot \sqrt{(393.5 \cdot 11)}, 2 \cdot 1940) \\ = 383.63 \text{ mm}$$

Wear Plate/Shell Stress ratio (4.15.26) [η]:

$$= \min(Sr/S, 1) \\ = \min(138/138, 1) \\ = 1.0000$$

Circumferential Stress at Saddle Base with Wear Plate (4.15.25) [σ_6, r]:

$$= -K5 \cdot Q \cdot k / (B1(t + \eta \cdot tr)) \\ = -0.7603 \cdot 12301 \cdot 0.1 / (383.63(11 + 1 \cdot 10)) \\ = -1.14 \text{ N./mm}^2$$

Circ. Comp. Stress at Horn of Saddle, $L \geq 8Rm$ (4.15.28) [σ_7, r]:

$$= -Q / (4(t + \eta \cdot tr)b1) - 3 \cdot K7 \cdot Q / (2(t + \eta \cdot tr)^2) \\ = -12301 / (4(11 + 1 \cdot 10)383.63) - 3 \cdot 0.05285 \cdot 12301 / (2(11 + 1 \cdot 10)^2) \\ = -25.43 \text{ N./mm}^2$$



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Results for Vessel Ribs, Web and Base:

Baseplate Length	Bplen	780.0000	mm
Baseplate Thickness	Bpthk	16.0000	mm
Baseplate Width	Bpwid	310.0000	mm
Number of Ribs (inc. outside ribs)	Nribs	3	
Rib Thickness	Ribtk	10.0000	mm
Web Thickness	Webtk	10.0000	mm
Web Location	Webloc	Center	
Saddle Design Temperature		21.1	°C
Saddle Yield Stress	Sy	246.0	N./mm ²
Height of Web at Center	Hw,c	525.0	mm
Friction Coefficient	mu	0.000	

Moment of Inertia of Saddle - Transverse Direction (90 degrees to long axis)

Inertia of Shell [ShellInertia]:

$$= ((1.56 \cdot \sqrt{(R \cdot t)} + \text{WearPlateWidth}) t^3) / 12$$

$$= ((1.56 \cdot \sqrt{(388 \cdot 11)} + 425) 11^3) / 12$$

$$= 5.844 \text{ cm}^4$$

Resolved Inertia for the Shell [I,shell]:

$$= \text{ShellInertia} + A_{\text{Shell}} (C1 - Y)^2$$

$$= 5.8444 + 57.961 (199.81 - 5.5)^2$$

$$= 21890.658 \text{ cm}^4$$

	B mm	D mm	Y mm	A cm ²	AY mm ³	I + AD ⁽²⁾ cm ⁴
Shell	526.9	11.0	5.5	58.0	31878.3	0.219E+05
Wearplate	425.0	10.0	16.0	42.5	68000.0	0.144E+05
Web	10.0	490.0	266.0	49.0	1303400.4	0.120E+05
BasePlate	310.0	16.0	519.0	49.6	2574240.0	0.505E+05
Totals	...	...	...	199.1	3977518.8	0.987E+05

Distance to Centroid [C1]:

$$= AY / A$$

$$= 3977518 / 199.06$$

$$= 199.814 \text{ mm}$$

Angle [β]:

$$= 180 - \text{Saddle Angle} / 2$$

$$= 180 - 120 / 2$$

$$= 120.0$$

Saddle Splitting Coefficient [K1]:

$$= (1 + \cos(\beta) - 0.5 \sin(\beta)^2) / (\pi - \beta + \sin(\beta) \cos(\beta))$$

$$= (1 + \cos(120) - 0.5 \sin(120)^2) / (\pi - 2.0944 + \sin(120) \cos(120))$$

$$= 0.2035$$

Saddle Splitting Force [Fh]:

$$= K1 \cdot Q$$

$$= 0.2035 \cdot 12301$$

$$= 2503.4470 \text{ Kg}$$

$$\text{Tension Stress, } St = (Fh / As) = 1.7400 \text{ N./mm}^2$$

$$\text{Allowed Stress, } Sa = 0.6 \cdot \text{Yield Str} = 147.5999 \text{ N./mm}^2$$

Saddle Splitting Dimension [d]:

$$= B - R \cdot \sin(\text{theta} / 2) / (\text{theta} / 2 \text{ in radians})$$

$$= 915 - 388 \cdot \sin(120 / 2) / 1.0472$$

$$= 594.127 \text{ mm}$$

$$\text{Bending Moment, } M = Fh \cdot d = 1487.3939 \text{ Kg-m.}$$



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Saddle Calcs: Ext. Pressure: Step: 25 9:41am Sep 9,2026

Bending Stress, $S_b = (M * C_1 / I) = 2.9515 \text{ N./mm}^2$
 Allowed Stress, $S_a = 2/3 * \text{Yield Str} = 163.9998 \text{ N./mm}^2$

Minimum Thickness of Baseplate per Moss:

$= \sqrt{(3(Q + \text{SaddleWt}) \text{BasePlateWidth} / (4 * \text{BasePlateLength} * \text{AllStress}))}$
 $= \sqrt{(3(12301 + 151.5)310 / (4 * 780 * 164))}$
 $= 14.898 \text{ mm}$

Calculation of Axial Load, Intermediate Values and Compressive Stress:

Web Length Dimension [Web Length]:
 $= 2 * \cos(90 - \text{Saddle Angle}/2) (\text{Inside Radius} + \text{Shell Thk} + \text{Wear Plate Thk})$
 $= 2 * \cos(90 - 120/2) (385 + 14 + 10)$
 $= 708.409 \text{ mm}$

Distance between Ribs [e]:
 $= \text{Web Length} / (\text{Nr ribs} - 1)$
 $= 708.41 / (3 - 1)$
 $= 354.204 \text{ mm}$

Baseplate Pressure Area [Ap]:
 $= e * \text{Bpwid} / 2$
 $= 354.2 * 310 / 2$
 $= 549.017 \text{ cm}^2$

Bearing Pressure [Bp]:
 $= (Q + \text{SaddleWt}) / (\text{BasePlateLength} * \text{BasePlateWidth})$
 $= 12452 / (780 * 310)$
 $= 5.150 \text{ Kg/cm}^2$

Axial Load [P]:
 $= A_p * B_p$
 $= 549.02 * 5.1498$
 $= 2827.310 \text{ Kg}$

Area of the Rib and Web [Ar]:
 $= \text{Rib Area} + \text{Web Area}$
 $= 27.1 + 17.71$
 $= 44.810 \text{ cm}^2$

Compressive Stress [Sc]:
 $= P / A_r$
 $= 2827.3 / 44.81$
 $= 6.188 \text{ N./mm}^2$

Check of Outside Ribs:

Inertia of Saddle, Outer Ribs - Longitudinal Direction

	B	D	Y	A	AY	Io
Rib+Web	10.0	281.0	...	28.1	...	0.185E+04

Rib dimension [D]:
 $= \text{Saddle Width} - \text{Web Thickness}$
 $= 281 - 10$
 $= 271.000 \text{ mm}$

Distance to Centroid from Datum [ytot]:
 $= AY / A$
 $= 0 / 44.81$
 $= 0.000 \text{ mm}$

Distance to Centroid [C1]:



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Saddle Calcs: Ext. Pressure: Step: 25 9:41am Sep 9, 2026

= Saddle Width / 2
= 281/2
= 140.500 mm

Radius of Gyration [r]:
= $\sqrt{\text{Total Inertia} / \text{Total Area}}$
= $\sqrt{1849/44.81}$
= 64.236 mm

Length of Outer Rib [L]:
= Saddle Height - $\cos(\theta/2) (\text{radius} + \text{shlthk} + \text{wpdthk}) - \text{bpthk}$
= 915 - $\cos(120/2) (385 + 14 + 10) - 16$
= 694.500 mm

Intermediate Term [Cc]:
= $\sqrt{(2 \cdot \pi^2 \cdot \text{Elastic Modulus} / \text{Yield Stress})}$
= $\sqrt{(2 \cdot \pi^2 \cdot 199943008/246)}$
= 126.667

Slenderness ratio [KL/r]:
= KL/r
= 1 • 694.5/64.236
= 10.812

Bending Moment [Rm]:
= $Fl / (2 \cdot B_{plen}) \cdot e \cdot L / 2$
= 1173.3 / $(2 \cdot 780) \cdot 354.2 \cdot 694.5/2$
= 92.507 Kg-m.

Compressive Allowable, $KL/r < Cc$ (10.812 < 126.67) per AISC E2-1 [Sca]:
= $(1 - (KL/r)^2 / (2 \cdot Cc^2)) F_y / (5/3 + 3 \cdot (KL/r) / (8 \cdot Cc) - (KL/r)^3 / (8 \cdot Cc^3))$
= $(1 - (10.812)^2 / (2 \cdot 126.67^2)) 246 / (5/3 + 3 \cdot (10.812) / (8 \cdot 126.67) - (10.812^3) / (8 \cdot 126.67^3))$
= 144.3 N./mm²

AISC Unity Check of Outside Ribs (must be ≤ 1)
= $S_c / S_{ca} + (R_m \cdot C_1 / I) / S_{ba}$
= 6.1876/144.3 + $(92.507 \cdot 140.5/18490034) / 164$
= 0.085

Check of Inside Ribs:

Inertia of Saddle, Inner Ribs - Axial Direction

	B	D	Y	A	AY	Io
Rib	10.0	271.0	0.0	27.1	0.0	0.185E+04
Web	354.2	10.0	0.0	35.4	0.0	2.95
Totals	...	...	...	62.5	...	0.185E+04

Distance to Centroid from Datum [ytot]:
= AY / A
= 0/62.52
= 0.000 mm

Distance to Centroid [C1]:
= Saddle Width / 2
= 281/2
= 140.500 mm

Length of Inner Rib [L]:
= Saddle Height - Outside Radius - Bpthk
= 915 - $\cos(409/2) (16 + 0 + 0) - 0$
= 490.000 mm

Radius of Gyration [r]:



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Saddle Calcs: Ext. Pressure:

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$$\begin{aligned}
 &= \sqrt{\text{Total Inertia} / \text{Total Area}} \\
 &= \sqrt{1851.9 / 62.52} \\
 &= 54.424 \text{ mm}
 \end{aligned}$$

Slenderness ratio [KL/r]:

$$\begin{aligned}
 &= KL/r \\
 &= 1 \cdot 490 / 54.424 \\
 &= 9.003
 \end{aligned}$$

Unit Force [Force,u]:

$$\begin{aligned}
 &= FI / (2 \cdot \text{Baseplate Length}) \\
 &= 1173.3 / (2 \cdot 780) \\
 &= 0.752 \text{ Kg/mm}
 \end{aligned}$$

Moment at base of inner Rib [Mbase,c]:

$$\begin{aligned}
 &= \text{Unit Force} \cdot e \cdot L \\
 &= 0.7521 \cdot 354.2 \cdot 490 \\
 &= 130.535 \text{ Kg-m.}
 \end{aligned}$$

Bending Stress due to Transverse Force and Weight Load [σ_B ,base,c]:

$$\begin{aligned}
 &= \text{Bending Moment} / \text{Section Modulus} \\
 &= 130.54 / 131806 \\
 &= 9.712 \text{ N./mm}^2
 \end{aligned}$$

Compressive Allowable, $KL/r < C_c$ ($9.0033 < 126.67$) per AISC E2-1 [Sca]:

$$\begin{aligned}
 &= (1 - (KL/r)^2 / (2 \cdot C_c^2)) F_y / (5/3 + 3 \cdot (KL/r) / (8 \cdot C_c) - (KL/r)^3 / (8 \cdot C_c^3)) \\
 &= (1 - (9.0033)^2 / (2 \cdot 126.67^2)) 246 / \\
 &\quad (5/3 + 3 \cdot (9.0033) / (8 \cdot 126.67) - (9.0033^3) / (8 \cdot 126.67^3)) \\
 &= 144.9 \text{ N./mm}^2
 \end{aligned}$$

AISC Unity Check of Inside Ribs (must be ≤ 1)

$$\begin{aligned}
 &= S_c / S_{ca} + (M_{base,c} \cdot C1/I) / S_{ba} \\
 &= 8.8189 / 144.91 + (130.54 \cdot 140.5 / 1851.9) / 164 \\
 &= 0.120
 \end{aligned}$$

Review notes about nozzle loadings and supports in the Warnings report.

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Nozzle Calcs.: N1-Tube Inlet

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Input, Nozzle Desc: N1-Tube Inlet

From: 20

Pressure for Reinforcement Calculations	P	5850.000	KPa.
Temperature for Internal Pressure	Temp	65	°C
Design External Pressure	Pext	103.42	KPa.
Temperature for External Pressure	Tempex	65	°C
Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	138.00	N./mm ²
Parent Allowable Stress At Ambient	Sva	138.00	N./mm ²
Inside Diameter of Cylindrical Shell	D	770.00	mm
Design Length of Section	L	1313.1666	mm
Shell Finished (Minimum) Thickness	t	22.0000	mm
Shell Internal Corrosion Allowance	c	3.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm
Distance from Bottom/Left Tangent		560.00	mm
User Entered Minimum Design Metal Temperature		-33.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-350 LF2 +IT	
Material UNS Number		K03011	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	138.00	N./mm ²
Allowable Stress At Ambient	Sna	138.00	N./mm ²
Diameter Basis (for tr calc only)		Inside	
Layout Angle		0.00	deg
Diameter		243.0000	mm
Size and Thickness Basis		Actual	
Actual Thickness	tn	15.0000	mm
Flange Material		SA-105	
Hub Height of Integral Nozzle	h	30.2071	mm
Height of Beveled Transition	L'	20.2072	mm
Hub Thickness of Integral Nozzle (tn or x+tp)		50.0000	mm
Corrosion Allowance	can	3.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	250.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	20.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	20.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm
ASME Code Weld Type per UW-16		a	
Flange Class		600	
Flange Grade		GR 1.1	

The Pressure Design option was Design Pressure + static head.



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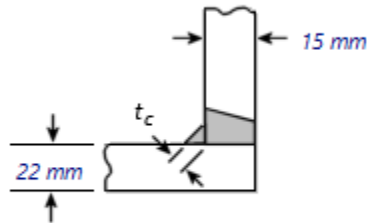
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Nozzle Calcs.: N1-Tube Inlet

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UW-16 (a)

Reinforcement CALCULATION, Description: N1-Tube Inlet

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation	243.000 mm.
Actual Thickness Used in Calculation	15.000 mm

Note:

Post Weld Heat Treatment is required for this nozzle and it was specified as being heat treated.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, t_r [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 5850 \cdot 388 / (138 \cdot 1 - 0.6 \cdot 5850)$
 $= 16.8784 \text{ mm}$

Reqd thk per UG-37(a) of Nozzle Wall, t_{rn} [Int. Press]
 $= P \cdot R / (S_n \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 5850 \cdot 124.5 / (138 \cdot 1 - 0.6 \cdot 5850)$
 $= 5.4159 \text{ mm}$

Required Nozzle thickness under External Pressure per UG-28 : 0.8424 mm

Intermediate Hub Nozzle Calculations:

Check to determine use of Sketch (e-1) or (e-2):

Height value from sketch (e-1) [te]:
 $= (\text{Hub Thickness} - \text{Neck Thickness}) / \tan(30)$
 $= 35 / 0.5773$
 $= 60.6218 \text{ mm}$

Hub Height was < 2.5 times Hub Thickness, use sketch UG-40 (e-1).

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	498.0000 mm
Parallel to Vessel Wall, opening length	d	249.0000 mm
Normal to Vessel Wall (Thickness Limit), no pad	T1np	47.5000 mm

Weld Strength Reduction Factor [fr1]:

$= \min(1, S_n / S_v)$
 $= \min(1, 138 / 138)$
 $= 1.000$

Weld Strength Reduction Factor [fr2]:

$= \min(1, S_n / S_v)$



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Nozzle Calcs.: N1-Tube Inlet

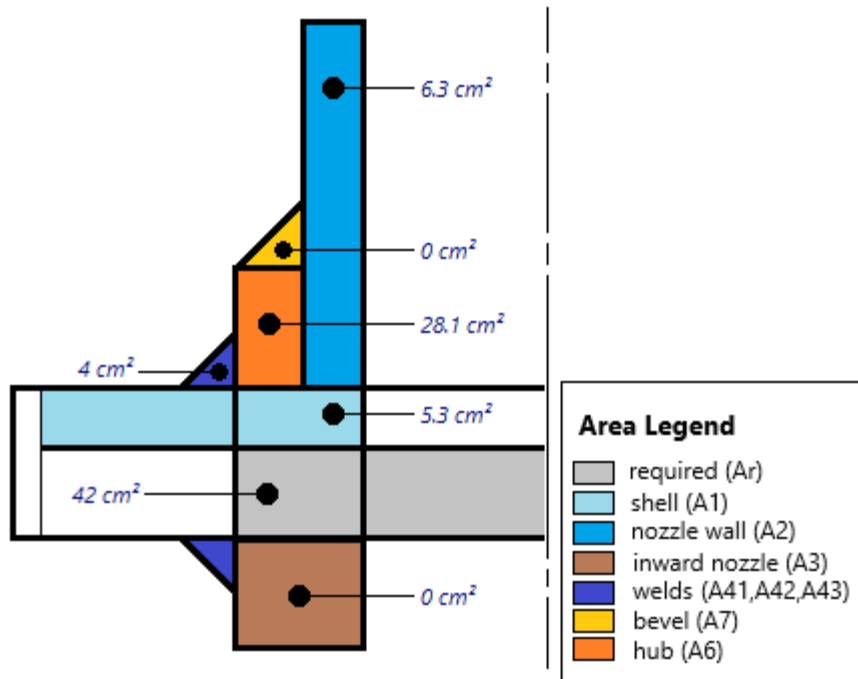
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= min(1, 138/138)
= 1.000

Weld Strength Reduction Factor [fr3]:

= min(fr2, fr4)
= min(1, 1)
= 1.000



Results of Nozzle Reinforcement Area Calculations: (cm²)

AREA AVAILABLE, A1 to A5		Design	External	Mapnc
Area Required	Ar	42.027	3.970	NA
Area in Shell	A1	5.283	39.370	NA
Area in Nozzle Wall	A2	6.255	10.600	NA
Area in Inward Nozzle	A3	0.000	0.000	NA
Area in Welds	A41+A42+A43	4.000	4.000	NA
Area in Element	A5	0.000	0.000	NA
Area in Hub + Bevel	A6	28.070	28.070	NA
TOTAL AREA AVAILABLE	Atot	43.608	82.040	NA

The Internal Pressure Case Governs the Analysis.

Nozzle Angle Used in Area Calculations

90.00 degs.

The area available without a pad is Sufficient.

Area Required [A]:

= (d • tr•F + 2 • tn • tr•F • (1-fr1)) UG-37(c)
= (249•16.878•1+2•47•16.878•1•(1-1))
= 42.027 cm²

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:

= d (E1•t - F•tr) - 2 • tn (E1•t - F•tr) • (1 - fr1)
= 249 (1 • 19 - 1 • 16.878) - 2 • 47



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Nozzle Calcs.: N1-Tube Inlet

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$$(1 \cdot 19 - 1 \cdot 16.878) \cdot (1 - 1) \\ = 5.283 \text{ cm}^2$$

Area Available in Nozzle Projecting Outward [A2]:

$$= (2 \cdot t_{np}) (t_n - t_{rn}) f_{r2} \\ = (2 \cdot 47.5) (12 - 5.4159) \cdot 1 \\ = 6.255 \text{ cm}^2$$

Area Available in Inward Weld + Outward Weld [A41 + A43]:

$$= W_o^2 \cdot f_{r2} + (W_i - c_{an}/0.707)^2 \cdot f_{r2} \\ = 20^2 \cdot 1 + (0)^2 \cdot 1 \\ = 4.000 \text{ cm}^2$$

Area Available in the Hub Section [A6]:

$$= (2 \cdot \min(t_{np}, h_o, \text{Hubht})) \cdot (\text{Hubtk} - t_n) \cdot f_{r2} \\ = (2 \cdot \min(47.5, 250, 30.207)) \cdot (50 - 15) \cdot 1 \\ = 28.070 \text{ cm}^2 \text{ Includes Bevel Area of } 6.925 \text{ cm}^2$$

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	$t_a = 8.4159 \text{ mm}$
Wall Thickness per UG16(b),	$tr_{16b} = 4.5000 \text{ mm}$
Wall Thickness, shell/head, internal pressure	$trb1 = 19.8784 \text{ mm}$
Wall Thickness	$tb1 = \max(trb1, tr_{16b}) = 19.8784 \text{ mm}$
Wall Thickness, shell/head, external pressure	$trb2 = 3.2909 \text{ mm}$
Wall Thickness	$tb2 = \max(trb2, tr_{16b}) = 4.5000 \text{ mm}$
Wall Thickness per table UG-45	$tb3 = 11.1026 \text{ mm}$

Determine Nozzle Thickness candidate [tb]:

$$= \min[tb3, \max(tb1, tb2)] \\ = \min[11.103, \max(19.878, 4.5)] \\ = 11.1026 \text{ mm}$$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

$$= \max(t_a, t_b) \\ = \max(8.4159, 11.103) \\ = 11.1026 \text{ mm}$$

Available Nozzle Neck Thickness = 15.0000 mm -> OK

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

Sustained	: 55.4,	Allowable	: 138.0 N./mm ²	Passed
Expansion	: 0.0,	Allowable	: 289.6 N./mm ²	Passed
Occasional	: 29.0,	Allowable	: 183.5 N./mm ²	Passed
Shear	: 17.2,	Allowable	: 96.6 N./mm ²	Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld, Curve: A

Govrn. thk, $t_g = 15$, $t_r = 5.4159$, $c = 3 \text{ mm}$, $E^* = 1$
 Thickness Ratio = $t_r \cdot E^* / (t_g - c) = 0.4513$, Temp. Reduction = 40 °C

Min Metal Temp. w/o impact per UCS-66, Curve A	4 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-36 °C

Nozzle-Shell/Head Weld (UCS-66(a)1(b)), Curve: B

Govrn. thk, $t_g = 22$, $t_r = 16.878$, $c = 3 \text{ mm}$, $E^* = 1$
 Thickness Ratio = $t_r \cdot E^* / (t_g - c) = 0.8883$, Temp. Reduction = 6 °C



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Nozzle Calcs.: N1-Tube Inlet Nozl: 9 9:41am Sep 9,2026

Min Metal Temp. w/o impact per UCS-66, Curve B -5 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -11 °C
 Min Metal Temp. w/o impact per UG-20(f) -29 °C

Gov. MDMT of the nozzle to shell joint welded assembly : -29 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c) -18 °C
 Flange MDMT with Temp reduction per UCS-66(b)(1)(-b) -42 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :

Design Pressure/Ambient Rating = 5850.00/10200.00 = 0.574

Weld Size Calculations, Description: N1-Tube Inlet

Intermediate Calc. for nozzle/shell Welds Tmin 19.0000 mm

Results Per UW-16.1:

Required Thickness Actual Thickness
 Nozzle Weld 6.0000 = Min per Code 14.1400 = 0.7 * Wo mm

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 5951 KPa.

Nozzle is O.K. for the External Pressure: 103.42 KPa.

The Drop for this Nozzle is : 40.3077 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 312.3077 mm

Input Echo, WRC297 Item 1, Description: N1-Tube Inlet :

Diameter Basis for Cylindrical Shell	ID	
Shell Internal Corrosion Allowance	3.0000	mm
Shell External Corrosion Allowance	0.0000	mm
Shell Diameter	770.000	mm
Shell Thickness	22.0000	mm
Shell Stress Concentration Factor	1.000	
Vessel Material	SA-516 70	
Vessel Cold S.I. Allowable	Smc 138.00	N./mm ²
Vessel Hot S.I. Allowable	Smh 138.00	N./mm ²

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Diameter Basis for Nozzle	ID	
Nozzle Corrosion Allowance	3.0000	mm
Nozzle Diameter	243.000	mm
Nozzle Thickness	50.0000	mm
Nozzle Stress Concentration Factor	1.000	
Nozzle Material	SA-350 LF2	
Nozzle Cold S.I. Allowable	SNmc 138.00	N./mm ²
Nozzle Hot S.I. Allowable	SNmh 138.00	N./mm ²

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Note:

External Forces and Moments in WRC 107/537 Convention:

These loads are assumed to be SUSained loads.

Design Internal Pressure	Dp	5850.00	KPa.
Radial Load	P	-2294.51	Kgf
Circumferential Shear	Vc	1657.15	Kgf
Longitudinal Shear	Vl	1657.15	Kgf



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Nozzle Calcs.: N1-Tube Inlet Nozl: 9 9:41am Sep 9, 2026

Circumferential Moment	Mc	1274.24	Kg-m.
Longitudinal Moment	Ml	1274.24	Kg-m.
Torsional Moment	Mt	1465.38	Kg-m.

Include Axial Pressure Thrust	No
Include Pressure Stress Indices per Div. 2	No
Local Loads applied at end of Nozzle/Attachment	No

*Warning - The ratio, Dn/Tn (6.860) is < 10
check the limitations of WRC bulletin 297.*

Stress Computations at the Edge of the Nozzle:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:

$$= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{R_{\text{mean}}(t - c_a)}$$

$$= 343.0 + 2 \cdot 1.65 \cdot \sqrt{397.5(22.0 - 3.0)}$$

$$= 629.787 \text{ mm}$$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter	(D) =	795.000 mm
Nozzle Outside Diameter	(d) =	343.000 mm
Vessel Thickness used	(T) =	19.000 mm
Nozzle Thickness used	(t) =	47.000 mm
T / t	=	0.404
d / t	=	10.000

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 2.791$$

Nr/P	=	0.061
Mr/P	=	0.111
M0/P	=	0.035
N0/P	=	0.080
MrD/Mc	=	0.268
NrDL/Mc	=	0.098
M0d/Mc	=	0.081
N0DL/Mc	=	0.084
MrD/Ml	=	0.094
NrDL/Ml	=	0.056
M0D/Ml	=	0.026
N0DL/Ml	=	0.093

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Membrane (P)	4	4	4	4
Outplane Bending (P)	13	-13	13	-13
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (Ml)	-9	-9	9	9
Outplane Bending (Ml)	-15	15	15	-15
Normal Pressure Stress	116	122	116	122
Outplane Stress Summary	109	120	159	107

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses parallel to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Inplane Membrane (P)	3	3	3	3
Inplane Bending (P)	41	-41	41	-41
Inplane Membrane (Mc)	0	0	0	0



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Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	-5	-5	5	5
Inplane Bending (ML)	-57	57	57	-57
Inplane Pressure Stress	58	58	58	58

Inplane Stress Summary	40	72	166	-30
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Vessel Stresses (N./mm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Shear stress normal to longitudinal plane)	Top Outside	Top Inside	Bottom Outside	Bottom Inside

Outplane Shear (Vc)	1	1	-1	-1
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	4	4	4	4

Shear Stress Summary	5	5	2	2
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Vessel Stresses (N./mm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Stress Intensities)	Top Outside	Top Inside	Bottom Outside	Bottom Inside

Two * Max Shear Stress	109	121	167	138
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Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stresses Normal to circumferential plane)	Left Outside	Left Inside	Right Outside	Right Inside

Outplane Membrane (P)	4	4	4	4
Outplane Bending (P)	13	-13	13	-13
Outplane Membrane (Mc)	-8	-8	8	8
Outplane Bending (Mc)	-48	48	48	-48
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	58	58	58	58

Outplane Stress Summary	18	90	133	9
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Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stresses parallel to circumferential plane)	Left Outside	Left Inside	Right Outside	Right Inside

Inplane Membrane (P)	3	3	3	3
Inplane Bending (P)	41	-41	41	-41
Inplane Membrane (Mc)	-9	-9	9	9
Inplane Bending (Mc)	-162	162	162	-162
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	116	122	116	122

Inplane Stress Summary	-10	237	333	-67
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Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Shear stress normal to circumferential plane)	Left Outside	Left Inside	Right Outside	Right Inside

Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	-1	-1	1	1
Torsional Shear (Mt)	4	4	4	4

Shear Stress Summary	2	2	5	5
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Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stress Intensities)	Cu	Cl	Du	Dl
	Left	Left	Right	Right
	Outside	Inside	Outside	Inside
Two * Max Shear Stress	29	237	333	78

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses in the hoop direction)	Au	Al	Bu	Bl
	Top	Top	Bottom	Bottom
	Outside	Inside	Outside	Inside
Hoop Membrane (P)	4	4	4	4
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	0	0	0	0
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	-9	-9	9	9
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	15	22	15	22
Hoop Stress Summary	11	18	30	37

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses Normal to pipe cross-section)	Au	Al	Bu	Bl
	Top	Top	Bottom	Bottom
	Outside	Inside	Outside	Inside
Axial Membrane (P)	0	0	0	0
Axial Bending (P)	4	-4	4	-4
Axial Membrane (Mc)	0	0	0	0
Axial Bending (Mc)	0	0	0	0
Axial Membrane (ML)	-4	-4	4	4
Axial Bending (ML)	-6	6	6	-6
Axial Pressure Stress	7	7	7	7
Axial Stress Summary	2	5	24	1

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE (Shear stress)	Au	Al	Bu	Bl
	Outside	Inside	Outside	Inside
Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	0	0	0	0
Shear due to Torsion	1	1	1	1
Shear Stress Summary	2	2	1	1

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE (Stress Intensities)	Au	Al	Bu	Bl
	Outside	Inside	Outside	Inside
Two * Max Shear Stress	11	18	30	37

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses in the hoop direction)	Cu	Cl	Du	Dl
	Left	Left	Right	Right
	Outside	Inside	Outside	Inside
Hoop Membrane (P)	4	4	4	4
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	-8	-8	8	8
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	0	0	0	0
Hoop Bending (ML)	0	0	0	0



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Hoop Pressure Stress	15	22	15	22
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Hoop Stress Summary	12	19	29	36
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Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses Normal to pipe cross-section)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
---	-----------------------	----------------------	------------------------	-----------------------

Axial Membrane (P)	0	0	0	0
Axial Bending (P)	4	-4	4	-4
Axial Membrane (Mc)	-4	-4	4	4
Axial Bending (Mc)	-21	21	21	-21
Axial Membrane (ML)	0	0	0	0
Axial Bending (ML)	0	0	0	0
Axial Pressure Stress	7	7	7	7

Axial Stress Summary	-12	20	39	-13
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Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Shear stress)	Cu Outside	Cl Inside	Du Outside	Dl Inside
---	---------------	--------------	---------------	--------------

Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	0	0	0	0
Shear due to Torsion	1	1	1	1

Shear Stress Summary	1	1	2	2
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Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stress Intensities	Cu Outside	Cl Inside	Du Outside	Dl Inside
--	---------------	--------------	---------------	--------------

Two * Max Shear Stress	25	21	39	50
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WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Vessel-Nozzle Junction (N./mm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	116	122	116	122	116	122	116	122
Circ. Pl (SUS)	-5	-5	13	13	-6	-6	12	12
Circ. Q (SUS)	-2	2	28	-28	-121	121	203	-203
Long. Pm (SUS)	58	58	58	58	58	58	58	58
Long. Pl (SUS)	-2	-2	8	8	-4	-4	12	12
Long. Q (SUS)	-16	16	98	-98	-35	35	61	-61
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	1	1	-1	-1	-1	-1	1	1
Shear Q (SUS)	4	4	4	4	4	4	4	4
Pm (SUS)	116.0	122.0	116.0	122.0	116.0	122.0	116.0	122.0
Pm+Pl (SUS)	111.0	117.0	129.0	135.0	110.0	116.0	128.0	134.0
Pm+Pl+Q (Total)	109.4	119.5	165.1	139.1	30.6	237.1	331.1	78.6

Vessel Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
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Pm (SUS)	122.00	138.00	Passed
Pm+Pl (SUS)	135.01	207.00	Passed
Pm+Pl+Q (TOTAL)	331.12	413.99	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Nozzle Stress Summation at Vessel-Nozzle Junction (N./mm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	15	22	15	22	15	22	15	22
Circ. Pl (SUS)	-5	-5	13	13	-4	-4	12	12
Circ. Q (SUS)	0	0	0	0	0	0	0	0
Long. Pm (SUS)	7	7	7	7	7	7	7	7
Long. Pl (SUS)	-4	-4	4	4	-4	-4	4	4
Long. Q (SUS)	-2	2	10	-10	-17	17	25	-25
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	0	0	0	0	0	0	0	0
Shear Q (SUS)	1	1	1	1	1	1	1	1
Pm (SUS)	15.0	22.0	15.0	22.0	15.0	22.0	15.0	22.0
Pm+Pl (SUS)	10.0	17.0	28.0	35.0	11.0	18.0	27.0	34.0
Pm+Pl+Q (Total)	10.1	17.1	28.1	35.0	25.1	20.4	36.1	48.0

Nozzle Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	22.00	138.00	Passed
Pm+Pl (SUS)	35.00	207.00	Passed
Pm+Pl+Q (TOTAL)	48.04	413.99	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

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Nozzle Calcs.: N2-Tube outlet

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Input, Nozzle Desc: N2-Tube outlet

From: 20

Pressure for Reinforcement Calculations	P	5857.547	KPa.
Temperature for Internal Pressure	Temp	65	°C
Design External Pressure	Pext	103.42	KPa.
Temperature for External Pressure	Tempex	65	°C
Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	138.00	N./mm ²
Parent Allowable Stress At Ambient	Sva	138.00	N./mm ²
Inside Diameter of Cylindrical Shell	D	770.00	mm
Design Length of Section	L	1313.1666	mm
Shell Finished (Minimum) Thickness	t	22.0000	mm
Shell Internal Corrosion Allowance	c	3.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm
Distance from Bottom/Left Tangent		560.00	mm
User Entered Minimum Design Metal Temperature		-33.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-350 LF2 +IT	
Material UNS Number		K03011	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	138.00	N./mm ²
Allowable Stress At Ambient	Sna	138.00	N./mm ²
Diameter Basis (for tr calc only)		Inside	
Layout Angle		180.00	deg
Diameter		243.0000	mm
Size and Thickness Basis		Actual	
Actual Thickness	tn	15.0000	mm
Flange Material		SA-105	
Hub Height of Integral Nozzle	h	30.2071	mm
Height of Beveled Transition	L'	20.2072	mm
Hub Thickness of Integral Nozzle (tn or x+tp)		50.0000	mm
Corrosion Allowance	can	3.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	250.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	20.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	20.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm
ASME Code Weld Type per UW-16		a	
Flange Class		600	
Flange Grade		GR 1.1	

The Pressure Design option was Design Pressure + static head.



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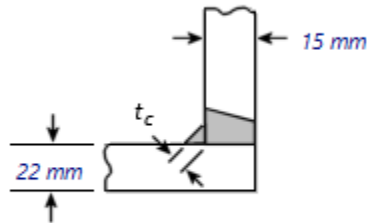
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UW-16 (a)

Reinforcement CALCULATION, Description: N2-Tube outlet

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation	243.000 mm.
Actual Thickness Used in Calculation	15.000 mm

Note:

Post Weld Heat Treatment is required for this nozzle and it was specified as being heat treated.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, t_r [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 5857.5 \cdot 388 / (138 \cdot 1 - 0.6 \cdot 5857.5)$
 $= 16.9008 \text{ mm}$

Reqd thk per UG-37(a) of Nozzle Wall, t_{rn} [Int. Press]
 $= P \cdot R / (S_n \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 5857.5 \cdot 124.5 / (138 \cdot 1 - 0.6 \cdot 5857.5)$
 $= 5.4231 \text{ mm}$

Required Nozzle thickness under External Pressure per UG-28 : 0.8424 mm

Intermediate Hub Nozzle Calculations:

Check to determine use of Sketch (e-1) or (e-2):

Height value from sketch (e-1) [te]:
 $= (\text{Hub Thickness} - \text{Neck Thickness}) / \tan(30)$
 $= 35 / 0.5773$
 $= 60.6218 \text{ mm}$

Hub Height was < 2.5 times Hub Thickness, use sketch UG-40 (e-1).

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	498.0000 mm
Parallel to Vessel Wall, opening length	d	249.0000 mm
Normal to Vessel Wall (Thickness Limit), no pad	T1np	47.5000 mm

Weld Strength Reduction Factor [fr1]:

$= \min(1, S_n / S_v)$
 $= \min(1, 138 / 138)$
 $= 1.000$

Weld Strength Reduction Factor [fr2]:

$= \min(1, S_n / S_v)$



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$$= \min(1, 138/138)$$

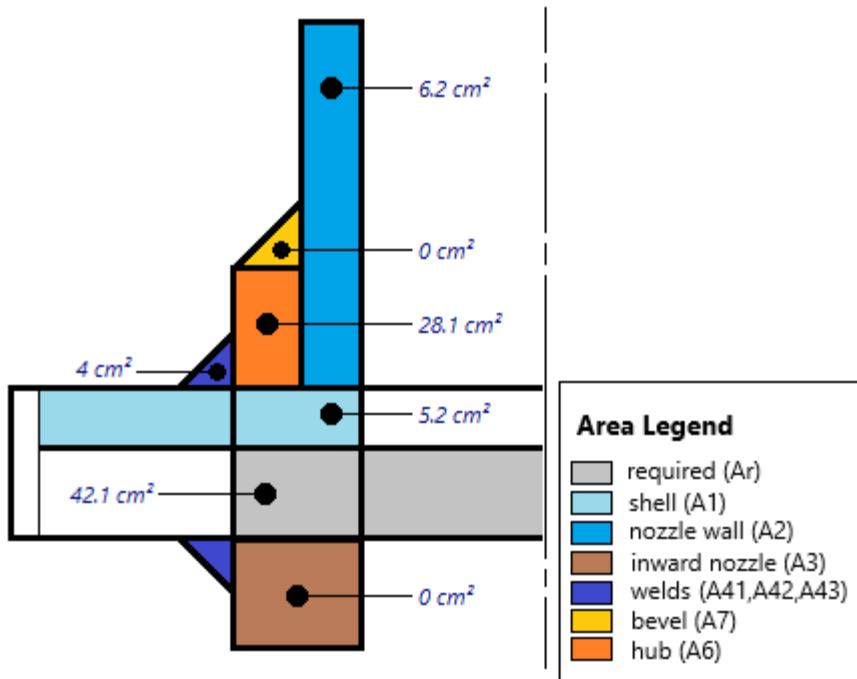
$$= 1.000$$

Weld Strength Reduction Factor [fr3]:

$$= \min(fr2, fr4)$$

$$= \min(1, 1)$$

$$= 1.000$$



Results of Nozzle Reinforcement Area Calculations: (cm²)

AREA AVAILABLE, A1 to A5		Design	External	Mapnc
Area Required	Ar	42.083	3.970	NA
Area in Shell	A1	5.227	39.370	NA
Area in Nozzle Wall	A2	6.248	10.600	NA
Area in Inward Nozzle	A3	0.000	0.000	NA
Area in Welds	A41+A42+A43	4.000	4.000	NA
Area in Element	A5	0.000	0.000	NA
Area in Hub + Bevel	A6	28.070	28.070	NA
TOTAL AREA AVAILABLE	Atot	43.546	82.040	NA

The Internal Pressure Case Governs the Analysis.

Nozzle Angle Used in Area Calculations

90.00 degs.

The area available without a pad is Sufficient.

Area Required [A]:

$$= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1)) \text{ UG-37(c)}$$

$$= (249 \cdot 16.901 \cdot 1 + 2 \cdot 47 \cdot 16.901 \cdot 1 \cdot (1 - 1))$$

$$= 42.083 \text{ cm}^2$$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:

$$= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$$

$$= 249(1 \cdot 19 - 1 \cdot 16.901) - 2 \cdot 47$$



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$$(1 \cdot 19 - 1 \cdot 16.901) \cdot (1 - 1) \\ = 5.227 \text{ cm}^2$$

Area Available in Nozzle Projecting Outward [A2]:

$$= (2 \cdot t_{np}) (t_n - t_{rn}) f_{r2} \\ = (2 \cdot 47.5) (12 - 5.4231) \cdot 1 \\ = 6.248 \text{ cm}^2$$

Area Available in Inward Weld + Outward Weld [A41 + A43]:

$$= W_o^2 \cdot f_{r2} + (W_i - c_{an}/0.707)^2 \cdot f_{r2} \\ = 20^2 \cdot 1 + (0)^2 \cdot 1 \\ = 4.000 \text{ cm}^2$$

Area Available in the Hub Section [A6]:

$$= (2 \cdot \min(t_{np}, h_o, \text{Hubht})) \cdot (\text{Hubtk} - t_n) \cdot f_{r2} \\ = (2 \cdot \min(47.5, 250, 30.207)) \cdot (50 - 15) \cdot 1 \\ = 28.070 \text{ cm}^2 \text{ Includes Bevel Area of } 6.925 \text{ cm}^2$$

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	$t_a = 8.4231 \text{ mm}$
Wall Thickness per UG16(b),	$tr_{16b} = 4.5000 \text{ mm}$
Wall Thickness, shell/head, internal pressure	$trb1 = 19.9008 \text{ mm}$
Wall Thickness	$tb1 = \max(trb1, tr_{16b}) = 19.9008 \text{ mm}$
Wall Thickness, shell/head, external pressure	$trb2 = 3.2909 \text{ mm}$
Wall Thickness	$tb2 = \max(trb2, tr_{16b}) = 4.5000 \text{ mm}$
Wall Thickness per table UG-45	$tb3 = 11.1026 \text{ mm}$

Determine Nozzle Thickness candidate [tb]:

$$= \min[tb3, \max(tb1, tb2)] \\ = \min[11.103, \max(19.901, 4.5)] \\ = 11.1026 \text{ mm}$$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

$$= \max(t_a, t_b) \\ = \max(8.4231, 11.103) \\ = 11.1026 \text{ mm}$$

Available Nozzle Neck Thickness = 15.0000 mm -> OK

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

Sustained	: 60.0,	Allowable	: 138.0 N./mm ²	Passed
Expansion	: 0.0,	Allowable	: 285.0 N./mm ²	Passed
Occasional	: 29.0,	Allowable	: 183.5 N./mm ²	Passed
Shear	: 17.2,	Allowable	: 96.6 N./mm ²	Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld, Curve: A

Govrn. thk, $t_g = 15$, $t_r = 5.4231$, $c = 3 \text{ mm}$, $E^* = 1$
 Thickness Ratio = $t_r \cdot E^* / (t_g - c) = 0.4519$, Temp. Reduction = 40 °C

Min Metal Temp. w/o impact per UCS-66, Curve A	4 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-36 °C

Nozzle-Shell/Head Weld (UCS-66(a)1(b)), Curve: B

Govrn. thk, $t_g = 22$, $t_r = 16.901$, $c = 3 \text{ mm}$, $E^* = 1$
 Thickness Ratio = $t_r \cdot E^* / (t_g - c) = 0.8895$, Temp. Reduction = 6 °C



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Min Metal Temp. w/o impact per UCS-66, Curve B -5 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -11 °C
 Min Metal Temp. w/o impact per UG-20(f) -29 °C

Gov. MDMT of the nozzle to shell joint welded assembly : -29 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c) -18 °C
 Flange MDMT with Temp reduction per UCS-66(b)(1)(-b) -42 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :

Design Pressure/Ambient Rating = 5857.55/10200.00 = 0.574

Weld Size Calculations, Description: N2-Tube outlet

Intermediate Calc. for nozzle/shell Welds Tmin 19.0000 mm

Results Per UW-16.1:

Required Thickness Actual Thickness
 Nozzle Weld 6.0000 = Min per Code 14.1400 = 0.7 * Wo mm

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 5943.4 KPa.

Nozzle is O.K. for the External Pressure: 103.42 KPa.

The Drop for this Nozzle is : 40.3077 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 312.3077 mm

Input Echo, WRC297 Item 1, Description: N2-Tube outlet:

Diameter Basis for Cylindrical Shell	ID	
Shell Internal Corrosion Allowance	3.0000	mm
Shell External Corrosion Allowance	0.0000	mm
Shell Diameter	770.000	mm
Shell Thickness	22.0000	mm
Shell Stress Concentration Factor	1.000	
Vessel Material	SA-516 70	
Vessel Cold S.I. Allowable	Smc 138.00	N./mm ²
Vessel Hot S.I. Allowable	Smh 138.00	N./mm ²

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Diameter Basis for Nozzle	ID	
Nozzle Corrosion Allowance	3.0000	mm
Nozzle Diameter	243.000	mm
Nozzle Thickness	50.0000	mm
Nozzle Stress Concentration Factor	1.000	
Nozzle Material	SA-350 LF2	
Nozzle Cold S.I. Allowable	SNmc 138.00	N./mm ²
Nozzle Hot S.I. Allowable	SNmh 138.00	N./mm ²

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Note:

External Forces and Moments in WRC 107/537 Convention:

These loads are assumed to be SUStained loads.

Design Internal Pressure	Dp	5857.55	KPa.
Radial Load	P	2294.51	Kgf
Circumferential Shear	Vc	1657.15	Kgf
Longitudinal Shear	Vl	1657.15	Kgf



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Circumferential Moment	Mc	1274.24	Kg-m.
Longitudinal Moment	Ml	1274.24	Kg-m.
Torsional Moment	Mt	1465.38	Kg-m.

Include Axial Pressure Thrust	No
Include Pressure Stress Indices per Div. 2	No
Local Loads applied at end of Nozzle/Attachment	No

*Warning - The ratio, Dn/Tn (6.860) is < 10
check the Limitations of WRC bulletin 297.*

Stress Computations at the Edge of the Nozzle:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:
 $= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{R_{\text{mean}}(t - c_a)}$
 $= 343.0 + 2 \cdot 1.65 \cdot \sqrt{397.5(22.0 - 3.0)}$
 $= 629.787 \text{ mm}$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter	(D) =	795.000 mm
Nozzle Outside Diameter	(d) =	343.000 mm
Vessel Thickness used	(T) =	19.000 mm
Nozzle Thickness used	(t) =	47.000 mm
T / t	=	0.404
d / t	=	10.000

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 2.791$$

Nr/P	=	0.061
Mr/P	=	0.111
M0/P	=	0.035
N0/P	=	0.080
MrD/Mc	=	0.268
NrDL/Mc	=	0.098
M0d/Mc	=	0.081
N0DL/Mc	=	0.084
MrD/Ml	=	0.094
NrDL/Ml	=	0.056
M0D/Ml	=	0.026
N0DL/Ml	=	0.093

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Membrane (P)	-4	-4	-4	-4
Outplane Bending (P)	-13	13	-13	13
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (Ml)	-9	-9	9	9
Outplane Bending (Ml)	-15	15	15	-15
Normal Pressure Stress	116	122	116	122
Outplane Stress Summary	73	137	124	124

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses parallel to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Inplane Membrane (P)	-3	-3	-3	-3
Inplane Bending (P)	-41	41	-41	41
Inplane Membrane (Mc)	0	0	0	0



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Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	-5	-5	5	5
Inplane Bending (ML)	-57	57	57	-57
Inplane Pressure Stress	58	58	58	58

Inplane Stress Summary	-49	147	75	44
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Vessel Stresses (N./mm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Shear stress normal to longitudinal plane)	Top Outside	Top Inside	Bottom Outside	Bottom Inside

Outplane Shear (Vc)	1	1	-1	-1
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	4	4	4	4

Shear Stress Summary	5	5	2	2
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Vessel Stresses (N./mm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Stress Intensities)	Top Outside	Top Inside	Bottom Outside	Bottom Inside

Two * Max Shear Stress	123	149	124	124
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Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stresses Normal to circumferential plane)	Left Outside	Left Inside	Right Outside	Right Inside

Outplane Membrane (P)	-4	-4	-4	-4
Outplane Bending (P)	-13	13	-13	13
Outplane Membrane (Mc)	-8	-8	8	8
Outplane Bending (Mc)	-48	48	48	-48
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	58	58	58	58

Outplane Stress Summary	-17	106	97	25
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Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stresses parallel to circumferential plane)	Left Outside	Left Inside	Right Outside	Right Inside

Inplane Membrane (P)	-3	-3	-3	-3
Inplane Bending (P)	-41	41	-41	41
Inplane Membrane (Mc)	-9	-9	9	9
Inplane Bending (Mc)	-162	162	162	-162
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	116	122	116	122

Inplane Stress Summary	-100	312	243	7
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Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Shear stress normal to circumferential plane)	Left Outside	Left Inside	Right Outside	Right Inside

Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	-1	-1	1	1
Torsional Shear (Mt)	4	4	4	4

Shear Stress Summary	2	2	5	5
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Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stress Intensities)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Two * Max Shear Stress	100	312	243	27

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses in the hoop direction)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Hoop Membrane (P)	-4	-4	-4	-4
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	0	0	0	0
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	-9	-9	9	9
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	15	22	15	22
Hoop Stress Summary	1	8	20	27

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses Normal to pipe cross-section)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Axial Membrane (P)	0	0	0	0
Axial Bending (P)	-4	4	-4	4
Axial Membrane (Mc)	0	0	0	0
Axial Bending (Mc)	0	0	0	0
Axial Membrane (ML)	-4	-4	4	4
Axial Bending (ML)	-6	6	6	-6
Axial Pressure Stress	7	7	7	7
Axial Stress Summary	-8	14	13	9

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE (Shear stress)	Au Outside	Al Inside	Bu Outside	Bl Inside
Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	0	0	0	0
Shear due to Torsion	1	1	1	1
Shear Stress Summary	2	2	1	1

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE (Stress Intensities)	Au Outside	Al Inside	Bu Outside	Bl Inside
Two * Max Shear Stress	10	15	20	27

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses in the hoop direction)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Hoop Membrane (P)	-4	-4	-4	-4
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	-8	-8	8	8
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	0	0	0	0
Hoop Bending (ML)	0	0	0	0



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Hoop Pressure Stress	15	22	15	22
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Hoop Stress Summary	2	9	19	26
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Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses Normal to pipe cross-section)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
---	-----------------------	----------------------	------------------------	-----------------------

Axial Membrane (P)	0	0	0	0
Axial Bending (P)	-4	4	-4	4
Axial Membrane (Mc)	-4	-4	4	4
Axial Bending (Mc)	-21	21	21	-21
Axial Membrane (ML)	0	0	0	0
Axial Bending (ML)	0	0	0	0
Axial Pressure Stress	7	7	7	7

Axial Stress Summary	-23	29	28	-5
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Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Shear stress)	Cu Outside	Cl Inside	Du Outside	Dl Inside
---	---------------	--------------	---------------	--------------

Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	0	0	0	0
Shear due to Torsion	1	1	1	1

Shear Stress Summary	1	1	2	2
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Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stress Intensities	Cu Outside	Cl Inside	Du Outside	Dl Inside
--	---------------	--------------	---------------	--------------

Two * Max Shear Stress	25	29	28	31
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WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Vessel-Nozzle Junction (N./mm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	116	122	116	122	116	122	116	122
Circ. Pl (SUS)	-13	-13	5	5	-12	-12	6	6
Circ. Q (SUS)	-28	28	2	-2	-203	203	121	-121
Long. Pm (SUS)	58	58	58	58	58	58	58	58
Long. Pl (SUS)	-8	-8	2	2	-12	-12	4	4
Long. Q (SUS)	-98	98	16	-16	-61	61	35	-35
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	1	1	-1	-1	-1	-1	1	1
Shear Q (SUS)	4	4	4	4	4	4	4	4
Pm (SUS)	116.0	122.0	116.0	122.0	116.0	122.0	116.0	122.0
Pm+Pl (SUS)	103.0	109.0	121.0	127.0	104.0	110.0	122.0	128.0
Pm+Pl+Q (Total)	123.4	149.9	123.2	125.1	99.1	313.0	243.2	28.2

Vessel Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
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Pm (SUS)	122.00	138.00	Passed
Pm+Pl (SUS)	128.02	207.00	Passed
Pm+Pl+Q (TOTAL)	313.04	413.99	Passed

*Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.*

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Nozzle Stress Summation at Vessel-Nozzle Junction (N./mm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	15	22	15	22	15	22	15	22
Circ. Pl (SUS)	-13	-13	5	5	-12	-12	4	4
Circ. Q (SUS)	0	0	0	0	0	0	0	0
Long. Pm (SUS)	7	7	7	7	7	7	7	7
Long. Pl (SUS)	-4	-4	4	4	-4	-4	4	4
Long. Q (SUS)	-10	10	2	-2	-25	25	17	-17
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	0	0	0	0	0	0	0	0
Shear Q (SUS)	1	1	1	1	1	1	1	1
Pm (SUS)	15.0	22.0	15.0	22.0	15.0	22.0	15.0	22.0
Pm+Pl (SUS)	3.0	9.0	20.0	27.0	3.0	10.0	19.0	26.0
Pm+Pl+Q (Total)	9.2	13.2	20.1	27.1	25.1	28.1	28.1	32.1

Nozzle Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	22.00	138.00	Passed
Pm+Pl (SUS)	27.00	207.00	Passed
Pm+Pl+Q (TOTAL)	32.06	413.99	Passed

*Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.*

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Input, Nozzle Desc: N3-Shell inlet

From: 40

Pressure for Reinforcement Calculations	P	2250.000	KPa.
Temperature for Internal Pressure	Temp	65	°C
Design External Pressure	Pext	103.42	KPa.
Temperature for External Pressure	Tempex	65	°C
Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	138.00	N./mm ²
Parent Allowable Stress At Ambient	Sva	138.00	N./mm ²
Inside Diameter of Cylindrical Shell	D	770.00	mm
Design Length of Section	L	9680.0000	mm
Shell Finished (Minimum) Thickness	t	14.0000	mm
Shell Internal Corrosion Allowance	c	3.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm
Distance from Bottom/Left Tangent		2612.17	mm
User Entered Minimum Design Metal Temperature		-33.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-350 LF2 +IT	
Material UNS Number		K03011	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	138.00	N./mm ²
Allowable Stress At Ambient	Sna	138.00	N./mm ²
Diameter Basis (for tr calc only)		Inside	
Layout Angle		0.00	deg
Diameter		243.0000	mm
Size and Thickness Basis		Actual	
Actual Thickness	tn	15.0000	mm
Flange Material		SA-105	
Hub Height of Integral Nozzle	h	60.0000	mm
Height of Beveled Transition	L'	49.0747	mm
Hub Thickness of Integral Nozzle (tn or x+tp)		100.0000	mm
Corrosion Allowance	can	3.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	300.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	20.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	14.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm
ASME Code Weld Type per UW-16		a	
Flange Class		300	
Flange Grade		GR 1.1	

The Pressure Design option was Design Pressure + static head.



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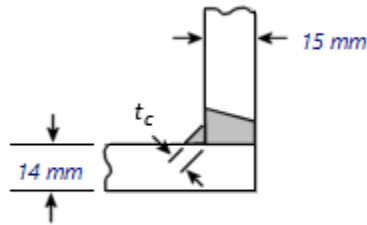
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UW-16 (a)

Reinforcement CALCULATION, Description: N3-Shell inlet

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation	243.000 mm.
Actual Thickness Used in Calculation	15.000 mm

Note:

Post Weld Heat Treatment is required for this nozzle and it was specified as being heat treated.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, t_r [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 2250 \cdot 388 / (138 \cdot 1 - 0.6 \cdot 2250)$
 $= 6.3891 \text{ mm}$

Reqd thk per UG-37(a) of Nozzle Wall, t_{rn} [Int. Press]
 $= P \cdot R / (S_n \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 2250 \cdot 124.5 / (138 \cdot 1 - 0.6 \cdot 2250)$
 $= 2.0501 \text{ mm}$

Required Nozzle thickness under External Pressure per UG-28 : 0.9135 mm

Intermediate Hub Nozzle Calculations:

Check to determine use of Sketch (e-1) or (e-2):

Height value from sketch (e-1) [te]:
 $= (\text{Hub Thickness} - \text{Neck Thickness}) / \tan(30)$
 $= 85 / 0.5773$
 $= 147.2244 \text{ mm}$

Hub Height was < 2.5 times Hub Thickness, use sketch UG-40 (e-1).

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	498.0000 mm
Parallel to Vessel Wall, opening length	d	249.0000 mm
Normal to Vessel Wall (Thickness Limit), no pad	T1np	27.5000 mm

Weld Strength Reduction Factor [fr1]:

$= \min(1, S_n / S_v)$
 $= \min(1, 138 / 138)$
 $= 1.000$

Weld Strength Reduction Factor [fr2]:

$= \min(1, S_n / S_v)$



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$$= \min(1, 138/138)$$

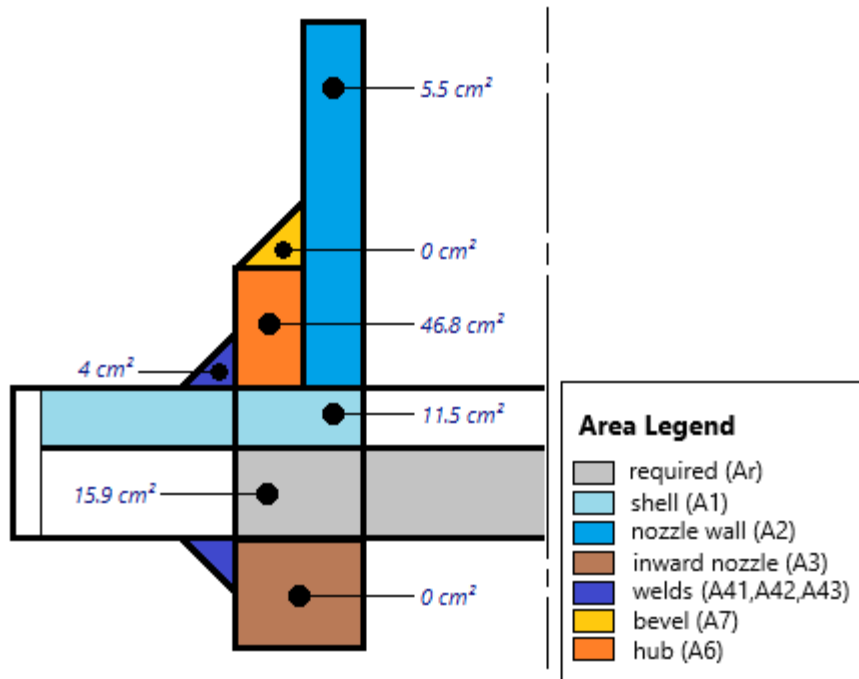
$$= 1.000$$

Weld Strength Reduction Factor [fr3]:

$$= \min(fr2, fr4)$$

$$= \min(1, 1)$$

$$= 1.000$$



Results of Nozzle Reinforcement Area Calculations: (cm²)

AREA AVAILABLE, A1 to A5		Design	External	Mapnc
Area Required	Ar	15.909	8.559	NA
Area in Shell	A1	11.481	10.273	NA
Area in Nozzle Wall	A2	5.472	6.098	NA
Area in Inward Nozzle	A3	0.000	0.000	NA
Area in Welds	A41+A42+A43	4.000	4.000	NA
Area in Element	A5	0.000	0.000	NA
Area in Hub	A6	46.750	46.750	NA
TOTAL AREA AVAILABLE	Atot	67.704	67.120	NA

The Internal Pressure Case Governs the Analysis.

Nozzle Angle Used in Area Calculations

90.00 degs.

The area available without a pad is Sufficient.

Area Required [A]:

$$= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1)) \text{ UG-37(c)}$$

$$= (249 \cdot 6.3891 \cdot 1 + 2 \cdot 97 \cdot 6.3891 \cdot 1 \cdot (1 - 1))$$

$$= 15.909 \text{ cm}^2$$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:

$$= d (E1 \cdot t - F \cdot tr) - 2 \cdot tn (E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$$

$$= 249 (1 \cdot 11 - 1 \cdot 6.3891) - 2 \cdot 97$$



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$$(1 \cdot 11 - 1 \cdot 6.3891) \cdot (1 - 1) \\ = 11.481 \text{ cm}^2$$

Area Available in Nozzle Projecting Outward [A2]:

$$= (2 \cdot t_{np}) (t_n - t_{rn}) f_{r2} \\ = (2 \cdot 27.5) (12 - 2.0501) \cdot 1 \\ = 5.472 \text{ cm}^2$$

Area Available in Inward Weld + Outward Weld [A41 + A43]:

$$= W_o^2 \cdot f_{r2} + (W_i - c_n / 0.707)^2 \cdot f_{r2} \\ = 20^2 \cdot 1 + (0)^2 \cdot 1 \\ = 4.000 \text{ cm}^2$$

Area Available in the Hub Section [A6]:

$$= (2 \cdot \min(t_{np}, h_o, \text{Hubht})) \cdot (\text{Hubtk} - t_n) \cdot f_{r2} \\ = (2 \cdot \min(27.5, 300, 60)) \cdot (100 - 15) \cdot 1 \\ = 46.750 \text{ cm}^2$$

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	$t_a = 5.0501 \text{ mm}$
Wall Thickness per UG16(b),	$tr_{16b} = 4.5000 \text{ mm}$
Wall Thickness, shell/head, internal pressure	$trb1 = 9.3891 \text{ mm}$
Wall Thickness	$tb1 = \max(trb1, tr_{16b}) = 9.3891 \text{ mm}$
Wall Thickness, shell/head, external pressure	$trb2 = 3.2909 \text{ mm}$
Wall Thickness	$tb2 = \max(trb2, tr_{16b}) = 4.5000 \text{ mm}$
Wall Thickness per table UG-45	$tb3 = 11.1026 \text{ mm}$

Determine Nozzle Thickness candidate [tb]:

$$= \min[tb3, \max(tb1, tb2)] \\ = \min[11.103, \max(9.3891, 4.5)] \\ = 9.3891 \text{ mm}$$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

$$= \max(t_a, t_b) \\ = \max(5.0501, 9.3891) \\ = 9.3891 \text{ mm}$$

Available Nozzle Neck Thickness = 15.0000 mm -> OK

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

Sustained	: 29.3	Allowable	: 138.0 N./mm ²	Passed
Expansion	: 0.0	Allowable	: 315.7 N./mm ²	Passed
Occasional	: 11.1	Allowable	: 183.5 N./mm ²	Passed
Shear	: 14.5	Allowable	: 96.6 N./mm ²	Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld, Curve: A

Govrn. thk, $t_g = 15$, $t_r = 2.0501$, $c = 3 \text{ mm}$, $E^* = 1$
 Thickness Ratio = $t_r \cdot E^* / (t_g - c) = 0.1708$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve A	4 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C

Nozzle-Shell/Head Weld (UCS-66(a)1(b)), Curve: B

Govrn. thk, $t_g = 14$, $t_r = 6.3891$, $c = 3 \text{ mm}$, $E^* = 1$
 Thickness Ratio = $t_r \cdot E^* / (t_g - c) = 0.5808$, Temp. Reduction = 24 °C



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Min Metal Temp. w/o impact per UCS-66, Curve B -18 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -42 °C
 Min Metal Temp. w/o impact per UG-20(f) -29 °C

Gov. MDMT of the nozzle to shell joint welded assembly : -42 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c) -18 °C
 Flange MDMT with Temp reduction per UCS-66(b)(1)(-b) -48 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :

Design Pressure/Ambient Rating = 2250.00/5100.00 = 0.441

Weld Size Calculations, Description: N3-Shell inlet

Intermediate Calc. for nozzle/shell Welds Tmin 11.0000 mm

Results Per UW-16.1:

Required Thickness Actual Thickness
 Nozzle Weld 6.0000 = Min per Code 14.1400 = 0.7 * Wo mm

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 3839.1 KPa.

Note: The MAWP of this junction was limited by the parent Shell/Head.

Nozzle is O.K. for the External Pressure: 103.42 KPa.

The Drop for this Nozzle is : 70.0988 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 384.0988 mm

Input Echo, WRC297 Item 1, Description: N3-Shell inlet:

Diameter Basis for Cylindrical Shell	ID		
Shell Internal Corrosion Allowance	3.0000	mm	
Shell External Corrosion Allowance	0.0000	mm	
Shell Diameter	770.000	mm	
Shell Thickness	14.0000	mm	
Shell Stress Concentration Factor	1.000		
Vessel Material	SA-516 70		
Vessel Cold S.I. Allowable	Smc	138.00	N./mm ²
Vessel Hot S.I. Allowable	Smh	138.00	N./mm ²

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Diameter Basis for Nozzle	ID		
Nozzle Corrosion Allowance	3.0000	mm	
Nozzle Diameter	243.000	mm	
Nozzle Thickness	100.0000	mm	
Nozzle Stress Concentration Factor	1.000		
Nozzle Material	SA-350 LF2		
Nozzle Cold S.I. Allowable	SNmc	138.00	N./mm ²
Nozzle Hot S.I. Allowable	SNmh	138.00	N./mm ²

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Note:

External Forces and Moments in WRC 107/537 Convention:

These loads are assumed to be SUSained loads.

Design Internal Pressure	Dp	2250.00	KPa.
Radial Load	P	-1912.11	Kgf



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Circumferential Shear	Vc	1274.73	Kgf
Longitudinal Shear	Vl	1274.73	Kgf
Circumferential Moment	Mc	891.97	Kg-m.
Longitudinal Moment	Ml	891.97	Kg-m.
Torsional Moment	Mt	1274.24	Kg-m.

Include Axial Pressure Thrust	No
Include Pressure Stress Indices per Div. 2	No
Local Loads applied at end of Nozzle/Attachment	No

*Warning - The ratio, Dn/Dv (.555) is > 0.5
check the limitations of WRC bulletin 297.*

*Warning - The ratio, Dn/Tn (4.430) is < 10
check the limitations of WRC bulletin 297.*

Stress Computations at the Edge of the Nozzle:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:

$$= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{R_{\text{mean}}(t - c_a)}$$

$$= 443.0 + 2 \cdot 1.65 \cdot \sqrt{393.5(14.0 - 3.0)}$$

$$= 660.111 \text{ mm}$$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter	(D) =	787.000 mm
Nozzle Outside Diameter	(d) =	443.000 mm
Vessel Thickness used	(T) =	11.000 mm
Nozzle Thickness used	(t) =	97.000 mm
T / t	=	0.113
d / t	=	10.000

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 4.761$$

Nr/P	=	0.046
Mr/P	=	0.070
M0/P	=	0.021
N0/P	=	0.033
MrD/Mc	=	0.244
NrDL/Mc	=	0.096
M0d/Mc	=	0.073
N0DL/Mc	=	0.061
MrD/Ml	=	0.054
NrDL/Ml	=	0.047
M0D/Ml	=	0.017
N0DL/Ml	=	0.032

Vessel Stresses (N/mm²):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Membrane (P)	5	5	5	5
Outplane Bending (P)	19	-19	19	-19
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (Ml)	-5	-5	5	5
Outplane Bending (Ml)	-16	16	16	-16
Normal Pressure Stress	78	80	78	80
Outplane Stress Summary	81	77	124	55

Vessel Stresses (N/mm²):

LONGITUDINAL PLANE (Stresses parallel to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
--	----------------------	---------------------	-------------------------	------------------------



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Nozzle Calcs.: N3-Shell inlet

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Inplane Membrane (P)	7	7	7	7
Inplane Bending (P)	65	-65	65	-65
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	-7	-7	7	7
Inplane Bending (ML)	-52	52	52	-52
Inplane Pressure Stress	39	39	39	39

Inplane Stress Summary	51	25	172	-64
------------------------	----	----	-----	-----

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE				
(Shear stress normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside

Outplane Shear (Vc)	1	1	-1	-1
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	3	3	3	3

Shear Stress Summary	5	5	2	2
----------------------	---	---	---	---

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE				
(Stress Intensities	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside

Two * Max Shear Stress	82	77	172	119
------------------------	----	----	-----	-----

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE				
(Stresses Normal to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside

Outplane Membrane (P)	5	5	5	5
Outplane Bending (P)	19	-19	19	-19
Outplane Membrane (Mc)	-9	-9	9	9
Outplane Bending (Mc)	-71	71	71	-71
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	39	39	39	39

Outplane Stress Summary	-17	86	144	-36
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Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE				
(Stresses parallel to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside

Inplane Membrane (P)	7	7	7	7
Inplane Bending (P)	65	-65	65	-65
Inplane Membrane (Mc)	-15	-15	15	15
Inplane Bending (Mc)	-239	239	239	-239
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	78	80	78	80

Inplane Stress Summary	-103	245	405	-201
------------------------	------	-----	-----	------

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE				
(Shear stress normal to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside

Outplane Shear (Vc)	0	0	0	0
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Nozzle Calcs.: N3-Shell inlet Nozl: 11 9:41am Sep 9,2026

Outplane Shear (V1)	-1	-1	1	1
Torsional Shear (Mt)	3	3	3	3

Shear Stress Summary	2	2	5	5
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Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stress Intensities)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
---	-----------------------	----------------------	------------------------	-----------------------

Two * Max Shear Stress	103	245	405	201
------------------------	-----	-----	-----	-----

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses in the hoop direction)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
---	----------------------	---------------------	-------------------------	------------------------

Hoop Membrane (P)	5	5	5	5
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	0	0	0	0
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	-5	-5	5	5
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	2	5	2	5

Hoop Stress Summary	2	5	12	15
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Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses Normal to pipe cross-section)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
--	----------------------	---------------------	-------------------------	------------------------

Axial Membrane (P)	0	0	0	0
Axial Bending (P)	0	0	0	0
Axial Membrane (Mc)	0	0	0	0
Axial Bending (Mc)	0	0	0	0
Axial Membrane (ML)	-1	-1	1	1
Axial Bending (ML)	0	0	0	0
Axial Pressure Stress	1	1	1	1

Axial Stress Summary	0	0	3	1
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Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE (Shear stress)	Au Outside	Al Inside	Bu Outside	Bl Inside
--------------------------------------	---------------	--------------	---------------	--------------

Shear due to (Vc)	0	0	0	0
Shear due to (V1)	0	0	0	0
Shear due to Torsion	0	0	0	0

Shear Stress Summary	0	0	0	0
----------------------	---	---	---	---

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE (Stress Intensities)	Au Outside	Al Inside	Bu Outside	Bl Inside
--	---------------	--------------	---------------	--------------

Two * Max Shear Stress	2	5	12	15
------------------------	---	---	----	----

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses in the hoop direction)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
--	-----------------------	----------------------	------------------------	-----------------------

Hoop Membrane (P)	5	5	5	5
Hoop Bending (P)	0	0	0	0



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Nozzle Calcs.: N3-Shell inlet

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Hoop Membrane (Mc)	-9	-9	9	9
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	0	0	0	0
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	2	5	2	5

Hoop Stress Summary	-2	0	17	20
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Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses Normal to pipe cross-section)	Cu	Cl	Du	Dl
	Left Outside	Left Inside	Right Outside	Right Inside

Axial Membrane (P)	0	0	0	0
Axial Bending (P)	0	0	0	0
Axial Membrane (Mc)	-1	-1	1	1
Axial Bending (Mc)	-2	2	2	-2
Axial Membrane (ML)	0	0	0	0
Axial Bending (ML)	0	0	0	0
Axial Pressure Stress	1	1	1	1

Axial Stress Summary	-1	2	5	0
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Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Shear stress)	Cu	Cl	Du	Dl
	Outside	Inside	Outside	Inside

Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	0	0	0	0
Shear due to Torsion	0	0	0	0

Shear Stress Summary	0	0	0	0
----------------------	---	---	---	---

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stress Intensities)	Cu	Cl	Du	Dl
	Outside	Inside	Outside	Inside

Two * Max Shear Stress	2	2	17	20
------------------------	---	---	----	----

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Vessel-Nozzle Junction (N./mm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	78	80	78	80	78	80	78	80
Circ. Pl (SUS)	0	0	10	10	-8	-8	22	22
Circ. Q (SUS)	3	-3	35	-35	-174	174	304	-304
Long. Pm (SUS)	39	39	39	39	39	39	39	39
Long. Pl (SUS)	0	0	14	14	-4	-4	14	14
Long. Q (SUS)	13	-13	117	-117	-52	52	90	-90
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	1	1	-1	-1	-1	-1	1	1
Shear Q (SUS)	3	3	3	3	3	3	3	3
Pm (SUS)	78.0	80.0	78.0	80.0	78.0	80.0	78.0	80.0
Pm+Pl (SUS)	78.0	80.0	88.0	90.0	70.0	72.0	100.0	102.0
Pm+Pl+Q (Total)	81.5	77.3	170.1	119.1	104.0	246.0	404.1	202.1



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Nozzle Calcs.: N3-Shell inlet

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Vessel Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	80.00	138.00	Passed
Pm+Pl (SUS)	102.02	207.00	Passed
Pm+Pl+Q (TOTAL)	404.06	413.99	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Nozzle Stress Summation at Vessel-Nozzle Junction (N./mm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	2	5	2	5	2	5	2	5
Circ. Pl (SUS)	0	0	10	10	-4	-4	14	14
Circ. Q (SUS)	0	0	0	0	0	0	0	0
Long. Pm (SUS)	1	1	1	1	1	1	1	1
Long. Pl (SUS)	-1	-1	1	1	-1	-1	1	1
Long. Q (SUS)	0	0	0	0	-2	2	2	-2
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	0	0	0	0	0	0	0	0
Shear Q (SUS)	0	0	0	0	0	0	0	0
Pm (SUS)	2.0	5.0	2.0	5.0	2.0	5.0	2.0	5.0
Pm+Pl (SUS)	2.0	5.0	12.0	15.0	2.0	1.0	16.0	19.0
Pm+Pl+Q (Total)	2.0	5.0	12.0	15.0	2.0	2.0	16.0	19.0

Nozzle Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	5.00	138.00	Passed
Pm+Pl (SUS)	19.00	207.00	Passed
Pm+Pl+Q (TOTAL)	19.00	413.99	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

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Nozzle Calcs.: N4-Shell outlet

Nozl: 12

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Input, Nozzle Desc: N4-Shell outlet

From: 70

Pressure for Reinforcement Calculations	P	2257.547	KPa.
Temperature for Internal Pressure	Temp	65	°C
Design External Pressure	Pext	103.42	KPa.
Temperature for External Pressure	Tempex	65	°C
Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	138.00	N./mm ²
Parent Allowable Stress At Ambient	Sva	138.00	N./mm ²
Inside Diameter of Cylindrical Shell	D	770.00	mm
Design Length of Section	L	9680.0000	mm
Shell Finished (Minimum) Thickness	t	14.0000	mm
Shell Internal Corrosion Allowance	c	3.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm
Distance from Bottom/Left Tangent		9872.17	mm
User Entered Minimum Design Metal Temperature		-33.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-350 LF2 +IT	
Material UNS Number		K03011	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	138.00	N./mm ²
Allowable Stress At Ambient	Sna	138.00	N./mm ²
Diameter Basis (for tr calc only)		Inside	
Layout Angle		180.00	deg
Diameter		243.0000	mm
Size and Thickness Basis		Actual	
Actual Thickness	tn	15.0000	mm
Flange Material		SA-105	
Hub Height of Integral Nozzle	h	50.9253	mm
Height of Beveled Transition	L'	49.0747	mm
Hub Thickness of Integral Nozzle (tn or x+tp)		100.0000	mm
Corrosion Allowance	can	3.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	300.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	20.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	14.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm
ASME Code Weld Type per UW-16		a	
Flange Class		300	
Flange Grade		GR 1.1	

The Pressure Design option was Design Pressure + static head.



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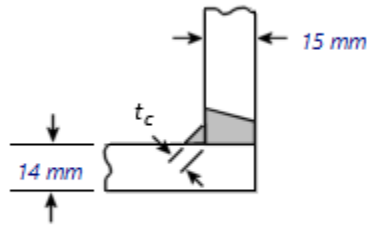
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Nozzle Calcs.: N4-Shell outlet

Noz1: 12 9:41am

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UW-16 (a)

Reinforcement CALCULATION, Description: N4-Shell outlet

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation	243.000 mm.
Actual Thickness Used in Calculation	15.000 mm

Note:

Post Weld Heat Treatment is required for this nozzle and it was specified as being heat treated.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, t_r [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 2257.5 \cdot 388 / (138 \cdot 1 - 0.6 \cdot 2257.5)$
 $= 6.4107 \text{ mm}$

Reqd thk per UG-37(a) of Nozzle Wall, t_{rn} [Int. Press]
 $= P \cdot R / (S_n \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 2257.5 \cdot 124.5 / (138 \cdot 1 - 0.6 \cdot 2257.5)$
 $= 2.0570 \text{ mm}$

Required Nozzle thickness under External Pressure per UG-28 : 0.9135 mm

Intermediate Hub Nozzle Calculations:

Check to determine use of Sketch (e-1) or (e-2):

Height value from sketch (e-1) [te]:
 $= (\text{Hub Thickness} - \text{Neck Thickness}) / \tan(30)$
 $= 85 / 0.5773$
 $= 147.2244 \text{ mm}$

Hub Height was < 2.5 times Hub Thickness, use sketch UG-40 (e-1).

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	498.0000 mm
Parallel to Vessel Wall, opening length	d	249.0000 mm
Normal to Vessel Wall (Thickness Limit), no pad	T1np	27.5000 mm

Weld Strength Reduction Factor [fr1]:

$= \min(1, S_n / S_v)$
 $= \min(1, 138 / 138)$
 $= 1.000$

Weld Strength Reduction Factor [fr2]:

$= \min(1, S_n / S_v)$



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Nozzle Calcs.: N4-Shell outlet

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$$= \min(1, 138/138)$$

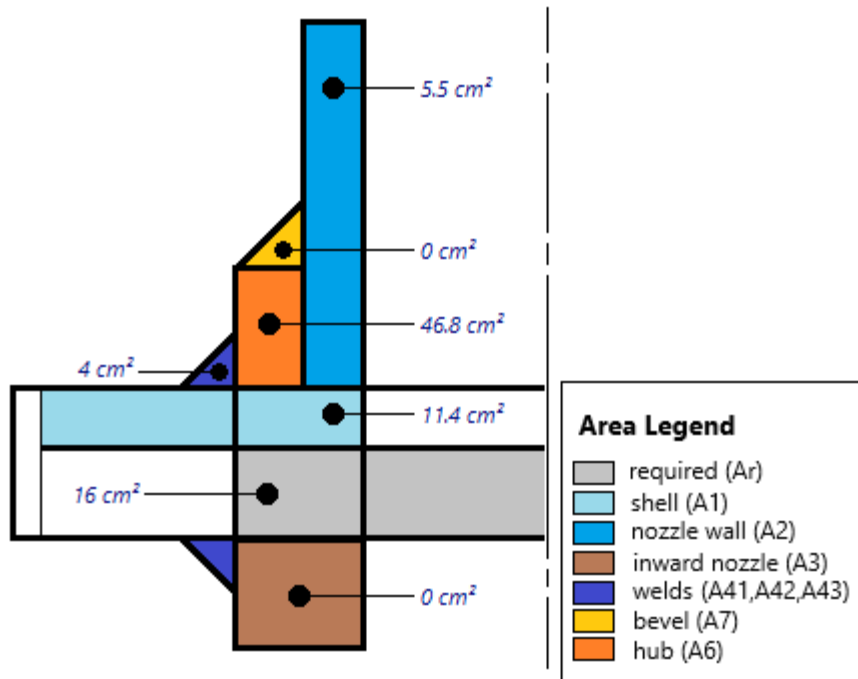
$$= 1.000$$

Weld Strength Reduction Factor [fr3]:

$$= \min(fr2, fr4)$$

$$= \min(1, 1)$$

$$= 1.000$$



Results of Nozzle Reinforcement Area Calculations: (cm²)

AREA AVAILABLE, A1 to A5		Design	External	Mapnc
Area Required	Ar	15.963	8.559	NA
Area in Shell	A1	11.427	10.273	NA
Area in Nozzle Wall	A2	5.469	6.098	NA
Area in Inward Nozzle	A3	0.000	0.000	NA
Area in Welds	A41+A42+A43	4.000	4.000	NA
Area in Element	A5	0.000	0.000	NA
Area in Hub	A6	46.750	46.750	NA
TOTAL AREA AVAILABLE	Atot	67.646	67.120	NA

The Internal Pressure Case Governs the Analysis.

Nozzle Angle Used in Area Calculations

90.00 degs.

The area available without a pad is Sufficient.

Area Required [A]:

$$= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1)) \text{ UG-37(c)}$$

$$= (249 \cdot 6.4107 \cdot 1 + 2 \cdot 97 \cdot 6.4107 \cdot 1 \cdot (1 - 1))$$

$$= 15.963 \text{ cm}^2$$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:

$$= d (E1 \cdot t - F \cdot tr) - 2 \cdot tn (E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$$

$$= 249 (1 \cdot 11 - 1 \cdot 6.4107) - 2 \cdot 97$$



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Nozzle Calcs.: N4-Shell outlet

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$$(1 \cdot 11 - 1 \cdot 6.4107) \cdot (1 - 1) \\ = 11.427 \text{ cm}^2$$

Area Available in Nozzle Projecting Outward [A2]:

$$= (2 \cdot t_{np}) (t_n - t_{rn}) \cdot fr_2 \\ = (2 \cdot 27.5) (12 - 2.057) \cdot 1 \\ = 5.469 \text{ cm}^2$$

Area Available in Inward Weld + Outward Weld [A41 + A43]:

$$= W_o^2 \cdot fr_2 + (W_i - c_n / 0.707)^2 \cdot fr_2 \\ = 20^2 \cdot 1 + (0)^2 \cdot 1 \\ = 4.000 \text{ cm}^2$$

Area Available in the Hub Section [A6]:

$$= (2 \cdot \min(t_{np}, h_o, \text{Hubht})) \cdot (\text{Hubtk} - t_n) \cdot fr_2 \\ = (2 \cdot \min(27.5, 300, 50.925)) \cdot (100 - 15) \cdot 1 \\ = 46.750 \text{ cm}^2$$

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta = 5.0570 mm
Wall Thickness per UG16(b),	tr16b = 4.5000 mm
Wall Thickness, shell/head, internal pressure	trb1 = 9.4107 mm
Wall Thickness	tb1 = max(trb1, tr16b) = 9.4107 mm
Wall Thickness, shell/head, external pressure	trb2 = 3.2909 mm
Wall Thickness	tb2 = max(trb2, tr16b) = 4.5000 mm
Wall Thickness per table UG-45	tb3 = 11.1026 mm

Determine Nozzle Thickness candidate [tb]:

$$= \min[tb3, \max(tb1, tb2)] \\ = \min[11.103, \max(9.4107, 4.5)] \\ = 9.4107 \text{ mm}$$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

$$= \max(ta, tb) \\ = \max(5.057, 9.4107) \\ = 9.4107 \text{ mm}$$

Available Nozzle Neck Thickness = 15.0000 mm -> OK

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

Sustained	: 29.4,	Allowable	: 138.0 N./mm ²	Passed
Expansion	: 0.0,	Allowable	: 315.6 N./mm ²	Passed
Occasional	: 11.2,	Allowable	: 183.5 N./mm ²	Passed
Shear	: 14.5,	Allowable	: 96.6 N./mm ²	Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld, Curve: A

Govrn. thk, tg = 15, tr = 2.057, c = 3 mm, E* = 1
Thickness Ratio = tr * E*/(tg - c) = 0.1714, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve A	4 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C

Nozzle-Shell/Head Weld (UCS-66(a)1(b)), Curve: B

Govrn. thk, tg = 14, tr = 6.4107, c = 3 mm, E* = 1
Thickness Ratio = tr * E*/(tg - c) = 0.5828, Temp. Reduction = 24 °C



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Nozzle Calcs.: N4-Shell outlet Nozl: 12 9:41am Sep 9,2026

Min Metal Temp. w/o impact per UCS-66, Curve B -18 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -42 °C
 Min Metal Temp. w/o impact per UG-20(f) -29 °C

Gov. MDMT of the nozzle to shell joint welded assembly : -42 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c) -18 °C
 Flange MDMT with Temp reduction per UCS-66(b)(1)(-b) -48 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
 Design Pressure/Ambient Rating = 2257.55/5100.00 = 0.443

Weld Size Calculations, Description: N4-Shell outlet

Intermediate Calc. for nozzle/shell Welds Tmin 11.0000 mm

Results Per UW-16.1:

Required Thickness Actual Thickness
 Nozzle Weld 6.0000 = Min per Code 14.1400 = 0.7 * Wo mm

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 3839.1 KPa.

Note: The MAWP of this junction was limited by the parent Shell/Head.

Nozzle is O.K. for the External Pressure: 103.42 KPa.

The Drop for this Nozzle is : 70.0988 mm
 The Cut Length for this Nozzle is, Drop + Ho + H + T : 384.0988 mm

Input Echo, WRC297 Item 1, Description: N4-Shell outle:

Diameter Basis for Cylindrical Shell	ID		
Shell Internal Corrosion Allowance	3.0000	mm	
Shell External Corrosion Allowance	0.0000	mm	
Shell Diameter	770.000	mm	
Shell Thickness	14.0000	mm	
Shell Stress Concentration Factor	1.000		
Vessel Material	SA-516 70		
Vessel Cold S.I. Allowable	Smc	138.00	N./mm ²
Vessel Hot S.I. Allowable	Smh	138.00	N./mm ²

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Diameter Basis for Nozzle	ID		
Nozzle Corrosion Allowance	3.0000	mm	
Nozzle Diameter	243.000	mm	
Nozzle Thickness	100.0000	mm	
Nozzle Stress Concentration Factor	1.000		
Nozzle Material	SA-350 LF2		
Nozzle Cold S.I. Allowable	SNmc	138.00	N./mm ²
Nozzle Hot S.I. Allowable	SNmh	138.00	N./mm ²

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Note:
 External Forces and Moments in WRC 107/537 Convention:
 These loads are assumed to be SUSTAINED loads.

Design Internal Pressure	Dp	2257.55	KPa.
Radial Load	P	-1912.11	Kgf



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Nozzle Calcs.: N4-Shell outlet

Nozl: 12

9:41am

Sep 9, 2026

Circumferential Shear	Vc	1274.73	Kgf
Longitudinal Shear	Vl	1274.73	Kgf
Circumferential Moment	Mc	891.97	Kg-m.
Longitudinal Moment	Ml	891.97	Kg-m.
Torsional Moment	Mt	1274.24	Kg-m.

Include Axial Pressure Thrust	No
Include Pressure Stress Indices per Div. 2	No
Local Loads applied at end of Nozzle/Attachment	No

*Warning - The ratio, Dn/Dv (.555) is > 0.5
check the limitations of WRC bulletin 297.*

*Warning - The ratio, Dn/Tn (4.430) is < 10
check the limitations of WRC bulletin 297.*

Stress Computations at the Edge of the Nozzle:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:

$$= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{R_{\text{mean}}(t - c_a)}$$

$$= 443.0 + 2 \cdot 1.65 \cdot \sqrt{393.5(14.0 - 3.0)}$$

$$= 660.111 \text{ mm}$$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter	(D) =	787.000 mm
Nozzle Outside Diameter	(d) =	443.000 mm
Vessel Thickness used	(T) =	11.000 mm
Nozzle Thickness used	(t) =	97.000 mm
T / t	=	0.113
d / t	=	10.000

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 4.761$$

Nr/P	=	0.046
Mr/P	=	0.070
M0/P	=	0.021
N0/P	=	0.033
MrD/Mc	=	0.244
NrDL/Mc	=	0.096
M0d/Mc	=	0.073
N0DL/Mc	=	0.061
MrD/Ml	=	0.054
NrDL/Ml	=	0.047
M0D/Ml	=	0.017
N0DL/Ml	=	0.032

Vessel Stresses (N/mm²):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Membrane (P)	5	5	5	5
Outplane Bending (P)	19	-19	19	-19
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (Ml)	-5	-5	5	5
Outplane Bending (Ml)	-16	16	16	-16
Normal Pressure Stress	78	80	78	80
Outplane Stress Summary	81	77	124	55

Vessel Stresses (N/mm²):

LONGITUDINAL PLANE (Stresses parallel to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
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Nozzle Calcs.: N4-Shell outlet

Nozl: 12 9:41am

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Inplane Membrane (P)	7	7	7	7
Inplane Bending (P)	65	-65	65	-65
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	-7	-7	7	7
Inplane Bending (ML)	-52	52	52	-52
Inplane Pressure Stress	39	39	39	39

Inplane Stress Summary	51	25	172	-64
------------------------	----	----	-----	-----

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE				
(Shear stress normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside

Outplane Shear (Vc)	1	1	-1	-1
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	3	3	3	3

Shear Stress Summary	5	5	2	2
----------------------	---	---	---	---

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE				
(Stress Intensities	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside

Two * Max Shear Stress	82	78	172	119
------------------------	----	----	-----	-----

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE				
(Stresses Normal to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside

Outplane Membrane (P)	5	5	5	5
Outplane Bending (P)	19	-19	19	-19
Outplane Membrane (Mc)	-9	-9	9	9
Outplane Bending (Mc)	-71	71	71	-71
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	39	39	39	39

Outplane Stress Summary	-17	86	145	-36
-------------------------	-----	----	-----	-----

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE				
(Stresses parallel to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside

Inplane Membrane (P)	7	7	7	7
Inplane Bending (P)	65	-65	65	-65
Inplane Membrane (Mc)	-15	-15	15	15
Inplane Bending (Mc)	-239	239	239	-239
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	78	80	78	80

Inplane Stress Summary	-103	245	405	-200
------------------------	------	-----	-----	------

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE				
(Shear stress normal to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside

Outplane Shear (Vc)	0	0	0	0
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Nozzle Calcs.: N4-Shell outlet Nozl: 12 9:41am Sep 9,2026

Outplane Shear (V1)	-1	-1	1	1
Torsional Shear (Mt)	3	3	3	3

Shear Stress Summary	2	2	5	5
----------------------	---	---	---	---

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stress Intensities)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
---	-----------------------	----------------------	------------------------	-----------------------

Two * Max Shear Stress	103	245	405	201
------------------------	-----	-----	-----	-----

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses in the hoop direction)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
---	----------------------	---------------------	-------------------------	------------------------

Hoop Membrane (P)	5	5	5	5
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	0	0	0	0
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	-5	-5	5	5
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	2	5	2	5

Hoop Stress Summary	2	5	12	15
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Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses Normal to pipe cross-section)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
--	----------------------	---------------------	-------------------------	------------------------

Axial Membrane (P)	0	0	0	0
Axial Bending (P)	0	0	0	0
Axial Membrane (Mc)	0	0	0	0
Axial Bending (Mc)	0	0	0	0
Axial Membrane (ML)	-1	-1	1	1
Axial Bending (ML)	0	0	0	0
Axial Pressure Stress	1	1	1	1

Axial Stress Summary	0	0	3	1
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Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE (Shear stress)	Au Outside	Al Inside	Bu Outside	Bl Inside
--------------------------------------	---------------	--------------	---------------	--------------

Shear due to (Vc)	0	0	0	0
Shear due to (V1)	0	0	0	0
Shear due to Torsion	0	0	0	0

Shear Stress Summary	0	0	0	0
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Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE (Stress Intensities)	Au Outside	Al Inside	Bu Outside	Bl Inside
--	---------------	--------------	---------------	--------------

Two * Max Shear Stress	2	5	12	15
------------------------	---	---	----	----

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses in the hoop direction)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
--	-----------------------	----------------------	------------------------	-----------------------

Hoop Membrane (P)	5	5	5	5
Hoop Bending (P)	0	0	0	0



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Nozzle Calcs.: N4-Shell outlet Nozl: 12 9:41am Sep 9,2026

Hoop Membrane (Mc)	-9	-9	9	9
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	0	0	0	0
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	2	5	2	5

Hoop Stress Summary	-2	0	17	20
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Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stresses Normal to pipe cross-section)	Left Outside	Left Inside	Right Outside	Right Inside

Axial Membrane (P)	0	0	0	0
Axial Bending (P)	0	0	0	0
Axial Membrane (Mc)	-1	-1	1	1
Axial Bending (Mc)	-2	2	2	-2
Axial Membrane (ML)	0	0	0	0
Axial Bending (ML)	0	0	0	0
Axial Pressure Stress	1	1	1	1

Axial Stress Summary	-1	2	5	0
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Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Shear stress)	Outside	Inside	Outside	Inside

Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	0	0	0	0
Shear due to Torsion	0	0	0	0

Shear Stress Summary	0	0	0	0
----------------------	---	---	---	---

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stress Intensities)	Outside	Inside	Outside	Inside

Two * Max Shear Stress	2	2	17	20
------------------------	---	---	----	----

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Vessel-Nozzle Junction (N./mm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	78	80	78	80	78	80	78	80
Circ. Pl (SUS)	0	0	10	10	-8	-8	22	22
Circ. Q (SUS)	3	-3	35	-35	-174	174	304	-304
Long. Pm (SUS)	39	39	39	39	39	39	39	39
Long. Pl (SUS)	0	0	14	14	-4	-4	14	14
Long. Q (SUS)	13	-13	117	-117	-52	52	90	-90
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	1	1	-1	-1	-1	-1	1	1
Shear Q (SUS)	3	3	3	3	3	3	3	3
Pm (SUS)	78.0	80.0	78.0	80.0	78.0	80.0	78.0	80.0
Pm+Pl (SUS)	78.0	80.0	88.0	90.0	70.0	72.0	100.0	102.0
Pm+Pl+Q (Total)	81.5	77.3	170.1	119.1	104.0	246.0	404.1	202.1



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 Nozzle Calcs.: N4-Shell outlet Nozl: 12 9:41am Sep 9,2026

Vessel Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	80.00	138.00	Passed
Pm+Pl (SUS)	102.02	207.00	Passed
Pm+Pl+Q (TOTAL)	404.06	413.99	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Nozzle Stress Summation at Vessel-Nozzle Junction (N./mm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	2	5	2	5	2	5	2	5
Circ. Pl (SUS)	0	0	10	10	-4	-4	14	14
Circ. Q (SUS)	0	0	0	0	0	0	0	0
Long. Pm (SUS)	1	1	1	1	1	1	1	1
Long. Pl (SUS)	-1	-1	1	1	-1	-1	1	1
Long. Q (SUS)	0	0	0	0	-2	2	2	-2
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	0	0	0	0	0	0	0	0
Shear Q (SUS)	0	0	0	0	0	0	0	0
Pm (SUS)	2.0	5.0	2.0	5.0	2.0	5.0	2.0	5.0
Pm+Pl (SUS)	2.0	5.0	12.0	15.0	2.0	1.0	16.0	19.0
Pm+Pl+Q (Total)	2.0	5.0	12.0	15.0	2.0	2.0	16.0	19.0

Nozzle Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	5.00	138.00	Passed
Pm+Pl (SUS)	19.00	207.00	Passed
Pm+Pl+Q (TOTAL)	19.00	413.99	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

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ASME TS Calc:

Case: 1 9:41a Sep 9, 2026

Input Echo, Tubesheet Number 1, Description: Fixed Tubesheet

Shell Data:

Main Shell Description: main shell-1

Shell Maximum Design Pressure	Psd,max	2250.00 KPa.
Shell Maximum Operating Pressure	Psox,max	2250.00 KPa.
Shell Minimum Operating Pressure	Psox,min	103.42 KPa.
Shell Thickness	ts	14.0000 mm
Shell Internal Corrosion Allowance	cas	3.0000 mm
Shell External Corrosion Allowance	casext	0.0000 mm
Inside Diameter of Shell	Ds	770.000 mm
Shell Circumferential Joint Efficiency	Esw	1.000
Shell Temperature for Internal Pressure	Ts	130.00 °C
Shell Material		SA-516 70

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Shell Material UNS Number		K02700
Shell Allowable Stress at Temperature	Ss	138.00 N./mm ²
Shell Allowable Stress at Ambient		138.00 N./mm ²

Channel Data:

Channel Description: LHS ch shell

Channel Type:		Cylinder
Channel Maximum Design Pressure	Ptd,max	5850.00 KPa.
Channel Maximum Operating Pressure	Ptox,max	5850.00 KPa.
Channel Minimum Operating Pressure	Ptox,min	0.00 KPa.
Channel Thickness	tc	22.0000 mm
Channel Internal Corrosion Allowance	cac	3.0000 mm
Channel External Corrosion Allowance	cacext	0.0000 mm
Inside Diameter of Channel	Dc	770.000 mm
Channel Design Temperature	TEMPC	65.00 °C
Channel Flange Matl		SA-516 70

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Channel Material UNS Number		K02700
Channel Allowable Stress at Temperature	Sc	138.00 N./mm ²
Channel Allowable Stress at Ambient		138.00 N./mm ²

Tube Data:

Number of Tube Holes	Nt	282
Tube Wall Thickness	et	2.1100 mm
Tube Outside Diameter	D	25.4000 mm
Total Straight Tube Length	Lt	10000.00 mm
Straight Tube Length (bet. inner tubsht faces) L		9680.00 mm
Design Temperature of the Tubes		130.00 °C
Tube Material		SA-334 6
Tube Material UNS Number		K03006
Is this a Welded Tube		No
Tube Material Specification used	Smls. & wld. tube	
Tube Allowable Stress at Temperature		118.00 N./mm ²
Tube Allowable Stress At Ambient		118.00 N./mm ²
Tube Yield Stress At design Temperature	Syt	216.40 N./mm ²
Tube Pitch (Center to Center Spacing)	P	33.8670 mm
Tube Layout Pattern		Triangular
Fillet Weld Leg	af	2.1100 mm
Groove Weld Leg	ag	2.1100 mm
Tube-Tubesheet Joint Weld Type		Full Strength
Method for Tube-Tubesheet Jt. Allowable		UW-20.3
Is the Tube Inset		No
Radius to Outermost Tube Hole Center	ro	362.479 mm



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ASME TS Calc:

Case: 1 9:41a Sep 9,2026

Largest Center-to-Center Tube Distance	U1	0.0000 mm
Length of Expanded Portion of Tube	ltx	0.0000 mm
Tube-side pass partition groove depth	hg	6.0000 mm
Area of Tube Layout	Ap	2578.1 cm ²
Perimeter of Tube Layout	Cp	2112.8 mm

Tubesheet Data:

Tubesheet TYPE: Fixed Tubesheet Exchanger, Conf B

Tubesheet Design Metal Temperature	T	130.00 °C
Tubesheet Material		SA-350 LF2

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Tubesheet Material UNS Number		K03011
Tubesheet Allowable Stress at Temperature	S	138.00 N./mm ²
Tubesheet Allowable Stress at Ambient	Tt	138.00 N./mm ²
Thickness of Tubesheet	h	160.0000 mm
Tubesheet Corr. Allowance (Shell side)	Cats	3.0000 mm
Tubesheet Corr. Allowance (Channel side)	Catc	3.0000 mm
Tubesheet Outside Diameter	A	1170.000 mm

Additional Data for Stepped Tubesheets:

Is the Tubesheet Stepped?		Yes
Is the Tubesheet Flat on Tubeside?		No
Step 1 Diameter on the Tubeside	dt1	846.00 mm
Step 1 Depth on the Tubeside	ht1	9.00 mm
Step 2 Diameter on the Tubeside	dt2	777.00 mm
Step 2 Depth on the Tubeside	ht2	7.00 mm
Is the Tubesheet Flat on Shellside?		Yes
Calculated Tubesheet Diameter per Div 2, 4.18.4		1170.00 mm

Note: Tubesheet diameter is now: 1170.000 mm Div 2, 4.18.4.

Area of the Untubed Lanes	AL	0.0 cm ²
---------------------------	----	---------------------

Additional Data for Fixed/Floating Tubesheet Exchangers:

Unsupported Tube Span under consideration	l	220.000 mm
Tube End condition corresponding to Span (l)	k	1.00
Ignore Radial Thermal Exp. effects (4.18.8/4.18.9)		Yes
Perform Differential Pressure Design		No
Run Multiple Load Cases		Yes
Shell Side Min. Design Pressure	Psd,min	-103.4190 KPa.
Mean Shell Metal Temp. along Shell len.	Tsm	0.00 °C
Mean Tube Metal Temp. along Tube length	Ttm	0.00 °C
Junction Stress Reduction option		Perform Plastic Calculation

Additional Data for Gasketed Tubesheets:

Tubesheet Gasket on which Side		Channel
Flange Outside Diameter	A	1170.000 mm
Flange Inside Diameter	B	770.000 mm
Flange Face Outside Diameter	Fod	839.000 mm
Flange Face Inside Diameter	Fid	770.000 mm
Gasket Outside Diameter	Go	828.000 mm
Gasket Inside Diameter	Gi	802.000 mm
Small end Hub thk.	g0	22.0000 mm
Large end Hub thk.	g1	33.0000 mm
Gasket Factor,	m	3.00
Gasket Design Seating Stress	y	68.95 N./mm ²
Flange Facing Sketch		Code Sketch 1a
Gasket Thickness	tg	3.1750 mm
Full face Gasket Flange Option		Program Selects



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ASME TS Calc:

Case: 1 9:41a Sep 9, 2026

Length of Partition Gasket	lp	804.000 mm
Width of Partition Gasket	wp	10.0000 mm
Partition Gasket Factor,	mPart	3.0000
Partition Gasket Design Seating Stress	yPart	68.95 N./mm ²
Partition Gasket Facing Sketch	Code Sketch	1a
Partition Gasket Column for Gasket Seating	Code Column	II

Bolting Information:

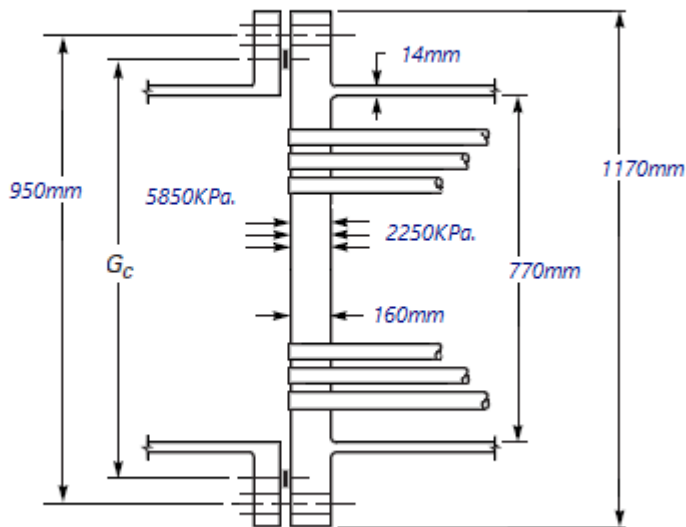
Diameter of Bolt Circle	C	950.000 mm
Nominal Bolt Diameter	dB	38.1000 mm
Thread Series	TEMA Thread Series	
Number of Bolts	n	36
Bolt Material		SA-193 B7
Bolt Allowable Stress At Temperature	Sb	172.00 N./mm ²
Bolt Allowable Stress At Ambient	Sa	172.00 N./mm ²
Weld between Flange and Shell/Channel		0.0000 mm

Note: The full bolt load $Sa \cdot Ab$, was used for all load cases.

Tubesheet Integral with	Shell	
Tubesheet Extended as Flange	Yes	
Thickness of Extended Portion of Tubesheet	Tf	128.0000 mm
Is Bolt Load Transferred to the Tubesheet	Yes	

Is Exchanger in Creep range (skip EP, Use 3S for Sps) No

Tubesheet Integral w/Shell, Gasketed w/Channel, Extended as a Flange



Configuration b:

ASME TubeSheet Results per Part UHX/4.18, 2025

Elasticity/Expansion Material Properties:

Shell	- TE-1	Carbon & Low Alloy Steels, Group 1
Shell	- TM-1	Carbon Steels with $C \leq 0.3\%$

Th. Exp. Coeff. Metal Temp. along Len	0.0 °C	0.0000112153 /°C
Elastic Mod. at Design Temperature	130.0 °C	0.19629E+09 KPa.
Th. Exp. Coeff. Metal Temp. at Tubsht	21.1 °C	0.0000115190 /°C
Elastic Mod. at Metal Temp. along Len	0.0 °C	0.20409E+09 KPa.



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ASME TS Calc:

Case: 1 9:41a Sep 9, 2026

Elastic Mod. at Ambient Temperature 21.1 °C 0.20270E+09 KPa.

Channel - TE-1 Carbon & Low Alloy Steels, Group 1
Channel - TM-1 Carbon Steels with C<= 0.3%

Th. Exp. Coeff. Metal Temp. at Tubsht 21.1 °C 0.0000115190 /°C
Elastic Mod. at Design Temperature 65.0 °C 0.20019E+09 KPa.
Elastic Mod. at Ambient Temperature 21.1 °C 0.20270E+09 KPa.

Tubes - TE-1 Carbon & Low Alloy Steels, Group 1
Tubes - TM-1 Carbon Steels with C<= 0.3%

Th. Exp. Coeff. Metal Temp. along Len 0.0 °C 0.0000112153 /°C
Elastic Mod. at Design Temperature 130.0 °C 0.19629E+09 KPa.
Elastic Mod. at Metal Temp. along Len 0.0 °C 0.20409E+09 KPa.
Elastic Mod. at Tubsht. Design Temp. 130.0 °C 0.19629E+09 KPa.
Elastic Mod. at Ambient Temperature 21.1 °C 0.20270E+09 KPa.

TubeSheet - TE-1 Carbon & Low Alloy Steels, Group 1
TubeSheet - TM-1 Carbon Steels with C> 0.3%

Elastic Mod. at Design Temperature 130.0 °C 0.19491E+09 KPa.
Elastic Mod. at Ambient Temperature 21.1 °C 0.20132E+09 KPa.

Note:

The Elasticity and Alpha values are taken from Tables in ASME II D.
Please insure these properties are consistent with the
type of Material for the tubes, shell, channel etc.

Tube Required Thickness under Internal Pressure (Tubeside pressure):

Thickness Due to Internal Pressure:
= $(P \cdot (D/2 - CAE)) / (S \cdot E + 0.4 \cdot P)$ per Appendix 1-1 (a)(1)
= $(5953.4 \cdot (25.4/2 - 0)) / (118 \cdot 1 + 0.4 \cdot 5953.4)$
= $0.6281 + 0.0000 = 0.6281$ mm

Tube Required Thickness under External Pressure (Shellside pressure) :

External Pressure Chart CS-2 at 130.00 °C
Elastic Modulus for Material 199943392.00 KPa.

Results for Max. Allowable External Pressure (Emawp):

TCA	ODCA	SLEN	D/T	L/D	Factor A	B
2.1100	25.40	9680.00	12.04	50.0000	0.0075908	122.73

EMAWP = $(4 \cdot B) / (3 \cdot (D/T)) = (4 \cdot 122.73) / (3 \cdot 12.038) = 13593$ KPa.

Results for Req'd Thickness for Ext. Pressure (Tca):

TCA	ODCA	SLEN	D/T	L/D	Factor A	B
0.6312	25.40	9680.00	40.24	50.0000	0.0006793	67.91

EMAWP = $(4 \cdot B) / (3 \cdot (D/T)) = (4 \cdot 67.913) / (3 \cdot 40.241) = 2250.1$ KPa.

Summary of Tube Required Thickness Results:

Total Required Thickness including Corrosion all.	0.6312 mm
Allowable Internal Pressure at Corroded thickness	20998.74 KPa.
Required Internal Design Pressure	5953.42 KPa.
Allowable External Pressure at Corroded thickness	13593.04 KPa.
Required External Design Pressure	2250.00 KPa.
Required Thickness due to Shell Side pressure	0.6312 mm

Detailed Results for load Case D3 un-corr. (Psd,max + Ptd,max)

Intermediate Calculations For Tubesheets Extended As Flanges:



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ASME TS Calc:

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Gasket Contact Width,	$N = (Goc - Gic) / 2$	13.000 mm
Basic Gasket Width,	$b0 = N / 2$	6.500 mm
Effective Gasket Width,	$b = \text{sqrt}(b0) * 2.5$	6.422 mm
Gasket Reaction Diameter,	$G = Go - 2.0 * b$	815.156 mm

Results for ASME Fixed Tubesheet Calculations for Configuration b:

Results for Tubesheet Calculations Original Thickness:

ASME Section VIII-2 Part 4.18.8.4 Step 1:

Tube expansion depth ratio $[\rho]$:

$= ltx / h$ (modified for corrosion if present)

$= 0/160$

$= 0.0000$ (must be $0 \leq \rho \leq 1$)

Effective tube hole diameter $[d*]$:

$= \max(dt - 2 * tt(EtT/E) * (StT/S) * (\rho); dt - 2 * tt)$

$= \max(25.4 - 2 * 2.11 (196289984/194911088) *$

$(118/138) * (0); 25.4 - 2 * 2.11)$

$= 25.4000$ mm

Equivalent outer tube limit circle diameter $[Do]$:

$= 2 * ro + dt$

$= 2 * 362.48 + 25.4$

$= 750.358$ mm

Determine the basic ligament efficiency for shear $[\mu]$:

$= (p - dt) / p$

$= (33.867 - 25.4) / 33.867$

$= 0.250$

Equivalent outer tube limit radius $[ao]$:

$= Do / 2$

$= 750.36/2$

$= 375.1790$ mm

Effective tube pitch $[p*]$:

$= p / \sqrt{ 1 - 4 * \min(AL * CNVfactor; 4 * Do * p) / (\pi * Do^2) }$

$= 33.867 / \sqrt{ 1 - 4 * \min(0 * 100 ; 4 * 750.36 * 33.867) / (\pi * 750.36^2) }$

$= 33.8670$ mm

Effective ligament efficiency for bending $[\mu*]$:

$= (p* - d*) / p*$

$= (33.867 - 25.4) / 33.867$

$= 0.2500$

Ratio $[ps]$:

$= as / ao$

$= 385/375.18$

$= 1.0262$

Ratio $[pc]$:

$= ac / ao$

$= 407.58/375.18$

$= 1.0864$

Compute parameter $[xt]$:

$= 1 - Nt((dt - 2 * tt) / (2 * ao))^2$

$= 1 - 282 ((25.4 - 2 * 2.11) / (2 * 375.18))^2$

$= 0.7753$



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Determine parameter [xs]:

$$= 1 - Nt / (2 \cdot a_o)^2$$

$$= 1 - 282 / (2 \cdot 375.18)^2$$

$$= 0.6769$$

Determine the value [h'g]:

$$= \max((h_g - CATC); 0) \text{ (For pressure only cases)}$$

$$= \max((6 - 0); 0)$$

$$= 6.000 \text{ mm}$$

ASME Section VIII-2 Part 4.18.8.4 Step 2:

Axial Shell Stiffness [Ks]:

$$= \pi \cdot t_s (D_s + t_s) E_s / L$$

$$= \pi \cdot 14 (770 + 14) 196289984 / 9680$$

$$= 71303.9062 \text{ Kg/mm}$$

Axial Tube Stiffness [Kt]:

$$= \pi \cdot t_t (D_t - t_t) E_t / L$$

$$= \pi \cdot 2.11 (25.4 - 2.11) 196289984 / 9680$$

$$= 319.2428 \text{ Kg/mm}$$

Stiffness Factor [Ks,t]:

$$= K_s / (N_t \cdot K_t)$$

$$= 71304 / (282 \cdot 319.24)$$

$$= 0.79203$$

Rigidity Ratio [J]:

$$= 1 / (1 + K_s / K_t)$$

$$= 1 / (1 + 71304 / 0)$$

$$= 1.0000 (= 1 \text{ if No Exp. Jt.})$$

Shell coefficient [βs]:

$$= ((12(1 - \nu_s^2))^{1/4}) / ((D_s + t_s) t_s)$$

$$= ((12(1 - 0.3^2))^{1/4}) / ((770 + 14) 14)$$

$$= 0.0174 \text{ 1/mm}$$

Shell coefficient [ks]:

$$= \beta_s \cdot E_s \cdot t_s^3 / (6(1 - \nu_s^2))$$

$$= 0.01735 \cdot 196289984 \cdot 14^3 / (6(1 - 0.3^2))$$

$$= 1711682688.0000 \text{ KPa} \cdot \text{mm}^2$$

Shell coefficient [λs]:

$$= 6(D_s + t_s) \cdot k_s / h^3 \cdot (1 + h \cdot \beta_s + (h \cdot \beta_s)^2 / 2)$$

$$= 6(770 + 14) \cdot 1711682688 / 160^3 \cdot (1 + 160 \cdot 0.01735 + (160 \cdot 0.01735)^2 / 2)$$

$$= 14998580.0000 \text{ KPa}$$

Shell coefficient [δs]:

$$= D_s^2 / (4 \cdot E_s \cdot t_s) \cdot (1 - D_s \cdot \nu_s / (2(D_s + t_s)))$$

$$= 770^2 / (4 \cdot 196289984 \cdot 14) \cdot (1 - 770 \cdot 0.3 / (2(770 + 14)))$$

$$= 0.45989E-01 \text{ mm/N./mm}^2$$

Intermediate parameters for Tubesheet Gasketed on the Channel Side:

βC, kc, λC, δC are all = 0

ASME Section VIII-2 Part 4.18.8.4 Step 3:

h/p = 4.724363 ; μ* = 0.250007 ; E* = 42221620 KPa.

E*/E = 0.216620 ; ν* = 0.396449

Note: As h/p (4.724) is > 2, data values for h/p = 2 were used.

Tube Bundle Stiffness Factor [Xa]:

$$= ((24(1 - \nu^*) N_t \cdot E_t \cdot t_t (d_t - t_t) a_o^2) / (E^* \cdot L \cdot H^3))^{1/4}$$

$$= ((24(1 - 0.3964^2) 282 \cdot 196289984 \cdot 2.11$$



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$$\begin{aligned} &= (25.4 - 2.11)375.18^2 / (42221620 \cdot 9680 \cdot 160^3)^{1/4} \\ &= 1.4666 \end{aligned}$$

Values from Table 4.18.3

$$\begin{aligned} Z_d &= 0.484783 ; Z_v = 0.171119 ; Z_m = 0.692649 \\ Z_a &= 0.542134E+00 ; Z_w = 0.171119 \end{aligned}$$

ASME Section VIII-2 Part 4.18.8.4 Step 4:

Diameter ratio [K]:

$$\begin{aligned} &= A / D_o \\ &= 1170 / 750.36 \\ &= 1.5593 \end{aligned}$$

Compute coefficient [F]:

$$\begin{aligned} &= (1 - \nu^*) / E^* \cdot (\lambda_s + \lambda_c + E \cdot \ln(K)) \\ &= (1 - 0.3964) / 42221620 \cdot (14998580 + 0 + 194911088 \cdot \ln(1.5593)) \\ &= 1.4521 \end{aligned}$$

Compute parameter [ϕ]:

$$\begin{aligned} &= (1 + \nu^*) F \\ &= (1 + 0.3964) 1.4521 \\ &= 2.0277 \end{aligned}$$

Compute parameter [Q1]:

$$\begin{aligned} &= (\psi_s - 1 - \phi \cdot Z_v) / (1 + \phi \cdot Z_m) \\ &= (1.0262 - 1 - 2.0277 \cdot 0.1711) / (1 + 2.0277 \cdot 0.6926) \\ &= -0.133419 \end{aligned}$$

Compute Parameter [Qz1]:

$$\begin{aligned} &= (Z_d + Q1 \cdot Z_w) / 2 \cdot X_a^4 \\ &= (0.4848 + -0.1334 \cdot 0.1711) / 2 \cdot 1.4666^4 \\ &= 1.0686 \end{aligned}$$

Compute parameter [Qz2]:

$$\begin{aligned} &= (Z_v + Q1 \cdot Z_m) / 2 \cdot X_a^4 \\ &= (0.1711 + -0.1334 \cdot 0.6926) / 2 \cdot 1.4666^4 \\ &= 0.1821 \end{aligned}$$

Compute parameter [U]:

$$\begin{aligned} &= (Z_w + (\psi_s - 1) Z_m) X_a^4 / (1 + \phi \cdot Z_m) \\ &= (0.1711 + (1.0262 - 1) 0.6926) 1.4666^4 / (1 + 2.0277 \cdot 0.6926) \\ &= 0.3641 \end{aligned}$$

ASME Section VIII-2 Part 4.18.8.4 Step 5:

Determine factor [γ_b]:

$$\begin{aligned} &= (G_c - C) / D_o \text{ (config b)} \\ &= (815.16 - 950) / 750.36 = -0.1797 \end{aligned}$$

Compute Parameter [γ]:

$$= 0.000 \text{ mm (For Pressure only cases)}$$

Calculate parameter [ω_s]:

$$\begin{aligned} &= \psi_s \cdot k_s \cdot \beta_s \cdot \delta S (1 + h \cdot \beta_s) \\ &= 1.0262 \cdot 1711682688 \cdot 0.01735 \cdot 0.045989152 (1 + 160 \cdot 0.01735) \\ &= 5293.1792 \text{ mm}^2 \end{aligned}$$

Calculate parameter [ω^*s]:

$$\begin{aligned} &= a o^2 (\psi_s^2 - 1) \cdot (\psi_s - 1) / 4 - \omega_s \\ &= 375.18^2 (1.0262^2 - 1) \cdot (1.0262 - 1) / 4 - 5293.2 \\ &= -5244.3223 \text{ mm}^2 \end{aligned}$$

Parameter [ω_c]:



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$$= \rho c \cdot k c [\delta C (C m \theta + \beta C + C q \Delta \cdot h \cdot \beta C^2) - \zeta c (C m \theta + h \cdot C q \theta \cdot h \cdot \beta C / 2)]$$

$$= 1.0864 \cdot 0 [0.000 (0 + 0 \cdot 0 \cdot 160) - 0.000 (0/160 + 0 \cdot 0/2)]$$

$$= 0.0000 \text{ mm}^2$$

Parameter [$\omega \cdot c$]:

$$= a o^2 [(\rho c^2 + 1) \cdot (\rho c - 1) / 4 - (\rho s - 1) / 2] - \omega c$$

$$= 375.18^2 [(1.0864^2 + 1) \cdot (1.0864 - 1) / 4 - (1.0262 - 1) / 2] - 0$$

$$= 4782.8521 \text{ mm}^2$$

Pressure [$P \cdot S$]:

= 0 For Pressure only cases or Configurations d,e,f,A,B,C,D

Pressure [$P \cdot C$]:

= 0 For Pressure only cases or Configurations b,c,d,B,C,D

ASME Section VIII-2 Part 4.18.8.4 Step 6:

Pressure [$P's$]:

$$= P_s \{ x_s + 2(1 - x_s) v_t + [2/Kst \cdot (Ds/Do)^2] v_s - [(\rho s^2 - 1) / (J \cdot Kst)] - [(1 - J) / (2J \cdot Kst)] [(D_j^2 - (Ds)^2) / Do^2] \}$$

$$= 2250 \{ 0.6769 + 2(1 - 0.6769) 0.3 + [2/0.792 \cdot (770/750.36)^2] 0.3 - [(1.0262^2 - 1) / (1 \cdot 0.792)] - [(1 - 1) / (2 \cdot 1 \cdot 0.792)] [(0^2 - (770)^2) / 750.36^2] \}$$

$$= 3603.3955 \text{ KPa.}$$

Pressure [$P't$]:

$$= [x_t + 2(1 - x_t) v_t + 1 / (J \cdot Kst)] P_t$$

$$= [0.7753 + 2(1 - 0.7753) 0.3 + 1 / (1 \cdot 0.792)] 5850$$

$$= 12710.3076 \text{ KPa.}$$

Pressure [P_y]:

$$= N_t \cdot K_t \cdot \gamma / (\pi \cdot a o^2)$$

$$= 282 \cdot 319.24 \cdot 0 / (3.1416 \cdot 375.18^2)$$

$$= 0.0000 \text{ KPa.}$$

Pressure [P_w]:

$$= -\gamma b \cdot U \cdot W \cdot / (2 \cdot \pi \cdot a o^2)$$

$$= -0.1797 \cdot 0.3641 \cdot 572319 / (2 \cdot \pi \cdot 375.18^2)$$

$$= 415.2552 \text{ KPa.}$$

Calculate the pressure [P_{Prim}]:

$$= - (U / a o^2) \cdot (\omega \cdot s \cdot P_s - \omega \cdot c \cdot P_t)$$

$$= - (0.3641 / 375.18^2) \cdot (-5244.3 \cdot 2250 - 4782.9 \cdot 5850)$$

$$= 102.9076 \text{ KPa.}$$

Calculate the pressure [P_w]:

$$= (U / a o^2) \cdot (\omega \cdot s \cdot P_s - \omega \cdot c \cdot P_t)$$

$$= (0.3641 / 375.18^2) \cdot (5293.2 \cdot 0 - 0 \cdot 0)$$

$$= 0.0000 \text{ KPa.}$$

Determine the effective pressure [P_e]:

$$= J \cdot Kst / (1 + J \cdot Kst (Qz1 + (\rho s - 1) Qz2)) \cdot (P's - P't + P_y + P_w + P_w + P_{\text{Prim}})$$

$$= 0.1000E+01 \cdot 0.792 / (1 + 1 \cdot 0.792 (1.0686 + (1.0262 - 1) 0.1821)) \cdot (3603.4 - 12710 + 0 + 0 + 415.26 + 102.91)$$

$$= -3676.7512 \text{ KPa.}$$

ASME Section VIII-2 Part 4.18.8.4 Step 7:

Determine factor [Q_2]:

$$= [((\omega \cdot s \cdot P_s - \omega \cdot c \cdot P_t) - (\omega \cdot s \cdot P_s - \omega \cdot c \cdot P_t)) + W \cdot \gamma b / (2 \cdot \pi)] / (1 + \phi \cdot Z_m)$$

$$= [((-5244.3 \cdot 2250 - 4782.9 \cdot 5850) -$$



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$$\begin{aligned} & (5293.2 \cdot 0 - 0 \cdot 0)) \cdot 0.000102 + \\ & 572319 \cdot -0.1797 / (2 \cdot \pi) / (1 + 2.0277 \cdot 0.6926) \\ & = -8494.6953 \text{ Kgf} \end{aligned}$$

Calculate factor [Q3]:

$$\begin{aligned} & = Q1 + 2 \cdot Q2 / (Pe \cdot ao^2) \\ & = -0.1334 + 2 \cdot -8494.7 / (-3676.8 \cdot 375.18^2) \\ & = 0.18850 \end{aligned}$$

Fm Value from Table 4.18.3 = 0.285592

Tubesheet Bending Stress - original thickness [σ]:

$$\begin{aligned} & = (1.5 \cdot Fm / \mu \cdot) \cdot (2 \cdot ao / (H - h'g))^2 \cdot Pe \\ & = (1.5 \cdot 0.2856 / 0.25) \cdot (2 \cdot 375.18 / (160 - 6))^2 \cdot -3676.8 \\ & = -149.5786 \text{ N./mm}^2 \end{aligned}$$

Allowable Tubesheet Bending Stress [σ allowed]:

$$= 1.5 \cdot S = 1.5 \cdot 138 = 207 \text{ N./mm}^2$$

Tubesheet Bending Stress - final thickness [σ_f]:

$$\begin{aligned} & = (1.5 \cdot Fm / \mu \cdot) \cdot (2 \cdot ao / (h - h'g))^2 \cdot Pe \\ & = (1.5 \cdot 0.275 / 0.25) \cdot (2 \cdot 375.18 / (130.11 - 6))^2 \cdot -3431.5 \\ & = -206.9926 \text{ N./mm}^2 \end{aligned}$$

Reqd Tubesheet Thickness, for Bending Stress (including CA) [HReqB]:

$$\begin{aligned} & = h + Cats + Catc \\ & = 130.11 + 0 + 0 \\ & = 130.1140 \text{ mm} \end{aligned}$$

ASME Section VIII-2 Part 4.18.8.4 Step 8:

Shear Stress check [τ limit]:

$$\begin{aligned} & = (2 \cdot \mu \cdot h / ao) \cdot \min(0.8S; 0.533Sy) \\ & = (2 \cdot 0.25 \cdot 160 / 375.18) \cdot 110.4 \\ & = 23.5410 \text{ N./mm}^2 \end{aligned}$$

Shear Stress is not required to be computed; [Pe] <= Tau limit

Tubesheet Shear Stress is probably low, use the following req. thk:

$$\begin{aligned} \text{Tubesheet thickness (Incl. Corr.)} &= 10.5411 \text{ mm} \\ \text{Tubesheet Shear Stress} &= 17.8009 \text{ N./mm}^2 \end{aligned}$$

Reqd Tubesheet Thickness for Given Loadings (including CA) [Hreqd] :

$$\begin{aligned} & = \max(HreqB; HreqS) \\ & = \max(130.11 ; 10.541) \\ & = 130.1140 \text{ mm} \end{aligned}$$

ASME Section VIII-2 Part 4.18.8.4 Step 9:

The F_{tmin} and F_{tmax} Coefficients from Table 4.18.4:

$$F_{tmin} = 0.7833, F_{tmax} = 1.196$$

First Extreme Tube Axial Stress from among all the tubes [σ_{t1}]:

$$\begin{aligned} & = ((Ps \cdot xs - Pt \cdot xt) - Pe \cdot F_{tmin}) / (Xt - Xs) \\ & = ((2250 \cdot 0.6769 - 5850 \cdot 0.7753) - (-3676.8) \cdot 0.7833) / \\ & \quad (0.7753 - 0.6769) \\ & = -1.3488 \text{ N./mm}^2 \end{aligned}$$

Second Extreme value of Tube Axial Stress from among all the tubes [σ_{t2}]:

$$\begin{aligned} & = ((Ps \cdot xs - Pt \cdot xt) - Pe \cdot F_{tmax}) / (Xt - Xs) \\ & = ((2250 \cdot 0.6769 - 5850 \cdot 0.7753) - (-3676.8) \cdot 1.196) / \\ & \quad (0.7753 - 0.6769) \\ & = 14.0677 \text{ N./mm}^2 \end{aligned}$$



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ASME TS Calc:

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Maximum Tube Axial Stress [$\sigma_t, \max$]:
 $= \max(|\sigma_{t,1}|; |\sigma_{t,2}|) = 14.068 \text{ N./mm}^2$

Allowable Tube Stress [σ_t, A]:
 $= S_{ot} = 117.9979 \text{ N./mm}^2$

Check for Buckling as some of the tubes are in compression.

Determine the Factor of Safety [F_s]:
 $= \max((3.25 - 0.25(Z_d + Q_3 \cdot Z_w)X_a^4); 1.25)$
 $= \max((3.25 - 0.25(0.4848 + 0.1885 \cdot 0.1711)1.4666^4); 1.25)$
 $= 2.0000 \text{ (Should be } \leq 2)$

Determine the factor [r_t]:
 $= (\sqrt{d_t^2 + (d_t - 2 \cdot t_t)^2})/4$
 $= (\sqrt{25.4^2 + (25.4 - 2 \cdot 2.11)^2})/4$
 $= 8.2680 \text{ mm}$

Determine the factor [C_t]:
 $= \sqrt{(2 \cdot \pi^2 \cdot E_t / S_{yt})}$
 $= \sqrt{(2 \cdot \pi^2 \cdot 196289984 / 216.4)}$
 $= 133.8137$

Determine the factor [F_t]:
 $= k \cdot L/r$
 $= 1 \cdot 220 / 8.268$
 $= 26.6087$

Buckling Allowable Stress [S_{tb}]:
 $= (S_{y,t} / F_s) \cdot (1 - F_t / (2 \cdot C_t))$
 $= (216.4 / 2) \cdot (1 - 26.609 / (2 \cdot 133.81))$
 $= 97.441 \text{ N./mm}^2 \text{ (Never greater than } S_{ot})$

Note: The Axial Compressive stress in tubes is within limits.

Largest tube-to-tubesheet Joint Load [W_t]:
 $= \sigma_{t, \max} \cdot \text{Tube Area}$
 $= 14.068 \cdot 1.5438$
 $= 221.46 \text{ Kg}$

Tube Metal Area [A_t]:
 $= \pi(d_o - t)t$
 $= \pi(25.4 - 2.11)2.11$
 $= 1.544 \text{ cm}^2$

Tube Weld Size Results per UW-20.3:

Tube Strength [F_t]:
 $= \pi \cdot t(d_o - t)S$
 $= \pi \cdot 2.11(25.4 - 2.11)118$
 $= 1857.579 \text{ Kg}$

Groove Weld Strength [F_g]:
 $= 0.85 \cdot \pi \cdot a_g(d_o + 0.67 \cdot a_g)S_w \text{ (but not } > F_t)$
 $= 0.85 \cdot \pi \cdot 2.11(25.4 + 0.67 \cdot 2.11)118$
 $= 1817.8311 \text{ Kg}$

Weld Strength Factor [f_w]:
 $= S / S_w \text{ where } S_w = \min(S \text{ and } S_t)$
 $= 118 / 118$
 $= 1.000$

Design Strength Ratio [f_d]:
 $= 1 \text{ for full strength welds}$



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ASME TS Calc:

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Fillet Weld Strength [Ff]:

$$= 0.55 \cdot \pi \cdot af (do + 0.67 \cdot af) Sw \text{ (but not } > Ft)$$

$$= 0.55 \cdot \pi \cdot 2.11 (25.4 + 0.67 \cdot 2.11) 118$$

$$= 1176.2437 \text{ Kgf}$$

Minimum Length of Weld [ar]:

$$= 2 \left(\sqrt{(0.75 \cdot do)^2 + 1.07 \cdot t \cdot (do - t) f_w \cdot f_d} - 0.75 \cdot do \right)$$

$$= 2 \left(\sqrt{(0.75 \cdot 25.4)^2 + 1.07 \cdot 2.11 \cdot (25.4 - 2.11) 1 \cdot 1} - 0.75 \cdot 25.4 \right)$$

$$= 2.667 \text{ mm}$$

Minimum Required Fillet Weld Leg [afr]: 1.3334 mm

Minimum Required Groove Weld Leg [agr]: 1.3334 mm

Tube-Tubesheet Jt allowable, 1857.6 is $\geq$ tube strength 1857.6 Kgf

Note: This tube-tubesheet joint is a Full Strength joint

ASME Section VIII-2 Part 4.18.8.4 Step 10:

Shell Axial Membrane Allowable Stress:

$$= S_s \cdot E_{sw} = 138 \cdot 1 = 138 \text{ N./mm}^2$$

Axial Membrane Stress in Shell [σ_s, m]:

$$= ao^2 / (Ds + ts) ts \cdot [Pe + (ps^2 - 1)(Ps - Pt)] + as^2 \cdot Pt / (Ds + ts) ts$$

$$= 375.18^2 / ((770 + 14) 14) \cdot [-3676.8 + (1.0262^2 - 1)(2250 - 5850)] + 385^2 \cdot 5850 / ((770 + 14) 14)$$

$$= 29.4025 \text{ N./mm}^2$$

ASME Section VIII-2 Part 4.18.8.4 Step 11:

Note:

For a given Shell thickness of 14 mm, the minimum Shell length adjacent to the tubesheet should be 186.888 mm

Shell Membrane Stress due to Joint Interaction [σ_s, m]:

$$= ao^2 / ((Ds + ts) ts) [Pe + (ps^2 - 1)(Ps - Pt)] + as^2 \cdot Pt / ((Ds + ts) ts)$$

$$= 375.18^2 / ((770 + 14) 14) [-3676.8 + (1.0262^2 - 1)(2250 - 5850)] + 385^2 \cdot 5850 / ((770 + 14) 14)$$

$$= 29.4025 \text{ N./mm}^2$$

Shell Bending Stress due to Joint Interaction [σ_s, b]:

$$= 6 \cdot ks / ts^2 \{ \beta_s [\delta_s \cdot Ps + as^2 \cdot P \cdot s / (Es \cdot ts)] + 6(1 - \nu^2) / (E \cdot h) (ao/h)^3 (1 + h \cdot \beta_s / 2) [Pe (Zv + Zm \cdot Q1) + 2/ao^2 \cdot Zm \cdot Q2] \}$$

$$= 6 \cdot 1711682688 / 14^2 \{ 0.01735 [0.045989152 \cdot 2250 + 385^2 \cdot 0 / (2.75E+09)] + 6(1 - 0.3964^2) / (42221620) (375.18/160)^3 (1 + 160 \cdot 0.01735/2) [-3676.8 (0.1711 + 0.6926 \cdot -0.1334) + 2/375.18^2 \cdot 0.6926 \cdot -8494.7] \}$$

$$= -120.2583 \text{ N./mm}^2$$

Shell Stress Summation vs. Allowable

$$\therefore |\sigma_s, m| + |\sigma_s, b| \leq 1.5 \cdot S_s$$

$$\therefore |29.402| + |-120.26| \leq 207 \text{ N./mm}^2$$

$$\therefore 149.66, \text{ must be } \leq 207 \text{ N./mm}^2$$

Computations completed for ASME tubesheet configuration: b

Stress/Force Summary for Loadcase D3 un-corr. (Psd,max + Ptd,max):

Stress Description	Actual	Allowable	Pass/Fail
Tubesheet Bend. Stress	149.6 <=	207.0 N./mm ²	Pass
Tubesheet Shear Stress	11.2 <=	110.4 N./mm ²	Pass



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ASME TS Calc:

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Maximum Tube Stress	14.1	<=	118.0 N./mm ²	Pass
Minimum Tube Stress (Buckling)	-1.3	<=	-97.4 N./mm ²	Pass
Maximum Force on any one Tube	221.5	<=	1857.6 Kgf	Pass
Axial Membrane Stress in Shell	29.4	<=	138.0 N./mm ²	Pass
Shell Stress (jt. inter.)	149.7	<=	207.0 N./mm ²	Pass

Thickness Results for Loadcase D3 un-corr. (Psd,max + Ptd,max):

Thickness (mm)	Required	Actual	P/F
Tubesheet Thickness :	130.114	160.000	Ok
Tube-Tubesheet Fillet Weld Leg :	1.333	2.110	Ok
Tube-Tubesheet Groove Weld Leg :	1.333	2.110	Ok

Fixed Tubesheet results per ASME Div. 2 Part 4.18.8 2025

Results for 16 Load Cases:

Case#	--Reqd. Thk. + CA Tbsht Extnsn	----	Tubesheet Bend Allwd	Stresses Shear Allwd	Case Type	Pass/ Fail
D1uc	145.562	79.948	177	207	16	110 Ps+Pt-Th D1 Ok
D2uc	39.897	...	26	207	5	110 Ps+Pt-Th D2 Ok
D3uc	130.114	...	150	207	11	110 Ps+Pt-Th D3 Ok
D4uc	9.930	...	1	207	...	110 Ps+Pt-Th D4 Ok
O1uc	60.000	...	163	444	16	110 Ps+Pt+Th O1 Ok
O2uc	67.439	...	82	444	5	110 Ps+Pt+Th O2 Ok
O3uc	59.907	...	139	444	11	110 Ps+Pt+Th O3 Ok
O4uc	61.584	...	64	444	1	110 Ps+Pt+Th O4 Ok
D1c	150.496	79.948	186	207	17	110 Ps+Pt-Th-c D1 Ok
D2c	37.230	...	27	207	5	110 Ps+Pt-Th-c D2 Ok
D3c	135.319	...	158	207	12	110 Ps+Pt-Th-c D3 Ok
D4c	11.935	...	1	207	...	110 Ps+Pt-Th-c D4 Ok
O1c	72.103	...	178	444	17	110 Ps+Pt+Th-c O1 Ok
O2c	75.312	...	89	444	5	110 Ps+Pt+Th-c O2 Ok
O3c	70.608	...	152	444	12	110 Ps+Pt+Th-c O3 Ok
O4c	70.546	...	71	444	1	110 Ps+Pt+Th-c O4 Ok
Max:	150.4962	79.948 mm	0.900	0.153	(Str. Ratio)	

Load Case Definitions:

[Ps & Pt]:

Shell-side and Tube-side Design or Operating Pressures derived from Psd,min Ptd,max, Psdx,min, Ptox,max etc. per the Load Case Tables

[(+)-Th]:

With or Without Thermal Expansion, Tt,mx & Ts,mx

[c]:

With or Without Corrosion Allowance

[D1, D2, D3]:

Design Load Cases using the Maximum and Minimum Design Pressures

[D4]:

Design Load Case using the Minimum (Vacuum) Pressures (if specified)

[O1, O2, O3, O4]:

Operating Load Cases using the Maximum and Minimum Operating Pressures and Operating Temperatures



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Shell Axial Membrane Stress Summary:

---- Shell Stresses ----					:	---- Shell Band Stress ----					:	Pass
Case#	Ten	Allwd	Cmp	Allwd	:	Ten	Allwd	Cmp	Allwd	:		Fail
D1uc	7	137	...	...		...	...	...	...			Ok
D2uc	21	137	...	...		...	...	...	...			Ok
D3uc	29	137	...	...		...	...	...	...			Ok
D4uc	1	137	-1	-117		...	...	...	...			Ok
O1uc	8	468	...	...		...	...	...	...			Ok
O2uc	24	468	...	...		...	...	...	...			Ok
O3uc	29	468	...	...		...	...	...	...			Ok
O4uc	2	468	...	...		...	...	...	...			Ok
D1c	8	137	...	...		...	...	...	...			Ok
D2c	27	137	...	...		...	...	...	...			Ok
D3c	37	137	...	...		...	...	...	...			Ok
D4c	1	137	-1	-114		...	...	...	...			Ok
O1c	10	468	...	...		...	...	...	...			Ok
O2c	30	468	...	...		...	...	...	...			Ok
O3c	37	468	...	...		...	...	...	...			Ok
O4c	3	468	...	...		...	...	...	...			Ok
Max RATIO												
	0.266		0.011			...			...			

Tube, Shell and Channel Stress Summary:

---- Tube Stresses ----					Tube Loads		Shell Stress		Channel Stress		Pass
Case#	Ten	Allwd	Cmp	Allwd	Ld	Allwd	Stress	Allwd	Stress	Allwd	Fail
D1uc	15	117	-2	-97	240	1857	241	468	...	...	Ok
D2uc	1	117	-1	-97	18	1857	130	206	...	...	Ok
D3uc	14	117	-1	-97	221	1857	150	206	...	...	Ok
D4uc	...	117	...	-97	1	1857	6	206	...	...	Ok
O1uc	15	235	-2	-97	239	3715	237	468	...	...	Ok
O2uc	5	235	-5	-97	72	3715	40	468	...	...	Ok
O3uc	14	235	-1	-97	221	3715	150	468	...	...	Ok
O4uc	5	235	-5	-97	85	3715	127	468	...	...	Ok
D1c	17	117	-3	-97	262	1857	259	468	...	...	Ok
D2c	1	117	-1	-97	24	1857	165	206	...	...	Ok
D3c	16	117	-2	-97	249	1857	143	206	...	...	Ok
D4c	...	117	...	-97	1	1857	8	206	...	...	Ok
O1c	17	235	-3	-97	261	3715	254	468	...	...	Ok
O2c	4	235	-4	-97	70	3715	35	468	...	...	Ok
O3c	16	235	-2	-97	249	3715	143	468	...	...	Ok
O4c	6	235	-6	-97	94	3715	135	468	...	...	Ok
Max RATIO											
	0.141		0.061		0.141		0.797				...

Summary of Thickness Comparisons for 16 Load Cases:

Thickness (mm)	Required	Actual	P/F
Tubesheet thickness :	150.496	160.000	Ok
Tubesheet thickness flanged extension :	79.948	128.000	Ok
Tube thickness :	0.631	2.110	Ok
Tube-tubesheet fillet weld leg :	1.333	2.110	Ok
Tube-tubesheet groove weld leg :	1.333	2.110	Ok

Min Shell length of thk, (14.000) adj. to tubesheet: 186.888 mm

Note: This is a full strength Tube to Tubesheet Joint.

Summary of Axial Differential Expansion between Shell and Tubes :

Due to Thermal Expansion Shell Extends by : +.000 mm
 Due to Pressure Shell Extends by : +.188 mm



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Due to Pressure + Thermal Shell Extends by : +.188 mm

Tubesheet MAWP used to Compute Hydrotest Pressure:

Stress / Force Condition	Tubeside MAWP	Ø shellside Stress Rat.	Shellside MAWP	Ø tubeside Stress Rat.
Tubesheet Bending Stress	7352.33	0.999	17211.47	1.000
Tubesheet Shear Stress	37672.34	1.000	51729.59	1.000
Tube Tensile Stress	43630.70	1.000	177110.28	1.000
Tube Compressive Stress	439709.78	1.000	192995.44	1.000
Tube-Tubesheet Joint load	43630.67	1.000	177102.11	1.000
Shell Stress (Axial, Junction)	7352.33	0.999	6397.10	1.000
Tube Pressure Stress	20998.73	1.000	13593.03	1.000
Tubesheet Extension Stress	8705.57	...	No Calc	No Calc
Minimum MAWP	7352.33		6397.10	

Tubesheet MAPnc used to Compute Hydrotest Pressure:

Stress / Force Condition	Tubeside MAPnc	Ø shellside Stress Rat.	Shellside MAPnc	Ø tubeside Stress Rat.
Tubesheet Bending Stress	7755.80	1.000	18027.96	1.000
Tubesheet Shear Stress	40013.38	1.000	52340.72	1.000
Tube Tensile Stress	47329.00	1.000	233728.59	1.000
Tube Compressive Stress	1525708.12	1.000	213199.16	1.000
Tube-Tubesheet Joint load	47328.91	1.000	233728.11	1.000
Shell Stress (Axial, Junction)	7755.80	0.497	9007.03	1.000
Tube Pressure Stress	20998.73	1.000	13593.03	1.000
Tubesheet Extension Stress	8705.57	...	No Calc	No Calc
Minimum MAPnc	7755.80		9007.03	

(*) Only the design load cases were analyzed to compute the MAWP for determining the test pressure.

Summary of Exchanger Pressure and Thermal Displacements: (mm)

Load Case#	Due to Pressure		Due to Thermal Exp.		Computed Delta Ms	
	UnCorroded	Corroded	UnCorroded	Corroded	UnCorroded	Corroded
1	0.1883	0.1884	...	...	...	...
2	-0.1638	-0.1639	...	...	...	...
3	0.0169	0.0169	...	...	...	...
4	0.0075	0.0075	...	...	...	...
5	0.1807	0.1808	...	...	...	...
6	-0.1638	-0.1639	...	...	...	...
7	0.0169	0.0169	...	...	...	...
8	...	...	...	...	...	...

Tubesheet MDMT Calculations:

Shell Side MDMT calculation:

Governing thickness on the shell side per figure UCS-66.3 (e):
= max(tubesheet thk/4, min(tubesheet thk, shell thickness))
= max(160/4, min(160, 14))
= 40.000 mm

Thickness Ratio = 0.7543, Temperature Reduction per Fig. UCS 66.1 = 14 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	11 °C
Min Metal Temp per UCS-66 and UCS-68(c), PWHT credit	-5 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-19 °C



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ASME TS Calc:

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Channel Side MDMT calculation:

Governing thickness for the channel side:

= tubesheet thickness/4

= 160/4

= 40.000 mm

Note:

This Material was specified as being an Impact Tested (Low Temperature) Material.

Impact Test Temperature provided per Specification -18 °C

where the MDMT reduction ratio per UCS 66 (b)(1)(b) is:

= max(pt/Tubeside MAPnc, ps/Shellside MAPnc), must be ≤ 1

= max(5850/7755.8, 2250/9007)

= 0.754

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PV Elite® Pressure Vessel Design Notes

Project Information	
Project Number	
Client	
Date	
Designer	
Design Code	
Additional Standards	

Design Conditions		
Internal Design Pressure:		
External Design Pressure:		
Design Int. Temperature:		
Design Ext. Temperature:		
Corrosion Allowance:		

Vessel Geometry		
Vessel Type		
Diameter		
Tangent-to-Tangent Length		
Head Type		
Number of Segments		

Material Specifications	
Shell:	
Head:	
Nozzle:	

Nozzle/Opening Details	ID/OD	Size	Rating	Matl.	Reinf.

Miscellaneous		
Support Type		
Number of Supports		
Support Material		
Seismic Code		
Wind Code		
Insulation		
Pressure Test		
Radiography		
Fatigue Analysis		

Analysis Notes: